
PDielec Documentation

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INTRODUCTION

The Python package, PDielec, calculates the infrared absorption characteristics of a crystalline material supported in a non-absorbing medium and the absorption of single crystals (thin films or thick slabs) of absorbing material at different orientations. The package also calculates powder and crystal Raman spectra from Raman tensors, with polarisation, collection geometry and optional NAC/EO corrections. See *Theory for Powder Raman*, *Theory for Crystal Layer Raman* and the *Version 10 migration* guide.

PDielec post-processes the output of solid-state quantum mechanical and molecular mechanics (MM) based codes such as VASP [1], CASTEP [2], CRYSTAL [3], Abinit [4], FHI-Aims [5], Quantum Espresso [6], Phonopy [7], and GULP [8]. The infrared absorption of crystalline insulator materials whose crystal size is small compared with the wavelength of the absorbing radiation is calculated using effective medium theory. The package is suited for the calculation of the complex, frequency-dependent permittivity and its associated absorption of infrared radiation for a finely ground crystalline material dispersed in a low-loss dielectric medium such as KBr or Polytetrafluoroethylene (PTFE). A particular feature of the program is its ability to take into account the constant permittivity of the supporting medium and the particle shape of the material of interest through an effective medium theory.

Using an effective medium method, the package calculates the internal electric field arising from different particle morphologies and calculates the resulting shift in absorption frequency and intensity arising from the coupling between a phonon and the internal field. The theory of the approach is described, followed by a description of the implementation within PDielec. For the specific case of a low concentration of spherical particles, calculations based on Mie scattering allow the exploration of particle size effects. A section providing several examples of its application is given.

For spherical, isotropic particles, embedded in a supporting matrix the Mie method is available to calculate the absorption of infrared radiation. The method uses the routines of PyMieScatt by B. Sumlin [9, 10].

The attenuated total reflection (ATR) spectrum can be calculated for a powdered material next to a high refractive index material such as diamond or germanium.

The calculation of the optical behaviour of a single crystal (thick slab or film) is performed using the pyGTM code available on GitHub [11]. This code implements a generalised transfer matrix method described by Passler et al. [12] and builds on previous publications [13, 14]. The algorithms used by pyGTM have been extended to include the scattering matrix method for light transmission and reflectance [15]

An interface to the vibAnalysis package of Philippe Teixeira [16] is provided through the vibanalysis command. The background to the theory of the analysis is given in a paper [17]. The package performs Vibrational Mode Decomposition (VMD) using three computational approaches; Vibrational Mode Projection, Vibrational Mode Linear Decomposition, and Vibrational Mode Automatic Relevance Determination.

OTHER SOFTWARE COMPONENTS

2.1 preader

preader is a ‘helper’ program that uses the underlying modules of PDielec to read output files and summarise the results of several MM/QM packages. The program can be used to perform some straightforward calculations. For instance, projection of any remaining centre-of-mass motion of the crystal can be performed to make sure that there are three zero frequencies. Also, the masses used in the calculation of the dynamical matrix can be altered. Unlike PDGui it is not necessary to have performed a full calculation of the dynamical matrix. In the majority of cases, preader will read geometry optimisation runs.

2.1.1 Command options

Examples of data sets for these packages are included with the distribution and can be found in the Examples/Package/preader directory. The program is run from the command line. There are several command options and these are summarised below. Some options may be repeated.

- **-program program_name**

Program_name can be “abinit”, “aims”, “castep”, “crystal”, “gulp”, “qe”, “experiment”, “phonopy”, “vasp”, “finite_field”, or “auto” and specifies the program which generated the results to be analysed

- **-neutral**

Impose neutrality on the Born charge matrices

- **-spectroscopy infrared|raman**

Select IR intensities (default) or Raman activities; Raman requires tensor data.

- **-nocalculation**

No calculations are performed. This results in a single line of output with just information from the program. If -eckart, -mass, -masses, -neutral or -crystal have -hessian crystal have been specified they will be ignored

- **-eckart**

Translational modes are projected out of the hessian before diagonalisation

- **-masses average**

The element mass_definition can be either “program”, “average” or “isotopic”, meaning that the masses used in the calculation of the frequencies are either taken from the QM program or are the average of the isotope abundancies or are the most abundant isotope mass.

- **-mass element mass**

The atomic mass of the element is set to mass. This can be used to explore the effect of isotope substitution on the calculated frequencies

2.1.2 Examples

```
preader -program vasp `find . -name OUTCAR` > results.csv
```

This reads all the VASP OUTCAR files in the current and any of its subdirectories and summarises the results to results.csv.

```
preader -program castep -eckart `find . -name \*.castep` > results.csv
```

This reads all the CASTEP output files in the current and any of its subdirectories and summarises the results to results.csv. For each file, the centre-of-mass motion of the crystal is projected. The results file contains both the unprojected and the projected results.

There are examples of preader being used in the Examples/'Package'/preader subdirectories of the distribution of PDielec.

2.2 aims-pdielec-born

To generate the Born charges from an FHI-Aims calculation of infrared vibrational frequencies and intensities, the aims-pdielec-born command reads the single point FHI-Aims calculations of forces and polarisation at the displaced geometries created by the get_vibrations.py script from the FHI-Aims distribution. The aims-pdielec-born writes a single file BORN_PDIELEC containing the Born charges for all atoms in the unit cell.

In the example given below the FHI-Aims command `python get_vibrations.py na2so42` was used to generate the perturbations.

```
aims-pdielec-born na2so42
```

The command line options are:

Command line option	Description
name	“name” is the string used in the get_vibrations.py command to identify the set of displacements.
-d delta -delta delta	Define the step used to generate the perturbed geometries the default delta is 0.0025
-p dir -permittivity dir	Set the directory name holding the calculation of permittivity the default is “Dielectric”
-h -help	Show some help information

2.3 phonopy-pdielec-born

The BORN file calculated by Phonopy holds the Born charges for the symmetry unique atoms in the unit-cell. To allow PDielec to use these Born charges the phonopy-pdielec-born command reads the phonopy.yaml and BORN files and writes out the full set of charges to the terminal. PDGui expects the Born charges to be in the file BORN_PDIELEC. An example of the use of the script is given below.

```
phonopy-pdielec-born BORN_PDIELEC
```

The command line options are:

Command line option	Description
name	Output the charges to a file called name
-input name -input name	Set the name of the Phonopy Born charge file the default is "BORN"
-output name -output name	Send the output to this file the default is to send the output to standard out
-yaml name -yaml name	Set the name of the phonopy.yaml file the default is "phonopy.yaml"
-symprec float -symprec float	Define the symmetry threshold the default 1.0E-5
-nosym	Do not use any symmetry
-h -help	Show some help information

2.4 vibanalysis

The vibanalysis command provides an interface the vibAnalysis package of Philippe Teixeira. This package provides Vibrational Mode Decomposition of the phonon modes using a variety of methods including: Vibrational Mode Projection (VMP), Vibrational Mode Linear Decomposition (VMLD) and Vibrational Mode Automatic Relevance Determination (VMARD). By default VMARD is used.

The package options can be determined using;

```
vibanalysis OUTCAR  
vibanalysis phonon.castep
```

which will analyse a VASP and a CASTEP phonon calculation output to produce a summary in the files OUTCAR.nma and phonon.nma respectively.

2.5 pdmake

Use `./pdmake` from the repository root, or the installed `pdmake` command. Running it without arguments prints help. Run named test suites from a checkout containing `Examples/` and `PDIELEC/Tests/`.

Table 2.1: Commands and options

Command or option	Behaviour
<code>tests</code> (or <code>test</code>)	Run the complete PDIELEC/Tests pytest suite first, including General, then the example regression suites.
<code>test-pytests</code>	Run the complete PDIELEC/Tests directory, equivalent to <code>python -m pytest PDIELEC/Tests</code> . New test directories are discovered automatically.
<code>test-pytest-powder_raman</code> , <code>test-pytest-crystal_raman</code> , <code>test-pytest-materials</code> , <code>test-pytest-calculator</code> , <code>test-pytest-unitcell</code> , <code>test-pytest-gtmcore</code> , <code>test-pytest-constants</code>	Run one configured pytest group.
<code>test-powder_ir</code> , <code>test-crystal_ir</code> , <code>test-powder_raman</code> , <code>test-crystal_raman</code> , <code>test-atr</code>	Run the corresponding example group.
<code>test-preader</code> , <code>test-p2cif</code> , <code>test-vibanalysis</code> , <code>test-pdgui</code>	Run the selected tool's example suite.
<code>benchmarks</code>	Run benchmark examples without reference comparison.
<code>pypi</code>	Remove existing build artifacts and build source and wheel archives using <code>python -m build</code> . Does not upload; unavailable on Windows.
<code>clean</code>	Remove generated regression outputs below the root directory. Review the working tree first; unavailable on Windows.
<code>install</code>	Legacy Unix script-copying installation. Prefer the package installation described in the installation guide.
<code>scripts DIRECTORY</code>	Choose the destination for the legacy install action (default <code>~/bin</code>).
<code>--root DIRECTORY</code>	Set the repository root used to locate tests and examples.
<code>--view command.pdmake</code>	Open a PDGui recipe interactively.
<code>--regenerate command.pdmake</code>	Replace reference outputs for the recipe; review differences first.
<code>--cpus N</code> , <code>--threads N</code> , <code>--threading</code>	Pass CPU count, thread count or threading selection to PDGui.
<code>--debug</code> , <code>--padding N</code>	Enable debugging or set the printed title width.
<code>--usesystem</code> , <code>-usesystem</code>	Use installed executables instead of local scripts.
<code>--directory</code> , <code>-directory</code>	Display directory names as test titles. This flag takes no value.

A recipe's first line is its title. Subsequent command lines invoke `pdgui`, `preader`, `p2cif` or `vibanalysis` with their arguments. Run recipes from the directory containing their input files; for example:

```
cd Examples/Powder_Raman/Castep
pdmake command.pdmake
pdmake --view command.pdmake
```

PDGui regression recipes load `script.py` and write `results.xlsx`. Use separate invocations for recipes and named suites when execution order matters.

Excel regression comparisons include the *Settings* worksheet, including Raman activity values and their unit headers, as well as the result sheets. A checked sheet present in only one of the reference and generated workbooks is reported as an error. Sheets absent from both are skipped, so infrared examples do not require Raman result sheets. *Main* and *Scenarios* remain excluded from the standard comparison. When running `checkexcel` directly, use `-settings` to include Settings or `-f` to include all three metadata sheets. Review reported differences before regenerating reference workbooks.

2.6 MM/QM Interfaces

The packages have interfaces to solid-state QM codes, VASP [1], CASTEP [2], CRYSTAL14 [3], Abinit [4], Quantum Espresso [6], FHI-Aims [5] and Phonopy [7]. In addition, an interface is available for GULP [8] which is a force field based solid-state code. Finally, an interface has been written to an ‘experiment’ file format which allows the preparation of a user-defined file specifying the permittivities and absorption frequencies. The origin of the dataset(s) used for processing is determined by a command line switch, `-program`. An outline of the interfaces to these codes is given here. The package used for the calculation is described by the `-program` option. In addition, a file name is given which contains the output to be processed by PDIELEC.

2.6.1 Raman input availability

Phonon data alone do not supply Raman tensors. The following examples show Raman-capable inputs; nonlinear susceptibility is additionally needed for an electro-optic correction. Availability depends on what was calculated and saved. The correction also needs the Born charges, optical permittivity, cell volume and compatible phonon modes. A Raman activity alone cannot replace a Raman tensor, and enabling the EO option cannot supply missing susceptibility data. Companion files must describe the same structure, atom order and Cartesian axes as the main calculation. See *Theory for Crystal Layer Raman* for the EO conventions.

Table 2.2: Raman inputs and companion data

Reader	Raman data	Nonlinear susceptibility
CASTEP	Mode derivatives or atomic polar tensors in output; matching phonon file.	Read when present in output.
CRYSTAL	TENS_RAMAN.DAT alongside the frequency output.	Susceptibility in output; optional CHI2.DAT takes precedence.
ABINIT	Raman response in output; see <code>Examples/Powder_Raman/AbInit</code> .	Read when present in output.
Quantum ESPRESSO	Phonon log and/or <code>tensors.xml</code> , with matching mode data.	Read from the log or XML when present.
VASP	Optional <code>Raman-Tensors.yaml</code> alongside the electronic/phonon output.	Use a complete finite-field JSON dataset for the EO workflow.
Finite field	JSON tensors or Cartesian dielectric derivatives with mode data.	Optional <code>chi2</code> in JSON; see the input-format reference below.

2.6.2 VASP

The name provided on the command line is an OUTCAR file. The OUTCAR is read by PDielec to determine the unit-cell, atomic masses, frequencies, normal modes, Born charge tensors and optical permittivity. The VASP run can be a DFPT or numerical calculation of the response. pdgui is able to parse the OUTCAR file for the information it needs, but it is recommended to use vasprun.xml as this file gives greater precision to the required variables. The repository includes VASP 5 and VASP 6 examples, including `Examples/Vasp/Vasp6`. Raman calculations can supply the optional companion file `Raman-Tensors.yaml`; see `Examples/Powder_Raman/Vasp`. This file contains mode Raman tensors and is required for Raman calculations through this reader; it does not supply the nonlinear susceptibility for EO. For a VASP finite-field calculation with EO, use the consolidated finite-field JSON reader with both Raman data and `chi2` in the documented internal units.

2.6.3 CASTEP

The name provided on the command line is the seedname for the calculation. The corresponding seedname.castep file in the current directory is read and processed to determine the unit-cell, atomic masses, optical permittivity and born charge tensors. The normal modes and their frequencies are determined from the seedname.phonon file. The CASTEP run needs to be a DFPT (phonon+efield) task. The repository includes CASTEP 24.1 and 25.12 Raman examples. The reader supports mode polarizability derivatives and the newer atomic polar tensors, together with nonlinear susceptibility when present in the output. Keep the matching `.castep` and `.phonon` files together. Raman tensor and nonlinear susceptibility tables must be included in the `.castep` output; no separate Raman or susceptibility companion file is needed.

2.6.4 CRYSTAL

The name on the command line is a file ending in `.out`, containing the output of a CRYSTAL run. The contents of this file alone are sufficient to provide the unit-cell, atomic masses, frequencies, normal modes and Born charge tensors. However, the number of significant figures for the normal modes is not sufficient for an accurate calculation and it is therefore recommended that the `HESSFREQ.DAT` and `BORN.DAT` files are also made available. If they are present in the directory containing the CRYSTAL output, it uses these files to calculate the Born charge tensors, frequencies and normal modes. The CRYSTAL calculation needs to be a frequency calculation (`FREQCALC`) with the infrared intensity (`INTENS`) selected. The default algorithm does not calculate the optical permittivity, so this needs to be provided on the command line. However, if the `CPHF` or `CPKS` algorithm is used for the frequency calculation, the optical permittivity is calculated and PDielec will automatically read it from the output file. By default CRYSTAL projects out the pure translational modes of the system before calculating the frequencies, this can also be done by the PDielec package. Small differences in the calculated frequencies between the CRYSTAL program and PDielec have been observed. These have been found to be due to a slightly different method for symmetrising the 2nd derivative matrix, because of this an optional directive “-hessian crystal” can be used to indicate that PDielec should use the same symmetrisation as CRYSTAL.

For Raman calculations, place `TENS_RAMAN.DAT` alongside the output file. This supplies the polarizability derivatives used to construct Raman tensors; printed Raman activities alone are insufficient.

For electro-optic corrections, the reader recognises the `FIRST HYPERPOLARIZABILITY (BETA) AND SECOND ELECTRIC SUSCEPTIBILITY (CHI(2))` table in the main output. Alternatively, place that table in an optional `CHI2.DAT` beside the output, for example when the nonlinear response was calculated separately. PDielec reads the `d(MKS)` column in pm/V and converts it using $\chi^{(2)} = 2d$. A successfully read `CHI2.DAT` overrides the susceptibility from the main output. It is redundant only when both sources contain the same tensor.

`Examples/Crystal_Raman/Crystal23` uses the susceptibility table embedded in `opt_raman.out`

without a CHI2.DAT companion. Its reference spreadsheet uses this embedded response for the EO correction. Examples/Powder_Raman/Crystal23 retains CHI2.DAT as an example of the optional companion format; its contents duplicate the embedded table, so it can be omitted without changing the susceptibility.

Versions of CRYSTAL which are known to be compatible with the package are CRYSTAL14, 17, 21 & 23 later versions may be compatible also.

2.6.5 ABINIT

The output file should come from a run containing three datasets. One to calculate the wavefunction at the optimised geometry, one to calculate the field perturbations and one to calculate the second derivatives. Examples of input files and output files are available with the distribution. The package should be compatible with AbInit 8 to 10. For Raman and EO, use an output containing the atomic susceptibility derivatives and the Non-linear optical susceptibility tensor dtable, as in Examples/Powder_Raman/AbInit/raman.abo. The reader constructs Raman tensors from these derivatives and the phonon modes. The example's DDB files and analysis.abo are used for the separate anaddb analysis; they are not automatically loaded as companions to raman.abo by PDielec.

2.6.6 QE

The output file is the dynamical matrix file, specified by “filedyn” in a run of the quantum espresso phonon package. Examples of input and output files are given in the PDielec distribution. It is assumed that this file ends with a “.dynG” extension. For Raman calculations, the reader supports tensors in the phonon log and companion tensors.xml. Keep matching dynamical-matrix and tensor files together; see Examples/Powder_Raman/QE for a complete dataset. Copy tensors.xml from the phonon calculation's _ph0/ph.save/ directory beside the input selected in PDielec. It is discovered automatically and its Raman data take precedence over the lower-precision log tables. For EO, the reader uses the log's Electro-optic tensor in cartesian axis block or the XML ELOP_TNS data, preferring the XML when present. A dynamical-matrix file alone does not supply these Raman and EO tensors; retain the tensor XML or the matching phonon log containing them. The latest version of QE for which the package has been tested is version 7.3.1. pdgui is capable of reading the output log file of pwscf and ph, but it is recommended to use the pwscf.xml and the .dynG file written by ph and pwscf.

2.6.7 FHI-Aims

The vibrational frequencies and intensities calculation uses FHI-Aims and the get_vibrations.py script which is distributed with it. After determining the optimised cell, the script is used to generate perturbed single point calculations of the forces and the polarization which are then used to provide data for numerical differentiation to form the hessian and the IR intensities. An example of the output from these calculations is provided in Examples/FHI-Aims/Na2SO42. In this example, PDGui requires the hessian.na2so42.dat, masses.na2so42.dat, geometry.in and the Dielectric/ directory. The Dielectric/ directory contains an FHI-Aims calculation of the electronic permittivity (ϵ_∞). More details of the calculations are provided in the example directory.

The any file with the “in” or “dat” extension will cause PDGui to assume the DFT package used to create the files in that directory was the FHI-Aims package. The current version of FHI-Aims which is supported is 250822.

2.6.8 PHONOPY

Phonopy calculates the dynamical matrix through numerical differentiation. It has interfaces to several programs. The second parameter for the `-program` directive is the PHONOPY interface that was used to calculate the forces. Typically these would be generated by performing;

```
phonopy -d -dim="1 1 1"
```

to calculate the displacements in a set of POSCAR-* files. After running VASP a single point VASP calculation for each displacement. The FORCE_SETS file can then be calculated using for example;

```
phonopy -f DISP-*/vasprun.xml
```

where the DISP-* directories are where the VASP calculation was performed. Finally, a dynamical is written out using;

```
phonopy -dim="1 1 1" -qpoints="0 0 0" -writedm
```

To calculate the infrared spectrum PDIELEC needs the Born charges for the atoms in the unit-cell and these can be calculated using the scripts provided by Phonopy. The Phonopy scripts (for instance `phonopy_vasp_born`) to write a file called BORN, which contains the symmetry unique Born charges. PDIELEC provides a script *phonopy-pdielec-born* which takes the BORN file and writes a file called BORN_PDIELEC containing the Born charges for all the atoms in the cell.

2.6.9 GULP

The name on the command line is a file ending in .gout, containing the output of a GULP run. The contents of this file alone are sufficient to provide the unit-cell, atomic masses, frequencies, normal modes, Born charge tensors and optical permittivity. Because GULP only writes out the Born charge matrices for the asymmetric unit, it is necessary to run a frequency calculation using P1 symmetry and a complete unit-cell. The keywords; `nosymm`, `phonon`, `intensity`, `eigen` and `cart` are recommended for the GULP calculation. In the case that no shells are used in the calculation the optical permittivity is not available in the output and it is necessary to provide it.

2.7 Experimental File Format

There is a way of reading in experimental or calculated permittivities and calculating the infrared optical behaviour of the material. This uses the *experimental* file format which by default is assumed to be associated with files with .exp extension. The available GUI features depend on the data supplied. Dielectric models support optical spectra; the additional vibrational blocks described below supply normal modes and Raman tensors for analysis, visualisation and Raman calculations. Select *Experiment* in the Main Tab or use `pdgui experiment material.exp`.

The general format of an experimental file is best shown by an example::

```
lattice
  5.027782
  0.866025403784439 -0.5000000000000000 0.0000000000000000
  0.0000000000000000 1.0000000000000000 0.0000000000000000
  0.0000000000000000 0.0000000000000000 1.097684415337773
species 2
  Si 0.0
  O 0.0
unitcell 9
  Si 0.0000000000000000 0.4772900000000000 0.3333333333333333
```

(continues on next page)

(continued from previous page)

```

Si 0.4772900000000000 0.0000000000000000 0.666666666666667
Si 0.5227100000000000 0.5227100000000000 0.0000000000000000
O 0.1607400000000000 0.7457030000000000 0.5373336700000000
O 0.5849630000000000 0.8392600000000000 0.8706670033333333
O 0.4150370000000000 0.2542970000000000 0.7959996633333333
O 0.7457030000000000 0.1607400000000000 0.4626663300000000
O 0.8392600000000000 0.5849630000000000 0.129332996666667
O 0.2542970000000000 0.4150370000000000 0.204000336666667

epsinf
2.296 0.0 0.0
0.0 2.296 0.0
0.0 0.0 2.334
# model taken from Winta, C. J., Wolf, M., & Paarmann, A. (2019) Physical
↪Review B, 99(14), 144308
fpsq
# 1.5K Data
# Omega(TO) Gamma(TO) Omega(LO) Gamma(LO) (<2 in Table is taken to be 1.0)
xx 6
# E Phonon modes
391.5 1.0 403.0 1.0
454.0 2.6 510.5 1.0
695.9 4.9 698.4 4.0
797.2 4.8 810.0 4.3
1063.7 6.1 1230.7 8.2
1157.2 6.2 1154.9 6.1
yy 6
391.5 1.0 403.0 1.0
454.0 2.6 510.5 1.0
695.9 4.9 698.4 4.0
797.2 4.8 810.0 4.3
1063.7 6.1 1230.7 8.2
1157.2 6.2 1154.9 6.1
zz 4
# A2 Phonon modes
360.7 1.0 384.8 1.0
497.9 3.1 553.6 2.8
773.7 5.4 789.9 6.3
1073.0 6.2 1238.7 12.4

```

The file starts with a definition of the lattice, the first number being a lattice constant followed by three lines specifying the a, b, and c directions of the unit-cell.

The next directive specifies the species and their masses, by default PDGui uses its own internal mass scheme, so unless otherwise required these masses will be overwritten.

The *unitcell* directive specifies the number of atoms in the unit-cell and their fractional coordinates.

The *epsinf* directive gives the values of the ϵ_{∞} tensor

Finally, a model for the specification of the frequency-dependent permittivity is given. In this case the four parameter semi-quantum model (FPSQ) is employed. Each diagonal component of the tensor is specified by the number of contributions and then each contribution provides the TO frequency and its γ

followed by the same for the LO frequency (all frequencies are in cm^{-1}). Full details of each permittivity model available are shown below.

2.7.1 Constant model (constant)

The constant model defines a frequency-independent permittivity. The data for such a model is shown below.:

```
constant
2.0+0.1j 0.0      0.0
0.0      2.0+0.1j 0.0
0.0      0.0      2.0+0.1j
```

This would specify an isotropic permittivity with some absorption.

2.7.2 FPSQ model (fpsq)

The FPSQ model defines a frequency-dependent permittivity using the Four Parameter Semi-Quantum model. An example of the data for such a model is shown below.:

```
fpsq
xx 6
# E Phonon modes
# Omega(TO) Gamma(TO) Omega(LO) Gamma(LO)
391.5      1.0      403.0      1.0
454.0      2.6      510.5      1.0
695.9      4.9      698.4      4.0
797.2      4.8      810.0      4.3
1063.7     6.1      1230.7     8.2
1157.2     6.2      1154.9     6.1
yy 6
# E Phonon modes
# Omega(TO) Gamma(TO) Omega(LO) Gamma(LO)
391.5      1.0      403.0      1.0
454.0      2.6      510.5      1.0
695.9      4.9      698.4      4.0
797.2      4.8      810.0      4.3
1063.7     6.1      1230.7     8.2
1157.2     6.2      1154.9     6.1
zz 4
# A2 Phonon modes
# Omega(TO) Gamma(TO) Omega(LO) Gamma(LO)
360.7      1.0      384.8      1.0
497.9      3.1      553.6      2.8
773.7      5.4      789.9      6.3
1073.0     6.2      1238.7     12.4
```

The model only allows for a diagonal permittivity tensor and each component of the tensor specified requires the number of terms in the expansion to the specified. Each component of the permittivity tensor is generated using the following formula;

$$\epsilon(\omega) = \epsilon_{\infty} \prod_j \frac{\Omega_{LO_j}^2 - \omega^2 - i\gamma_{LO_j}\omega}{\Omega_{TO_j}^2 - \omega^2 - i\gamma_{TO_j}\omega}$$

2.7.3 Drude-Lorentz model (drude-lorentz)

The drude-lorentz model defines a frequency-dependent permittivity. An example of the data for such a model describing MgO is shown below.:

```
drude-lorentz
xx 2
# T Phonon modes
# Frequency      Strength  Gamma
  413.7           1050.0   22.2
  652.2            90.0   64.6
yy 2
  413.7           1050.0   22.2
  652.2            90.0   64.6
zz 2
  413.7           1050.0   22.2
  652.2            90.0   64.6
```

The model only allows for a diagonal permittivity tensor and each component of the tensor specified requires the number of terms in the expansion to be specified. Each component of the permittivity tensor is generated using the following formula;

$$\epsilon(\omega) = \epsilon_{\infty} + \sum_j \frac{S_j^2}{\Omega_j^2 - \omega^2 - i\sigma_j\omega}$$

Where S_j is the strength of the phonon mode, Ω_j its frequency and σ_j the width of the mode.

2.7.4 Tabulated permittivity

The *interpolate* models read a frequency followed by real and imaginary parts of the permittivity. Frequencies are in cm^{-1} and should increase strictly; provide enough points for interpolation and cover the range to be plotted. Finish each block with `end` or `&end`. Commas or spaces separate values.

Table 2.3: Interpolation columns

Directive	Columns after the frequency
<code>interpolate</code>	Real and imaginary parts of an isotropic permittivity.
<code>interpolate_3</code>	Real/imaginary pairs for xx, yy and zz, in that order.
<code>interpolate_6</code>	Real/imaginary pairs for xx, yy, zz, xy, xz and yz, in that order. The off-diagonal entries are copied to their transposed positions.

For example, an isotropic table is written as:

```

interpolate
100.0  2.10  0.01
200.0  2.15  0.02
300.0  2.20  0.03
400.0  2.25  0.02
end

```

A tabulation supplies the permittivity itself, not an oscillator strength. The *constant* model instead takes three rows of three complex numbers, using Python’s *j* notation for the imaginary unit. The *fpsq* and *drude-lorentz* models above require three consecutive sections, *xx*, *yy* and *zz*, including a count for each. Their frequencies, damping parameters and Drude-Lorentz strength parameters are in cm^{-1} . The Drude-Lorentz strength is squared in [the model expression](#); it is not an integrated intensity.

2.7.5 Vibrational and Raman data in experimental files

An experimental file can also supply frequencies, normal modes, Born charges and Raman tensors. This makes it useful for importing results from a program which PDielec does not read directly. The contents determine the available features: a dielectric model alone supports optical calculations, whereas normal-mode analysis and animation require eigenvectors. Raman spectra require Raman tensors; polar-mode corrections additionally need the structure, masses, complete normal modes, Born charges and optical permittivity.

Put the lattice, species and unit cell first, followed by `epsinf` (or its synonym `static`), then the following blocks. Keep identical atom and mode ordering in all blocks. Use a complete set of $3N$ modes for a cell of N atoms, including acoustic and silent modes. List modes in ascending frequency order and use negative frequencies for imaginary modes.

Table 2.4: Vibrational blocks

Header	Following data
<code>frequencies M</code>	M lines, each starting with a frequency in cm^{-1} . An optional scalar strength supplies all three diagonal oscillator strengths; three strengths supply <i>xx</i> , <i>yy</i> and <i>zz</i> . Omitting strengths sets them to zero. These are oscillator-strength tensor entries in atomic units (squared electron charge divided by electron mass), not Raman activities. When complete modes and Born charges are supplied the reader calculates oscillator strengths from them.
<code>born_charges N</code>	Three rows of three real numbers per atom, giving the Born effective charge tensor in electron-charge units. Use the atom order of <code>unitcell</code> . An optional atom label before each tensor is ignored.
<code>normal_modes M</code>	N rows of <i>x</i> , <i>y</i> , <i>z</i> components per mode. These are mass-weighted eigenvectors, normalised so that the sum of squared components is one. They must be mutually orthogonal and consistent with the supplied masses.
<code>raman_tensors M</code> <code>epsilon</code>	Three rows of three real numbers per mode, giving $R_\epsilon = \sqrt{V} \partial \epsilon_\infty / \partial Q$ in $(\text{\AA}/\text{amu})^{1/2}$. There is no frequency line or mode-number line between tensors.

The optional unit word after `raman_tensors M` is significant. `epsilon` stores the tensor directly. `castep`, the default when the word is omitted, multiplies the input by 4π : its input is the polarizability-volume derivative divided by $\sqrt{V}$. `skelton` multiplies the polarizability-volume derivative in $\text{\AA}^2/\sqrt{\text{amu}}$ by $4\pi/\sqrt{V}$. Use explicit `epsilon` when exporting PDielec tensors. Do not insert Raman activities in place of tensor components.

Blank lines and comment lines beginning with # in the first column are permitted between data rows. Supply exactly the declared number of rows; interpolation blocks need their terminator. Lattice vectors are multiplied by the scale on the line after `lattice` and the resulting lengths are in Angstrom. `unitcell` coordinates are fractional. `species` masses are in amu. The reader also accepts CP2K-style `&CELL` and `&COORD` blocks, as illustrated by the existing `Examples/Experiment/cp2k` examples.

The experimental reader preserves supplied normal modes; changing masses does not provide the same eigenvector recalculation as an ab initio dynamical matrix. Use masses and eigenvectors from the same calculation. For general mass/Eckart reprojection, use the JSON format below with force constants or a complete reference mode set. The experimental format currently has no `chi2` input block; use JSON to supply the nonlinear response for EO.

Two downloadable examples are provided: a constant dielectric and ZnO with frequencies, modes, Born charges and Raman tensors. The latter was converted from the CASTEP ZnO example; it is not measured data. To open it, select *Experiment* in the *Main Tab*, or run:

```
pdgui experiment zno-raman.exp -spectroscopy powder_raman
```

The `.exp` format describes a material. It is different from the two-column experimental spectrum read by the *Fitter Tab*.

2.8 Finite-Difference and Finite-Field JSON Format

The *Finite_field* input program reads a consolidated JSON description of a structure and its vibrational and Raman response. A finite-displacement or finite-field workflow using VASP, Quantum ESPRESSO, CASTEP or another program can write this common format. PDIELEC reads the results; it does not launch the displaced or finite-field electronic-structure calculations.

Choose *Finite field* and *Finite field (.json)** in the *Main Tab* file selector, or use:

```
pdgui finite_field zno-raman.json -spectroscopy crystal_raman
```

Recognised schema names are exactly `pdielect-finite-field-raman-v1`, `pdielect-r_epsilon-v1` and `pdielect-vASP-consolidated-raman-results-v1`. The spelling and case are part of these identifiers. An unfamiliar schema produces a warning and the reader attempts to read the fields below. Automatic program detection recognises schema strings beginning with `pdielect-` and containing `raman` or `r_epsilon`; specify `finite_field` explicitly for other schema names. JSON uses double-quoted keys, ordinary arrays and numbers, and `true/false` booleans. Comments and complex-number literals are not accepted.

2.8.1 Structure and reference response

The following is a minimal structural/optical example. It contains no phonons or Raman response and is intended to illustrate the required structure keys.

```
{
  "schema": "pdielect-finite-field-raman-v1",
  "structure": {
    "lattice_angstrom": [
      [
        5.0,
        0.0,
        0.0
      ]
    ]
  }
}
```

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```
    ],
    [
        0.0,
        5.0,
        0.0
    ],
    [
        0.0,
        0.0,
        5.0
    ]
],
"symbols": [
    "Si"
],
"positions_fractional": [
    [
        0.0,
        0.0,
        0.0
    ]
]
},
"epsilon_infinity": [
    [
        2.0,
        0.0,
        0.0
    ],
    [
        0.0,
        2.0,
        0.0
    ],
    [
        0.0,
        0.0,
        2.0
    ]
]
}
```

Table 2.5: Structure and reference fields

Key	Meaning
structure.lattice_angstrom	Required 3 by 3 array: rows are the a, b and c lattice vectors in Angstrom.
structure.symbols	Required list of N element symbols, one per atom.
structure.positions_fractional	N by 3 fractional coordinates. Takes precedence over structure.positions_cartesian_angstrom, the alternative N by 3 Cartesian coordinates in Angstrom. One must be present.
structure.volume_angstrom3	Optional cell volume. Defaults to the absolute determinant of the lattice. If supplied it must agree with the lattice, since Raman normalisation and electrostatic corrections use this volume.
program_masses_amu	Optional N masses or a dictionary keyed by element symbol. Takes precedence over masses_amu, then the mass_amu entries of an atoms list. Masses default to PDielec average masses when absent.
epsilon_infinity	3 by 3 relative optical permittivity, dimensionless. If absent at the top level, read from reference.epsilon_infinity; otherwise defaults to zeros. Supply physical values for optical/NAC calculations.
born_effective_charges	N by 3 by 3 Born effective charge tensors in electron-charge units, in structure order. The reference block is the fallback. Missing tensors default to zero, eliminating the corresponding polar coupling.
chi2	Optional real 3 by 3 by 3 nonlinear susceptibility in PDielec's internal, cell-dependent EO convention. This array is stored directly; it is not a raw pm/V tensor or a Voigt 3 by 6 d-tensor.

For a physical susceptibility in pm/V the internal chi2 is

$$\tilde{\chi}_{ijl}^{(2)} = \chi_{ijl}^{(2)} [\text{pm/V}] \frac{\sqrt{\text{amu}/m_e} e 10^8}{4\pi\epsilon_0 \sqrt{V [\text{Angstrom}^3]}}.$$

Convert a reported $d = \chi^{(2)}/2$ tensor to $\chi^{(2)}$ first. The meaning and limitations of this electronic response are explained in *Theory for Crystal Layer Raman*. A non-zero array alone is insufficient for EO: the polar correction also needs the Born charges, optical permittivity, normal modes and the relevant scenario option enabled.

2.8.2 Modes, force constants and dielectric derivatives

modes is a list of objects, one per mode. frequency_cm-1 is a real frequency; imaginary: true makes a positive supplied frequency negative. Modes are sorted by this signed frequency. Each mode may contain an N by 3 mass_weighted_eigenvector and a 3 by 3 raman_tensor. Eigenvectors must be orthonormal and refer to the declared program masses. Raman tensors use $R_e = \sqrt{V} \partial \epsilon_\infty / \partial Q$, including $\sqrt{V}$, in $(\text{Angstrom}/\text{amu})^{1/2}$. All modes must contain a given array for that array set to be loaded. Provide the complete $3N$ mode set for recalculation and mode mixing.

Unweighted force constants may replace or supplement the supplied modes. The first available representation in the following list is used:

1. force_constants_matrix_sym_eV_per_angstrom2: a $3N$ by $3N$ matrix.

2. `force_constants_matrix_eV_per_angstrom2`: the same shape.
3. `force_constants_eV_per_angstrom2`: an N by 3 by N by 3 array, indexed by atom, Cartesian component, atom, Cartesian component.
4. `cartesian_displacements`: central finite-difference force records.

Matrix row and column order is atom 0 x, y, z, atom 1 x, y, z, and so on. The reader symmetrises the matrix, converts eV/Angstrom² to atomic units, and diagonalises it using the active masses. This recalculates frequencies and eigenvectors, rather than retaining conflicting supplied frequencies. The active masses initially use PDielec average masses; the *Settings Tab* can select the original program masses instead.

Cartesian dielectric derivatives are supplied as `deps_dr`, an N by 3 by 3 by 3 array with indices atom, displacement component, dielectric row, dielectric column, in inverse Angstrom. The fallbacks, in order, are `diagnostics.deps_dr` and dielectric central differences in `cartesian_displacements`. With active modes u_{mak} the reader forms

$$R_{ij}^{(m)} = \sqrt{V} \sum_{ak} \frac{u_{mak}}{\sqrt{M_a}} \frac{\partial \epsilon_{ij}}{\partial r_{ak}}.$$

These derivatives take precedence over supplied mode Raman tensors during reprojection. When only projected tensors are supplied, also include reference eigenvectors so their phases and mode mixing can be tracked. Without reference eigenvectors or derivatives, the reader retains tensors only when masses are unchanged and Eckart projection is off; unsupported reprojection raises an error. Such a tensor-only dataset is not a substitute for a complete mode set.

For central differences, each `cartesian_displacements` entry contains zero-based `atom` and `cart` (0=x, 1=y, 2=z), a non-zero positive `delta_angstrom`, and either or both pairs of arrays:

- `forces_plus_eV_per_angstrom` and `forces_minus_eV_per_angstrom`: N by 3 forces at the positive and negative displacement. The force constant is the negative force difference divided by twice the displacement.
- `epsilon_plus` and `epsilon_minus`: 3 by 3 optical permittivities at those displacements. Their difference divided by twice the displacement gives `deps_dr`.

Supply all 3N Cartesian displacements for each response being reconstructed. Missing displacements and inconsistent array shapes raise errors. Keep all structures, responses, masses and array orderings consistent; schema recognition is not a full validation of the physical dataset.

Download the minimal JSON or the complete E0-capable ZnO JSON. The ZnO file contains structure, masses, reference response, `chi2`, modes, Raman tensors and Cartesian dielectric derivatives reconstructed from the CASTEP mode tensors. It is a format-conversion example, not a claim of a new finite-difference calculation. Its provenance is stored in the file. `Sphinx/tools/make_format_examples.py` regenerates these examples and checks that they can be read. Existing finite-difference workflow data can also be found in `Examples/Powder_Raman/Finite_difference`. Reader regression recipes are in `Examples/Finite_difference/preader` and `Examples/Finite_difference/preader_raman`.

2.9 The Format of the Materials' Database

The materials' database stores some basic information about the supporting materials often used in THz and Infrared spectroscopy. The file *PDielec/MaterialsDatabase.xlsx* can be opened and inspected to see the format of the entries. The first sheet gives some information about the structure of the spreadsheet. Each tab, except the first, represents a material and the tab name can be referred to in the *Support matrix* pulldown of the *PowderScenarioTab*.

In Columns G&H of a material sheet are, in arbitrary order: an entry, a density, a temperature; low and high frequency values, the number of points in a frequency-dependent entry and a reference. Column G holds the directive and column H holds its value For the case that the unit-cell of the material is required (for anisotropic materials for instance), the unit-cell can be provided either in terms of a , b , c , α , β and γ or as cell vectors.

The entry type may be one of: “Constant refractive index”, “Constant permittivity”, “Tabulated refractive index” or “Tabulated permittivity”, “Drude-Lorentz” or “FPSQ”.

For columns A to F the first row provides a label for the data in each column. The label is not used by PDGui but can be used to describe the contents of the column. The contents of these columns depend upon the entry type.

For tabulated entries, column A must have the frequency in cm^{-1} and in ascending order. Column B is not used by PDGui but because refractive indices are often tabulated with micron wavelengths, this column may be used to allow conversion to cm^{-1} .

Summary of columns used for different Entry modes

Entry	A:	B:	C:	D:	E:	F:
Constant permittivity			ϵ_r	ϵ_i		
Constant refractive index			ν	κ		
Tabulated permittivity	$\nu(\text{ cm}^{-1})$		ϵ_r	ϵ_r		
Tabulated refractive index	$\nu(\text{ cm}^{-1})$		ν	κ		
FPSQ	xx/yy/zz	ϵ_∞	Ω_{TO}	γ_{TO}	Ω_{LO}	γ_{LO}
Drude-Lorentz	xx/yy/zz	ϵ_∞	Ω_j	S_j	σ_j	

Summary of columns G and H

Directive (G)	Contents (H)
Entry:	One of; • ‘tabulated refractive index’ • ‘tabulated permittivity’ • ‘constant refractive index’ • ‘constant refractive index’ • ‘Drude-Lorentz’ • ‘FPSQ’
Density:	Density in g/ml
Temperature:	Temperature of measurement
Low:	The lowest valid frequency
High:	The highest valid frequency
NPoints:	The number of points in a tabulation
Reference:	The reference for the data
a_vector	3 numbers specifying the a-direction
b_vector	3 numbers specifying the b-direction
c_vector	3 numbers specifying the c-direction
a:	The length of a-direction
b:	The length of b-direction
c:	The length of c-direction
alpha:	Cell angle alpha
beta:	Cell angle beta
gamma:	Cell angle gamma

2.10 The Format of the Output Spreadsheet

The Excel spreadsheet, which can be written by the program, contains details of the system being analysed, a list of the scenario settings and tables of absorption, permittivity, reflection and transmission. It also includes mode frequencies, IR intensities and Raman activities when Raman data are available. Result sheets are conditional on the scenario types present: powder IR/ATR, crystal IR, powder Raman and crystal Raman each supply their corresponding sheets. Spectrum frequencies are exported in cm^{-1} . Plotting-tab spectrum renormalisation is a display operation and is not applied by the spreadsheet exporter. The Settings sheet contains base mode data, rather than the optional NAC/EO diagnostics displayed in the frequency table.

The spreadsheet is divided into different sheets;

- **Main**

This stores the information concerning the file that is being analysed and the program associated with the output. A list of the frequencies read in from the output file is given too.

- **Settings**

The information here comes from the Settings Tab: masses, permittivities and the frequencies that will be used in subsequent calculations. These frequencies can include corrections to the dynamical matrix to project out translational modes, corrections to the Born charges to ensure they sum to zero and changes to the masses of the atoms. Each

mode has an inclusion flag and a linewidth (Sigma), followed by its frequency, IR intensity, integrated molar absorption and absorption maximum. The Raman columns give total, parallel and perpendicular activities, labelled Raman `R_total`, Raman `R_parallel` and Raman `R_perp`. The activity values use the units selected by *Raman activity units* in the Settings Tab: $\text{\AA}^4/\text{amu}$ for polarizability-volume activities or $\text{\AA}/\text{amu}$ for the internal convention. The column headers indicate the selected units. These activities describe the individual modes, rather than the Raman spectrum calculated for a particular scenario.

- **Analysis**

The analysis of the vibrational modes into molecular, internal and external contributions is summarised here.

- **Scenarios**

A complete list of the settings for every scenario used in the calculation is given.

- **Powder Molar Absorption (cells)**

The molar absorption in moles of unit-cell is given as a function of frequency for each powder scenario. The units are $\text{L}\cdot\text{mol}^{-1}\cdot\text{cm}^{-1}$.

- **Powder Absorption**

The absorption for each powder scenario is tabulated as a function frequency. The units are cm^{-1}

- **Powder Real Permittivity**

The real component of the complex permittivity for each powder scenario is tabulated as a function frequency.

- **Powder Imaginary Permittivity**

The imaginary component of the complex permittivity for each powder scenario is tabulated as a function frequency.

- **Powder ATR Reflectance**

The extinction coefficient associated with the attenuated total reflectance for each powder scenario is tabulated as a function frequency. See [Calculation of ATR Extinction Spectra](#) for more details.

- **Crystal R_p**

The P polarised component of the reflectance for each single crystal scenario is tabulated as a function of frequency.

- **Crystal R_s**

The S polarised component of the reflectance for each single crystal scenario is tabulated as a function of frequency.

- **Crystal T_p**

The P polarised component of the transmittance for each single crystal scenario is tabulated as a function of frequency.

- **Crystal T_s**

The S polarised component of the transmittance for each single crystal scenario is tabulated as a function of frequency.

- **Real Crystal Permittivity**

The real components of the crystal permittivity tensor are tabulated as a function of frequency.

- **Imag Crystal Permittivity**

The imaginary components of the crystal permittivity tensor are tabulated as a function of frequency.

- **Powder Molar Absorption (mols) and Powder Molar Absorption (atoms)**

The selected molecular or atomic molar definition adds the corresponding sheet, in $\text{L}\cdot\text{mol}^{-1}\cdot\text{cm}^{-1}$, alongside the unit-cell sheet.

- **Crystal A_p and Crystal A_s**

Dimensionless p- and s-polarised absorptance for crystal infrared scenarios, alongside reflectance and transmittance.

- **Powder Raman and Crystal Raman**

Raman spectra for the corresponding scenarios, in arbitrary intensity units, as a function of Raman shift in cm^{-1} . These include scenario-dependent geometry and line broadening; they are distinct from the Settings-sheet mode activities in $\text{\AA}^4/\text{amu}$ or $\text{\AA}/\text{amu}$.

INSTALLATION

3.1 Requirements and installation choices

PDielec 10 requires **Python 3.10 or later**. Install it in a dedicated conda or Python virtual environment so that its dependencies are kept together. The GUI uses **PyQt6** through QtPy; the 3D viewer also requires working OpenGL support from the operating system and graphics driver.

Choose a published package for normal use, or a Git checkout to run the examples, develop the code, or use changes not yet published. The version available on [PyPI](#) or [conda-forge](#) may differ from the version described by the development documentation. Check the installed version with:

```
python -c "from PDielec import __version__; print(__version__)"
```

Package installations provide the Python modules, GUI and command-line entry points. Obtain the example datasets and regression tests from the Git repository; they are not included in the installed wheel. Use a checkout matching the installed version when comparing with reference results.

3.2 Conda

With conda installed, create and activate an environment. On Windows, use a terminal configured for conda, such as the Miniconda prompt:

```
conda create --name pdielec --channel conda-forge pdielec
conda activate pdielec
pdgui
```

Update to the version available on that channel with:

```
conda activate pdielec
conda update --channel conda-forge pdielec
```

For development from source, create an environment containing Python and pip, then follow the Git checkout instructions below. For example:

```
conda create --name pdielec-dev --channel conda-forge python=3.12 pip
conda activate pdielec-dev
```

Python 3.12 is an example environment choice, not the minimum requirement. The package installation step below installs its declared dependencies.

3.3 PyPI

Use Python 3.10 or later to create a virtual environment. On Linux or macOS:

```
python3 -m venv pdielec-env
source pdielec-env/bin/activate
```

On Windows, the equivalent commands in Command Prompt are:

```
python -m venv pdielec-env
pdielec-env\Scripts\activate.bat
```

If you already activated a conda environment, use that environment instead. In the activated environment, install and start PDielec:

```
python -m pip install --upgrade pip
python -m pip install pdielec
pdgui
```

To update a PyPI installation, use `python -m pip install --upgrade pdielec`. Always run pip through the Python interpreter of the intended environment. See the [Python Packaging installation guide](#) for environment setup on other shells.

If a dependency cannot be installed from a wheel, check that it provides a wheel for your operating system, processor architecture and Python version. Use a supported Python version with matching wheels, or install the build tools required by that dependency. PDielec itself is a Python package, but several scientific and GUI dependencies contain compiled components.

3.4 Installing from GitHub

Install Git and activate a Python environment as above. From a directory where you want to keep the source, run:

```
git clone https://github.com/JohnKendrick/PDielec.git
cd PDielec
python -m pip install -e .
pdgui
```

The editable installation reads dependencies from `pyproject.toml` and creates commands such as `pdgui`, `preader` and `pdmake` in the active environment. It works from a checkout on Linux, macOS or Windows without manually copying scripts or relying on Windows symbolic links. Activate the same environment whenever using those commands.

To install a fixed copy instead of an editable checkout, use `python -m pip install ..`. With an editable installation, Python source changes take effect directly. After updating a checkout with `git pull`, repeat `python -m pip install -e .` to pick up dependency or entry-point changes as well.

3.4.1 Checking an installation

Check dependency consistency and launch the GUI:

```
python -m pip check
pdgui
```

From the repository root, run the pytest suites with:

```
pdmake test-pytests
```

Run pytest followed by the example regression suites with:

```
pdmake tests
```

On Linux or macOS, `./pdmake` explicitly selects the wrapper in the current checkout. These tests need the checkout's `Examples` and `PDielec/Tests` directories. Normal regression runs compare generated results with references; `--regenerate` replaces references and is not an installation check.

3.5 Package dependencies

The authoritative list is `[project].dependencies` in `pyproject.toml`. Pip installs these automatically when installing PDielec. The accompanying `requirements.txt` lists the same dependencies for environments that need an explicit requirements file:

```
dill
imageio
imageio-ffmpeg
matplotlib
numpy
openpyxl
psutil
PyOpenGL
PyQt6
pytest
PyYAML
QtPy
scikit-learn
scipy
setuptools
spglib
termcolor
XlsxWriter
```

QtPy is an abstraction layer; it does not replace the PyQt6 binding. Pytest is currently a declared dependency because PDielec includes test-running commands. A separate `mk1` installation is not required by PDielec.

3.5.1 Optional tools

Phonopy: install `phonopy` to use `phonopy-pdielec-born`, which calls the Phonopy API to expand Born charges. It is not in the core dependency list:

```
python -m pip install phonopy
```

Documentation: from the repository root, install the documentation tools:

```
python -m pip install -r Sphinx/requirements.txt
```

This includes Sphinx, its configured extensions and theme, and `sphinx-autobuild` for live previews. With Make available, `make livehtml` in `Sphinx/` starts a preview that updates when sources change. A direct cross-platform alternative, run from the repository root, is:

```
sphinx-autobuild -b html Sphinx Sphinx/_build/html
```

For a static preview, use `sphinx-build -b html Sphinx Sphinx/_build/html`. The custom `make html` target also copies built HTML into `docs/`. Screenshot regeneration needs a working Qt/OpenGL display; see `Sphinx/tools/README.md` for the capture and input-validation tools.

PDF manual: install the additional Python converter, together with a TeX installation providing `pdflatex` and `latexmk`, and the system `rsvg-convert` command (provided by `librsvg`). From the repository root, run:

```
python -m pip install -r Sphinx/requirements-pdf.txt
make -C Sphinx pdf
```

Alternatively, if you are already inside `Sphinx/`, run:

```
make pdf
```

This builds the manual and copies it to `PDielec.pdf` in the repository root after a successful build. The intermediate manual remains at `Sphinx/_build/latex/PDielec.pdf`; `make latexpdf` builds only that copy. SVG figures are converted to vector PDF files automatically in the build directory. HTML continues to use the original SVG files and does not require the PDF converter or TeX. Unicode mappings for scientific symbols are applied only to the LaTeX output.

Release builds: install the build frontend before creating archives:

```
python -m pip install build
python -m build
```

The build environment installs the backend declared in `pyproject.toml`, including `setuptools>=77.0.3` for SPDX license metadata. `pdm make pypi` also builds archives on supported platforms, but removes previous build output first. Building archives does not upload them to PyPI. Maintainers can install `twine` separately to check or upload a release.

3.6 GitHub - PDielec directory structure

Paths below are relative to the repository root:

- `PDielec/` contains the readers and numerical routines; `PDielec/GUI/` contains `PDGui`.
- `PDielec/Tests/` contains pytest regression tests.
- `Examples/` contains input/output datasets, regression recipes and helper scripts.
- `Sphinx/` contains the documentation sources; `docs/` contains published HTML.
- `pyproject.toml` defines dependencies, package data and installed command entry points.

3.7 Examples

The catalogue below describes the tracked example families. A directory with `command.pdmake` contains a regression recipe. PDGui recipes normally use `script.py` and compare a generated `results.xlsx` with `results.ref.xlsx`. Reader and analysis recipes instead use CSV, CIF or normal-mode-analysis references. Supporting files and notebooks are not all standalone regression tests, and `pdmake tests` runs the configured suites, not every file below `Examples/`.

From an example directory containing a PDGui recipe, run:

```
pdmake command.pdmake
pdmake --view command.pdmake
```

The second command opens the saved scenario interactively. From the repository root, `pdmake tests` runs `pytest` first and then the configured example suites. `pdmake benchmarks` runs the benchmark selection without reference comparison; see [Performance](#). Use `--regenerate` only after reviewing differences.

The small `Crystal_Raman/AbInit` and `Crystal_Raman/Finite_field` directories currently contain provenance notes only. Use `Powder_Raman/AbInit` and `Powder_Raman/Finite_difference` for the corresponding complete datasets. The CASTEP 25.12 directory includes both a crystal Raman recipe and data used by the atomic Raman reader tests.

Table 3.1: Examples available in the repository

Directory	Input workflow	or Purpose
ATR/AlAs	AbInit	Maxwell Garnett calculation of the ATR spectrum of an ellipsoid along [001]. The incident angle varies from 0 to 80 degrees.
ATR/Na2SO42	Vasp	Maxwell-Garnett calculation of the ATR spectrum, changes the S polarisation component from 0 to 100%
ATR/Na2SO42_fit	Vasp	Maxwell-Garnett calculation of the ATR spectrum, an example of fitting the spectrum to experiment
AbInit/AlAs	AbInit	Average permittivity and Maxwell-Garnett calculation of sphere, plate and ellipsoid
AbInit/BaTiO3	AbInit	Average permittivity and Maxwell-Garnett calculations of sphere, plate and ellipsoid, using average isotope masses
AbInit/Na2SO42	AbInit	Average permittivity and Maxwell-Garnett calculations of Na2(SO4)2, sphere, plate and ellipsoid, using program-defined masses
AbInit/preader	AbInit	AbInit preader test
Castep/AsparticAcid	Castep	Average permittivity and Maxwell-Garnett calculations of sphere, plate and ellipsoid, using program-defined masses
Castep/Bubbles	Castep	Maxwell-Garnett calculation showing the effect of air bubbles at 24% volume fraction and 30 micron radius
Castep/Castep17	Castep	Castep 17, Maxwell-Garnett sphere and plates with 3 surfaces
Castep/Isoleucine	Castep	Maxwell-Garnett sphere

continues on next page

Table 3.1 – continued from previous page

Directory	Input or workflow	Purpose
Castep/MgO	Castep	Comparison of MG, Bruggeman and AP methods changing shapes and volume fractions
Castep/Na2SO42	Castep	Comparison of MG and Bruggeman, for needle, ellipsoid and plate shapes
Castep/preader	Castep	Castep preader test
Crystal/Leucine	Crystal	Comparison of MG, plates and ellipsoids
Crystal/Na2SO42	Crystal	Comparison of MG for needle, ellipsoid and plate shapes
Crystal/Na2SO42_C17	Crystal	Comparison of MG for needle, ellipsoid and plate shapes, reading output from Crystal 17
Crystal/Quartz	Crystal	Comparison of MG for needle, ellipsoid and plate shapes
Crystal/ZnO	Crystal	PDGui Crystal of ZnO (CPHF); PDGui Crystal of ZnO (default settings); PDGui Crystal of ZnO (NOECKART settings)
Crystal/preader	Crystal	Crystal preader test
Crystal_Raman/Castep	Crystal Raman	PDGui Castep Crystal Raman of ZnO
Crystal_Raman/Castep-25.12	Crystal Raman	ZnO crystal Raman scenarios and CASTEP 25.12 atomic polar tensor reader checks.
Crystal_Raman/Crystal23	Crystal Raman	CRYSTAL23 ZnO crystal Raman scenarios, including optical geometry and coherence.
Experiment/AlN	Experiment	Aluminium Nitride multi-layer system including SiC and Si
Experiment/Forsterite	Experiment	Single crystal calculations of a thick slab, for a, b and c axis alignments with polarisation direction. Uses FPSQ model for permittivity.
Experiment/Mayerhofer	Experiment	Example of a Drude Lorentz model permittivity
Experiment/Sapphire	Experiment	Sapphire example and test of the materials database
Experiment/constant	Experiment	Example of a constant permittivity with loss
Experiment/cp2k	Experiment	PDGui experiment Single Crystal for cysteine form3
Experiment/drude-lorentz	Experiment	A Drude-Lorentz model for MgO, varying the angle of incidence
Experiment/fpsq	Experiment	An FPSQ model for Quartz, showing polarisation on along different axes and different incident angles.
Experiment/interpolation	Experiment	An example of an interpolation model
FHI-Aims/Na2SO42	FHI-Aims	Powder calculation for spheres, including a comparison with VASP results
FHI-Aims/preader	FHI-Aims	FHI-Aims preader test
Finite_difference/preader	Finite-field JSON	Finite-difference VASP ZnO preader infrared (Eckart)
Finite_difference/preader_raman	Finite-field JSON	Finite-difference VASP ZnO preader Raman
Gulp/Na2SO42	Gulp	Maxwell-Garnett and Bruggeman on needle, ellipsoid and plate

continues on next page

Table 3.1 – continued from previous page

Directory	Input or workflow	Purpose
Gulp/calcite	Gulp	Maxwell-Garnett method on Sphere and Plate
Gulp/preader	Gulp	Gulp preader test
Helper/Helper	Python API	Python helper API demonstration; inspect test_helper.py for its workflow.
Helper/Jupyter	Python API	Jupyter notebooks and supporting datasets for API workflows.
Mie/MgO	Castep	Mie method with varying volume fractions and sphere sizes
Mie/MgO_lognormal	Castep	Mie method with varying volume fractions and sphere size distributions
P2Cif	CIF export	Testing p2cif
Phonopy/Al2O3	Phonopy	Primitive and standard cell example of using Phonopy
Phonopy/Na2SO42	Phonopy	Maxwell-Garnett and Bruggeman method for needle, ellipsoid and plate shapes, with varying volume fractions
Phonopy/QE	Phonopy/QE	Powder and single crystal Phonopy example using QE
Phonopy/Vasp	Phonopy/Vasp	Powder and single crystal Phonopy example using Vasp
Phonopy/ZnO	Phonopy	Maxwell-Garnett and Bruggeman method for needle, ellipsoid and plate shapes
Phonopy/preader	Phonopy	Phonopy preader test
Powder_Raman/AbInit	Powder Raman	PDGui Abinit powder Raman of ZnO
Powder_Raman/Castep	Powder Raman	PDGui Castep powder Raman of ZnO
Powder_Raman/Crystal23	Powder Raman	PDGui Crystal powder Raman of ZnO
Powder_Raman/Finite_difference	Powder Raman	VASP ZnO finite-difference JSON; powder Raman with EO disabled and enabled.
Powder_Raman/QE	Powder Raman	PDGui QE powder Raman of ZnO
Powder_Raman/Vasp	Powder Raman	PDGui Vasp powder Raman of ZnO
QE/Cocaine	Quantum Espresso	Maxwell-Garnett sphere, using QE 4.1
QE/Na2SO42	Quantum Espresso	Maxwell-Garnett and Bruggeman on needle, ellipsoid and plate, using QE 5.1
QE/Na2SO42-v7	Quantum Espresso	Maxwell-Garnett and Bruggeman on needle, ellipsoid and plate, using QE 7.3.1
QE/Urea	Quantum Espresso	Maxwell-Garnett and single crystal, using QE 7.3.1
QE/ZnO	Quantum Espresso	Maxwell-Garnett and Bruggeman on needle, ellipsoid and plate, using QE 5.4.0
QE/preader	QE	QE preader test
SingleCrystal/Bi2Se3	Vasp	Single crystal example of thick slab, angle of incidence varies from 0 to 90
SingleCrystal/Bi2Se3_film	Vasp	Single crystal example of thin film, angle of incidence varies from 0 to 90

continues on next page

Table 3.1 – continued from previous page

Directory	Input workflow	or	Purpose
SingleCrystal/ L-alanine	SingleCrystal		PDGui Single crystal L-alanine test
SingleCrystal/MgO	SingleCrystal		PDGui Castep of MgO Single Crystal coherent
SingleCrystal/ ScatteringMatrix	SingleCrystal		PDGui Single crystal scattering matrix test
SizeEffects/BaTiO3	Abinit		Exploration of size effects in Bruggeman effective medium theory
SizeEffects/MgO	Castep		Exploration of size effects in Bruggeman and Maxwell-Garnett effective medium theories
SizeEffects/ZnO	Vasp		Exploration of size effects in Maxwell-Garnett effective medium theory
Vasp/F-Apatite	Vasp		Maxwell-Garnett, sphere plates and needles, using Vasp 5.3.5
Vasp/Na2SO42	Vasp		Maxwell-Garnett and Bruggeman, needle, plate and needle, using Vasp 5.3.5
Vasp/Na2SO42_v	Vasp		PDGui Vasp for Na2(SO4)2 v dependent matrix permittivities
Vasp/Urea	Vasp		Powder and single crystal exampl, using Vasp 5.4.4
Vasp/Vasp6	Vasp		The DFT calculation were performed using VASP 6.4.2, with thanks to Dr David Santos-Carballal for performing the DFT calculation
Vasp/ZnO	Vasp		Maxwell-Garnett and Bruggeman, needle, plate and needle, mass fraction, using Vasp 5.3.5
Vasp/preader	Vasp		Vasp preader test
VibAnalysis/ AsparticAcid	Mode analysis		Vibanalysis Castep for L-Aspartic acid
VibAnalysis/BaTiO3	Mode analysis		Vibanalysis Abinit for BaTiO3
VibAnalysis/Cocaine	Mode analysis		Vibanalysis QE for cocaine salt
VibAnalysis/ F-Apatite	Mode analysis		Vibanalysis Vasp for F-Apatite
VibAnalysis/ Isoleucine	Mode analysis		Vibanalysis Castep for Isoleucine
VibAnalysis/Leucine	Mode analysis		Vibanalysis Crystal for Leucine

VERSION 10 MIGRATION

Version 10 adds powder and crystal Raman scenarios. Before reusing an older script or saved scenario, check the following changes.

4.1 Python scripts

Public methods and functions formerly written in camelCase have been renamed to snake_case throughout the numerical modules, readers and GUI. Older scripts calling those names need updating. Use the current API reference and the `script.py` files in the examples as templates; do not assume every rename can be obtained by a mechanical text replacement. Save a new settings script from the GUI after checking an older calculation.

4.2 Crystal Raman geometry

`Collection angle = -1` selects automatic collection. On the superstrate side this now means retro-backscattering; on the substrate side it means collinear forward scattering. Saved automatic scenarios can therefore change at nonzero incidence. Normal-incidence automatic scenarios are unchanged. To retain specular reflection, explicitly set the collection angle equal to the incidence angle. Other negative collection angles are explicit signed angles, not automatic-selection sentinels.

`Approximate ES` defaults to False. When enabled, it evaluates the reciprocal detector field once at the laser frequency instead of at each Stokes frequency. It reuses the incident field only when the detector optical system and signed angle match. Review these settings before comparing old and new spectra; see *Theory for Crystal Layer Raman* and *PDGui Settings Reference*.

4.3 Spreadsheets and regression references

Settings-sheet Raman activities now follow the activity units selected in the Settings frequency table, with units in the column headers. `pdmake` checks this sheet as well as result sheets and reports a checked sheet missing from one workbook. Review differences before regenerating reference workbooks. See *Other Software Components* for the conditional Raman and infrared output sheets.

PDGui provides a graphical front end to PDielec. It reads a material's structure and response from a QM/MM calculation or a user-supplied input file, and calculates infrared, ATR or Raman spectra. The interface is a notebook with *Main*, *Settings*, *Scenario*, *Plotting*, *Analysis*, *3D Viewer* and *Fitter* tabs. Additional scenarios appear as additional tabs.

5.1 Starting PDGui

PDGui can be run without arguments, or with an output file. It makes a best guess at the program from the filename and associated files. For example:

```
pdgui
pdgui OUTCAR
pdgui raman.castep -spectroscopy crystal_raman
pdgui raman.castep -spectroscopy powder_raman
pdgui experiment material.exp
pdgui finite_field zno-raman.json -spectroscopy crystal_raman
pdgui qe run1.dynG
```

Use `qe` as the program name when explicitly selecting that reader. A program name can also be supplied with `-program`. For a saved session use `pdgui -script script.py`. To calculate and write a spreadsheet without remaining in the GUI, use:

```
pdgui -script script.py -spreadsheet results.xlsx -nosplash -exit
```

Table 5.1: Common command-line options

Option	Description
-spectroscopy type	Initial spectroscopy: <code>powder_infrared</code> (default), <code>crystal_infrared</code> , <code>powder_raman</code> or <code>crystal_raman</code> . Select <i>Powder ATR</i> in the Settings Tab or a script.
-program name	Explicit input program, for example <code>castep</code> , <code>experiment</code> or <code>finite_field</code> .
-script filename	Read a Python settings script.
-spreadsheet filename	Write results to an Excel file with an <code>.xlsx</code> extension.
-cpus N	Number of workers; zero selects the available physical cores.
-threading	Use threads instead of the default worker processes.
-nosplash	Suppress the splash window.
-exit	Exit after the script and requested spreadsheet output.
-debug	Enable diagnostic logging.

The figures below use the CASTEP wurtzite ZnO calculation in `Examples/Crystal_Raman/Castep`. Its output includes Raman tensors and a non-zero nonlinear optical susceptibility, allowing the electro-optic (EO) correction to be demonstrated. The Raman examples use 532 nm excitation, 298 K and a Lorentzian half-width of 5 cm⁻¹. These are example settings; individual control defaults are described below. The Fitter uses the existing sodium-sulphate ATR example with an experimental spectrum.

5.2 Main Tab

The *Main Tab* selects the calculation to analyse and the files used for saving results and settings.

Choose the *QM/MM Program* from Abinit, Aims, Castep, Crystal, Experiment, Finite field, Gulp, Phonopy, Quantum Espresso, Vasp or PDGui. Enter the output filename in *Analyse this output file*, or select it with *File manager*. The *PDGui* choice reads a saved Python session. Some readers also use files beside the selected output, so keep the calculation’s companion files together.

Experiment reads material descriptions in the *Experimental File Format*. *Finite field* reads the consolidated *Finite-Difference and Finite-Field JSON Format*, including results from finite-difference workflows. These are material inputs; the experimental spectrum used for fitting is selected in the Fitter Tab.

Excel spread sheet sets the `.xlsx` destination. *Save results* writes the calculated results and settings. *Script filename* sets the Python file written by *Save settings*. Reload that file to reproduce the session; the calculation input and any materials database referenced by it must still be available.

The lower text areas report the unit-cell vectors in Å and the frequencies read from the input. They are outputs, not editable model parameters. Frequencies after mass changes and other corrections are shown in the Settings Tab.

5.3 Settings Tab

The *Settings Tab* controls the spectroscopy and the common vibrational data used by every scenario.

Spectroscopy type selects *Powder Infrared*, *Powder ATR*, *Powder Raman*, *Crystal Infrared* or *Crystal Raman*. Changing it replaces the scenarios with one scenario of the selected type. Save a session before changing spectroscopy if the existing scenarios will be needed again.

Main Settings Scenario 1 Plotting Analysis 3D Viewer Fitter

QM/MM Program

Analyse this output file

Excel spread sheet

Script filename

Unit-cell (Angstrom) from raman.castep

```
a = 2.83953 -1.63940 0.00000
b = 0.00000 3.27881 0.00000
c = 0.00000 0.00000 5.30308
```

Frequencies from raman.castep

```
-0.027
-0.022
-0.016
108.638
108.643
266.714
334.566
367.101
367.101
398.155
398.157
501.089
```

Fig. 5.1: The Main Tab for the ZnO calculation.

Main
Settings
Scenario 1
Plotting
Analysis
3D Viewer
Fitter

Spectroscopy type: Crystal Raman

Apply Eckart conditions? ☒

Apply Born charge neutrality? ☐

Atomic mass definition: Average natural abundance

Atomic masses

	O	Zn
	15.999400	65.380000

Lorentzian width (σ): 5.0

Raman activity units: Polarizability volume ($\text{\AA}^3/\text{amu}$)

Optical permittivity:

	x	y	z
x	5.3296	-0.0000	-0.0000
y	-0.0000	5.3296	0.0000
z	-0.0000	0.0000	5.0633

Symmetrise optical permittivity? ☒

NAC Options: ☒ Apply NAC x: 0 y: 0 z: 1 ☒ Include EO correction ($\chi^{(2)}$)

Raman polar simulator: Raman polar plot

	Sigma (cm ⁻¹)	TO freq (cm ⁻¹)	LO freq (cm ⁻¹)	Intensity (Debye ² / $\text{\AA}^2/\text{amu}$)	Integrated Molar Absorption (L/mole/cm ²)	Absorption maximum (L/mole/cm)	Raman total ($\text{\AA}^3/\text{amu}$)	Raman $\parallel$ ($\text{\AA}^3/\text{amu}$)	Raman $\perp$ ($\text{\AA}^3/\text{amu}$)	
1	☐	5.0	-0.0000	0.0002	0.0000	0.00	0.00	0.005160	0.002949	0.002212
2	☐	5.0	0.0000	0.0000	0.0000	0.00	0.00	0.008073	0.004613	0.003460
3	☐	5.0	0.0000	-0.0000	0.0000	0.00	0.00	0.005199	0.002971	0.002228
4	☒	5.0	108.6457	108.6457	0.0000	0.00	0.00	0.476529	0.272302	0.204227
5	☒	5.0	108.6503	108.6503	0.0000	0.00	0.00	0.476529	0.272302	0.204227
6	☐	5.0	266.7339	266.7339	0.0000	0.00	0.00	0.000000	0.000000	0.000000
7	☒	5.0	334.5700	510.0862	17.3767	73426.86	9349.00	225.694736	137.749...	87.945...
8	☒	5.0	367.0424	367.0424	16.6191	70225.87	8941.44	0.000311	0.000179	0.000132
9	☒	5.0	367.0430	367.0430	16.6190	70225.42	8941.38	0.000228	0.000130	0.000098
10	☒	5.0	398.1586	398.1586	0.0000	0.00	0.00	2.014557	1.151176	0.863382
11	☒	5.0	398.1604	398.1604	0.0000	0.00	0.00	2.014557	1.151176	0.863382
12	☐	5.0	501.0892	501.0892	0.0000	0.00	0.00	0.000000	0.000000	0.000000

Fig. 5.2: The Settings Tab with NAC and EO diagnostics enabled for ZnO.

Apply Eckart conditions? projects out the three translational modes of the dynamical matrix and is on by default. *Apply Born charge neutrality?* removes the sum of the Born-charge tensors and is off by default. These controls require the corresponding data from the reader.

Atomic mass definition offers average natural-abundance masses (the default), program masses, most-common-isotope masses or individually entered masses. The *Atomic masses* table is editable and uses amu. With a dynamical matrix, changing masses recalculates modes and frequencies. The input-format reference explains the limitations when only precomputed modes or Raman tensors are supplied.

Lorentzian width (σ) sets the common half-width at half maximum in cm^{-1} , initially 5. Individual mode widths can be changed in the lower table. The *Optical permittivity* table contains the dimensionless electronic background tensor. Edit its elements when a value must be supplied or replaced. *Symmetrise optical permittivity?*, on by default for electronic-structure readers, averages it over the crystal point group. Experimental input leaves this off by default. Reading a new input resets input-dependent masses and permittivity.

For Raman calculations, *Raman activity units* chooses polarizability-volume activities in $\text{\AA}^4/\text{amu}$ or activities in the internal R_ϵ convention in $\text{\AA}/\text{amu}$. This changes the displayed activities, not the physical normalisation of the tensors used by scenarios. The Raman activity columns in the exported *Settings* worksheet use the same selected units, indicated in their column headers.

The *NAC Options* row controls the mode-table diagnostics. *Apply NAC* fills the LO-frequency column using the direction specified by x, y and z in the crystal Cartesian frame; initially it is off and the direction is (0, 0, 1). Use a non-zero direction. *Include EO correction* adds the nonlinear response to the diagnostic Raman tensors when the required data are present. It is initially off. These controls are separate from *Layer NAC mode* and *Include electro-optic term* in a Crystal Raman scenario. *Raman polar plot* opens the *Raman Polar Window*.

The lower table has one row per mode. The checkbox beside *Sigma* determines whether the mode contributes to subsequent spectra; greyed values indicate unselected modes. The columns show TO frequency, optional LO frequency, IR intensity in $\text{Debye}^2/\text{\AA}^2/\text{amu}$, integrated molar absorption, absorption maximum, and total, parallel and perpendicular Raman activities where present. The headers give the units. The IR quantities and the Raman activities are different measures of response; a Raman-active mode can have zero IR intensity.

To copy values from the frequency and intensity table, select the required cells and press Ctrl+C. The selected values are copied as tab-separated text, with one line per table row, ready to paste into a spreadsheet or text editor. This also works for the Raman activity columns when they are shown.

5.4 Scenario Tabs

A scenario defines the sample and measurement conditions. *Add another scenario* copies the current scenario, allowing one variable to be changed for comparison. *Delete this scenario* removes it. The *legend* field names its curve in plots. All scenarios in a session use the spectroscopy selected in the Settings Tab; use that tab to change the spectroscopy.

Scenario types: *Powder Infrared* · *Powder ATR* · *Crystal Infrared* · *Powder Raman* · *Crystal Raman*.

5.4.1 Powder Infrared Scenarios

A powder scenario describes crystallites dispersed in a supporting material. The particle shape and the permittivities determine the effective optical response.

Open materials' database selects an Excel materials database; the following line reports the current file. *Support matrix* selects an entry. Its adjacent read-only description reports the kind of dielectric model

Main
Settings
Scenario 1
Plotting
Analysis
3D Viewer
Fitter

Spectroscopy type: Powder Infrared

Apply Eckart conditions? ☒

Apply Born charge neutrality? ☐

Atomic mass definition: Average natural abundance

Atomic masses

	O	Zn
	15.999400	65.380000

Lorentzian width (σ): 5.0

Optical permittivity:

	x	y	z
x	5.3296	-0.0000	-0.0000
y	-0.0000	5.3296	0.0000
z	-0.0000	0.0000	5.0633

Symmetrise optical permittivity? ☒

NAC Options: ☐ Apply NAC x: 0 y: 0 z: 1

		Sigma (cm ⁻¹)	TO freq (cm ⁻¹)	LO freq (cm ⁻¹)	Intensity (Debye ² /Å ² /amu)	Integrated Molar Absorption (L/mole/cm ²)	Absorption maximum (L/mole/cm)	Raman total (Å ⁴ /amu)	Raman (Å ⁴ /amu)	Raman ⊥ (Å ⁴ /amu)
1	☐	5.0	-0.0000	—	0.0000	0.00	0.00	0.005344	0.003054	0.002290
2	☐	5.0	0.0000	—	0.0000	0.00	0.00	0.007780	0.004446	0.003334
3	☐	5.0	0.0000	—	0.0000	0.00	0.00	0.005307	0.003033	0.002275
4	☒	5.0	108.6457	—	0.0000	0.00	0.00	0.476529	0.272302	0.204227
5	☒	5.0	108.6503	—	0.0000	0.00	0.00	0.476529	0.272302	0.204227
6	☐	5.0	266.7339	—	0.0000	0.00	0.00	0.000000	0.000000	0.000000
7	☒	5.0	334.5700	—	17.3767	73426.86	9349.00	225.694740	137.749...	87.945...
8	☒	5.0	367.0424	—	16.6191	70225.87	8941.44	0.000312	0.000179	0.000132
9	☒	5.0	367.0430	—	16.6190	70225.42	8941.38	0.000224	0.000128	0.000096
10	☒	5.0	398.1586	—	0.0000	0.00	0.00	2.014557	1.151176	0.863382
11	☒	5.0	398.1604	—	0.0000	0.00	0.00	2.014557	1.151176	0.863382
12	☐	5.0	501.0892	—	0.0000	0.00	0.00	0.000000	0.000000	0.000000

Fig. 5.3: The same input with Powder Infrared selected. Raman-specific controls are hidden.

Open materials' database: Open materials' database

Current materials' database: PDIElec/MaterialsDataBase.xlsx

Support matrix: ptfe Constant permittivity

Support density: 2.200

Support permittivity: 2.000 0.000

% Air void volume fraction: 0.0

Air void radius (μm): 30.000

% Mass fraction of dielectric: 21.658531

% Volume fraction of dielectric: 10.000000

Method: Maxwell-Garnett

Particle radius (μm): 0.000100

Particle sigma (μm): 0.000000

Particle shape: Sphere

Unique direction [abc]: 0 0 1

Ellipsoid a/b eccentricity: 1.000000

Powder scenario legend: Scenario 1

Add another scenario Delete this scenario

Fig. 5.4: The Powder Infrared scenario, using the default PTFE support.

and, for tabulated entries, the frequency range. *Support density* is in g/cm^3 . *Support permittivity* shows real and imaginary parts. Editing supported matrix values creates a manual description; frequency-dependent database entries should be used within their stated range.

The mass and volume fractions are displayed as percentages. Editing one recalculates the other using the material densities. They describe the amount of dielectric crystallite in the mixture. *% Air void volume fraction* and *Air void radius* describe additional air inclusions; the radius is in μm . The air-void fraction defaults to zero.

Method selects Maxwell-Garnett (default), Bruggeman, Averaged Permittivity or Mie. The first two are effective-medium treatments, Averaged Permittivity averages the response, and Mie includes finite-size spherical scattering. *Particle radius* and *Particle sigma* are in μm ; the latter describes the size-distribution parameter, not the Lorentzian mode width. Their availability depends on the method. Use a small-particle effective-medium treatment only where its assumptions apply; see [Theory for Infrared Absorption in Powders](#).

Particle shape offers Sphere, Needle, Plate and Ellipsoid. A sphere needs no orientation. A needle or ellipsoid uses *Unique direction [abc]* for its axis; a plate uses the corresponding (hkl) surface normal. *Ellipsoid a/b* is the axial aspect ratio, with one giving a sphere. Direction and ratio controls are enabled only for the relevant shape.

Scenario types: [Powder Infrared](#) · [Powder ATR](#) · [Crystal Infrared](#) · [Powder Raman](#) · [Crystal Raman](#).

5.4.2 Powder ATR Scenarios

Powder ATR uses the particle, support-matrix and effective-medium controls described under [Powder Infrared Scenarios](#), with additional controls for the ATR measurement geometry.

The *Powder ATR* choice adds the ATR-material refractive index, the incident angle in degrees, and the s-polarisation fraction. A fraction of zero is pure p polarisation, one is pure s, and 0.5 is an equal mixture. These controls specify the ATR optical geometry; they do not change the particle orientation. See [Calculation of ATR Extinction Spectra](#) for the calculation.

Scenario types: [Powder Infrared](#) · [Powder ATR](#) · [Crystal Infrared](#) · [Powder Raman](#) · [Crystal Raman](#).

5.4.3 Crystal Infrared Scenarios

A crystal scenario describes a stack of planar layers between a superstrate and substrate. The *Dielectric layer* uses the calculation loaded in Main; other materials are selected from the database.

Single crystal methodology selects Transfer matrix or Scattering matrix. The default Scattering matrix is useful for avoiding numerical problems with strongly absorbing or thick layers. *Global azimuthal angle* rotates the stack about its surface normal; *Angle of incidence* measures the incoming beam from that normal. Both angles are in degrees and initially zero.

Under *Layer information*, each row represents a layer. *Material* opens the layer editor. *Thickness* and *Units* set the finite-layer thickness; the semi-infinite superstrate and substrate do not have an editable optical thickness. H, K and L specify the crystalline surface normal, and *Azimuthal* rotates that layer about the normal. These orientation fields do not apply to isotropic materials.

Options selects the layer's optical coherence treatment. *Coherent* retains phase interference; *Incoherent (intensity)* combines intensities, *Incoherent (phase averaging)* averages optical phases, and *Incoherent (non-reflective)* describes a thick non-reflective medium. These optical options are distinct from Raman depth and layer coherence. The up/down arrows move a layer within the stack; the cross deletes it. *New layer...* adds a material selected from the database. Check the layer order after making changes.

Main	Settings	Scenario 1	Plotting	Analysis	3D Viewer	Fitter
Open materials' database	Open materials' database					
Current materials' database	PDIElec/MaterialsDataBase.xlsx					
Support matrix	ptfe Constant permittivity					
Support density	2.200					
Support permittivity	2.000 0.000					
% Air void volume fraction	0.0					
Air void radius (μm)	30.000					
% Mass fraction of dielectric	21.658531					
% Volume fraction of dielectric	10.000000					
Method	Maxwell-Garnett					
Particle radius (μm)	0.000100					
Particle sigma (μm)	0.000000					
Particle shape	Sphere					
Unique direction [abc]	0 0 1					
Ellipsoid a/b eccentricity	1.000000					
ATR material refractive index	4.000					
ATR incident angle	45.0					
ATR S polarisation fraction	0.500					
Powder scenario legend	Scenario 1					
Add another scenario			Delete this scenario			

Fig. 5.5: The Powder ATR scenario adds the measurement geometry.

Main
Settings
Scenario 1
Plotting
Analysis
3D Viewer
Fitter

Open materials' database
Open materials' database

Current materials' database
PDielec/MaterialsDataBase.xlsx

Single crystal methodology
Scattering matrix

Global azimuthal angle
0.00

Angle of incidence
0.00

Layer information

Material	Thickness	Units	H	K	L	Azimuthal	Options	Move
air	1.000	um	0	0	0	0.00	Coherent	
Dielectric layer	1.000	um	0	0	1	0.00	Coherent	
air	1.000	um	0	0	0	0.00	Coherent	
New layer...								

Crystal IR Scenario legend
Scenario 1

Add another scenario
Delete this scenario

Fig. 5.6: The Crystal Infrared scenario for air / ZnO / air.

The Layer Editor

Press a *Material* button to edit the corresponding layer.

Layer type:

ZnO layer

Material:

Dielectric layer

Thickness: um

hkl: Azimuthal:

Lab frame information

crystal a-axis in lab frame: 2.83953, -1.63940, 0.00000
 crystal b-axis in lab frame: 0.00000, 3.27881, 0.00000
 crystal c-axis in lab frame: 0.00000, 0.00000, 5.30308

Fig. 5.7: The layer editor for crystalline ZnO.

Layer type and *Material* identify the layer being edited. *Thickness* and its unit selector describe the finite layer. For a crystal, *hkl* and *Azimuthal* set the surface and rotation. The *Lab frame information* output shows the lattice vectors in laboratory coordinates and their relation to the surface. It is not a second set of editable lattice parameters.

The laboratory Z-axis is normal to the surface and the incidence plane is XZ. p-polarised light lies in this plane; s-polarised light is perpendicular to it. The relationship is described in [Crystal & Laboratory Coordinate Frames](#). *OK* accepts the layer changes and *Cancel* discards them.

Scenario types: *Powder Infrared* · *Powder ATR* · *Crystal Infrared* · *Powder Raman* · *Crystal Raman*.

5.4.4 Powder Raman Scenarios

The *Powder Raman* scenario averages scattering over crystallite orientations and includes the particle's local optical fields. It requires Raman tensors from the loaded calculation. The theory is described in [Theory for Powder Raman](#).

The database, support material, density, complex permittivity, mass/volume fractions and particle-shape controls have the meanings given for powder infrared scenarios. The example uses 10% volume fraction. *Laser wavelength* is in nm, initially 785; the figures use 532. *Raman laser polarisation* selects VV (parallel, the default), VH or HV (crossed), or Unpolarised. *Temperature* is in K, initially 298, and determines the thermal population factor.

Include electro-optic term adds the finite-particle EO contribution to the Raman tensors using the particle's depolarisation field. It is off by default and explicitly enabled here. The input must contain nonlinear susceptibility as well as the response needed for the particle phonon correction. Enabling the checkbox cannot supply missing data. The CASTEP ZnO example contains this response, and switching EO off gives a different spectrum.

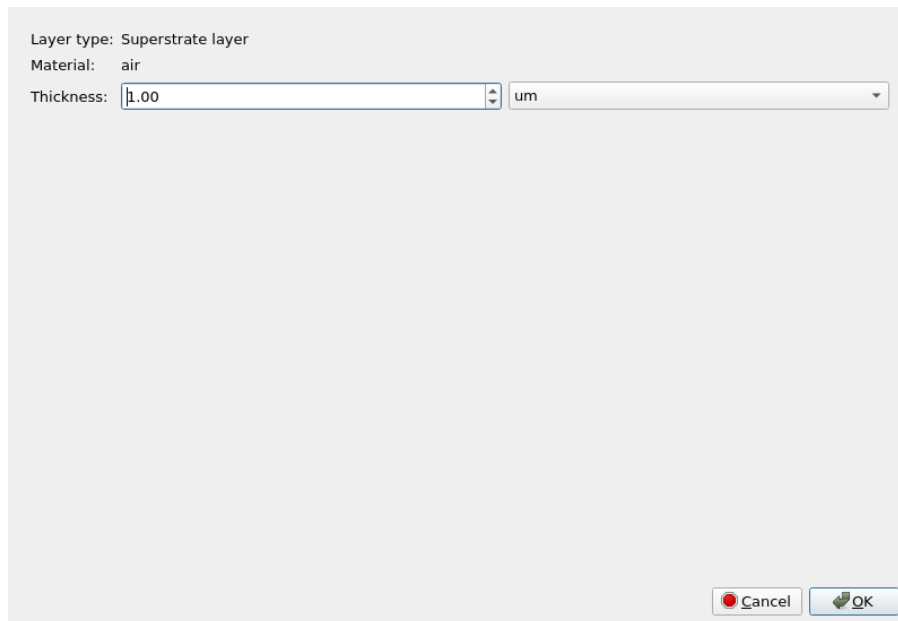


Fig. 5.8: An isotropic material does not require crystallographic orientation controls.

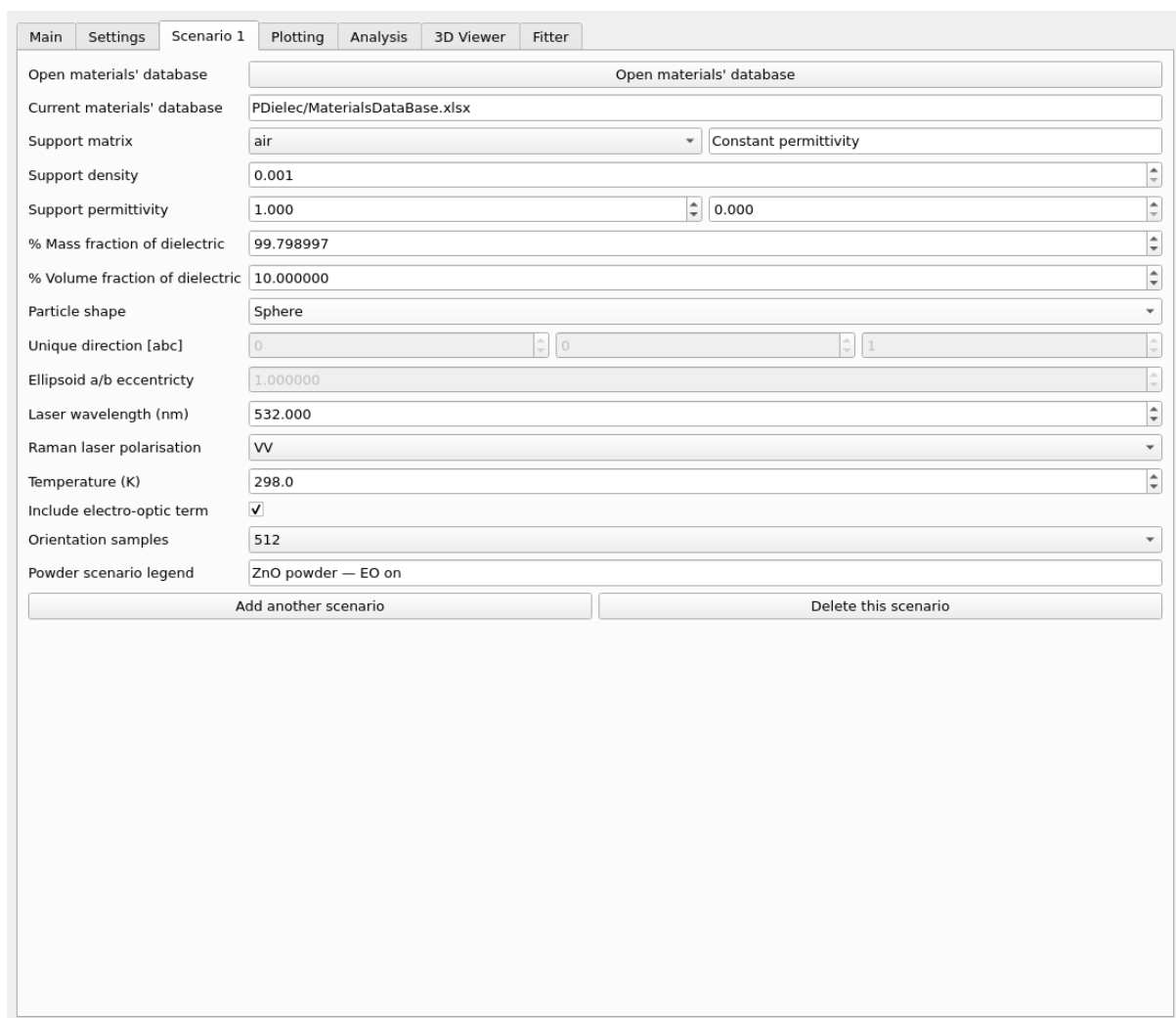


Fig. 5.9: ZnO spheres in air with the EO correction enabled.

Orientation samples selects the number of numerical orientations for non-spherical particles, from 4 to 4096, initially 512. Increase it to check convergence of an anisotropic particle average. Spheres do not need this numerical sampling. Particle shape can affect both the mode frequencies and the Raman response, so a change in shape is not simply a rescaling of peaks.

The screenshot shows the 'Scenario 1' settings in the PDielec software. The 'Particle shape' is set to 'Ellipsoid', which has enabled the 'Unique direction [abc]' controls (a, b, c) and the 'Ellipsoid a/b eccentricity' control. Other settings include 'Support matrix' as 'air', 'Support density' as 0.001, 'Support permittivity' as 1.000, '% Mass fraction of dielectric' as 99.798997, '% Volume fraction of dielectric' as 10.000000, 'Laser wavelength (nm)' as 532.000, 'Raman laser polarisation' as 'VV', 'Temperature (K)' as 298.0, 'Include electro-optic term' checked, 'Orientation samples' as 512, and 'Powder scenario legend' as 'ZnO powder — EO on'.

Fig. 5.10: Selecting Ellipsoid enables the unique-axis and aspect-ratio controls.

Scenario types: *Powder Infrared* · *Powder ATR* · *Crystal Infrared* · *Powder Raman* · *Crystal Raman*.

5.4.5 Crystal Raman Scenarios

The *Crystal Raman* scenario calculates laser and reciprocal scattered fields in a layered sample and combines them with the Raman tensors. The common optical method and incidence controls are followed by Raman controls, the layer table, and the azimuthal sweep controls at the bottom.

Laser wavelength is in nm (default 532) and *Temperature* is in K (default 298). *Integration density* sets the depth quadrature density in points/ μm . The default is 20 and the example uses 40; doubling it to 80 changes this spectrum negligibly. Convergence depends on the layer thickness and optical fields, so check it for a different sample.

Polarisation contains the incident (*in*) and detected (*out*) choices. The defaults are p/p; s is perpendicular to the incidence plane and Unpolarised detection sums the two detector channels as intensities. Alternatively enter *Porto* notation and press *Apply*. This sets a normal-incidence geometry, surface di-

Main
Settings
Scenario 1
Plotting
Analysis
3D Viewer
Fitter

Open materials' database
Open materials' database

Current materials' database
PDielec/MaterialsDataBase.xlsx

Single crystal methodology
Scattering matrix

Global azimuthal angle
0.00

Angle of incidence
0.00

Laser wavelength (nm)
532.0

Temperature (K)
298.0

Integration density (pts/ μm)
40.000

Polarisation
in p out p Porto x(yy)x
Apply

Collection side
superstrate

Collection angle ($^\circ$, -1 = auto)
-1.0

Depth coherence
Incoherent intensity

Layer combination
Incoherent intensities

Include electro-optic term
☒

Layer NAC mode
Snell's law (EO)

Layer information

Material	Thickness	Units	H	K	L	Azimuthal	Options	Move
air	1.000	μm	0	0	0	0.00	Coherent	
Dielectric layer	1.000	μm	0	0	1	0.00	Coherent	
air	1.000	μm	0	0	0	0.00	Coherent	
New layer...								

Azimuthal sweep

Sweep points
36

Plot azimuthal sweep

Crystal Raman Scenario legend
ZnO crystal — EO on

Add another scenario
Delete this scenario

Fig. 5.11: A 1 μm ZnO layer with p/p backscattering and Snell's law (EO).

rection and polarisations; the selectors then show *porto*. For example, $x(yy)-x$ denotes backscattering with y-in/y-out. The common shorthand $x(yy)x$ is also interpreted as backscattering; $x(yy)+x$ explicitly requests forward scattering. Compound polarisations use comma-separated expressions, for example $z(x+y, x+y)z$.

Collection side chooses superstrate (backscattering, default) or substrate (forward scattering). *Collection angle* is in degrees. The default -1 is automatic: retro-backscattering on the superstrate side or collinear forward scattering on the substrate side. An explicit angle equal to the incidence angle specifies specular reflection. The distinction matters at oblique incidence.

Depth coherence selects Coherent amplitude or Incoherent intensity. The former adds amplitudes through a layer before squaring; it requires an appropriate phase-coherent phonon model. The latter integrates local intensities. *Layer combination* separately specifies Incoherent intensities or Coherent amplitudes between active layers. Incoherent depth forces incoherent layer combination and disables that selector. The default depth choice is Coherent amplitude and the default layer combination is Incoherent intensities. The introductory example uses incoherent intensity at both levels. Thin optical layers alone do not establish coherent spontaneous Raman emission.

Include electro-optic term is on by default for crystal Raman. It uses the input nonlinear susceptibility to correct the Raman tensor of a polar mode. The *Layer NAC mode* chooses the phonon-frequency treatment:

Table 5.2: Layer NAC choices

Choice	Meaning
TO	Use bulk transverse-optic frequencies without the polar correction. The EO term is not applied in this mode.
Snell's law	Use the macroscopic scattering geometry to determine the phonon wavevector for the non-analytic correction. This is the default.
Dominant mode	Use the dominant internal optical eigenmode to determine that direction.
All modes	Resolve incident/scattered internal propagation-channel pairs, allowing more than one corrected frequency for an original TO mode.

The last three choices acquire an (*EO*) suffix when the nonlinear response is available and the EO option is enabled. The selected example therefore shows *Snell's law (EO)*. These phonon corrections are separate from solving optical propagation with the transfer/scattering matrix. See [Theory for Crystal Layer Raman](#) for the assumptions and equations.

Modal pair combination offers Group q channels (default), Incoherent pairs and Coherent all pairs. Group q channels combines amplitudes for the same phonon final state before summing intensities over different states. The other two choices are diagnostic alternatives. *Modal-pair final-state model* distinguishes Bulk phase matched from Local incoherent for incoherent-depth, grouped-q calculations. In the bulk model *Modal pair q-angle tolerance*, initially 90° , controls the accepted directions relative to the external momentum transfer. It is inactive for Local incoherent. Controls are enabled only where their combination is meaningful; a hidden All modes control does not change the Snell's-law calculation.

The layer-table columns and editor are shared with Crystal Infrared. Raman response comes from the active dielectric layer; a database optical material does not acquire Raman tensors merely by being inserted into the stack.

Under *Azimuthal sweep*, *Sweep points* specifies equally spaced angles over $0-360^\circ$, excluding the repeated endpoint, initially 36. *Plot azimuthal sweep* calculates the selected geometry at those angles and opens the [Azimuthal Sweep Window](#). Run a spectrum first to define the frequency axis. Each new sweep opens a separate result window.

Main
Settings
Scenario 1
Plotting
Analysis
3D Viewer
Fitter

Open materials' database
Open materials' database

Current materials' database
PDielec/MaterialsDataBase.xlsx

Single crystal methodology
Scattering matrix

Global azimuthal angle
0.00

Angle of incidence
0.00

Laser wavelength (nm)
532.0

Temperature (K)
298.0

Integration density (pts/ μm)
40.000

Polarisation
in p out p Porto x(yy)x Apply

Collection side
superstrate

Collection angle ($^\circ$, -1 = auto)
-1.0

Depth coherence
Incoherent intensity

Layer combination
Incoherent intensities

Include electro-optic term
☒

Layer NAC mode
All modes (EO)

Modal pair combination
Group q channels

Modal-pair final-state model
Bulk phase matched

Modal pair q-angle tolerance
90.00 $^\circ$

Layer information

Material	Thickness	Units	H	K	L	Azimuthal	Options	Move
air	1.000	um	0	0	0	0.00	Coherent	↑ ↓ ✕
Dielectric layer	1.000	um	0	0	1	0.00	Coherent	↑ ↓ ✕
air	1.000	um	0	0	0	0.00	Coherent	↑ ↓ ✕
New layer...								

Azimuthal sweep

Sweep points
36

Plot azimuthal sweep

Crystal Raman Scenario legend
ZnO crystal — EO on

Add another scenario
Delete this scenario

Fig. 5.12: The additional controls shown for All modes (EO).

Scenario types: *Powder Infrared* · *Powder ATR* · *Crystal Infrared* · *Powder Raman* · *Crystal Raman*.

5.5 Plotting Tab

The *Plotting Tab* displays the calculated scenarios together.

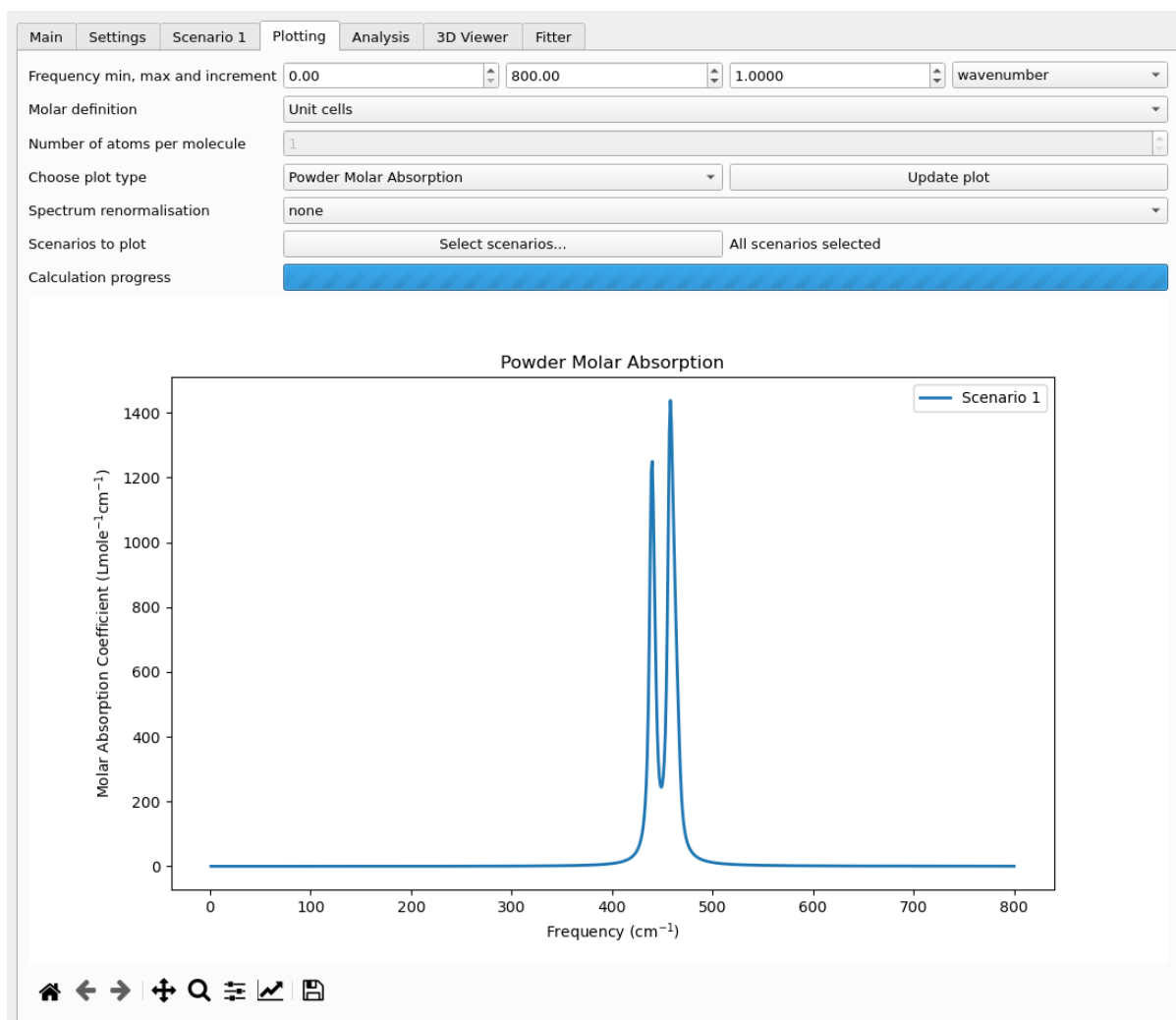


Fig. 5.13: The Plotting Tab for Powder Infrared.

Frequency min, max and increment set the calculation grid. The adjacent unit selector offers wavenumber, THz, GHz and wavelength units from Å to metres. Zero is not a valid wavelength. For Raman, use wavenumber to show Raman shifts unambiguously. A sufficiently small increment is needed to resolve narrow peaks.

Molar definition chooses Unit cells, Atoms or Molecules for molar absorption; *Number of atoms per molecule* supplies the molecular conversion. These controls do not set the normalisation of a Raman tensor. *Choose plot type* offers quantities appropriate to the spectroscopy: powder absorption or permittivity, ATR response, crystal reflectance/transmittance/absorptance, or Raman intensity.

Spectrum renormalisation offers none, max=1 and area=1. Crystal Raman also offers layer depth to divide by active-layer depth. Use none when comparing absolute calculated scales between scenarios; independently normalising curves can obscure an EO or thickness effect. Raman intensities are reported in arbitrary units, not calibrated experimental cross sections.

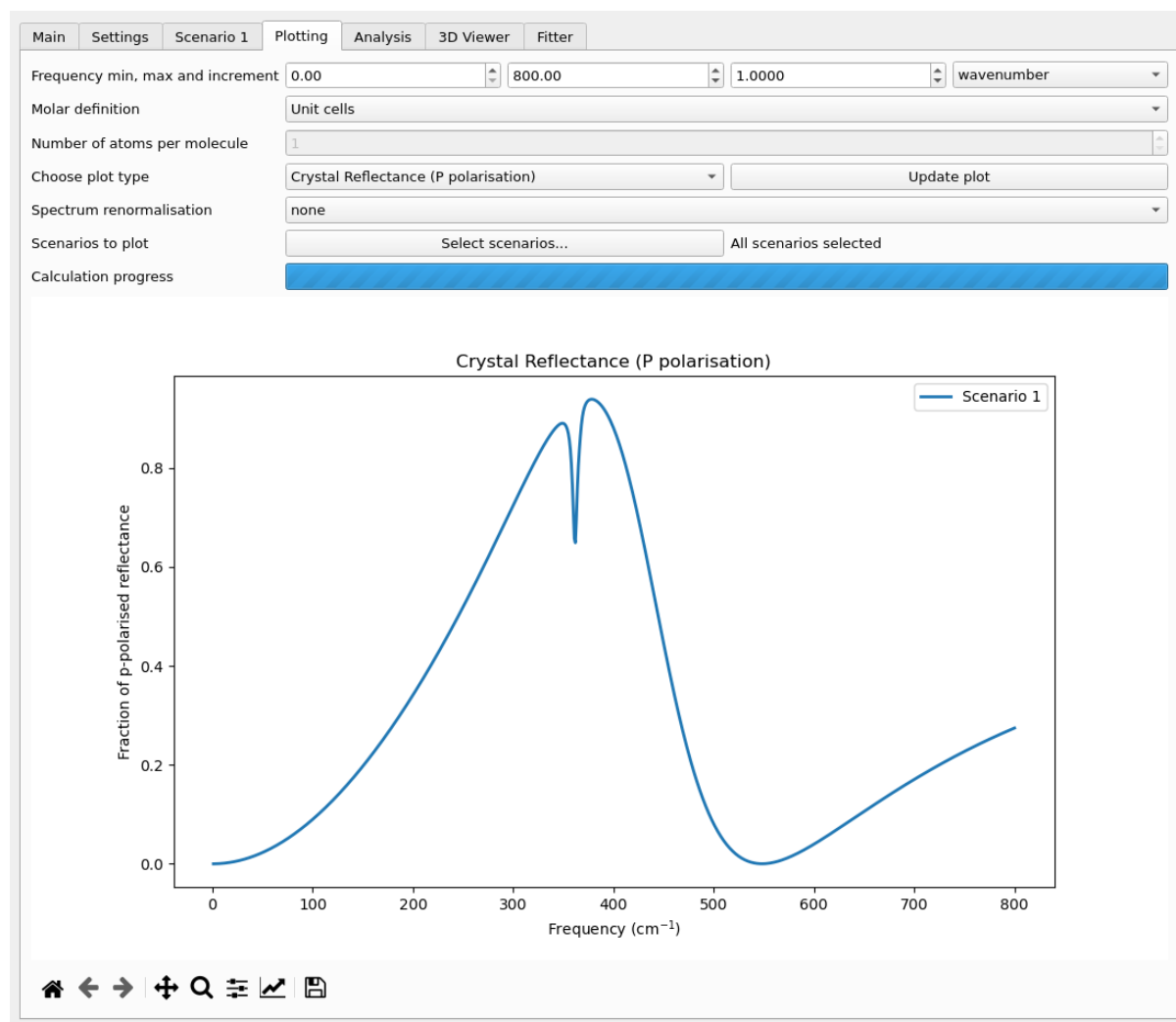


Fig. 5.14: The Plotting Tab for Crystal Infrared.

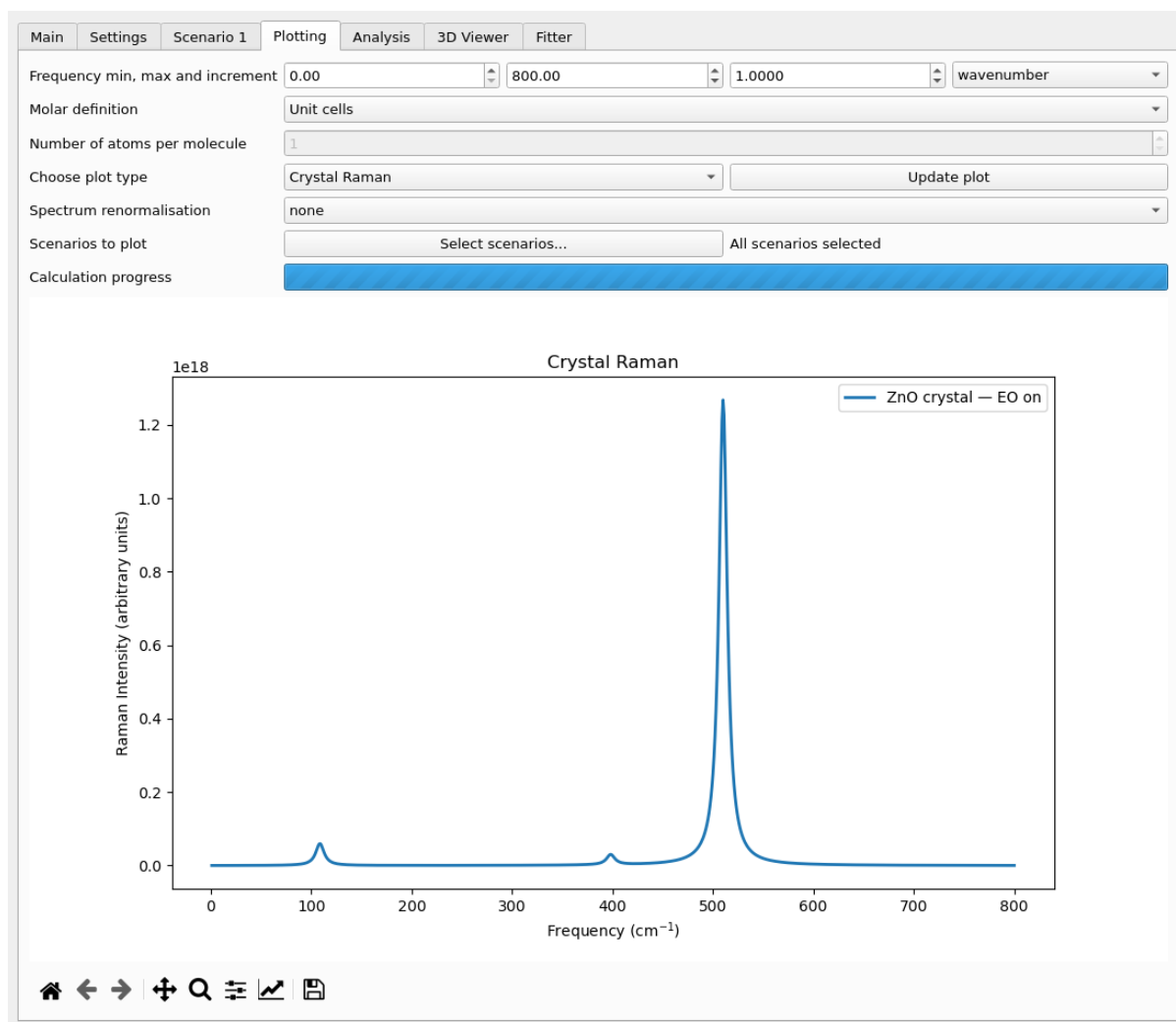


Fig. 5.15: The ZnO Crystal Raman spectrum with EO enabled.

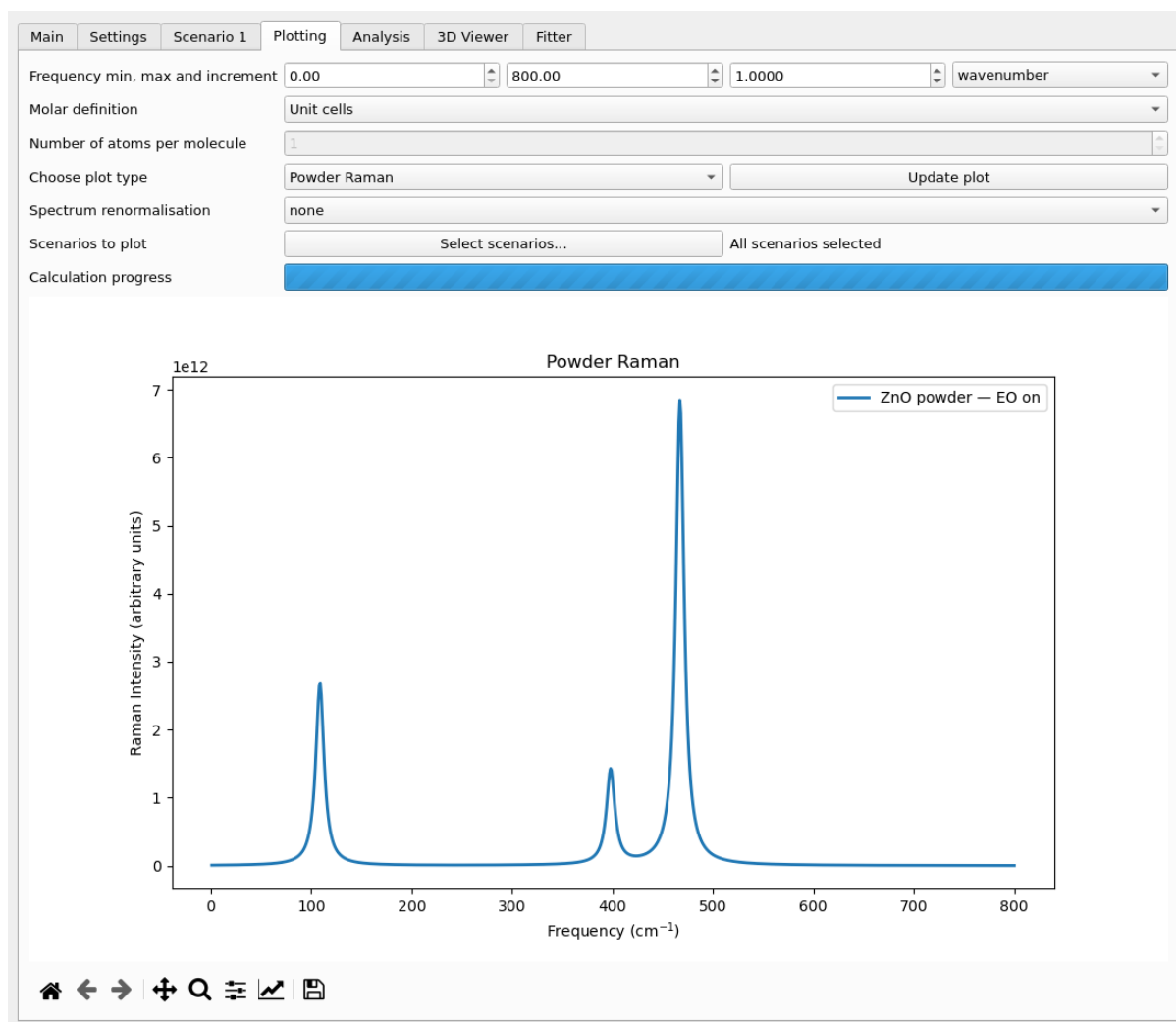


Fig. 5.16: The corresponding Powder Raman spectrum.

Select scenarios... opens a table with a *Plot* checkbox, scenario number and legend for each scenario. *OK* applies the selection; *Cancel* keeps it. The text beside the button reports how many are selected. *Update plot* refreshes the calculations and figure, with *Calculation progress* reporting completion. The Matplotlib toolbar provides Home, back/forward view history, pan, zoom, subplot/curve configuration and image saving; these are display operations and do not change the scenario physics.

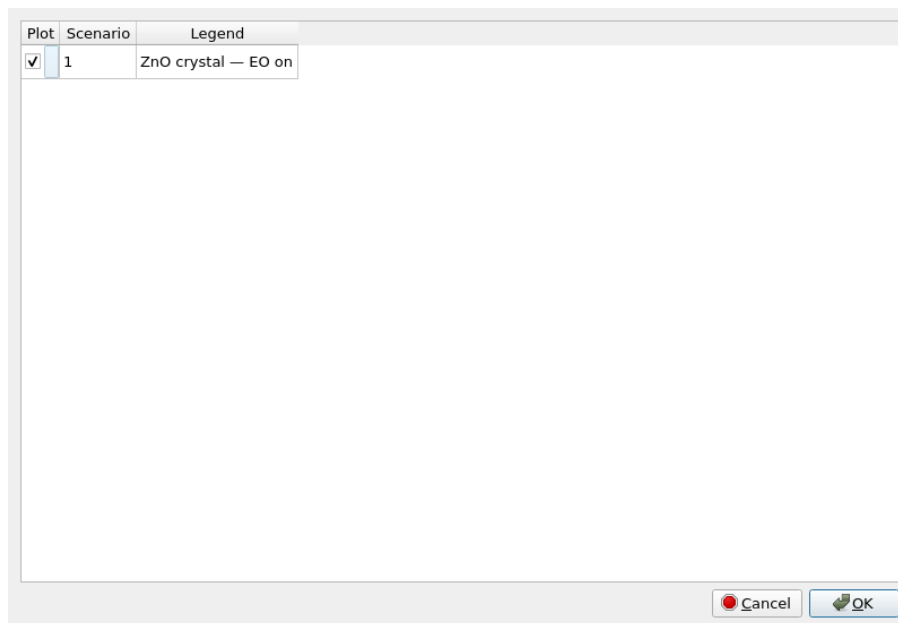


Fig. 5.17: Choosing which scenarios appear in the plot.

5.6 Analysis Tab

The *Analysis Tab* decomposes normal-mode motion into internal and external components or molecular contributions. It needs atomic positions and normal modes, rather than a permittivity-only input.

Minimum frequency and *Maximum frequency* select modes in cm^{-1} . *Bonding scale and tolerance* identify connected atoms using a distance cutoff of $\text{scale} \times (\text{radius } i + \text{radius } j) + \text{tolerance}$; tolerance and *Atomic radii* are in Å. The defaults are scale 1.1 and tolerance 0.1. Edit the individual radii in the table where a different bonding description is needed. *Number of molecules found* is an output of that connectivity analysis. In an extended crystal a connected group should not automatically be interpreted as an isolated molecule.

Bar width (default 0.5) controls the appearance of the bars and *Plot title* sets the heading. *Choose the plot type* switches between Internal vs External and Molecular Composition. Neither option changes the phonon frequencies. The figure toolbar has the same navigation and saving actions as Plotting. See *Phonon Analysis* for the interpretation of the decomposition.

5.7 3D Viewer Tab

The *3D Viewer* displays atomic structure and phonon displacements using the bonding information from Analysis.

Select phonon mode selects a numbered mode; the disabled *Frequency* box reports its frequency in cm^{-1} . *Choose the plot type* selects Animation, Arrows or No arrows or animation. Mouse movement and the wheel rotate and magnify the view. The *Settings* strip contains the following tabs; use its arrows to reach those outside the available width.

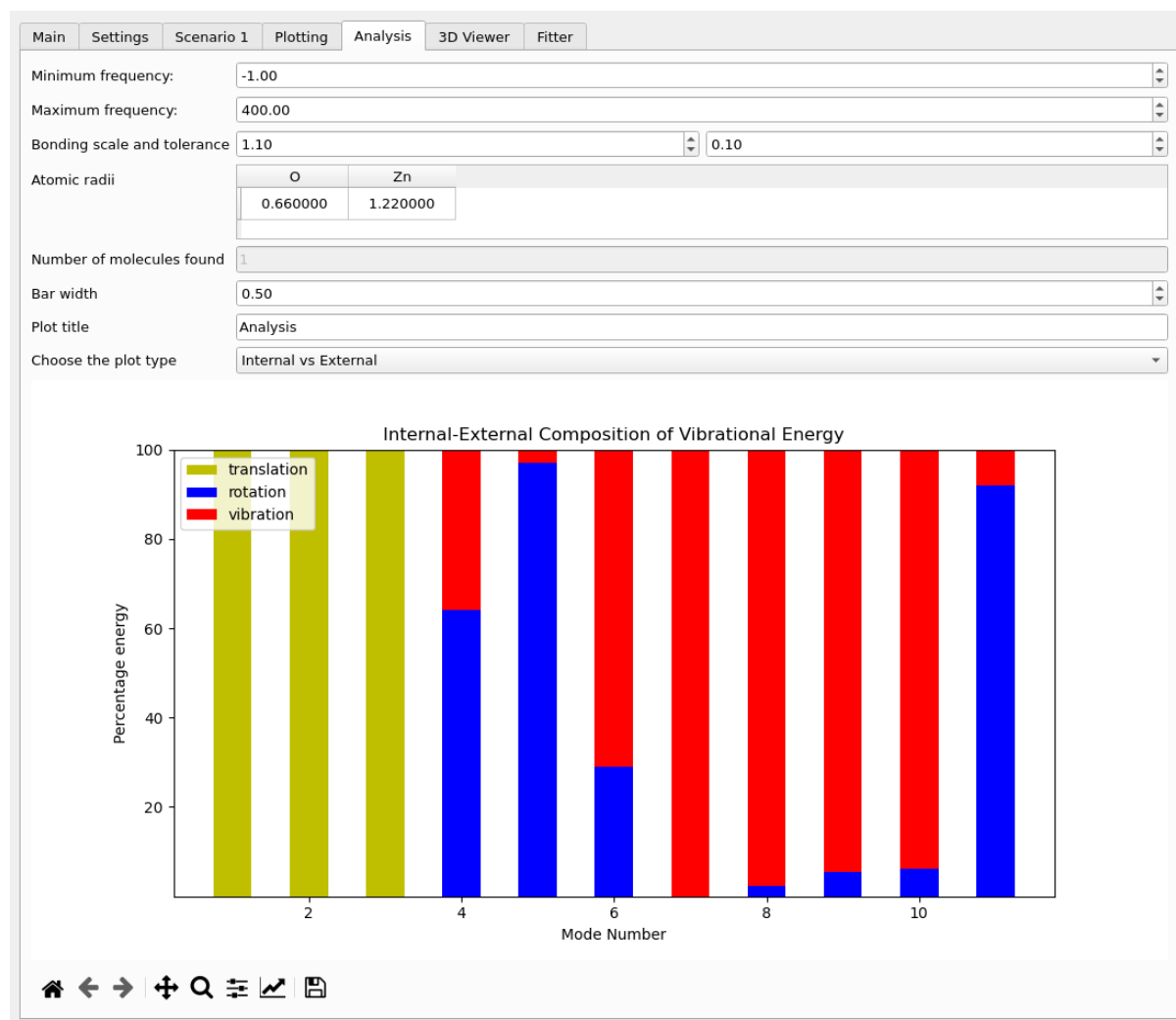


Fig. 5.18: The Analysis Tab for ZnO.

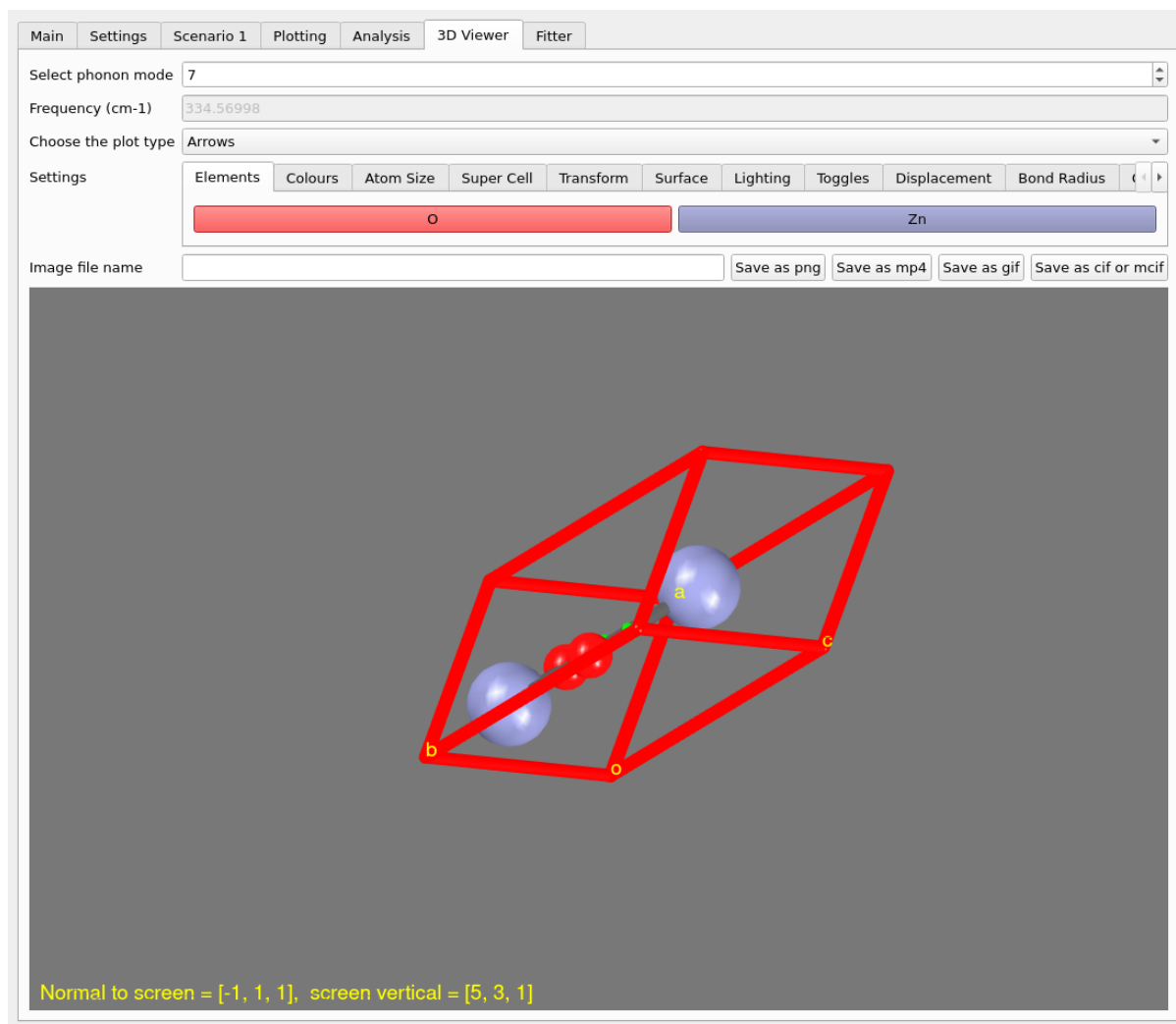


Fig. 5.19: The 3D Viewer showing a ZnO phonon.

Table 5.3: Viewer settings

Tab	Description
Elements	Element buttons open colour selectors for the atomic spheres.
Colours	Separate colours for background, cell outline, bonds and displacement arrows.
Atom Size	Scale factor multiplying the covalent radii for displayed sphere sizes.
Super Cell	Integer repetitions along a, b and c of the original DFT cell.
Transform	Open the cell-transformation matrix editor described below.
Surface	hkl chooses a surface normal and uvw an in-plane direction for the view.
Lighting	Switch light sources 0–7 on or off.
Toggles	Turn cell labels, the cell outline, the orientation indicator, bonds and atoms on or off using the selector.
Displacement	Maximum displayed phonon displacement; controls visual scale, not a thermal displacement amplitude.
Bond Radius	Radius of displayed bond cylinders in Å.
Cell Radius	Radius of the cell-outline cylinders in Å.
Arrow Radius	Radius of displacement arrows in Å.

Image file name supplies the output path. *Save as png* writes a still image; *Save as mp4* and *Save as gif* write animation; *Save as cif* or *mcif* exports the structure or mode information according to the extension. These are outputs of the viewer, not replacements for the calculation input.

5.7.1 The Transformation Window

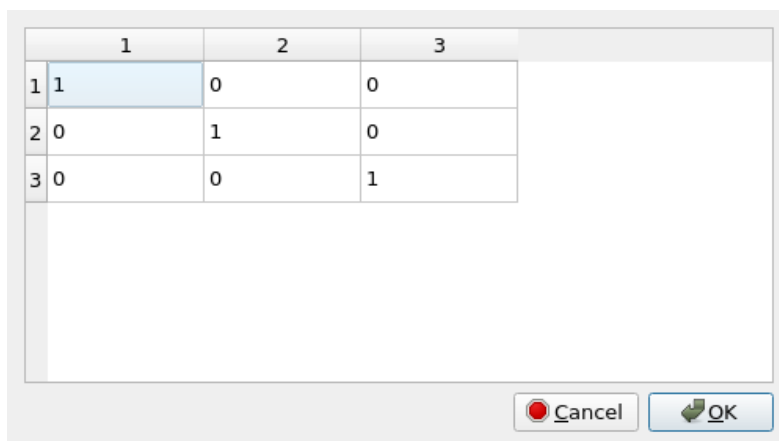


Fig. 5.20: The cell-transformation editor.

Each row of the 3 by 3 table gives a new lattice vector as a linear combination of the old lattice vectors. The identity matrix leaves the cell unchanged. Entries accept numerical expressions such as fractions. *OK* accepts the transformation and *Cancel* discards it. The viewer transformation changes the displayed cell, not the DFT calculation that supplied the modes.

5.8 Fitter Tab

The *Fitter Tab* compares a calculated scenario with an experimental spectrum and can adjust selected linewidths and an optional frequency scale.

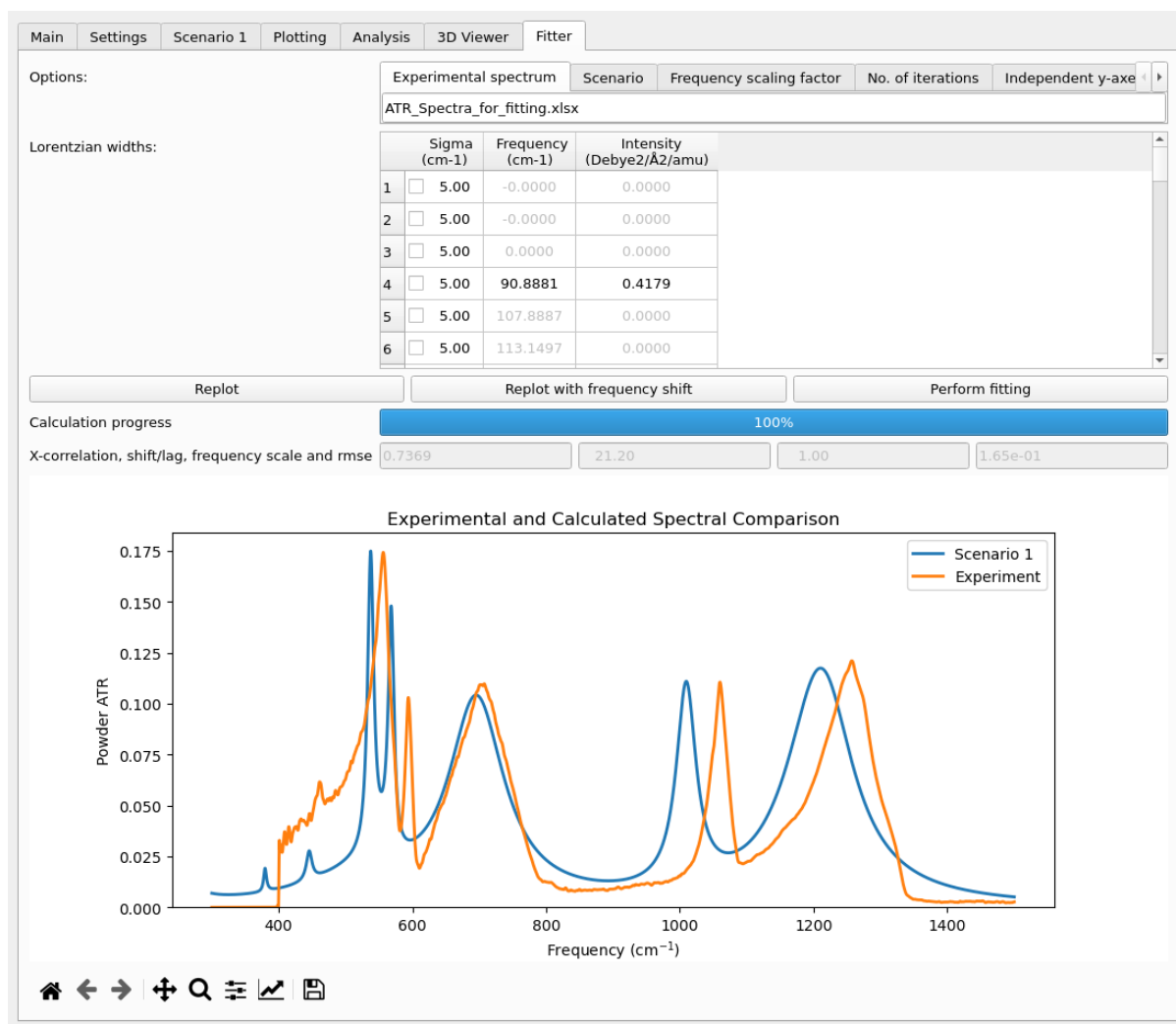


Fig. 5.21: The existing sodium-sulphate ATR fitting example.

The *Options* tabs provide the controls below. The experimental file is an `.xlsx` spreadsheet or `.csv` file with frequency in cm^{-1} in the first column and measured signal in the second, with at most one heading row. This is distinct from an *Experiment* material file. The example uses `Examples/ATR/Na2S042_fit/ATR_Spectra_for_fitting.xlsx` and its matching calculation; it does not use an invented ZnO experimental spectrum.

Table 5.4: Fitter options

Tab	Description
Experimental spec-trum	Path of the measured data. Press Return after entering it to read it.
Scenario	Select the calculated scenario to compare. Its plotted quantity must match the experimental signal.
Frequency scaling fac-tor	Multiplier for calculated frequencies, initially 1.0.
No. of iterations	Optimisation iteration limit, initially 20.
Independent y-axes	Use separate scales for calculated and experimental signals; initially on. Inspect the scales before comparing peak heights.
Fitting type	Minimise x-correlation maximises the correlation through its objective; Minimise spectral difference minimises the spectral error.
Optimise scaling	Include the frequency scaling factor in optimisation; initially off.
Spectral difference threshold	Signal threshold for the spectral-difference objective, initially 0.05.
Baseline removal?	Apply baseline correction to the experimental data; initially off.
HP Filter Lambda	Base-10 logarithm of the Hodrick-Prescott filter parameter, initially 7.

In *Lorentzian widths*, edit sigma values in cm^{-1} and check those to include in the fit. Frequency and intensity columns identify the transitions; they are not independent fitted parameters. *Replot* recalculates the comparison. *Replot with frequency shift* additionally applies the lag that maximises the cross-correlation. *Perform fitting* starts optimisation and changes to an interrupt control while running. The progress bar reports calculation progress.

The four read-only diagnostics are cross-correlation, shift/lag in cm^{-1} , frequency scale and root-mean-square error. Inspect the resulting spectrum as well as these numbers. The fitting uses the frequency range from Plotting; its toolbar controls the view and exports images.

5.9 Raman Polar Window

The *Raman polar plot* button in Settings opens a tensor-level polarisation simulator. It is useful for examining selection rules without constructing an optical layer stack.

The *Raman-active modes* table contains an Include checkbox, mode number, TO frequency, LO frequency when NAC data are available, and total Raman activity in the units chosen in Settings. *Select all active modes* and *Clear selection* change the plotted set.

Light direction Rz gives the propagation/rotation axis and *Zero-angle pol. Pza* the initial incident polarisation, both as Cartesian vectors. Pza should be perpendicular to Rz; changing Rz adjusts Pza accordingly. The read-only *Orthogonality* value is their dot product and should be zero. *Phonon $\hat{q}$ direction* independently sets the crystal-frame direction for the LO and EO corrections. *Sync $\hat{q}$ to Rz* copies the light axis, appropriate to the backscattering approximation.

Include EO correction (χ^2 available) is shown when nonlinear response is available and is initially checked. It is this window's own control, not the scenario checkbox. The LO column likewise uses

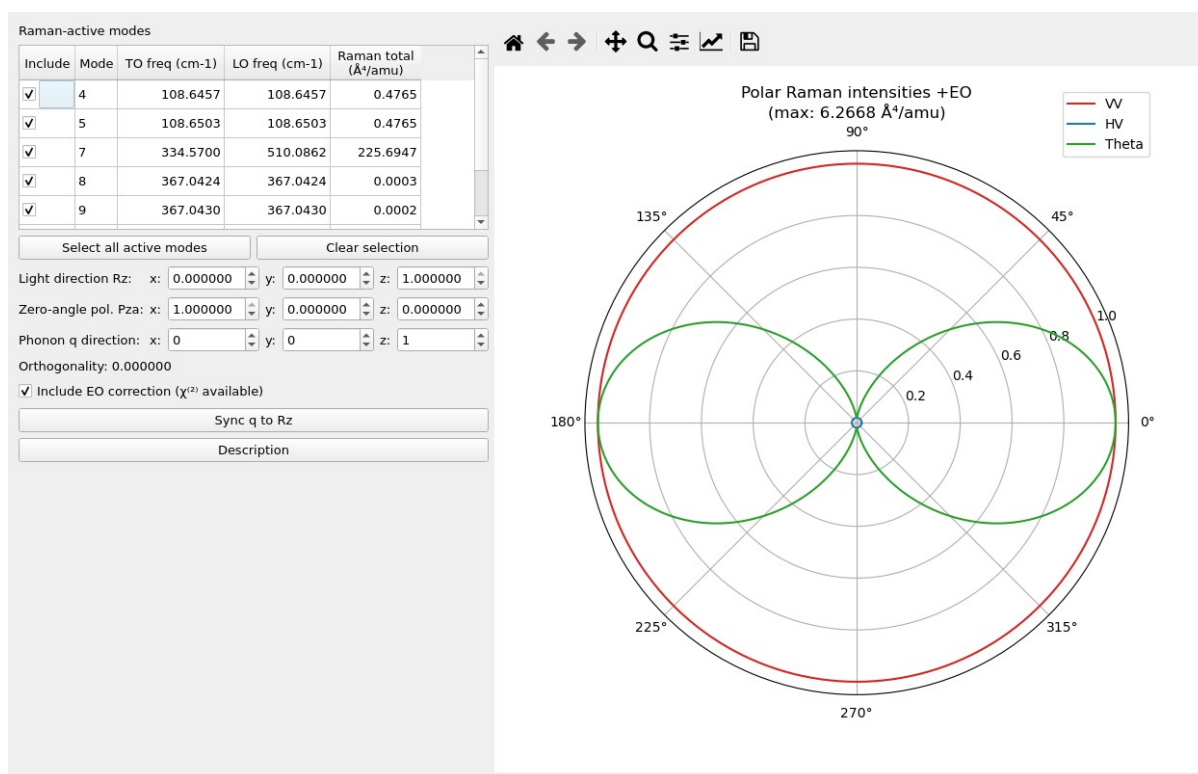


Fig. 5.22: The ZnO Raman polar simulator with the nonlinear response available.

this window's phonon direction. The three curves show parallel VV, crossed HV and a fixed-incident, rotating-analyser geometry. They show tensor-level angular response, without the scenario's layer interference or depth integration. The toolbar saves and navigates the plot. *Description* explains the axes and geometries.

5.10 Azimuthal Sweep Window

The Crystal Raman *Plot azimuthal sweep* button opens a result window using the scenario's layer stack, optical fields, NAC and EO settings. This is a different calculation from the tensor-level polar simulator.

Plot type offers Polar, Colour map, Waterfall and Line. *Mode* chooses an individual mode or All modes (integrated). *Mode display* selects Individual eigenmodes or Group near-degenerate. Grouping reveals *Tol* (cm^{-1}), initially 2.0, which sets the frequency tolerance. Grouping sums the intensities of nearly degenerate modes, avoiding an interpretation that depends on an arbitrary choice of eigenvectors within a degenerate subspace.

Polar and Line show angular intensities. Colour map and Waterfall show the spectral dependence across angles. Switching the plot presentation does not rerun the scenario; to change physical settings, return to the scenario and run a new sweep. The toolbar navigates and saves the figure.

5.11 Saving Settings and Scripting

Save settings on Main writes a Python session file. The script accesses the notebook as `self.notebook` and the setting dictionaries of its tabs. A saved session records the chosen data and options; the display-only controls of auxiliary windows should be set in those windows when required.

The [PDGui Settings Reference](#) lists every tab's dictionary keys, accepted values and their effects, includ-

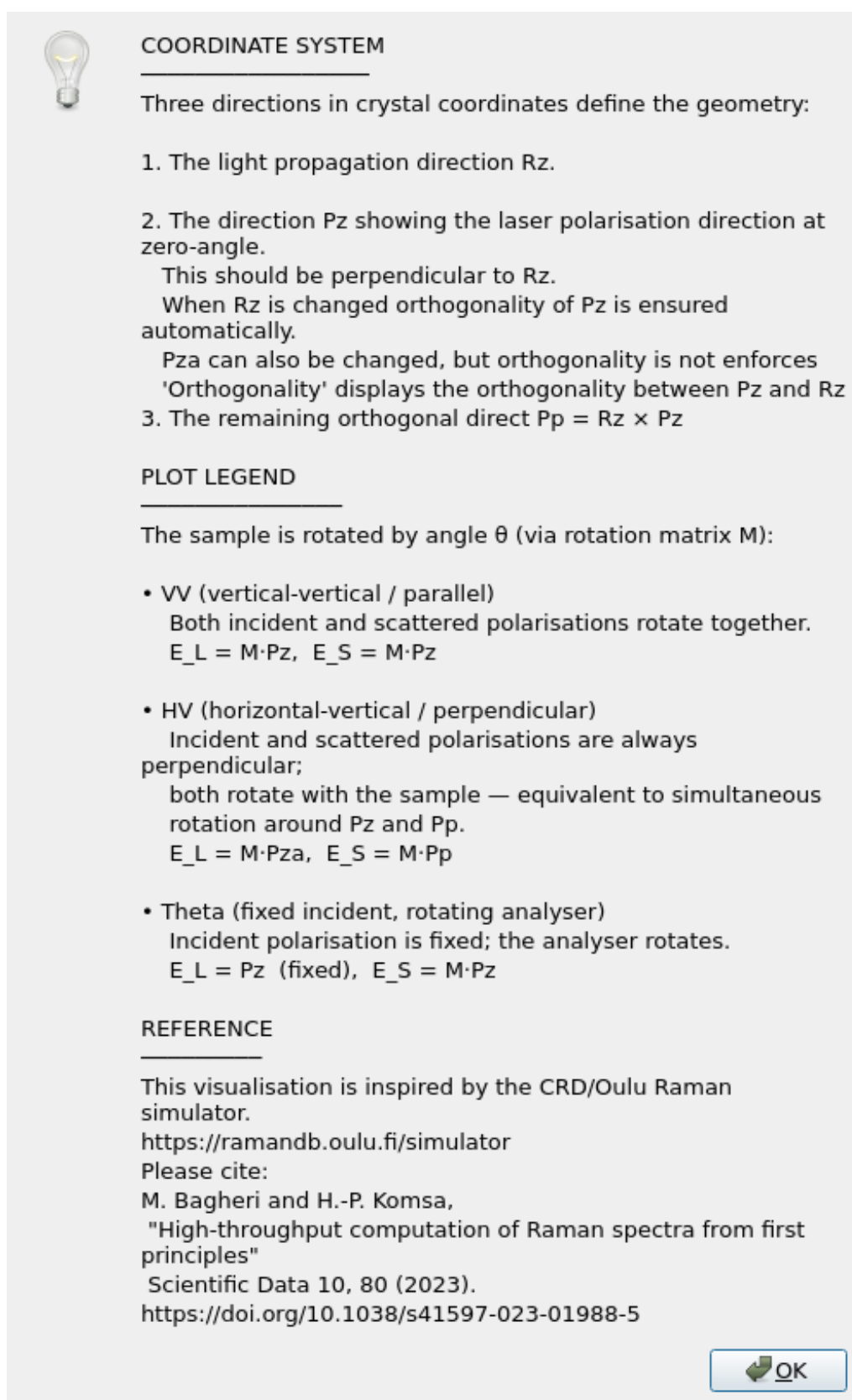


Fig. 5.23: The description window; close it to return to the simulator.

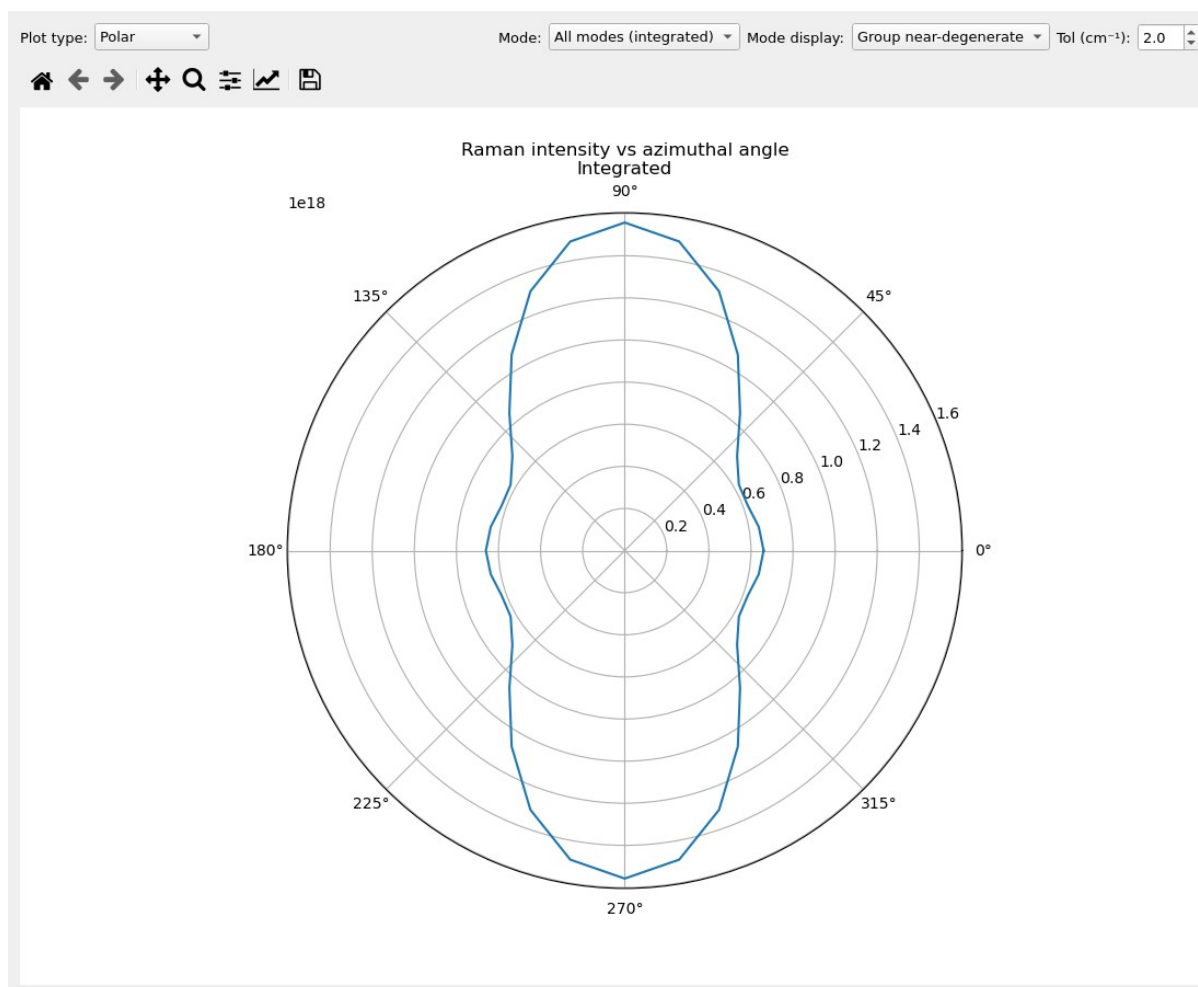


Fig. 5.24: A ZnO (100) azimuthal sweep with mode grouping enabled. This surface was chosen to show the variation under rotation, unlike the (001) main example.

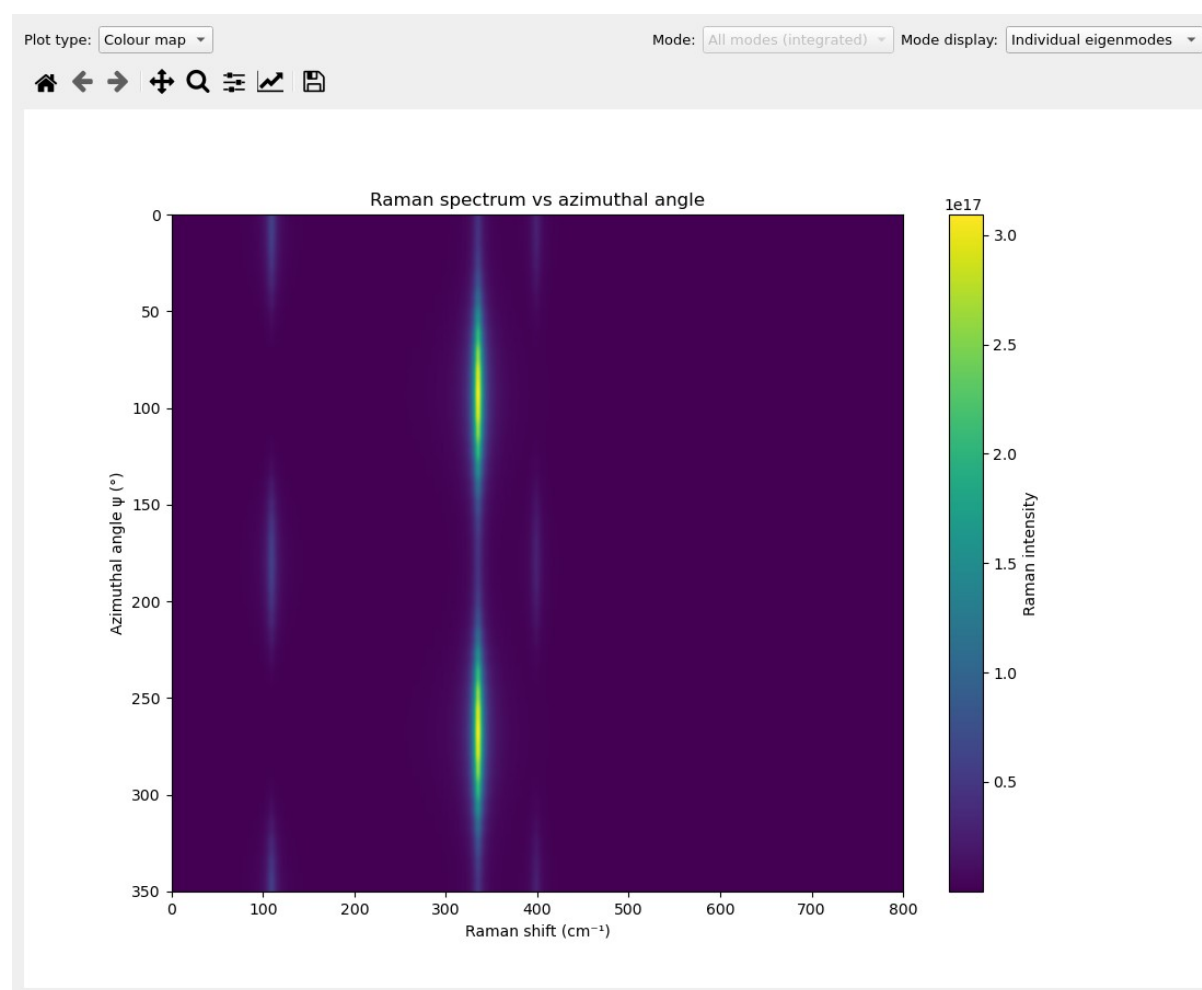


Fig. 5.25: The spectral colour-map view of the same sweep.

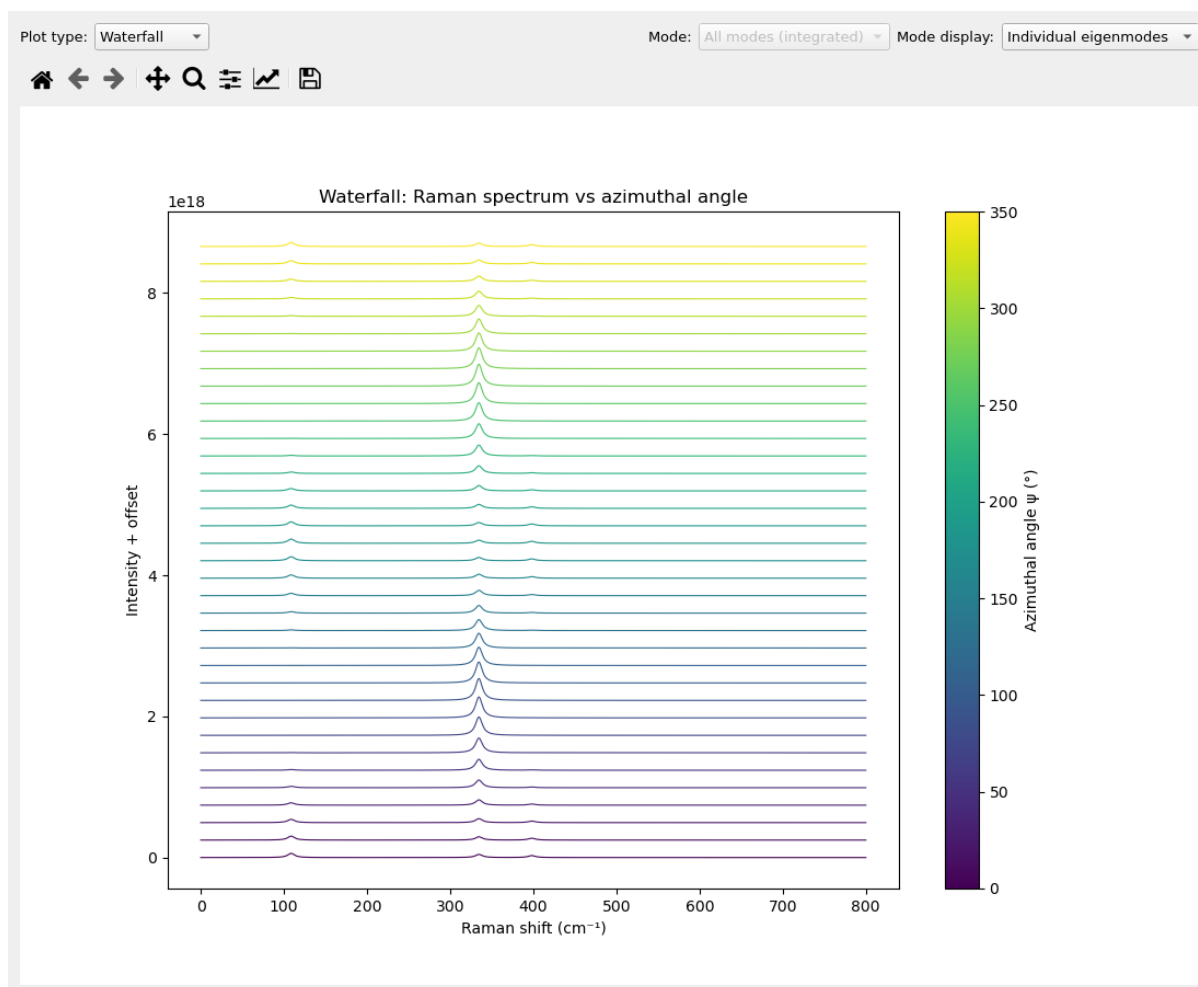


Fig. 5.26: The waterfall view.

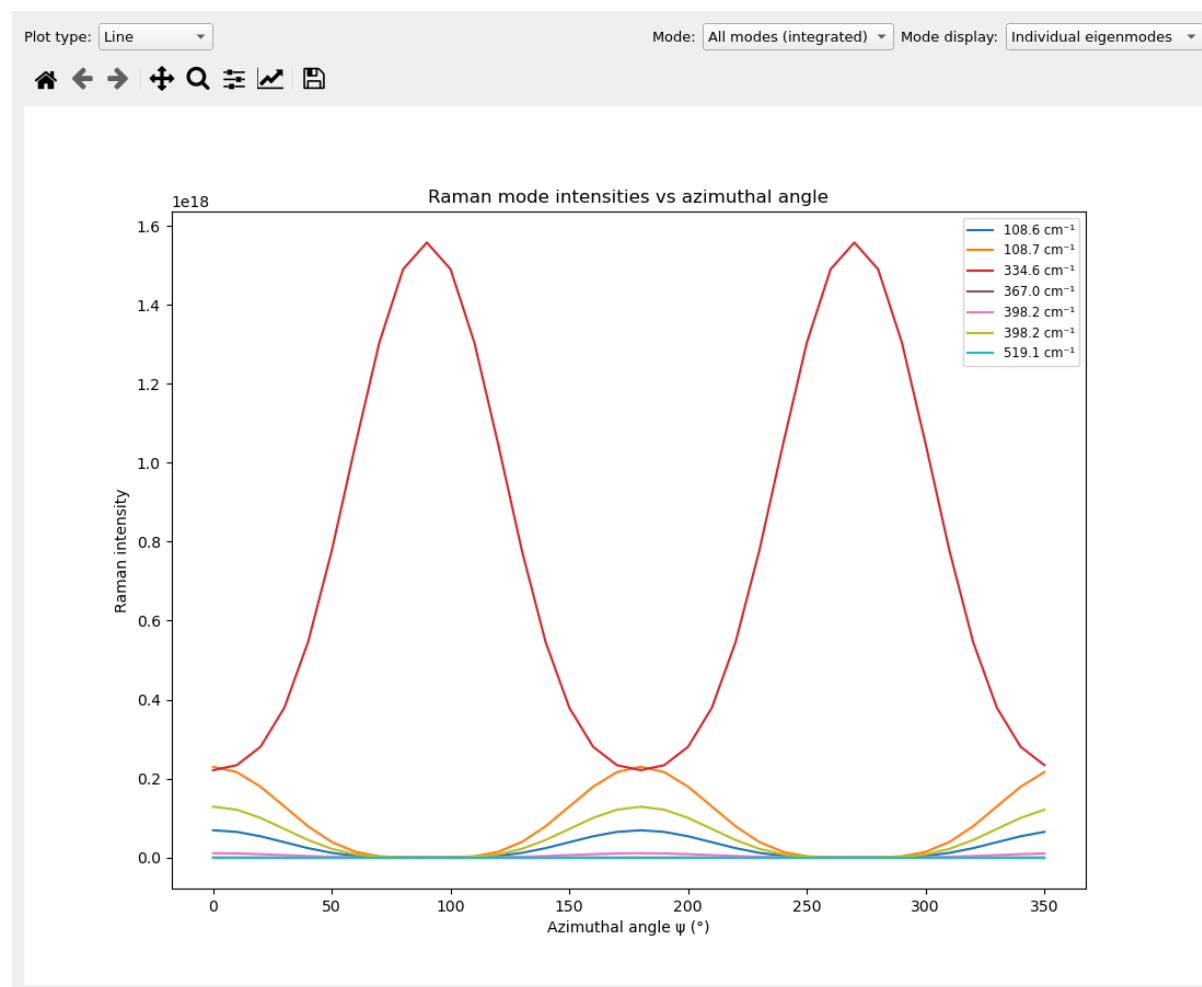


Fig. 5.27: The line view.

ing settings available only through scripts.

Two advanced Crystal Raman settings are available through scripts while being absent from the scenario GUI:

```
tab = self.notebook.scenarios[0]
tab.settings['Coalesce equivalent layers'] = True
tab.settings['Approximate ES'] = False
```

Coalesce equivalent layers, default True, merges adjacent equivalent coherent layers before Raman field integration. Disabling it is useful for diagnosing artificial internal-boundary effects. *Approximate ES*, default False, evaluates the reciprocal detector field once at the laser frequency when set True, instead of at each mode's Stokes frequency. It retains the resolved collection geometry and reuses the incident field only when the optical system and signed angle match. It changes the optical approximation and should be recorded when reporting results.

The powder Raman wavelength in nm is stored as `Raman laser wavelength`. The crystal NAC selector uses `Layer NAC mode` values `none`, `geometry`, `dominant_mode` and `modal_pairs` for TO, Snell's law, Dominant mode and All modes respectively. Use the saved script as the starting point rather than guessing visible-label spellings. In an interactive script, request a refresh after changing settings.

5.12 Performance

PDGui can distribute calculations over worker processes. Use `-cpus N` to limit the number of workers, and `-threading` where a thread-based worker pool is needed. The environment variables `PDIEC_NUM_PROCESSORS`, `PDIEC_THREADING` and `PDIEC_NUM_THREADS` also configure parallelism. Choose settings appropriate to the available memory and calculation size; more workers need not make a small calculation faster.

5.13 Additional settings references

The settings views show each viewer and fitter settings tab separately, so controls hidden by the tab strip in an overview remain visible. The settings reference documents the dictionary keys used in Python session scripts.

5.13.1 Viewer and Fitter Settings Views

These snapshots show the individual settings tabs described in *PDGui*. Use the arrows at the right of a tab strip to reach the remaining tabs. The viewer uses ZnO; the Fitter uses the sodium-sulphate ATR example.

Viewer: Elements



Fig. 5.28: The Elements settings tab.

Viewer: Colours



Fig. 5.29: The Colours settings tab.

Viewer: Atom Size



Fig. 5.30: The Atom Size settings tab.

Viewer: Super Cell



Fig. 5.31: The Super Cell settings tab.

Viewer: Transform

Viewer: Surface

Viewer: Lighting

Viewer: Toggles

Viewer: Displacement

Viewer: Bond Radius

Viewer: Cell Radius

Viewer: Arrow Radius

Fitter: Experimental spectrum

Fitter: Scenario

Fitter: Frequency scaling factor

Fitter: No. of iterations

Fitter: Independent y-axes

Fitter: Fitting type

Fitter: Optimise scaling

Fitter: Spectral difference threshold

Fitter: Baseline removal

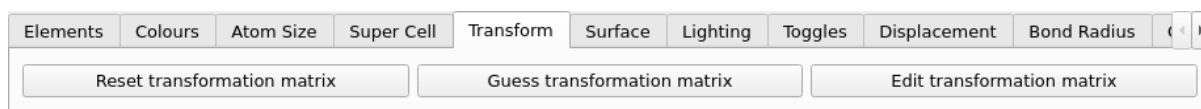


Fig. 5.32: The Transform settings tab.

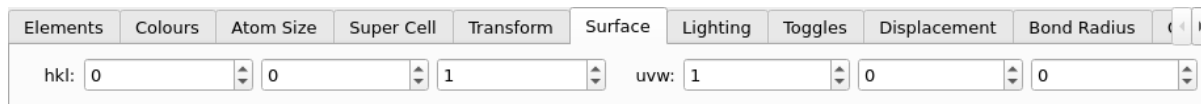


Fig. 5.33: The Surface settings tab.

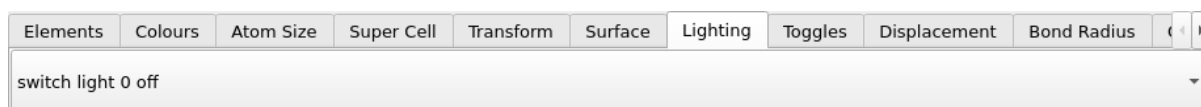


Fig. 5.34: The Lighting settings tab.

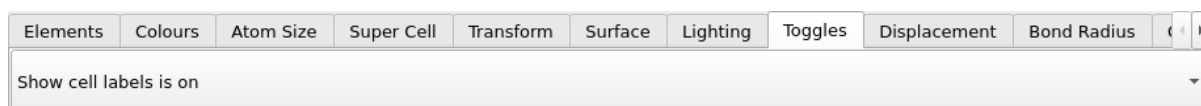


Fig. 5.35: The Toggles settings tab.



Fig. 5.36: The Displacement settings tab.



Fig. 5.37: The Bond Radius settings tab.

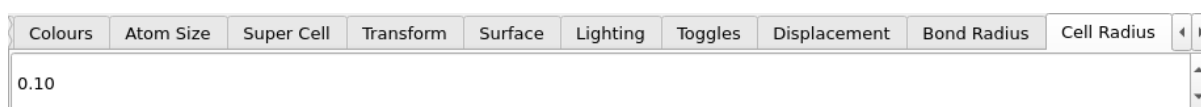


Fig. 5.38: The Cell Radius settings tab.

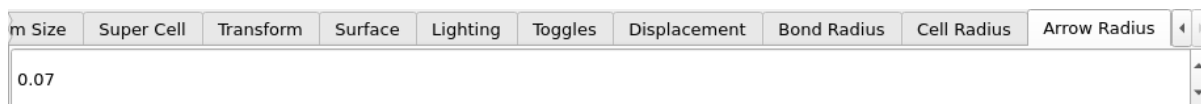


Fig. 5.39: The Arrow Radius settings tab.

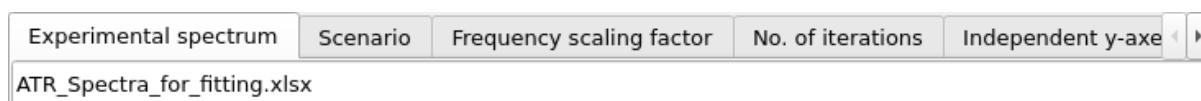


Fig. 5.40: The Experimental spectrum settings tab.

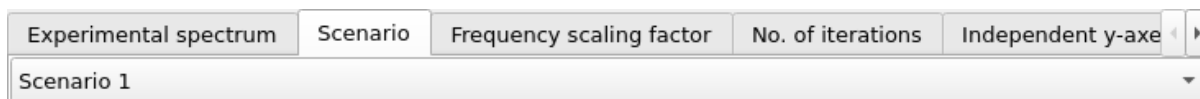


Fig. 5.41: The Scenario settings tab.

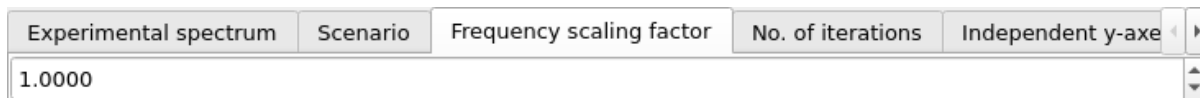


Fig. 5.42: The Frequency scaling factor settings tab.

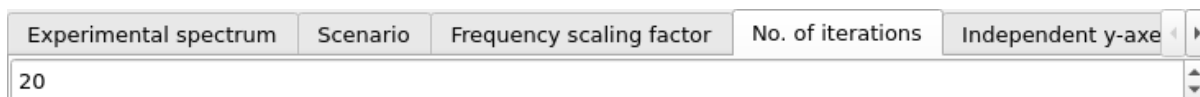


Fig. 5.43: The No. of iterations settings tab.

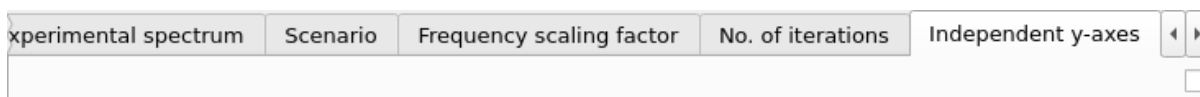


Fig. 5.44: The Independent y-axes settings tab.

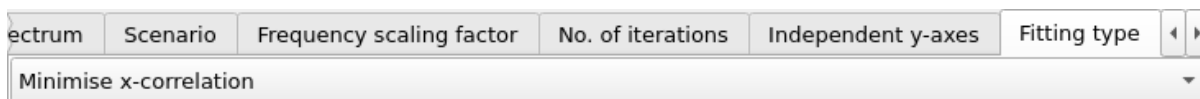


Fig. 5.45: The Fitting type settings tab.

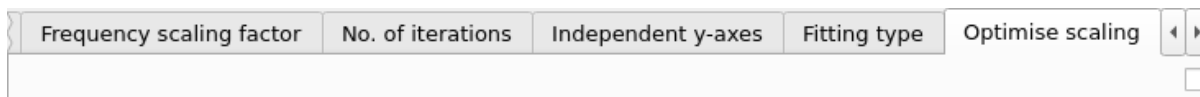


Fig. 5.46: The Optimise scaling settings tab.

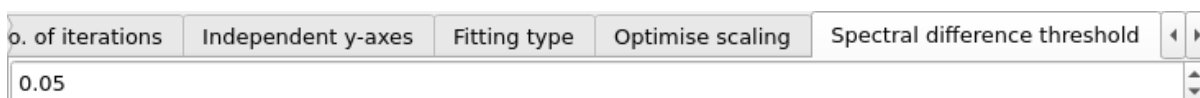


Fig. 5.47: The Spectral difference threshold settings tab.

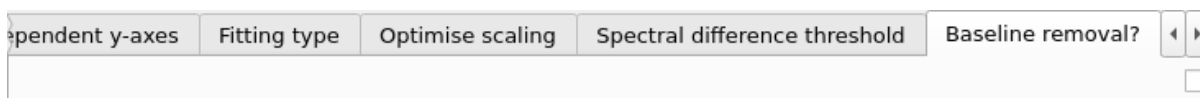


Fig. 5.48: The Baseline removal settings tab.

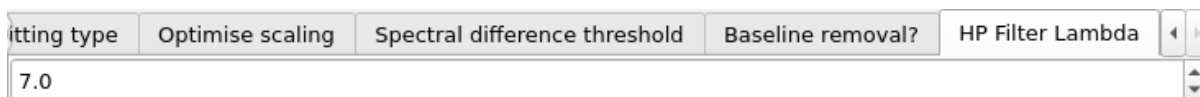


Fig. 5.49: The HP Filter Lambda settings tab.

Fitter: HP Filter Lambda

5.13.2 PDGui Settings Reference

This reference describes the keys in each tab's settings dictionary. Use these exact names, including spaces and capitalisation, in a saved PDGui Python script. The *PDGui* guide describes the corresponding windows.

Save settings on the Main Tab provides a working starting point. For example:

```
tab = self.notebook.settingsTab
tab.settings['Sigma value'] = 5.0

tab = self.notebook.scenarios[0]
tab.settings['Legend'] = 'ZnO, parallel polarisations'

tab = self.notebook.plottingTab
tab.settings['Minimum frequency'] = 100.0
tab.settings['Maximum frequency'] = 800.0
tab.settings['Frequency increment'] = 1.0
```

Run the saved script with `pdgui -script script.py`. Scenario indices start at zero. Set up the required spectroscopy and scenario type before editing its settings; changing a type string alone does not reconstruct a tab.

Values below are Python values: use True and False for switches, quoted strings for choices, and lists for arrays. Defaults refer to a newly created tab and can be replaced by the input file or a restored session. Entries marked *derived*, *legacy* or *reserved* are included because they can occur in the dictionary, but are not independent calculation controls.

Main Tab

Use `self.notebook.mainTab.settings`.

Program

Input reader: 'Abinit', 'Aims', 'Castep', 'Crystal', 'Experiment', 'Finite_field', 'Gulp', 'Phonopy', 'Quantum espresso', 'Vasp' or 'PDGui'. The last choice reads a saved session. See *Other Software Components* for supported inputs, including *Experimental File Format* and *Finite-Difference and Finite-Field JSON Format*.

Menu labels, saved reader names and CLI options are distinct:

Table 5.5: Reader names

Menu label	Saved settings['Program']	CLI -program value (pdgui and preader)
Quantum Espresso	'Quantum espresso'	qe
Finite field	'Finite_field'	finite_field

Output file name

Name of the calculation output to read. Saved sessions use the notebook's working directory for associated files; retain companion input files required by the reader.

Excel file name

Destination spreadsheet name, ending in `.xlsx`. An empty string means no spreadsheet has been selected.

Script file name

Filename used by *Save settings* for the Python session script.

Compatibility mode

Platform metadata, normally 'Linux' or 'Window'. The current tab records this value but does not use it to select a calculation method.

Settings Tab

Use `self.notebook.settingsTab.settings`. These settings are common to all scenarios. See [Theory for Infrared Absorption in Powders](#) for the dielectric response from normal modes and [Theory for Powder Raman](#) for Raman tensors and activities.

Spectroscopy type

'Powder Infrared', 'Powder ATR', 'Crystal Infrared', 'Powder Raman' or 'Crystal Raman'. Selects the family of calculations and available scenario controls. Use a saved session of the required type or the Settings Tab spectroscopy selector to establish matching scenarios.

Eckart flag

Boolean, default True. Projects out translational motion when rebuilding the dynamical matrix, helping to enforce the acoustic modes.

Neutral Born charges

Boolean, default False. Enforces charge neutrality on the Born effective charges before calculating the dielectric response.

Sigma value

Common Lorentzian half-width in cm^{-1} , default 5. Supplies the mode broadening; individual linewidths are held in `tab.sigmas_cm1`.

Mass definition

'average' uses average atomic masses; 'program' uses masses from the calculation; 'isotope' uses the isotope mass table; 'gui' uses user-entered masses from `tab.masses_dictionary`. Default 'average'. Mass changes affect the normal modes and their response.

Optical permittivity

The dimensionless high-frequency dielectric tensor as a 3×3 nested list. Initially None; normally obtained from the reader. Used for the dielectric background, local fields and NAC calculations. See [Theory for Infrared Absorption in Powders](#) and [Theory for Crystal Layer Raman](#).

Optical permittivity edited

Boolean recording a manual tensor override. Set True when supplying Optical permittivity explicitly so that refresh and session saving preserve the edited tensor.

Symmetrise optical permittivity

Boolean, default True. Applies the available crystal symmetry to the optical tensor; False retains the unsymmetrised reader tensor.

Raman activity units

'polarizability' displays activities in $\text{\AA}^4/\text{amu}$; 'epsilon' uses the internal `R_epsilon` activity convention in $\text{\AA}/\text{amu}$. This changes the Settings table and Raman polar display, not the physical scenario spectrum. See [Theory for Powder Raman](#) for the tensor conventions.

The selected-mode mask, individual linewidths, masses and NAC diagnostic controls are not additional settings[] keys. Saved scripts may also contain assignments to tab attributes such as `modes_selected`, `sigmas_cm1` and `masses_dictionary`; retain those assignments when reproducing a session.

Scenario Tabs: Common Settings

Use `self.notebook.scenarios[index].settings`. The following keys are inherited by powder and crystal scenarios.

Legend

Text used to identify the scenario in plot legends and output.

Scenario type

Records the scenario family. This is session metadata; use the notebook's scenario creation and spectroscopy controls to change the type.

Materials database

Path to the Excel materials database, normally `MaterialsDataBase.xlsx`. Provides support matrices and optical layer materials. See the materials database description in *Other Software Components*.

Powder Infrared and Powder ATR

Use a powder scenario's `settings` dictionary. Most particle and support settings are also present in Powder Raman, with the qualifications below. See *Theory for Infrared Absorption in Powders* for effective-medium and particle-shape theory.

Matrix

Support-material name in the materials database, initially 'ptfe'. 'Material defined manually' selects a user-defined constant support. Powder Raman also accepts 'none': this uses the bulk TO Raman response without support local-field or particle NAC/EO corrections.

Matrix density

Support density in g/cm^3 . Normally supplied by the database; a positive manual value is used to convert between mass and volume fractions.

Matrix permittivity

Dimensionless constant support permittivity for a manually defined material; may be complex. Database materials can instead provide a frequency-dependent response.

Mass fraction

Crystallite mass fraction between zero and one, not a percentage. The corresponding volume fraction is calculated using the densities.

Volume fraction

Crystallite volume fraction between zero and one, not a percentage. Controls the effective-medium mixture and scales the powder Raman response. Default 0.1.

Mass or volume fraction

'mass' or 'volume' identifies which fraction is the independent input when converting the other; default 'volume'.

Bubble volume fraction

Fraction of additional air voids, between zero and one; default 0. Used in the powder infrared mixture calculation.

Bubble radius

Air-void radius in μm , default 30. Used by the infrared bubble correction when the bubble fraction is non-zero.

Effective medium method

'Maxwell-Garnett' or 'Bruggeman' selects an effective-medium model; 'Averaged Permittivity' averages the crystallite response; 'Mie' includes finite-size spherical scattering. These choices govern powder infrared/ATR, not the separate Raman tensor calculation. See *Theory for Infrared Absorption in Powders*.

Particle shape

'Sphere', 'Needle', 'Plate' or 'Ellipsoid'. Determines the depolarisation tensor and hence local fields and particle response.

Particle size(mu)

Particle radius in μm , default 0.0001. Relevant to finite-size infrared scattering, particularly Mie; not a Raman mode linewidth.

Particle size distribution sigma(mu)

Width parameter for the infrared particle-size distribution, default 0 for a single size. This is the lognormal distribution parameter, not a Lorentzian frequency width.

Ellipsoid a/b

Axial aspect ratio for an ellipsoid; default 1 gives a sphere.

Unique direction - h

First component of the particle's unique direction.

Unique direction - k

Second component of the particle's unique direction.

Unique direction - l

Third component of the particle's unique direction. Together the three keys specify a direct-lattice [abc] axis for a needle/ellipsoid or an (hkl) surface normal for a plate. The default is [0, 0, 1]; spheres do not use this direction. See *Theory for Infrared Absorption in Powders*.

ATR material refractive index

Real refractive index of the ATR prism, default 4. Powder ATR only; see *Calculation of ATR Extinction Spectra*.

ATR theta

ATR incidence angle in degrees, default 45.

ATR S polarisation fraction

Fraction of s-polarised illumination: 0 is pure p, 1 is pure s, and 0.5 is an equal mixture. Used to combine ATR responses.

Powder Raman

These keys supplement the common scenario and powder particle settings. See *Theory for Powder Raman* for orientation averaging, internal fields, particle-corrected phonons and the electro-optic contribution.

Raman laser wavelength

Laser wavelength in nm. A positive number, default 785; determines the excitation and Stokes frequency factors. Do not supply a frequency in Hz or cm^{-1} .

Raman laser polarisation

'VV' selects parallel polarisations, 'VH' and 'HV' select crossed polarisations, and

'Unpolarised' selects the unpolarised response. Default 'VV'.

Raman temperature

Temperature in kelvin, default 298. Controls the Stokes thermal population factor.

Raman orientation samples

Number of numerical orientations for non-spherical particles. The GUI offers 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096; default 512. Increase to check convergence. Spheres use an analytic average.

Raman electro-optic term

Boolean, default False for a new powder scenario. Includes the electro-optic contribution when particle correction data and $\chi^{(2)}$ are available. Has no effect for `Matrix = 'none'`. See [Theory for Powder Raman](#).

Crystal Infrared and Crystal Raman: Layers

Use a crystal scenario's settings dictionary. See [Theory for Single Crystal Infrared Optics](#) for the optical stack, orientation conventions and coherent/incoherent propagation, and [Theory for Crystal Layer Raman](#) for Raman fields.

Global azimuthal angle

Rotation of the stack about its surface normal, in degrees; default 0.

Angle of incidence

Incident angle from the surface normal in the superstrate, in degrees; default 0. See [Crystal & Laboratory Coordinate Frames](#).

Mode

'Scattering matrix' (default) or 'Transfer matrix' for the infrared optical solver. Scattering matrices avoid numerical problems that transfer matrices can encounter in strongly absorbing thick layers. See [Theory for Single Crystal Infrared Optics](#).

Layer material names

List of database material names, in order from superstrate to substrate. 'Dielectric layer' denotes the material read from the calculation. The initial stack is ['air', 'Dielectric layer', 'air'].

Layer hkl

One three-component (hkl) surface normal per layer. Used for anisotropic layers; isotropic layers do not need a crystal orientation.

Layer azimuthals

One in-plane rotation in degrees per layer, applied with its surface orientation and the global azimuthal angle.

Layer thicknesses

One thickness per layer, in the corresponding `Layer thickness units`. The first and last layers are semi-infinite ports, so their stored thicknesses do not represent finite films.

Layer thickness units

List of length-unit strings: 'ang' (Å), 'nm', 'um', 'mm' or 'cm'. The GUI offers the latter four; scripts can also use 'ang'. Keep the list aligned with the thicknesses.

Layer incoherent options

One optical propagation choice per layer: 'Coherent' retains phase; 'Incoherent (intensity)' propagates intensities; 'Incoherent (phase averaging)' averages sampled

phases; 'Incoherent (phase cancelling)' removes propagation phase; and 'Incoherent (non-reflective)' treats a non-reflecting thick layer. See *Theory for Single Crystal Infrared Optics*. These are optical layer choices, distinct from the Raman Depth coherence and Layer combination.

Layer dielectric flags

Boolean list marking layers that use the loaded dielectric material. Keep these flags consistent with Layer material names; the initial three-layer stack uses [False, True, False].

All seven layer lists must have the same length. Saved sessions restore the layer objects from these lists; live layer edits should use the layer editor so that both the objects and their serialised settings remain consistent.

Percentage average incoherence

Phase range as a percentage of a full 2π cycle, default 100. Used for layers with phase averaging.

Number of average incoherence samples

Number of phase samples per phase-averaged layer, default 10. Multiple such layers require combinations of their sampled phases.

Percentage partial incoherence

Strength of the infrared partial-incoherence sampling, default 0 (disabled). Randomly perturbs the active crystal thickness, orientation and incidence angle; this is separate from phase averaging.

Partially incoherent samples

Number of random samples for that infrared average, default 20.

Filter kernel size

Savitzky–Golay smoothing window for the partially incoherent infrared spectrum. Default 1 disables smoothing; an enabled window must be odd, exceed the polynomial order, and fit within the spectrum length.

Filter polynomial size

Polynomial order for that smoothing filter, default 3. Relevant only when the kernel size is greater than two.

Frequency units

Legacy stored value, initially 'wavenumber'. The current crystal tab does not consume it; select display units with Plotting's singular Frequency unit key.

Crystal Raman: Scattering and Coherence

These settings supplement the crystal layer settings. See *Theory for Crystal Layer Raman* for the scattering geometry, reciprocal fields, NAC/EO corrections and the different coherence choices.

Laser wavelength nm

Positive laser wavelength in nm, default 532.

Incident polarisation

'p' or 's' selects the incident channel; default 'p'. Mixed coherent polarisations can be represented by a two-component complex Jones vector in p/s order, as produced by a saved Porto setup.

Detected polarisation

'p' or 's' selects an analyser, while 'unpolarised' sums the two detected intensities. Default 'p'. A two-component complex p/s Jones vector specifies a coherent analyser combination. Summing unpolarised intensities is different from summing field amplitudes.

Temperature K

Temperature in kelvin, default 298, used in the Stokes population factor.

GL point density

Gauss–Legendre integration-point density per μm of active material, default 20. Increase to check depth-integration convergence.

Collection side

'superstrate' collects backscattered light on the incident side; 'substrate' collects forward-scattered light. Default 'superstrate'.

Collection angle

Collection angle in degrees. -1 (default) requests automatic geometry; other values specify a signed angle explicitly. Automatic collection uses minus the incidence angle on the superstrate side (retro-backscattering) and plus the incidence angle on the substrate side (forward collection). An explicit positive angle equal to incidence selects the historical specular-reflection geometry on the superstrate side.

Layer combination

'Incoherent intensities' (default) sums the layer intensities; 'Coherent amplitudes' sums complex layer amplitudes before squaring. Coherent layer combination is unavailable with incoherent depth integration.

Depth coherence

'Coherent amplitude' (default) integrates amplitudes over depth before squaring. 'Incoherent intensity' integrates local intensities. This is a Raman emission choice, not the optical layer's propagation coherence.

Approximate ES

Script-only boolean, default False. If True, evaluates the reciprocal detector field once at the laser frequency instead of at each mode's Stokes frequency, retaining the resolved collection geometry. The incident field is reused only when the optical system and signed angle match. This changes the optical approximation; see *Theory for Crystal Layer Raman*.

Coalesce equivalent layers

Script-only boolean, default True. Merges adjacent equivalent coherent layers before Raman integration. False retains the original layer boundaries and is useful for subdivision checks.

Raman electro-optic term

Boolean, default True for crystal scenarios. Includes $\chi^{(2)}$ in the NAC-corrected Raman tensors when those data are available. Disabling it retains the NAC frequency correction without the EO tensor contribution.

Layer NAC mode

'none' uses bulk TO modes; 'geometry' (default, displayed as *Snell's law*) obtains phonon momentum from the macroscopic geometry; 'dominant_mode' uses the dominant internal Berreman mode; 'modal_pairs' (*All modes*) resolves the internal optical mode pairs. GUI labels may append (*EO*); that suffix is not part of the stored value.

Modal pair combination

Used with Layer NAC mode = 'modal_pairs'. 'Group q channels' (default) combines amplitudes sharing a phonon momentum and detector channel. 'Incoherent pairs' squares pairs separately for comparison with older calculations. 'Coherent all pairs' is a diagnostic option that mixes distinct phonon final states. See *Theory for Crystal Layer Raman*.

Modal pair final-state model

For incoherent-depth modal-pair calculations, 'Bulk phase matched' (default) selects internal

momenta relative to the external transfer. 'Local incoherent' assigns local field components to the externally selected phonon state without the q-angle filter.

Modal pair q-angle tolerance

Maximum angle in degrees between internal and external momentum transfer, from 0 to 90. Default 90 accepts the forward hemisphere in the bulk phase-matched model. Inactive for local incoherent depth.

Modal pair include zero q

Reserved/legacy entry, initially None. It is retained in the dictionary but is not read by the current scenario calculation; it does not control zero-momentum channel selection.

Azimuthal sweep points

Positive integer number of angles over 0–360°, excluding the repeated endpoint; default 36. Used when opening an azimuthal sweep.

Porto notation

Stored normal-incidence geometry text, initially empty. For example, 'x(yy)x' denotes the backscattering shorthand and 'x(yz)+x' explicit forward scattering. Applying the Porto entry in the GUI sets the layer normal, angles, collection side and polarisations. Preserve these accompanying values in a script; assigning the text alone is not a replacement for applying the geometry. See *Crystal Raman Scenarios*.

Phonon boundary correction

Legacy key from older sessions. With Layer NAC mode = 'none', the old values 'NAC', 'slab-environment' and 'slab' map to 'geometry'. Use Layer NAC mode in new scripts.

Plotting Tab

Use `self.notebook.plottingTab.settings`.

Minimum frequency

Lower calculation limit in cm^{-1} , default 1. Stored in wavenumbers regardless of the chosen display unit.

Maximum frequency

Upper calculation limit in cm^{-1} , default 200.

Frequency increment

Calculation-grid spacing in cm^{-1} , default 0.2. Use sufficiently small spacing to resolve the selected mode linewidths.

Frequency unit

Display unit: 'wavenumber' (cm^{-1}), 'THz', 'GHz', or the wavelength units 'ang', 'nm', 'um', 'mm', 'cm', 'm'. Wavelength is reciprocal to frequency; the calculation-grid settings above remain in cm^{-1} .

Molar definition

'Unit cells' (default), 'Atoms' or 'Molecules' sets the amount used to normalise powder molar absorption. See *Theory for Infrared Absorption in Powders*.

Number of atoms

Number of atoms per molecule for Molar definition = 'Molecules'; default 1. Used with the unit-cell atom count and volume to calculate the molecular concentration.

Plot type

Exact curve name. Powder infrared offers 'Powder Molar Absorption', 'Powder Absorption', 'Powder Real Permittivity' and 'Powder Imaginary Permittivity'.

Powder ATR additionally offers 'Powder ATR'. Raman uses 'Powder Raman' or 'Crystal Raman'. Crystal infrared offers 'Crystal Reflectance (P polarisation)', 'Crystal Reflectance (S polarisation)', 'Crystal Transmittance (P polarisation)', 'Crystal Transmittance (S polarisation)', 'Crystal Absorbance (P polarisation)' and 'Crystal Absorbance (S polarisation)'. Preserve the historical spelling *Absorbance* in scripts. See *Theory for Infrared Absorption in Powders*, *Calculation of ATR Extinction Spectra*, *Theory for Single Crystal Infrared Optics*, *Theory for Powder Raman* and *Theory for Crystal Layer Raman* for the corresponding observables.

Spectrum renormalisation

'none' retains the spectrum's units; 'max=1' divides each curve by its peak; 'area=1' divides by its integrated area. 'layer depth' is available for Crystal Raman and divides by the total active-layer depth. These are plotting normalisations.

Scenarios to plot

List of booleans, one per scenario in notebook order, for example [True, False, True]. Controls visibility. An initially empty list is expanded by the tab to match the scenarios.

concentration

Derived molar concentration in mol/L for the selected molar definition. Recalculated from the cell volume, atom count and molar definition; set those inputs rather than this value directly.

cell concentration

Derived unit-cell concentration in mol/L, used in molar normalisation.

Analysis Tab

Use `self.notebook.analysisTab.settings`. See *Phonon Analysis* for the internal/external and molecular decomposition of modes.

Radii

List of covalent radii in Å, in the tab's element order. Initially None; populated from the element table and editable thereafter.

Minimum frequency

Lower frequency shown in the analysis plot, in cm^{-1} ; default -1.

Maximum frequency

Upper frequency shown in the analysis plot, in cm^{-1} ; default 400.

title

Analysis plot title, default 'Analysis'. This key is lower case.

Covalent radius scaling

Multiplier on the sum of the atomic covalent radii when deciding bonds; default 1.1. Changes molecular grouping and therefore mode analysis.

Bonding tolerance

Additional distance in Å added to the scaled sum of radii when deciding bonds; default 0.1. Bonding information is also used by the Viewer.

Bar width

Width of the mode bars in the analysis plot, default 0.5.

3D Viewer Tab

Use `self.notebook.viewerTab.settings`. These are display settings; they do not change the calculated spectrum. Colours use `[red, green, blue, alpha]` with integer components from zero to 255.

Selected mode

One-based phonon number to display/animate; initially 4. This differs from zero-based Python array and scenario indices.

Atom scaling

Multiplier on covalent radii for atom spheres, default 0.5.

Maximum displacement

Maximum displayed displacement in Å, default 1. Scales the vibration for viewing rather than changing the normal-mode eigenvector.

Bond colour

Bond-cylinder RGBA colour, default `[80, 80, 80, 255]`.

Bond radius

Bond-cylinder radius in Å, default 0.1.

Cell colour

Unit-cell edge RGBA colour, default `[255, 0, 0, 255]`.

Cell radius

Radius of the cell-edge cylinders in Å, default 0.1.

Text colour

Cell-label RGBA colour, default `[255, 255, 0, 255]`.

Text size

Cell-label font size, stored as a string; default '24'.

Background colour

Viewport background RGBA colour, default `[120, 120, 120, 255]`.

Arrow colour

Displacement-arrow RGBA colour, default `[0, 255, 0, 255]`.

Arrow radius

Displacement-arrow radius in Å, default 0.07.

Number of phase steps

Number of displacement steps for the animation, default 41. Must be odd; the Viewer increases an even value by one.

Super Cell

Three positive integer repetitions along the cell vectors, for example `[2, 2, 1]`. Default `[1, 1, 1]`.

Transform

3×3 lattice transformation, normally stored as nested lists of strings. Default identity. Changes the displayed cell; see *The Transformation Window*.

Primitive transform

Derived transformation obtained from the reader when the primitive-cell action is used. Normally preserve the value saved by that action.

hkl

Three-component surface normal used to orient the view, default (0, 0, 1). See *Crystal & Laboratory Coordinate Frames*.

uvw

Three-component direct-lattice direction used to orient the view, default (1, 0, 0).

Element palette

'Jmol' (default) or 'Vesta' selects the element colour table; other names fall back to the built-in elemental palette.

Element colours

Optional list of RGBA colours in the Viewer species order, overriding the palette for those species. Initially None.

Toggle states

Five booleans in this order: cell labels, cell edges, orientation axes, bonds, atoms. Default [True, True, True, True, True].

Fitter Tab

Use `self.notebook.fitterTab.settings`. The Fitter compares an experimental spectrum with the selected calculated spectrum; see the Fitter section in *PDGui* for the fitting workflow and experimental spectrum columns.

Experimental file name

Filename of the measured spectrum to compare, initially empty. This is a frequency/intensity spectrum, distinct from an `.exp` material input.

Plot title

Comparison plot title, initially 'Experimental and Calculated Spectral Comparison'.

Fitting type

'Minimise x-correlation' selects the correlation-based objective; 'Minimise spectral difference' selects the spectral RMSE objective. The optimisation adjusts selected linewidths and, optionally, frequency scale.

Number of iterations

Optimisation iteration control, default 20.

Frequency scaling factor

Multiplier on calculated frequencies in the comparison, default 1. Does not rescale the underlying normal modes in the input reader.

Optimise frequency scaling

Boolean, default False. Includes the frequency scaling factor among the variables adjusted during fitting.

Independent y-axes

Boolean, default True. Plots experimental and calculated spectra on independent y axes; False uses a shared intensity axis.

Spectral difference threshold

Threshold applied after intensity normalisation when calculating the spectral difference, default 0.05. Values below it are set to zero in the comparison objective.

Baseline removal

Boolean, default `False`. Applies a Hodrick–Prescott background removal to the resampled experimental spectrum.

HPFilter lambda

Base-10 logarithm of the Hodrick–Prescott smoothing parameter, default 7 (a coefficient of 10^7). Used when baseline removal is enabled.

Scenario index

Zero-based index of the scenario being fitted; initially the last scenario in the notebook.

Spectrum scaling

Legacy boolean, initially `False`. The current Fitter stores this value but does not apply it in the plotting or fitting calculation.

Spectrum scaling factor

Legacy multiplier, initially 1. Retained with `Spectrum scaling`; it is not an active intensity-scaling control in the current Fitter.

THEORY FOR INFRARED ABSORPTION IN POWDERS

The theory underlying the Python package PDielec is described. PDielec calculates the infrared absorption characteristics of a crystalline material supported in a non-absorbing medium. PDielec post processes solid-state quantum mechanical and molecular mechanical calculations of the phonons and dielectric response of the crystalline material. The molecular and solid-state quantum mechanical (QM) calculations of response properties, such as the frequencies and intensities of infrared (IR) and terahertz (THz) radiation absorption, have become generally available in many molecular and solid-state computer programs. A common approach is to assume the harmonic approximation and calculate the mass-weighted force constant matrix (for molecules) or the dynamical matrix at the gamma point (for periodic solids). Diagonalisation of the matrix gives the frequencies for absorption and the normal modes (molecules) or phonon displacements (periodic solids). The calculation of the absorption intensity for each mode requires the calculation of the change in dipole moment caused by the displacement of the atoms for that mode. For solids where there is a large separation of charge, there can be a large coupling between a phonon mode and the internal field within a particle resulting from its morphology.

PDielec post processes the output of solid-state quantum mechanical and molecular mechanics (MM) based codes such as; VASP [1], CASTEP [2], CRYSTAL [3], Abinit [4], Quantum Espresso [6], Phonopy [7], and GULP [8], to predict the infrared absorption of crystalline insulator materials whose crystal size is small compared with the wavelength of the absorbing radiation. The package is suited for the calculation of the complex, frequency-dependent permittivity and its associated absorption of infrared radiation for a finely ground crystalline material dispersed in a low-loss dielectric medium such as KBr or Polytetrafluoroethylene (PTFE). A particular feature of the program is its ability to take into account the constant permittivity of the supporting medium and the particle shape of the material of interest through an effective medium theory. The documentation outlines the theory used by the program and gives some examples of the application of the program for ionic and molecular materials.

Equations (6.1) and (6.2) describe Beer-Lambert's law [18] where α is the (decadic) absorption coefficient (usually given in cm^{-1}), I and I_0 are the intensities after and before absorption respectively and d is the path length.

$$\frac{I}{I_0} = 10^{-\alpha d} \quad (6.1)$$

$$\log \left(\frac{I}{I_0} \right) = -\alpha d \quad (6.2)$$

It is common, especially in the chemistry community, when reporting infrared spectra to use a decadic molar absorption coefficient (a), which has units of $\text{L}\cdot\text{mol}^{-1}\cdot\text{cm}^{-1}$. The relationship between the absorption coefficient and the molar absorption coefficient [18] is;

$$\alpha = aC \quad (6.3)$$

where C is the concentration of the absorbing species.

6.1 Molecular approach to absorption intensity

For molecules, the transition intensity I_k of the k^{th} vibrational mode (calculated from the change in dipole moment along the mode displacement) can be converted to an integrated molar absorption coefficient, A_k , which can then be more readily compared with experiment. The theory for this is described by Wilson, Decius and Cross [19] and results in expressions such as those given in equations (6.4) and (6.5). The first expression shows the relationship between the integrated molar absorption coefficient and the transition intensity and uses the number of molecules per unit volume (N), the velocity of light (c) and the degeneracy of the mode (g_k). The second expression shows the appropriate conversion factors if the units for the integrated molar absorption coefficient are $\text{L}\cdot\text{mol}^{-1}\cdot\text{cm}^{-2}$ ($1 \text{ L}\cdot\text{mol}^{-1}\cdot\text{cm}^{-2} = 0.01 \text{ km}\cdot\text{mol}^{-1}$) and the units for the transition intensity are $\text{D}^2\text{\AA}^{-2}\text{amu}^{-1}$, where D represents the Debye unit of dipole moment and amu is an atomic mass unit. The factor $\log_e 10$ arises due to the choice of a decadic Beer's law.

$$A_k = \frac{N\pi}{3c^2 \log_e 10} g_k I_k \quad (6.4)$$

$$A_k = \frac{Na\pi}{3000c^2 2.302585} g_k I_k = 4225.6 I_k \quad (6.5)$$

The derivation of the above expressions assumes that the rotational levels are not quantised and that the vibrational levels are thermally occupied according to a Boltzmann distribution. To use the calculated molecular intensities to predict a spectrum it is usual to assume [19] that each transition is associated with a Lorentzian line shape with a full width at half maximum (FWHM) of σ_k . It is common, when reporting comparisons between theoretical and experimental spectra, to assume that the line widths are the same for all modes [20, 21]. Recent work on terahertz absorption in crystalline pentaerythritol tetranitrate (PETN) using molecular dynamics calculations [22] in combination with the direct calculation of the cubic anharmonic couplings of the normal modes [22] has shown that the FWHM of the intense absorptions may vary between 10 and 25 cm^{-1} . Assuming a Lorentzian line shape, the molar absorption coefficient for the k^{th} mode at wavenumber, $\bar{\nu}_k$, can be written as a function of frequency or wavenumber ($\bar{\nu}$);

$$a_k(\bar{\nu}) = \frac{2A_k}{\pi} \frac{\sigma_k}{4(\bar{\nu} - \bar{\nu}_k)^2 + \sigma_k^2} \quad (6.6)$$

$$a_k^{max} = \frac{2A_k}{\pi\sigma_k} \quad (6.7)$$

The maximum height of the Lorentzian, a_k^{max} depends upon the value of σ_k . As can be seen in Equation (6.8), the choice of normalisation for the Lorentzian means that integration of the molar absorption coefficient over wavenumber returns the integrated molar absorption coefficient and a sum over all the bands provides the total molar absorption coefficient $a^{mol}(\bar{\nu})$ as a function of wavenumber, calculated from the intensities of each band. Equation (6.10) shows the relationship between the absorption and the molar absorption coefficients. C is the concentration usually expressed in $\text{mol}\cdot\text{L}^{-1}$.

$$A_k = \int a_k(\bar{\nu}) d\bar{\nu} \quad (6.8)$$

$$a^{mol}(\bar{\nu}) = \sum_k a_k(\bar{\nu}) \quad (6.9)$$

$$\alpha^{mol}(\bar{\nu}) = C a^{mol}(\bar{\nu}) \quad (6.10)$$

A comment should be made about the various units which can be used for these quantities. A common unit for the transition intensity is $\text{D}^2\text{\AA}^{-2}\text{amu}^{-1}$, and another is $\text{km}\cdot\text{mol}^{-1}$. However, it should be pointed out that strictly speaking the latter unit refers to the integrated molar absorption coefficient as defined above in Equation (6.8) and therefore relies on the assumptions made in its derivation. ($1 \text{ D}^2\text{\AA}^{-2}\text{amu}^{-1}$ is equivalent to $42.256 \text{ km}\cdot\text{mol}^{-1}$).

6.2 Solid state approach to absorption intensity

The optical properties of a solid are determined by its complex, frequency-dependent relative permittivity, $\bar{\epsilon}(\bar{\nu})$, and in particular the real and imaginary components, $\bar{\kappa}$ and $\bar{n}$, of the complex refractive index, $\bar{N}(\bar{\nu})$ where;

$$\bar{N}(\bar{\nu})^2 = \bar{\epsilon}(\bar{\nu}) \quad (6.11)$$

$$\bar{N}(\bar{\nu}) = \bar{n}(\bar{\nu}) + i\bar{\kappa}(\bar{\nu}) \quad (6.12)$$

The intensity of absorption is given by the imaginary component of the refractive index which for an isotropic material is [23];

$$I = I_0 e^{-\frac{4\pi\bar{\kappa}(\bar{\nu})d}{\lambda}} \quad (6.13)$$

$$I = I_0 e^{-4\pi\bar{\nu}\bar{\kappa}(\bar{\nu})d}$$

$$-ln\left(\frac{I}{I_0}\right) = 4\pi\bar{\nu}\bar{\kappa}(\bar{\nu})d \quad (6.14)$$

$$-log\left(\frac{I}{I_0}\right) = 4\pi\bar{\nu}\bar{\kappa}(\bar{\nu})d \cdot log(e)$$

Comparison with the definition of the absorption coefficient from Beer-Lambert's law, Equation (6.1), and using Equation (6.10) gives;

$$\alpha^{sol}(\bar{\nu}) = 4\pi\bar{\nu}\bar{\kappa}(\bar{\nu})log(e) \quad (6.15)$$

$$a^{sol}(\bar{\nu}) = \frac{\alpha^{sol}(\bar{\nu})}{C} \quad (6.16)$$

Since the refractive index is dimensionless, the absorption coefficient, α^{sol} is specified in cm^{-1} . The superscripts 'sol,' for solid, and 'mol,' for molecular, are used here to distinguish between the two methods of calculating the absorption (α) and molar absorption coefficients (a). In the calculation of the imaginary component of the refractive index it is necessary to choose the solution which gives a positive value. This is consistent with the Kramers-Kronig relationship between the real and imaginary components [24].

To calculate the relationship between absorption and molar absorption coefficients it is necessary to know the concentration. For solid-state calculations, the required unit is; moles of unit cells per liter. One of the drawbacks of this molar absorption coefficient unit is that the number of molecules in a unit cell can change depending on whether a supercell, primitive or non-primitive unit cell is being used. A more natural unit would be to use a mole of formula units, or a mole of molecules. To aid comparison between such calculations PDielec can calculate concentration in both moles of atoms and moles of molecules. However for the rest of this paper Equation (6.17) will be used, where V is the volume of the unit cell, and therefore the concentration C is moles of unit cell/liter.

$$C = \frac{f \cdot 1000\text{cm}^3}{VN_a} \quad (6.17)$$

The volume fraction, f , of the dielectric material in a supporting matrix of non-absorbing material is included in the expression for the concentration as it will be useful when the theory for mixtures is developed.

For a periodic system, the permittivity tensor can be calculated as a sum over Lorentz oscillators, incorporating an imaginary loss component through the damping factor σ_k [25]. The frequencies of the oscillators are the transverse optic (TO) phonon frequencies of the system.

$$\bar{\epsilon}(v) = \bar{\epsilon}_\infty + \frac{4\pi}{V} \sum_k \frac{\bar{S}_k}{\nu_k^2 - \nu^2 - i\sigma_k\nu} \quad (6.18)$$

$$\bar{\bar{S}}_k = \bar{\mathbf{Z}}_k \bar{\mathbf{Z}}_k^T \quad (6.19)$$

$$\bar{\mathbf{Z}}_k = \sum_a \bar{\mathbf{Z}}^a \bar{\mathbf{U}}^a \mathbf{a}_k \quad (6.20)$$

$$\begin{aligned} \bar{\bar{D}} \bar{\mathbf{U}}_k &= \Lambda_k \bar{\mathbf{U}}_k \\ \nu_k^2 &= \Lambda_k \end{aligned} \quad (6.21)$$

$$I_k = \text{tr}(\bar{\bar{S}}_k) \quad (6.22)$$

V is the volume of the unit cell, $\bar{\bar{S}}_k$ is the dipole oscillator strength tensor for the k_{th} transition, with a TO frequency of ν_k and $\bar{\bar{\epsilon}}_\infty$ the optical permittivity tensor, which represents the electronic contribution to the permittivity. The intensity of a transition, I_k , is given by the trace of the oscillator strength tensor, Equation (6.22). The damping factor σ_k removes any discontinuities at the TO frequencies. Since the oscillator strengths and phonon frequencies can be calculated routinely in solid-state quantum mechanical packages, the calculation of the frequency-dependent complex permittivity using Equation (6.18) is straightforward.

In some cases, using Equations (6.19) and (6.20), PDielec calculates the oscillator strengths from the Born charge matrix for atom a and its contributions to the k_{th} phonon mode [25]. As shown in Equation (6.21), at the Γ point the k_{th} phonon mode is described by the eigenvector, $\bar{\mathbf{U}}_k$, and eigenvalue, Λ_k , of the mass-weighted, dynamical matrix, $\bar{\bar{D}}$, which is a $3N \times 3N$ matrix, where N is the number of atoms in the unit cell. The eigenvalues are the squared frequencies of the phonon modes, Equation (6.21). The displacement of each atom in the k_{th} is proportional to $m_a^{-1/2}$, where m_a is the mass of atom a . The dynamical matrix has $3N$ eigenvectors and eigenvalues, of which three should be zero due to translational invariance. If there are any negative eigenvalues the system is unstable to some displacement and therefore not at an energy minimum.

For ionic systems, it is common practice in solid-state QM and MM programs to include a long wavelength, non-analytic correction to the mass-weighted dynamical matrix at the Γ point, which describes the coupling of the longitudinal optic (LO) modes to the induced field resulting from the vibration. This may be written for atoms s and t and their Cartesian components α and β as [25];

$$\left(\bar{\bar{D}}_{\mathbf{q} \rightarrow 0}^{\text{LO}}\right)_{s,\alpha;t,\beta} = \left(\bar{\bar{D}}\right)_{s,\alpha;t,\beta} + \frac{4\pi}{V\sqrt{M_s M_t}} \frac{\left(\bar{\mathbf{q}}^T \bar{\bar{\mathbf{Z}}}_s\right)_\alpha \left(\bar{\mathbf{q}}^T \bar{\bar{\mathbf{Z}}}_t\right)_\beta}{\bar{\mathbf{q}}^T \cdot \bar{\bar{\epsilon}}_\infty \cdot \bar{\mathbf{q}}} \quad (6.23)$$

The mass weighting has been incorporated through the mass of the atoms, M_s and M_t . The correction depends upon the direction, $\bar{\mathbf{q}}$, that the long wave-length limit is approached. Diagonalisation of the corrected matrix gives the squared frequencies of $N-1$ LO modes and $2N-2$ TO modes, Equation (6.21). In some of the examples given below the LO frequencies will be given for comparison with the TO frequencies.

6.3 Effect of particle shape on infrared absorption

It has long been recognised that, especially for ionic materials, the local field within a crystal and its coupling with the transverse optical phonons has an important effect on the position and intensity of the absorption. Fröhlich [26] was one of the first to point out that the frequency of absorption of a small ionic sphere embedded in a low dielectric medium is shifted to lie between the transverse and longitudinal optical frequencies of the material making up the sphere.

In the development of the theory used in PDielec an important assumption is that the particle size of the crystallites in the sample is small compared with the wavelength of light. Using this approach Genzel and Martin [27] were able to explain the observed infrared absorption of small spheres of MgO crystallites and the effect of the permittivity of the supporting medium on the spectrum. Studies of the infrared absorption

by small particles of $\alpha\text{-Fe}_2\text{O}_3$ using an effective medium theory and an absorption/scattering theory [28, 29] showed that it was necessary to adjust not only the damping factors in Equation (6.18) but also the permittivity of the matrix and the volume fraction of the dielectric medium to fit the experimental data. The volume fraction was used to account for aggregation effects as the volume fraction increased. It was also shown that effective medium theories were only applicable to particles smaller than the wavelength of light. For larger particles, the scattering from the particles becomes increasingly important.

More recently Balan and others in a series of papers [30, 31, 32] used density functional calculations together with an effective medium theory to calculate the infrared absorption of several minerals incorporating information about the crystallite shape. In an experimental and theoretical study of irradiated kaolinite [32] it was shown that exposure to radiation resulted in shifts in the infrared spectrum which could be accounted for by increasing the polarisability of the particles through an increase in the optical permittivity tensor.

The underlying theory adopted by PDielec is based on similar premises to the work described above, namely that the dielectric response of small spherical, ellipsoidal, slab-like or needle-like crystallites randomly distributed in a non-absorbing medium such as PTFE, KBr or Nujol, is the same as that of an effective medium material whose frequency-dependent dielectric response can be calculated from the frequency-dependent permittivity tensor of the crystal (as calculated by solid-state QM or MM calculations), the shape of the crystallites and the permittivity of the non-absorbing medium (taken to be a constant over the frequency range of interest).

The development of the theory reported here closely follows the work of Sihvola [33]. It will be assumed that the inclusion particles, which may be non-isotropic, ellipsoidal (including spherical, needle-like and plate-like), are randomly orientated in an embedding, non-absorbing medium such as PTFE, KBr or Nujol. It should be emphasized that, whilst PDielec can take account of particle shape, particle and matrix permittivity many additional aspects of infrared absorption that need to be considered when comparing calculated and experimental results. Most notable of these are; the coupling between phonons and mobile electrons or holes (so-called phonon-polariton coupling) [34], the scattering which starts to dominate as the particles get larger [32] and the agglomeration of particles as the volume fraction increases.

6.3.1 The polarisability of an isolated particle

Figure Fig. 6.1 shows a schematic of the field and polarisation inside an inclusion with non-isotropic permittivity $\bar{\epsilon}_i$ embedded in a supporting medium with permittivity, ϵ_e . The internal field within the inclusion is indicated by $\bar{E}_i$ the external, applied field is indicated by $\bar{E}_e$ and the induced polarisation in the inclusion is shown by $\bar{P}$.

The figure shows the field and polarisation inside an inclusion with non-isotropic permittivity $\bar{\epsilon}_i$ embedded in a supporting medium with permittivity ϵ_e . The internal field within the inclusion is indicated by $\bar{E}_i$, the external, applied field is indicated by $\bar{E}_e$ and the induced polarisation in the inclusion is shown by $\bar{P}$.

The electric field internal to the inclusion gives rise to a polarisation density which is no longer necessarily aligned with the field because the material is non-isotropic. The polarisation density in the inclusion can be expressed as the tensor product of the permittivity contrast between the inclusion and the supporting medium and the (as yet unknown) internal field.

$$\bar{P} = (\bar{\epsilon}_i - \epsilon_e \bar{\mathbf{1}}) \bar{E}_i \quad (6.24)$$

For any ellipsoidal shape (including sphere, slab and needle) with volume V , the polarisation density throughout the particle is uniform and integrating over all space gives the field-induced dipole moment of the inclusion, $\bar{p}$.

$$\bar{p} = V \bar{P} = V (\bar{\epsilon}_i - \epsilon_e \bar{\mathbf{1}}) \bar{E}_i \quad (6.25)$$

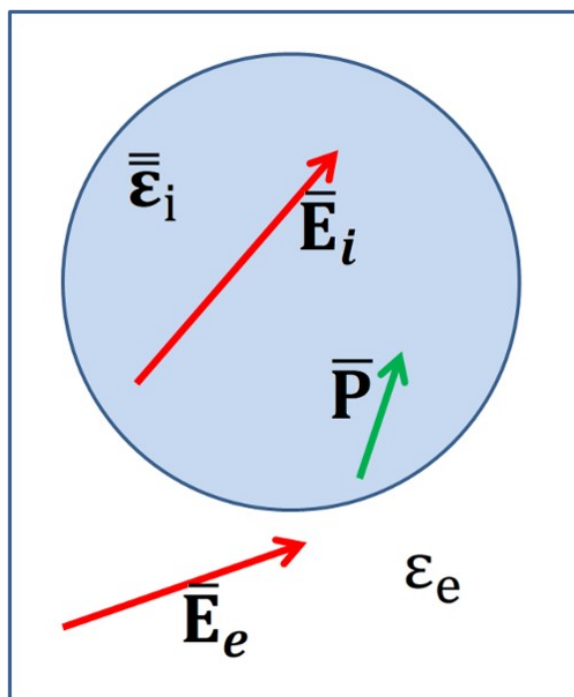


Fig. 6.1: Schematic showing the various terms used to define the effective medium of an inclusion

The dipole and the external field, $\bar{E}_e$, are related by the polarisability tensor, $\bar{\alpha}$.

$$\bar{p} = \bar{\alpha} \bar{E}_e \quad (6.26)$$

Equations (6.25) and (6.26) allow the determination of the polarisability, once the field internal to the inclusion has been expressed in terms of the shape of the inclusion and its permittivity. The polarisation within the inclusion gives rise to a depolarisation field, $\bar{E}_d$, which depends on the shape of the inclusion through the symmetric and unit-trace depolarisation tensor, $\bar{L}$.

$$\bar{E}_d = -\frac{1}{\epsilon_e} \bar{L} \cdot \bar{P} \quad (6.27)$$

The internal field is the sum of the external field and the depolarisation field.

$$\bar{E}_i = \bar{E}_e + \bar{E}_d \quad (6.28)$$

The depolarisation matrix is determined from the shape of the crystallite. For an ellipse, slab or needle, a unique axis [hkl] using the crystal axes of the unit cell is specified. PDielec transforms these to a cartesian coordinate system using the unit cell lattice vectors. In the case of a slab morphology, the unique direction is normal to the surface specified by its Miller indices (hkl). The definitions of the various depolarisation tensors are indicated in Table below.

Table 6.1: Definitions used for the depolarisation tensor

Shape	Depolarisation Tensor
Sphere	$\bar{L} = \frac{1}{3} (\bar{V}_1 \bar{V}_1^T + \bar{V}_2 \bar{V}_2^T + \bar{V}_3 \bar{V}_3^T)$
Slab	$\bar{L} = \bar{V}_1 \bar{V}_1^T$
Needle	$\bar{L} = \frac{1}{2} (\bar{V}_2 \bar{V}_2^T + \bar{V}_3 \bar{V}_3^T)$
Ellipsoid	$\bar{L} = a \bar{V}_1 \bar{V}_1^T + b \bar{V}_2 \bar{V}_2^T + b \bar{V}_3 \bar{V}_3^T$

The three directions defined by $\bar{V}_1$, $\bar{V}_2$ and $\bar{V}_3$ are mutually orthogonal cartesian vectors calculated from [hkl] for an ellipse or needle and (hkl) for a slab. In the case of a slab, needle or ellipsoid, $\bar{V}_1$ defines the unique direction and the other vectors are orthogonal to it. For the case of an ellipsoid, the parameters a and b in the Table depend on the ratio, z , of the unique axis length to the length of an axis perpendicular to it [33].

For $z > 1$ the ellipsoid is prolate;

$$e = \sqrt{1 - z^{-2}}, \quad a = \frac{(1 - e^2)}{2e^3} \left(\log \frac{1 + e}{1 - e} - 2e \right), \quad b = \frac{1}{2} (1 - a) \quad (6.29)$$

For $z < 1$ the ellipsoid is oblate

$$e = \sqrt{z^{-2} - 1}, \quad a = \frac{(1 + e^2)}{e^3} (e - e), \quad b = \frac{1}{2} (1 - a) \quad (6.30)$$

The depolarisation tensor acts as a projection or screening operator describing the effect of the geometry of the inclusion on the depolarisation field which results from its polarisation. For instance, in the case of a needle, only polarisation perpendicular to the needle axis contributes to the depolarising field, whilst for a slab only polarisation perpendicular to the slab face may contribute. Similarly, for a sphere, all directions contribute and so the depolarisation matrix is diagonal with a value of 1/3 for each diagonal element, as the trace of the depolarisation tensor must be 1. It follows from Equations (6.24), (6.27) and (6.28) that;

$$\bar{E}_i = \bar{E}_e - \frac{1}{\varepsilon_e} \bar{L} \left(\bar{\varepsilon}_i - \varepsilon_e \bar{1} \right) \bar{E}_i \quad (6.31)$$

Rearrangement allows the internal field of the inclusion to be expressed in terms of the known permittivities, the shape of the inclusion and the external field.

$$\bar{E}_i \left(\varepsilon_e \bar{1} + \bar{L} \left(\bar{\varepsilon}_i - \varepsilon_e \bar{1} \right) \right) = \varepsilon_e \bar{E}_e \quad (6.32)$$

$$\bar{E}_i = \varepsilon_e \bar{E}_e \left(\varepsilon_e \bar{1} + \bar{L} \left(\bar{\varepsilon}_i - \varepsilon_e \bar{1} \right) \right)^{-1} \quad (6.33)$$

Substituting the internal field expression, (6.31), into Equation (6.25) for the dipole moment and requiring the dipole moments calculated using the polarisation density to equal those calculated from the polarisability allows the polarisability to be written;

$$\bar{\alpha} = V \varepsilon_e \left(\bar{\varepsilon}_i - \varepsilon_e \bar{1} \right) \left(\varepsilon_e \bar{1} + \bar{L} \left(\bar{\varepsilon}_i - \varepsilon_e \bar{1} \right) \right)^{-1} \quad (6.34)$$

Although it has not been specified explicitly the permittivity of the inclusion, and therefore the polarisability tensor, are frequency dependent through the oscillator strengths of each phonon mode contributing to the permittivity according to Equation (6.18). The calculation of the complex, frequency-dependent polarisability tensor of the composite material is the key step in the calculation of its effective permittivity.

6.3.2 The effective permittivity of a mixture

To extend this approach to include the effect of a number of inclusions we need to introduce the concept of an effective permittivity, $\bar{\varepsilon}_{eff}$, which describes the behaviour of an average field, $\langle \bar{E} \rangle$, where the angle brackets indicate an average over a volume of the composite material. It is required that the average electric flux density $\langle \bar{D} \rangle$ is the same in the effective medium as in the composite medium;

$$\langle \bar{D} \rangle = \bar{\varepsilon}_{eff} \langle \bar{E} \rangle = \varepsilon_e \langle \bar{E} \rangle + \langle \bar{P} \rangle \quad (6.35)$$

The averaging is necessary because the polarisation within a given inclusion has an effect on the field in other inclusions. The local field in the cavity left by a single inclusion embedded in the average polarisation field is given by;

$$\bar{\bar{E}}_L = \langle \bar{\bar{E}} \rangle + \frac{1}{\varepsilon_e} \bar{\bar{L}} \langle \bar{\bar{P}} \rangle \quad (6.36)$$

The local field ‘excites’ the inclusion resulting in a dipole moment $\mathbf{p}_{mix}$ that is related to the polarisation through the number density of inclusions, n , and through the polarisability of the inclusion, which is already known from Equation (6.34).

$$\langle \bar{\bar{P}} \rangle = n \bar{\bar{p}}_{mix} = n \langle \bar{\bar{\alpha}} \bar{\bar{E}}_L \rangle \quad (6.37)$$

The angle brackets around the product of the polarisability and the local field indicate that it is necessary to average the polarisation according to the distribution of alignments of inclusions. In this work, it will be assumed that the inclusions are randomly aligned. Substituting the expression for the local field, Equation (6.36), gives;

$$\langle \bar{\bar{P}} \rangle = \left(\bar{\bar{1}} - \frac{n \langle \bar{\bar{\alpha}} \bar{\bar{L}} \rangle}{\varepsilon_e} \right)^{-1} n \langle \bar{\bar{\alpha}} \rangle \langle \bar{\bar{E}} \rangle \quad (6.38)$$

6.3.3 Mixing rules

There are many mixing rules which have been proposed to describe the homogenization of composite materials and a lot of work has been done in comparing their accuracy. Here two methods will be used. The first and the most commonly used method is the Maxwell-Garnett mixing rule [33]. Indeed this has been implied by the use of Equation (6.35) to define the effective permittivity. The other commonly used method is the Bruggeman mixing rule [33], which differs substantially in the way the two components of the system are treated. It is usually stated that the Maxwell-Garnett mixing rule is good for low-volume fractions of the inclusion and the Bruggeman approach should be better for higher-volume fractions [35]. In addition to these mixing rules one other approach will be described, namely the Averaged Permittivity (AP) mixing rule, which calculates the absorption spectrum ignoring the effects of the internal field on the absorption and can therefore be used as an indicator of the shifts in frequency and intensity which have occurred owing to the effect of particle shape.

6.3.4 Maxwell-Garnett mixing rule

The Maxwell-Garnett approach for treating the properties of heterogeneous mixtures assumes that the average field and the average flux density result from volume fraction weighted sums. Substituting Equation (6.38) into Equation (6.35) gives the Maxwell-Garnett effective permittivity;

$$\bar{\bar{\varepsilon}}_{mg} = \bar{\bar{1}} + \left(\bar{\bar{1}} - \frac{n \langle \bar{\bar{\alpha}} \bar{\bar{L}} \rangle}{\varepsilon_e} \right)^{-1} n \langle \bar{\bar{\alpha}} \rangle \quad (6.39)$$

The fact that the polarisability tensor has a volume term in it (Equation (6.34)) means that the terms in Equation (6.39) containing $n \bar{\bar{\alpha}}$ depend on the volume fraction f . Although written as a tensor, because the assumption has been made that the inclusions are randomly orientated, the effective permittivity has to be diagonal with equal complex values. Since the polarisability is complex and frequency dependent, the effective permittivity is also and its calculation using Equations (6.39) and (6.34) need to be calculated over the frequency range of interest.

6.3.5 Bruggeman mixing rule

In the Maxwell-Garnett mixing formalism there is a distinction between the inclusion and the supporting medium which results in an asymmetry in the treatment of the two species in the mixture. Instead, the Bruggeman mixing rule assumes that each species is polarised against the background of the effective medium and therefore the polarisation in the two components cancels each other out;

$$\langle \bar{\mathbf{P}}_1 \rangle + \langle \bar{\mathbf{P}}_2 \rangle = 0 \quad (6.40)$$

where the components are now labelled 1 and 2 rather than external and internal. The polarisation for species 1 and 2 with a number density of species represented by n_1 and n_2 can be obtained from the polarisability of the species (Equation (6.40));

$$\langle \bar{\mathbf{P}}_1 \rangle = n_1 \langle \bar{\alpha}_1 \rangle \bar{\mathbf{E}} \quad (6.41)$$

Substituting Equation (6.41) into Equation (6.40) leads to the requirement that;

$$n_1 \langle \bar{\alpha}_1 \rangle + n_2 \langle \bar{\alpha}_2 \rangle = 0 \quad (6.42)$$

Taking Equation 19 and generalising it for species i , (where i is 1 or 2) embedded in an effective permittivity given by $\bar{\epsilon}_{br}$;

$$\bar{\alpha}_i = V_i \bar{\epsilon}_{br} (\bar{\epsilon}_i - \bar{\epsilon}_{br}) \left(\bar{\epsilon}_{br} + \bar{\mathbf{L}} (\bar{\epsilon}_i - \bar{\epsilon}_{br}) \right)^{-1} \quad (6.43)$$

Equation (6.42) has to be solved for $\bar{\epsilon}_{br}$. Since the systems considered here are isotropic with random inclusions, a solution has to be found for a complex value of the Bruggeman permittivity at each frequency considered. An issue in the use of Equation (6.43) is that the same depolarisation matrix is being used for both species, which is not always appropriate. The solution of Equation (6.42) can be achieved either by iteration or by casting the equation as a minimisation problem. The iterative approach implemented in PDielec involves repeated application of (6.44) until convergence [36]. The starting point for the iterations is taken as the Maxwell-Garnett solution for the first frequency and then the solution at the previous frequency is used to start the next iteration.

$$\bar{\epsilon}_{br} = \frac{f_1 \bar{\epsilon}_1 \left[\bar{\mathbf{1}} + \bar{\mathbf{L}} (\bar{\epsilon}_1 - \bar{\epsilon}_{br}) \right]^{-1} + f_2 \bar{\epsilon}_2 \left[\bar{\mathbf{1}} + \bar{\mathbf{L}} (\bar{\epsilon}_2 - \bar{\epsilon}_{br}) \right]^{-1}}{f_1 \left[\bar{\mathbf{1}} + \bar{\mathbf{L}} (\bar{\epsilon}_1 - \bar{\epsilon}_{br}) \right]^{-1} + f_2 \left[\bar{\mathbf{1}} + \bar{\mathbf{L}} (\bar{\epsilon}_2 - \bar{\epsilon}_{br}) \right]^{-1}} \quad (6.44)$$

Although the Bruggeman permittivity is written here as a tensor, the polarisabilities in Equation (6.42) have to be averaged over the random orientation of the inclusions and therefore the homogenized material is isotropic with a single complex value for the diagonal tensor. Also, as with the Maxwell-Garnett mixing rule, since the polarisability is complex and frequency dependent, the effective permittivity is also, and its calculation using Equation (6.39) needs to be performed over the frequency range of interest.

The choice between using the Bruggeman or Maxwell-Garnett model is often governed by the assumption that the Maxwell-Garnett model works well at low concentrations and the Bruggeman model works better at higher concentrations. Work by Karkkainen *et al.*, using a finite difference method for random mixtures of non-absorbing materials, indicated that the Bruggeman approximation works best when there is some clustering of the inclusions and the Maxwell Garnett model works best when there is no clustering [37].

The Bruggeman solution has been shown to be unphysical in certain circumstances [38]. In particular when the real components of the permittivity have different signs and when the absolute values of the real components are much larger than those of the imaginary components. Unfortunately, it may be that these conditions will apply to modelling infrared absorption. As a result, only a few of the examples discussed below will include results using the Bruggeman mixing rule; the majority will use the Maxwell-Garnett mixing rule.

6.3.6 Averaged-Permittivity mixing rule

It is useful to be able to compare the effective medium theories with the absorption predicted using no shape information, that is using only the TO frequencies.

$$\bar{\bar{\epsilon}}_{TO} = f \langle \bar{\bar{\epsilon}}_i \rangle + (1 - f) \epsilon_e \quad (6.45)$$

Equation (6.45) defines an isotropic permittivity which can be used to calculate such an absorption coefficient. The angle brackets indicate an average of orientation. This mixing rule provides a useful comparison between the absorption calculated without any shape effects and that calculated including shape effects using the other mixing rules presented above. At low concentrations, the peak positions of the AP mixing rule will be at the TO frequencies.

6.4 Particle size effects

Meier and Wokaun [39] outlined an approach to treating large (metal) spherical particles, where particle size is incorporated in terms up to 3rd order in the wave vector k . Using Equations (6.24) and (6.28) we can write;

$$\bar{\mathbf{P}} = \left(\bar{\bar{\epsilon}}_i - \epsilon_e \bar{\mathbf{1}} \right) \left(\bar{\mathbf{E}}_e + \bar{\mathbf{E}}_d \right) \quad (6.46)$$

$$\bar{\mathbf{E}}_d = - \frac{\left(1 - x^2 - i \frac{2}{3} x^3 \right)}{\epsilon_e} \bar{\mathbf{L}} \bar{\mathbf{P}} \quad (6.47)$$

$$x = ak = \frac{2\pi a}{\lambda}$$

Here a is the radius of the spherical particle and x is the dimensionless ‘size’ of the particle with respect to the wavelength of the incident light. The first term relating the depolarisation field to the polarisability is the same as that used above in Equation (6.27). The second term is a dynamic depolarisation term and the third term is a radiation damping correction to the electrostatic solution [39].

A slightly different, but related, approach is presented by Shivola [33]. Starting from Equation (6.31), a size-dependent term $G(x)$ is introduced as indicated by the work of Peltoniemi [40];

$$G(x) = G_1(x) + G_2(x)$$

$$G_1(x) = \frac{2}{3} \left[(1 + ix) e^{-ix} - 1 \right] \quad (6.48)$$

$$G_2(x) = \left(1 + ix - \frac{7}{15} x^2 - i \frac{2}{15} x^3 \right) e^{-ix} - 1$$

The modified equation for the relationship between the internal field and the external field based on equation (6.31) becomes;

$$\bar{\mathbf{E}}_i = \bar{\mathbf{E}}_e - \frac{(1 - G(x))}{\epsilon_e} \bar{\mathbf{L}} \left(\bar{\bar{\epsilon}}_i - \epsilon_e \bar{\mathbf{1}} \right) \bar{\mathbf{E}}_i \quad (6.49)$$

This leads to a modified equation for the polarisability of spherical particles;

$$\bar{\bar{\alpha}}(x) = V \epsilon_e \left(\bar{\bar{\epsilon}}_i - \epsilon_e \bar{\mathbf{1}} \right) \left(\epsilon_e \bar{\mathbf{1}} + (1 - G(x)) \bar{\mathbf{L}} \left(\bar{\bar{\epsilon}}_i - \epsilon_e \bar{\mathbf{1}} \right) \right)^{-1} \quad (6.50)$$

Using the modified, sized dependent polarisability all the Bruggeman and Maxwell mixing schemes can be implemented in a way that incorporates size effects. Generally, the onset of changes in the calculated absorption is an indication that size effects are important and should be treated properly.

6.5 Mie Theory of Light Scattering

For spherical particles that are comparable in size to the wavelength of light, the theory developed by Mie and described fully by van de Hulst [23] can be used. Unfortunately, this theory is only applicable to spherical, isotropic, particles where the separation between the particles is large compared with the wavelength of light.

PDielec applies Mie theory using the Python library PyMieScatt [9]. To treat anisotropic systems, PDielec first calculates the Maxwell-Garnett permittivity for the anisotropic sphere embedded with random orientation in the supporting matrix. The program then calculates the required permittivity of an isotropic sphere which would give the same Maxwell-Garnett permittivity as the anisotropic system. The Mie scattering is then performed on this isotropic sphere.

An alternative method that diagonalises the real permittivity and then transforms the complex permittivity into the new basis was found to cause anomalous absorptions at the TO frequencies and so the method is no longer used. However, the theory is given here in case it ever has to be revived.

$$\begin{aligned}\bar{\bar{U}}^T \bar{\bar{\epsilon}}^{real} \bar{\bar{U}} &= \bar{\bar{\epsilon}}^{diagonal} \\ \bar{\bar{\epsilon}}^{new} &= \bar{\bar{U}}^T \bar{\bar{\epsilon}}^{full} \bar{\bar{U}}\end{aligned}\quad (6.51)$$

The core of Mie theory is the calculation of efficiency factors $S1(\theta)$ and $S2(\theta)$ (page 125 in reference [23]). These factors depend upon the size of the particle relative to the wavelength of the light (x) and the refractive index of infrared active material (m_1) relative to that of the supporting medium (m_2). Both refractive indices are scalars, m_1 is complex as light is absorbed by the material and m_2 is assumed to be real.

For each of the three diagonal elements of $\bar{\bar{\epsilon}}^{new}$ the relative refractive index, m is calculated;

$$\begin{aligned}m_1 &= \bar{\bar{\epsilon}}_{w,w}^{new} \\ m &= \frac{m_1}{m_2} \\ x &= \frac{2\pi a}{\lambda} \\ \lambda &= \frac{\lambda_{vac}}{m_2}\end{aligned}\quad (6.52)$$

The size parameter depends upon the refractive index of the supporting medium because the wavelength of light is smaller than its vacuum value in a medium with a refractive index greater than one. The subscript w indicates one of the three directions x, y or z in the basis which diagonalises the real permittivity tensor.

Using the routines available in PyMieScatt [9] the efficiency factor $S(0)$, where $S(0) = S1(0) = S2(0)$, is calculated from m and x . The extinction (this includes absorption and scattering) along with retardation are reflected in the overall complex refractive index, page 129 in reference [23];

$$\begin{aligned}\tilde{m}_w &= m_2 (1 + iS(0) 2\pi N k^{-3}) \\ N &= \frac{f}{V_{sphere}} \\ V_{sphere} &= \frac{4}{3}\pi a^3 \\ k &= \frac{2\pi}{\lambda}\end{aligned}\quad (6.53)$$

Here N is the number density of particles and k is the wave-vector of the light in the surrounding medium. The subscript w indicates that there are three values for the effective refractive index, depending on which

diagonal value of the permittivity tensor is taken. There is a change in sign from the expressions given by Hulst, this is to make the imaginary component compatible with the effective permittivities. Once the effective refractive index is known in each direction the effective permittivity can be calculated using the one-third rule [41];

$$\begin{aligned}\tilde{m}_{eff} &= \frac{1}{3} \sum_w \tilde{m}_w \\ \varepsilon_{eff} &= \tilde{m}_{eff}^2\end{aligned}\tag{6.54}$$

This approach to taking anisotropy into account when the embedded particles are anisotropic but randomly oriented is approximate but has been shown to have a reasonably wide range of application [41].

6.6 Calculation of ATR Extinction Spectra

Attenuated total reflectance spectroscopy (ATR) uses a medium of high refractive index contrast to modify the reflectance and transmission of the infrared radiation. Balan has reported the calculation of the ATR spectrum of apatite [42] and chrysotile [43]. A similar approach is adopted by the PDielec package. Denoting the angle of incidence by θ and the refractive index of the high index, non-absorbing material by n_0 and that of the absorbing material by n_s , the Fresnel amplitudes of reflection of the parallel polarised component (r_p) and perpendicular polarised component (r_s) is given as follows; [44]

$$\begin{aligned}r_s &= \frac{-n_0 \cos \theta + \sqrt{n_s^2 - n_0^2 \sin^2 \theta}}{n_0 \cos \theta + \sqrt{n_s^2 - n_0^2 \sin^2 \theta}} \\ r_p &= \frac{\frac{n_s^2}{n_0} \cos \theta + \sqrt{n_s^2 - n_0^2 \sin^2 \theta}}{\frac{n_s^2}{n_0} \cos \theta + \sqrt{n_s^2 - n_0^2 \sin^2 \theta}}\end{aligned}\tag{6.55}$$

Using f_p and f_s as the fractions of P and S polarised incident light, the extinction is expressed in absorbance units as;

$$\begin{aligned}e_s &= -\log_{10}(r_s^* r_s) \\ e_p &= -\log_{10}(r_p^* r_p) \\ e_{total} &= -\log_{10}(f_p r_p^* r_p + f_s r_s^* r_s)\end{aligned}\tag{6.56}$$

The approach adopted by Balan and in the PDielec package is to use the calculated effective permittivity to determine the (complex) refractive index of the material being studied. ATR measurements are taken with compressed powders of the material, so the appropriate effective medium can be determined from a high volume fraction of the crystal in air.

6.7 Scattering Effects due to Air Inclusions

A common method for taking a terahertz spectrum is to dilute the active dielectric material in a transparent material such as polyethylene or PTFE spheres and then compress the composite material. This leads to the presence of large air inclusions, or bubbles in the sample material. The size of these inclusions can vary in size from 10 to 100 microns and is therefore capable of scattering terahertz radiation. To account for this the effective dielectric constant of the supporting matrix can include these scattering effects using Waterman-Truett scattering theory [45].

$$K = \sqrt{k^2 \left[1 + \frac{2\pi N f(0)}{k^2}\right]^2 - \left[\frac{2\pi N f(\pi)}{k^2}\right]^2}\tag{6.57}$$

Where N is the number density of voids, K is the effective wavenumber of the wave in the material, k is the wavenumber of the wave in the support without air inclusions and $f(0)$ and $f(\pi)$ are the forward and backward amplitude scattering factors.

$$f(\theta) = \frac{i}{k} S(\theta) \quad (6.58)$$

The scattering factors are determined using Mie theory, where it is assumed that there are spherical voids in the matrix, which are large enough to scatter light.

$$\begin{aligned} m_2 &= \frac{K\lambda}{2\pi} \\ \epsilon_e &= m_2^2 \end{aligned} \quad (6.59)$$

The effective permittivity of the matrix with voids is then used in the calculation of the effective medium theory for the dielectric material. The calculation proceeds as it did in the case with no scattering, only now a complex permittivity is used to describe the supporting matrix.

6.8 Removing Baseline Artefacts from Experiment

In the process of reproducing experimental spectra using calculated information, it is often necessary to remove any baseline artefacts from the experimental data. A relatively straightforward approach is adopted by *PDielec* using a Hodrick-Prescott filter. The approach is also called asymmetric least squares smoothing.

If there is a data sequence y , sampled at regular intervals we want to determine the baseline sequence y such that x and y are as close as possible to each other with the requirement that x is as smooth as required.

It can be shown [46] that this can be achieved by an iterative solution for x of ;

$$(\bar{\bar{W}} + \lambda \bar{\bar{D}}^T \bar{\bar{D}})x = \bar{\bar{W}}y \quad (6.60)$$

where $\bar{\bar{W}}$ is a diagonal matrix of weights modified during the iterations to make values of x lie below those of y and $\bar{\bar{D}}$ is a 3rd order differencing matrix. Since the differencing matrix is very sparse, sparse matrix routines are used to solve the equations.

In *PDGui* the value of λ is supplied as $\log \lambda$.

THEORY FOR SINGLE CRYSTAL INFRARED OPTICS

The *SingleCrystal* tab enables the calculation of the optical behaviour of a layer of crystalline films. The methods used are based on the pyGTM code available on GitHub [11]. This code implements a general transfer matrix method described by Passler et al. [12] and that builds on previous publications [13, 14]. The code has been modified substantially to model incoherent and coherent systems. Also a scattering matrix has been implemented as described in the PyLlama software documentation [15]

A summary of the theory underlying the transfer and scattering matrix methods is given to aid in the understanding of the methods implemented and the range of their applications.

In general PDGui can handle multilayered frequency-dependent films, sandwiched between a superstrate and a substrate. For the sake of simplicity, the following will describe a system with a single dielectric (which is defined from the calculated permittivity from a DFT calculation or an experimental file). The media will be indexed 0, 1 and 2 for the superstrate, dielectric and substrate respectively. A schematic illustrating this is shown in Fig. 7.1. To aid understanding of the input and output and the limitations of the methods used, a brief summary of the transfer matrix method is given first, specific to its application in PDGui. This will be followed by a description of the scattering matrix approach as implemented in the package. The treatment of light transmission and reflection in incoherent media will also be outlined.

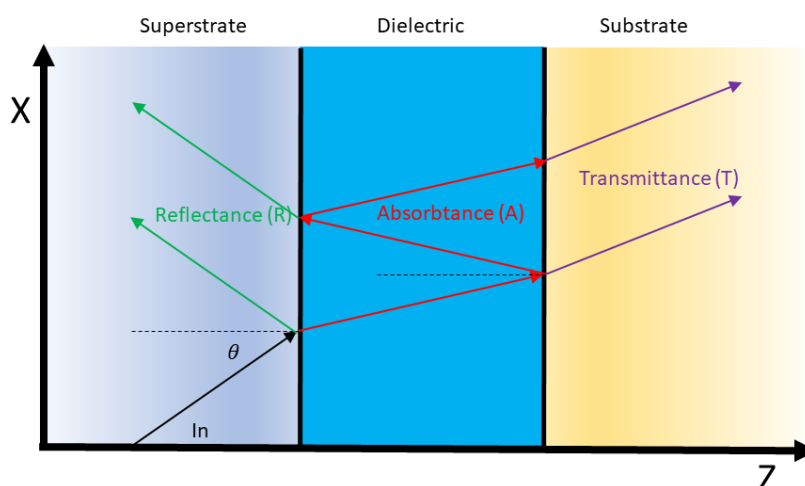


Fig. 7.1: Layers used to calculate Transmittance, Reflectance and Absorbance

7.1 The Transfer Matrix

At an interface, the tangential (in-plane) electric and magnetic fields (given by a vector $\bar{F}_i$) match in both media. This requires that the 4 amplitudes (vector $\bar{A}_i$) of the electric and magnetic fields, forward and backward, s- and p- polarised, in medium i are related through a 4x4 dynamical matrix $\bar{D}_i$.

$$\bar{F}_i = \bar{D}_i \bar{A}_i \quad (7.1)$$

Requiring that the in-plane fields match in both media at an interface gives;

$$\begin{aligned} \bar{F}_{i-1} &= \bar{F}_i \\ \bar{D}_{i-1} \bar{A}_{i-1} &= \bar{D}_i \bar{A}_i \\ \bar{A}_{i-1} &= \bar{D}_{i-1}^{-1} \bar{D}_i \bar{A}_i \end{aligned} \quad (7.2)$$

Within a given, homogeneous, layer the mode amplitudes will change according to the propagation through the medium described by an diagonal exponential matrix $\bar{P}_i$. Thus the amplitudes on the left side of a layer are related to those on the right side by;

$$\bar{A}_{i,left} = \bar{P}_i \bar{A}_{i,right} \quad (7.3)$$

The total transfer matrix for the 3 media considered (superstrate, dielectric and substrate; 0, 1 and 2 respectively) is therefore;

$$\bar{T} = \bar{D}_0^{-1} (\bar{D}_1 \bar{P}_1 \bar{D}_1^{-1}) \bar{D}_2 \quad (7.4)$$

The elements of the total 4x4 transfer matrix $\bar{T}$ can be used to determine the total reflectance and transmittance for each s- and p- mode.

For the multi-layer case it is straightforward to generalise this expression by writing the transfer matrix, $\bar{T}_i$, for the i 'th layer as:

$$\bar{T}_i = \bar{D}_i \bar{P}_i \bar{D}_i^{-1} \quad (7.5)$$

The total transfer matrix for the whole system consisting of a superstrate, N layers and a substrate can then be written:

$$\bar{T} = \bar{D}_0^{-1} \left(\prod_{i=1}^N \bar{T}_i \right) \bar{D}_{N+1} \quad (7.6)$$

The role of the total transfer matrix is to convert the field amplitudes on the left-hand side (the incident amplitudes) to those on the right-hand side (the transmitted amplitudes).

7.2 The Scattering Matrix

Unfortunately, transfer matrix theory is known to exhibit serious numerical instabilities when the layers become thick. This is mainly due to the fact that the forward waves (which move from left to right) are defined as the layers in the transfer matrix are being added from right to left. The forward moving wave is growing in amplitude as the layers are being added and this leads to all kinds of instabilities. The scattering matrix method overcomes this problem by providing a matrix for each layer which couples the incoming waves (two from the left and two from the right) to the outgoing waves. In this way it is possible to describe waves which are always decreasing in amplitude. To demonstrate the difference between the approaches the transfer matrix connects fields in one layer to those in the next layer:

$$\begin{bmatrix} \bar{F}_{i+1}^+ \\ \bar{F}_{i+1}^- \end{bmatrix} = \bar{T}_i \begin{bmatrix} \bar{F}_i^+ \\ \bar{F}_i^- \end{bmatrix} \quad (7.7)$$

The scattering matrix for this is written;

$$\begin{bmatrix} \bar{\mathbf{F}}_i^- \\ \bar{\mathbf{F}}_{i+1}^+ \end{bmatrix} = \bar{\mathbf{S}}_i \begin{bmatrix} \bar{\mathbf{F}}_i^+ \\ \bar{\mathbf{F}}_{i+1}^- \end{bmatrix} \quad (7.8)$$

The fields on the right-hand side represent light entering the layer and the fields on the left-hand side represent light leaving the layer. The fields in these expressions are two component vectors representing the p- and s- fields. The superscripts ‘+’ and ‘-’ indicate forward and backward moving light respectively. It should be noted that this definition is slightly different to the definition used by PyLlama.

The scattering matrix is more complicated to handle than the transfer matrix. Scattering matrices between different layers can be combined using the Redheffer product [47]

$$\bar{\mathbf{S}} = \bar{\mathbf{S}}_0 \otimes \dots \bar{\mathbf{S}}_i \otimes \bar{\mathbf{S}}_{i+1} \otimes \dots \bar{\mathbf{S}}_N \quad (7.9)$$

The transfer matrix method in PDGui is provided by a modified version of the pyGTM [11]. A full description of the method is provided by Passler et al. [12]. The scattering matrix method was implemented within the pyGTM framework using the algorithms suggested in PyLlama [15], with some modifications.

7.3 Incoherence

Because the above approaches use amplitudes there is always the possibility that a mismatch in phase for a reflected or transmitted wave will result in interference and therefore oscillations (etalons) in intensity as the frequency is changed. Experimentally, for thick crystals, it is known that the presence of defects in the crystal and at the interfaces, and differences in thickness cause this coherence to be lost and instead incoherent light transmission is observed and the etalons are lost. PDielec implements several schemes treating incoherence.

The “Incoherent (intensity)” method creates a transfer matrix like that in (7.4), but instead of amplitudes, uses intensities for the transfer matrix of the dielectric. This method has not been implemented for the scattering matrix formalism although such an approach has been implemented in FSRStools [48] a Python implementation of Yeh’s transfer matrix method [49]. The theory is described by Beck [50] and by Katsidis and Siapkias [51]. This approach has been incorporated in PDGui by modifying the total transfer matrix as follows;

$$\bar{\mathbf{T}}_{int} = | \bar{\mathbf{D}}_0^{-1} \bar{\mathbf{D}}_1 |^2 | \bar{\mathbf{P}}_1 |^2 | \bar{\mathbf{D}}_1^{-1} \bar{\mathbf{D}}_2 |^2 \quad (7.10)$$

The use of intensities instead of amplitudes ensures that the phase information is lost. The actual implementation is appropriate for a multi-layer system where all dielectric layers are treated as being incoherent films, whatever their thickness.

The “Incoherent (phase averaging)” method, which is implemented for both transfer and scattering matrix formalisms, instead averages over the phase of the propagating matrix ($\bar{\mathbf{P}}_i$) for the backward travelling waves in the incoherent material.

$$\bar{\mathbf{P}}_i = \begin{bmatrix} e^{-ik_0 d_i q_i^1} & 0 & 0 & 0 \\ 0 & e^{-ik_0 d_i q_i^2} & 0 & 0 \\ 0 & 0 & e^{-ik_0 d_i q_i^3} & 0 \\ 0 & 0 & 0 & e^{-ik_0 d_i q_i^4} \end{bmatrix} \quad (7.11)$$

The layer thickness is d_i , $\bar{q}_i$ are the wavevectors of the 4 modes. The backward travelling waves are modified by adding a phase between 0 and 2π . Early implementations of this method chose the phase at random and the final results were averaged over many phases [52]. It has been shown [53] that a more efficient procedure is to average over an equally spaced range of phases. In PDGui 10 equally

spaced phases are used and the full range of possible phase variation is considered. In principle it is also possible to treat some partial incoherence by limiting the range of phases considered.

A third method for treating incoherence, called “Incoherent (non-reflective)” is also implemented for both transfer and scattering matrix methods. This method sets the amplitude of the internally reflected wave to zero. This has the effect of artificially increasing the absorptance in the layer.

For all methods, the absorptance (A) is defined from the reflectance (R) and transmittance (T) as below;

$$A = 1 - R - T \quad (7.12)$$

7.4 Comparison of computational approaches

Fig. 7.2 shows a comparison of the approaches discussed above applied to MgO. The DFT calculations were performed by Castep. The incident angle is 45° and the film thickness is $10\mu\text{m}$. The calculations were performed using the transfer matrix method. The ‘Coherent’ curve shows large amplitude oscillations before and after the Reststrahlen region of absorption. The ‘Incoherent (phase averaging)’ and ‘Incoherent (intensity)’ are similar at high frequencies, but at low frequencies the intensity calculation shows higher reflectance.

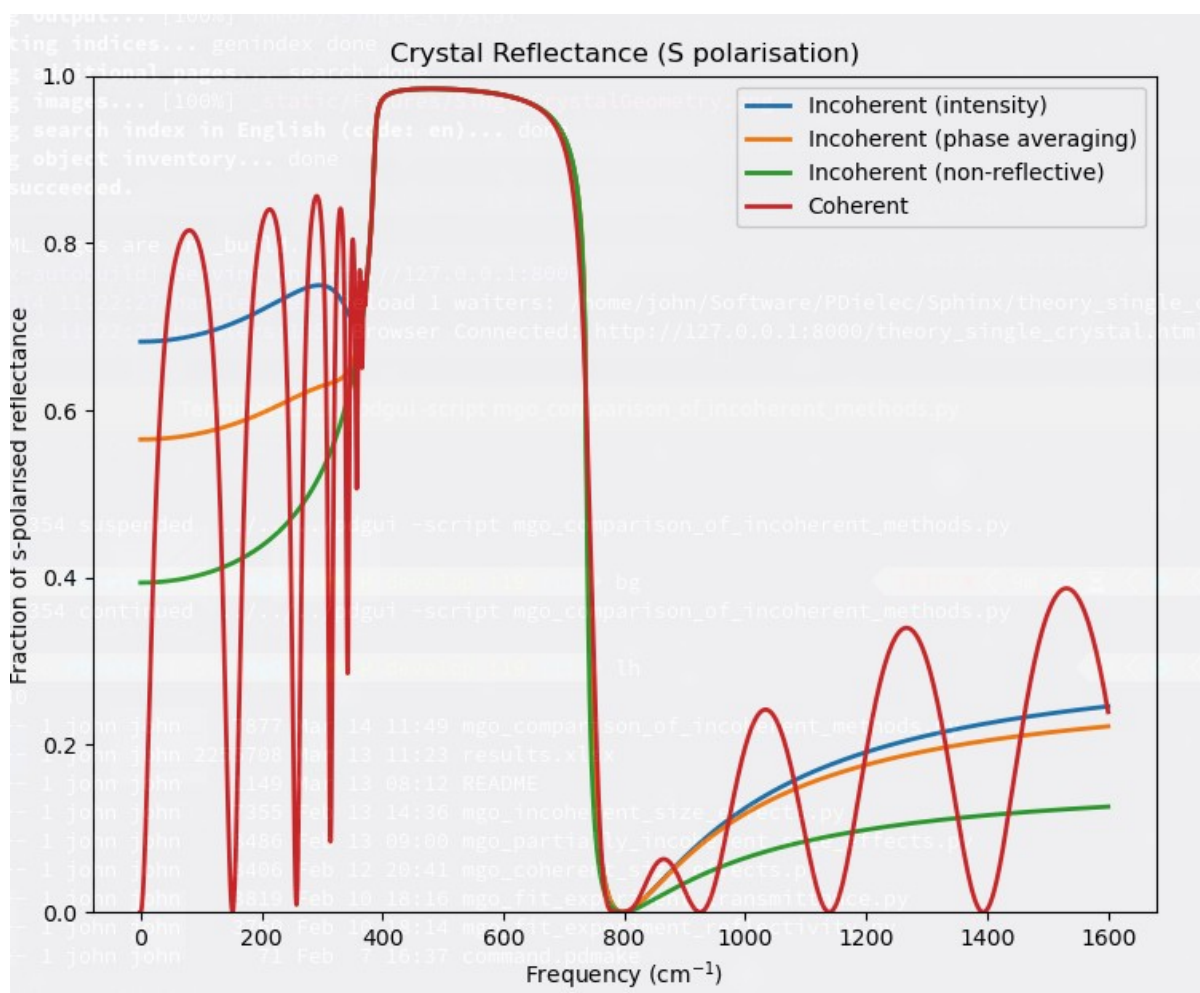


Fig. 7.2: Comparison of Incoherent Methods

With the transfer matrix method, it is common to see numerical problems in regions where the extinction coefficient is high and the dielectric is thick. The numerical problems manifest themselves with either with sudden spikes in reflectance or with a ‘Warning exponential overflow occurred’ message.

These problems manifest themselves acutely in calculations on l-alanine. The DFT calculations were performed by Crystal. Fig. 7.3 shows the predicted p-polarised reflection for a 1mm film of the (001) surface with an azimuthal angle of 90° .

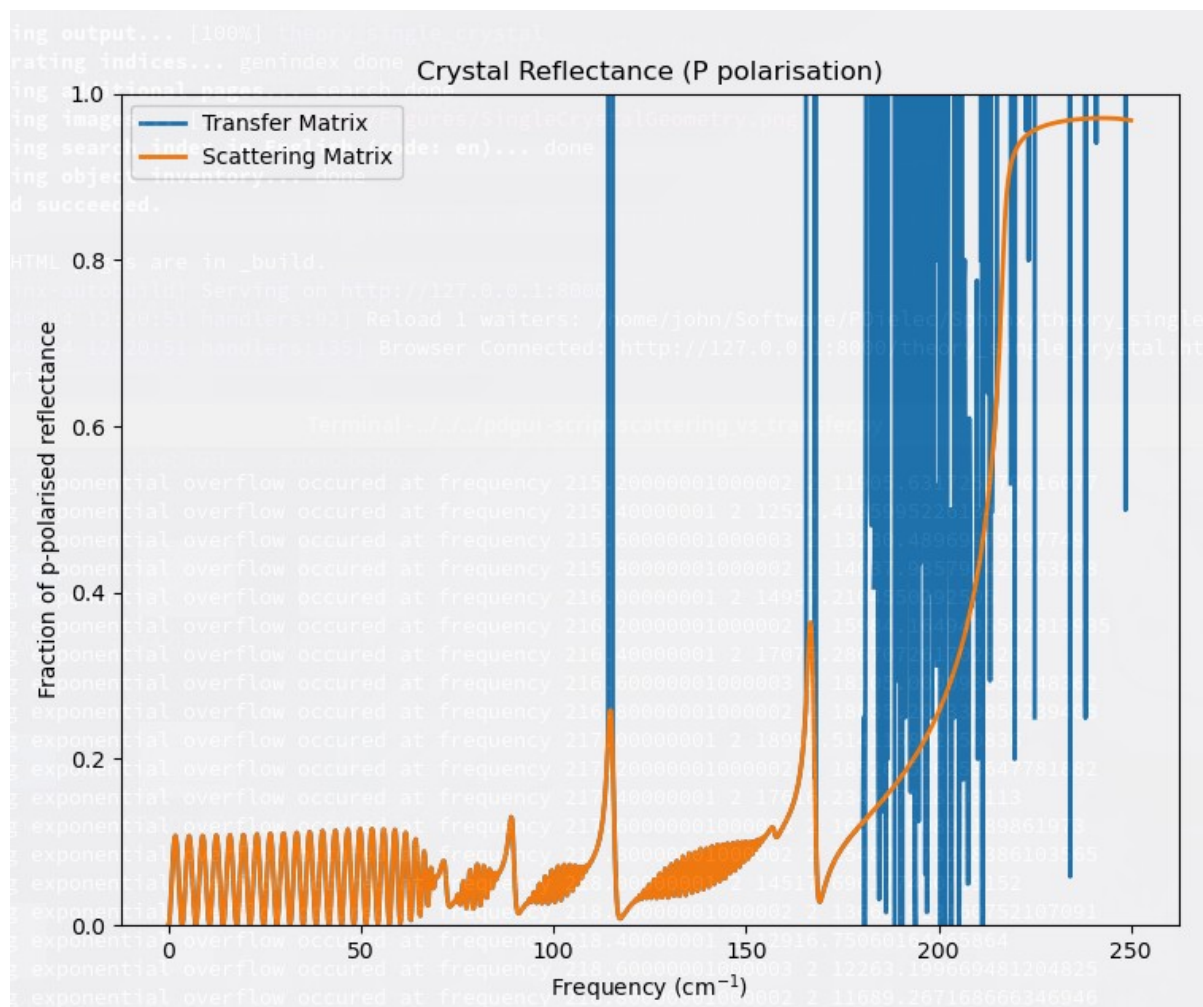


Fig. 7.3: Comparison of Transfer and Scattering Matrix and Methods

The transfer and scattering matrix methods agree well with each other upto about 114 cm^{-1} . Above 180 cm^{-1} the transfer matrix method is completely unstable, while the scattering matrix method works well.

7.5 Crystal & Laboratory Coordinate Frames

There are two coordinate systems to consider. The first is the laboratory coordinate system (X,Y,Z). It is assumed that the normal to the crystal surface is aligned with the Z-axis and that the incident radiation is in the X-Z plane. p-polarised light is polarised parallel to this plane and s-polarised perpendicular (senkrecht) to this plane. The laboratory frame is illustrated in Fig. 7.4.

The other coordinate system of interest is that of the crystal. The calculation of the permittivity tensor is performed in the crystal coordinate system (x,y,z). The crystal plane is defined by a set of Miller indices (hkl) and the normal to the crystal surface is rotated to align with the laboratory Z-axis. Finally in the laboratory frame the crystal may be rotated around the Z-axis by the azimuthal angle ϕ . The precise definition of the azimuthal angle is somewhat arbitrary as it depends on the crystal unit-cell definition, but PDGui provides information of the details of the relationship between the crystal axes and the laboratory frame. The arrangement described is illustrated in the Figure below where E_s and E_p are

the field directions of the s- and p- polarised incident light respectively. Information about the crystal coordinates relative to the laboratory frame is given in the GUI.

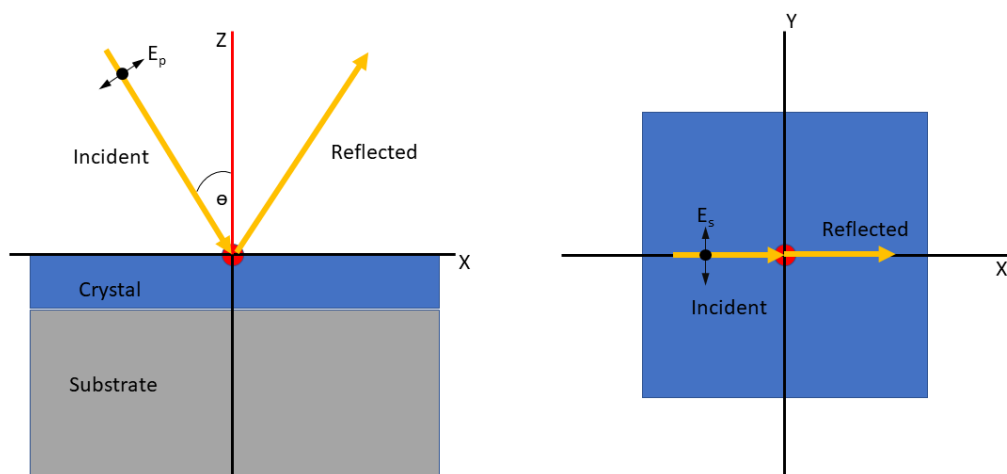


Fig. 7.4: Definition of single crystal laboratory coordinates in PDGui

THEORY FOR POWDER RAMAN

The *Powder Raman* calculation predicts non-resonant Raman spectra from Raman tensors calculated by a solid-state quantum mechanical program. The calculation follows the Placzek approximation, in which the intensity of a first-order Stokes line is determined by the derivative of the polarisability with respect to the normal-mode coordinate. For a mode m , incident laser frequency ν_L , scattered frequency $\nu_S = \nu_L - \nu_m$, incident polarisation $\mathbf{e}_L$, and detected polarisation $\mathbf{e}_S$, the scattering strength is proportional to

$$S_m = \frac{\nu_S^4}{c^4} \frac{n(\nu_m) + 1}{8\pi^2 \nu_m} \left| \mathbf{e}_S^T \bar{\bar{\mathbf{R}}}^{(m)} \mathbf{e}_L \right|^2 \quad (8.1)$$

where $n(\nu_m)$ is the Bose-Einstein occupation factor. PDielec reports Raman intensities in arbitrary units, so the most important outputs are the relative mode strengths and the changes introduced by particle shape, local fields, polarisation geometry and linewidth.

8.1 Raman Tensor

Every Raman output reader returns the bulk dielectric-derivative tensor

$$\bar{\bar{\mathbf{R}}}_\epsilon^{(m)} = \sqrt{V_{cell}} \frac{\partial \bar{\bar{\epsilon}}_\infty}{\partial Q_m} \quad (8.2)$$

where V_{cell} is in $\text{\AA}^3$ and Q_m is in $\text{\AA}$ times the square root of amu. The tensor units are $(\text{\AA}/\text{amu})^{1/2}$. The reader already includes $\sqrt{V_{cell}}$; the powder and crystal calculations apply no additional cell-volume factor. A source polarizability-volume derivative $R_\alpha = \partial[V_{cell}(\epsilon - I)/(4\pi)]/\partial Q_m$ is converted on read using $R_\epsilon = 4\pi R_\alpha/\sqrt{V_{cell}}$. Activities in $\text{\AA}^4/\text{amu}$ are display quantities, obtained by multiplying the internal activities by $V_{cell}/(16\pi^2)$.

For particles whose Raman tensor does not depend on shape, the powder average can be written in terms of rotational invariants of $\bar{\bar{\mathbf{R}}}$. For one mode,

$$\begin{aligned} \alpha &= \frac{1}{3} (R_{xx} + R_{yy} + R_{zz}) \\ \bar{\gamma} &= \frac{1}{2} (\bar{\bar{\mathbf{R}}} + \bar{\bar{\mathbf{R}}}^T) - \alpha \bar{\mathbf{1}} \\ \bar{\kappa} &= \frac{1}{2} (\bar{\bar{\mathbf{R}}} - \bar{\bar{\mathbf{R}}}^T) \end{aligned} \quad (8.3)$$

and

$$\begin{aligned} \alpha^2 &= \alpha \alpha^* \\ \gamma^2 &= \frac{3}{2} \sum_{ij} \gamma_{ij} \gamma_{ij}^* \\ \kappa^2 &= \frac{3}{2} \sum_{ij} \kappa_{ij} \kappa_{ij}^* \end{aligned} \quad (8.4)$$

The usual parallel (VV), crossed (VH), and unpolarised powder strengths are then

$$\begin{aligned} I_{VV} &\propto 45\alpha^2 + 4\gamma^2 \\ I_{VH} &\propto 3\gamma^2 + 5\kappa^2 \\ I_{total} &\propto 45\alpha^2 + 7\gamma^2 + 5\kappa^2 \end{aligned} \quad (8.5)$$

For non-chiral, non-magnetic Raman scattering the antisymmetric term is normally zero.

8.2 Local Field Correction

For powder infrared calculations PDielec treats each crystallite as a small inclusion in a non-absorbing matrix. The powder Raman calculation uses the same macroscopic picture when local-field effects are requested. The particle is assumed to be much smaller than the laser wavelength, so the field inside the particle is uniform. For an inclusion with optical permittivity $\bar{\epsilon}_i$, a matrix permittivity ϵ_e , and a depolarisation tensor $\bar{\bar{L}}$, the internal field is

$$\mathbf{E}_i = \bar{\bar{N}} \mathbf{E}_e \quad (8.6)$$

with

$$\bar{\bar{N}} = \left[\bar{\bar{1}} + \frac{1}{\epsilon_e} \bar{\bar{L}} \left(\bar{\epsilon}_i - \epsilon_e \bar{\bar{1}} \right) \right]^{-1} \quad (8.7)$$

For a sphere $\bar{\bar{L}}$ has diagonal elements of $1/3$. Other ellipsoidal shapes use the same depolarisation tensor as the powder infrared effective-medium theory.

For a reciprocal dielectric, the effective bulk Raman tensor includes one incident and one scattered optical local-field factor:

$$\bar{\bar{R}}_{eff}^{(m)} = \bar{\bar{N}}(\nu_S)^T \left[\bar{\bar{R}}_{\epsilon,0}^{(m)} + \Delta \bar{\bar{R}}_{\epsilon,EO}^{(m)} \right] \bar{\bar{N}}(\nu_L) \quad (8.8)$$

The transpose is ordinary, including for complex reciprocal permittivities. The current powder implementation uses the same optical permittivity at both frequencies. R_{eff} retains the bulk R_ϵ normalization; it is not an extensive particle dipole-polarizability derivative.

8.3 Particle Frequencies

In ionic crystals the particle boundary conditions can also shift the vibrational frequencies. The phonon-induced polarisation is

$$\mathbf{P}_{ph} = \frac{1}{V} \bar{\bar{Z}}^{mw} \mathbf{x} \quad (8.9)$$

where $\bar{\bar{Z}}^{mw}$ is the mass-weighted Born charge tensor and $\mathbf{x}$ is the vector of mass-weighted atomic displacements. The depolarisation field generated by this polarisation applies a restoring force to the ions. This gives an effective particle dynamical matrix

$$\bar{\bar{D}}^{particle} = \bar{\bar{D}}^{TO} + \frac{1}{\epsilon_0 \epsilon_e V} \left(\bar{\bar{Z}}^{mw} \right)^T \bar{\bar{N}}_{bg} \bar{\bar{L}} \bar{\bar{Z}}^{mw} \quad (8.10)$$

where $\bar{\bar{D}}^{TO}$ is the transverse-optic dynamical matrix and $\bar{\bar{N}}_{bg}$ is the internal-field tensor constructed from the background permittivities. Diagonalising $\bar{\bar{D}}^{particle}$ gives particle-mode frequencies and eigenvectors appropriate to the selected crystallite shape.

In the SI expression above Born charges are in coulombs and masses are in kilograms. The implementation uses charges in electrons, masses in electron-mass units, volume in Bohr³, and the prefactor $4\pi/(\epsilon_e V)$. The same electrostatic kernel $K = N_{bg}L/\epsilon_e$ determines the particle EO correction:

$$\Delta R_{\epsilon,EO,ij}^{(m)} = -8\pi \sum_l \tilde{\chi}_{ijl}^{(2)} (K Z^{mw} u_m)_l.$$

Here $\tilde{\chi}^{(2)}$ is the cell-dependent internal reader value, not the raw susceptibility in pm/V. Its conversion is specified in *Theory for Crystal Layer Raman*. Both the mechanical Raman tensor and the mode charge must use the same particle eigenvector. No bulk propagation direction is needed for this ellipsoid boundary-value model. It describes a quasistatic particle in a lossless scalar host; it is not a bulk directional LO/TO powder average. `Matrix=none` bypasses particle and optical-field corrections; use a host with permittivity one to describe an isolated particle in vacuum.

8.4 Orientation Averaging

For a crystallite in orientation g , the particle Raman tensor in the laboratory frame is

$$\bar{\bar{R}}_{lab}^{(m)}(g) = g \bar{\bar{R}}_{eff}^{(m)} g^T \quad (8.11)$$

The powder strength is the orientational average over all rotations,

$$\langle S_m \rangle_{powder} = \int_{SO(3)} S_m(g) dg \quad (8.12)$$

For spherical particles this average reduces to the invariant expressions above. For more general particle tensors, PDielec samples the rotation group and averages the fixed laboratory polarisation geometry over many crystallite orientations.

If Born-charge or Hessian data are unavailable, PDielec retains the supplied TO frequencies and Raman tensors and omits the particle frequency/EO correction. Optical local fields still apply. For fixed relative shape/crystal axes in an isotropic host, the effective tensor rotates as one rank-two tensor, so the analytic invariants also supply the nonspherical fallback. Missing polar data therefore do not imply zero Raman scattering.

Both analytic and sampled paths exclude nonfinite, nonpositive and above-laser mode frequencies before evaluating the Stokes weight, as well as modes below the acoustic cutoff. This does not remove unstable frequencies from the reader.

The broadened Raman spectrum is formed as a sum over active modes,

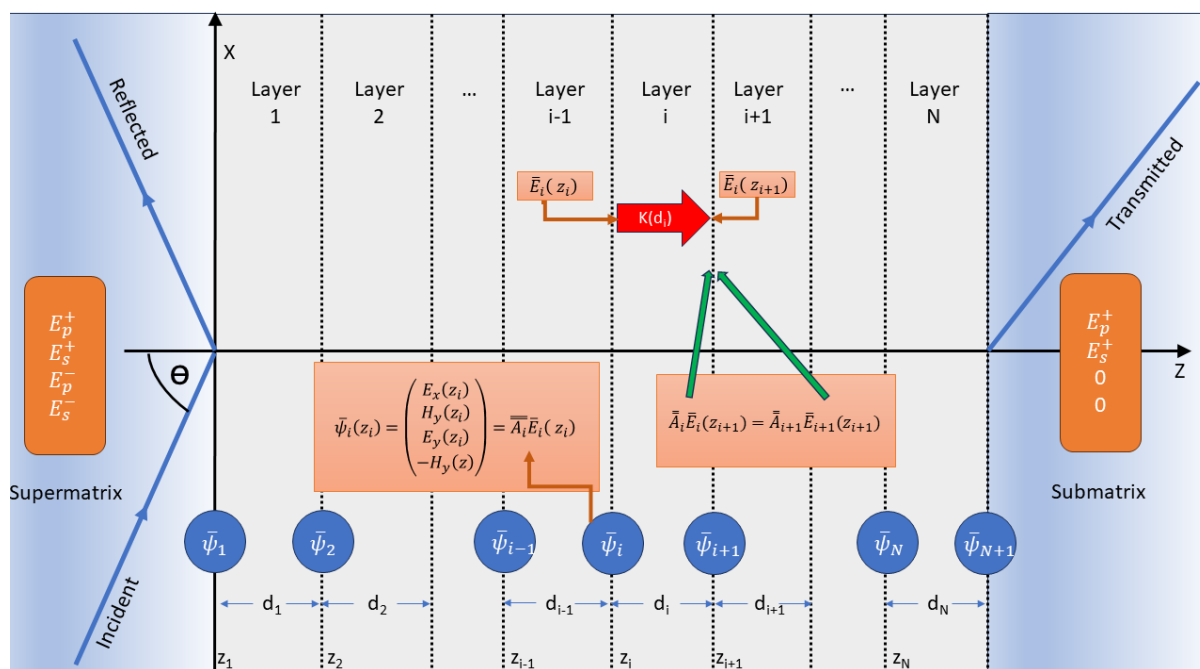
$$I(\Delta\nu) = \sum_m \langle S_m \rangle_{powder} \frac{\sigma_m}{(\Delta\nu - \nu_m)^2 + \sigma_m^2} \quad (8.13)$$

where σ_m is the Lorentzian half-width at half maximum.

The *Crystal Raman* calculation models Raman scattering from a crystal slab or a multilayer stack. It combines the Raman tensors and phonon frequencies from a lattice-dynamical calculation with the optical electric fields calculated by the transfer matrix or scattering matrix methods described in *Theory for Single Crystal Infrared Optics*.

The method is intended for non-resonant Raman scattering from homogeneous Raman-active layers. The incident laser field and the reciprocal scattered field may be modified by refraction, reflection, absorption and interference within the stack. These effects are included by evaluating the electric fields throughout each Raman-active layer and integrating the Raman source over the layer thickness.

The crystal layer uses the same laboratory frame as the single-crystal infrared calculation. The layer surface lies in the XY plane and the surface normal is the laboratory Z direction. The incident radiation lies in the XZ plane; p-polarised light is polarised in this plane and s-polarised light is polarised along Y .



The laser is incident with angle θ_L . The detector may be on the superstrate side for backscattering or on

the substrate side for forward scattering. The detected scattered radiation has frequency

$$\nu_S = \nu_L - \nu_m \quad (9.1)$$

for Stokes scattering from mode m .

9.2 Layer Raman Amplitude

At a point z_s inside a Raman-active layer the scattering amplitude of mode m is

$$A_m(z_s) \propto \mathbf{E}_S(z_s, \nu_S)^T \bar{\bar{\mathbf{R}}}_{lab}^{(m)} \mathbf{E}_L(z_s, \nu_L) \quad (9.2)$$

where $\mathbf{E}_L$ is the local electric field produced by the incident laser, $\mathbf{E}_S$ is the reciprocal field associated with the detected scattered channel, and $\bar{\bar{\mathbf{R}}}_{lab}^{(m)}$ is the Raman tensor rotated into the laboratory frame. The transpose on $\mathbf{E}_S$ is not a Hermitian conjugate; this is the usual reciprocity overlap used for Raman scattering from layered optical systems.

For a homogeneous layer ℓ , the layer amplitude is obtained by integrating through the layer thickness,

$$A_{\ell m} = \int_{z_\ell}^{z_{\ell+1}} \mathbf{E}_S(z, \nu_S)^T \bar{\bar{\mathbf{R}}}_{\ell,lab}^{(m)} \mathbf{E}_L(z, \nu_L) dz \quad (9.3)$$

PDielec evaluates this integral numerically using Gauss-Legendre quadrature points inside each Raman-active layer.

9.3 Optical Fields

The fields $\mathbf{E}_L$ and $\mathbf{E}_S$ are calculated from the same transfer matrix or scattering matrix machinery used for single-crystal infrared spectra. At optical laser frequencies the phonon contribution to the permittivity is not included; the high-frequency optical permittivity of each material is used for propagation through the stack.

The incident field $\mathbf{E}_L(z, \nu_L)$ is calculated in the original multilayer system at the laser frequency. The scattered field is calculated at the shifted frequency ν_S . For backscattering the reciprocal field is launched from the superstrate side. For forward scattering it is launched from the substrate side, which is implemented by using the reversed stack and mapping each integration point to the corresponding position in that reversed system.

With the script option `Approximate ES=True`, the reciprocal detector field is instead evaluated once at the laser frequency and reused for all modes. The resolved collection geometry is retained. The incident field is reused only when the optical system and signed angle match those of the detector field. The default is `False`, which uses each mode's Stokes frequency.

The detected channel can be p-polarised, s-polarised, or unpolarised. In the unpolarised case the p- and s-detected intensities are summed incoherently.

9.4 Raman Tensor Orientation

Reader tensors use $R_\epsilon = \sqrt{V_{cell}} \partial \epsilon_\infty / \partial Q_m$, with volume in $\text{\AA}^3$ and Q_m in $\text{\AA}$ times the square root of amu. They already contain the cell-volume normalization and enter `RamanLayer` directly. Raw Gaussian susceptibility derivatives require $\partial \epsilon = 4\pi \partial \chi$; SI dimensionless susceptibility instead obeys $\partial \epsilon = \partial \chi_{SI}$.

The Raman tensor from the quantum mechanical calculation is defined in the crystal coordinate frame. If $\bar{\bar{G}}$ maps vectors from the crystal frame to the laboratory frame,

$$\mathbf{v}_{lab} = \bar{\bar{G}} \mathbf{v}_{crystal} \quad (9.4)$$

then the laboratory Raman tensor is

$$\bar{\bar{R}}_{lab}^{(m)} = \bar{\bar{G}} \bar{\bar{R}}_{crystal}^{(m)} \bar{\bar{G}}^T \quad (9.5)$$

The rotation includes the surface orientation, specified by the layer Miller indices, and the azimuthal rotation about the laboratory Z axis. This allows the Raman intensity to be calculated as a function of azimuthal angle.

9.5 Combining Layers and Modes

For one mode, the layer amplitudes may be combined coherently or incoherently. Coherent combination sums amplitudes before squaring,

$$I_m \propto \left| \sum_{\ell} A_{\ell m} \right|^2 \quad (9.6)$$

whereas incoherent combination sums the layer intensities,

$$I_m \propto \sum_{\ell} |A_{\ell m}|^2 \quad (9.7)$$

This layer summation option is distinct from the *Coherent* or *Incoherent* choice in the layer table. The layer-table option controls optical propagation through that layer: whether the transfer/scattering matrix calculation retains phase interference from internal reflections, averages over phase, or treats a thick layer approximately. It therefore changes the optical fields $\mathbf{E}_L$ and $\mathbf{E}_S$ used in the Raman overlap.

The *Layer combination* option controls only how Raman amplitudes from different Raman-active layers are combined after those optical fields have been calculated. Use *Incoherent intensities* when separate active layers should contribute independent intensities, and use *Coherent amplitudes* only when Raman emission from those layers is expected to retain a fixed phase relationship and interfere.

The *Depth coherence* option is a third, more local coherence choice. It controls how Raman sources are combined through the thickness of each individual Raman-active layer. *Coherent amplitude* integrates the complex source amplitude before squaring. This assumes a common phonon final state with a specified spatial phase; small film thickness alone does not establish that assumption:

$$I_{\ell m} \propto \left| \int_{\ell} \mathbf{E}_S(z, \nu_S)^T \bar{\bar{R}}_{\ell, lab}^{(m)} \mathbf{E}_L(z, \nu_L) dz \right|^2. \quad (9.8)$$

Incoherent intensity integrates the local intensity:

$$I_{\ell m} \propto \int_{\ell} \left| \mathbf{E}_S(z, \nu_S)^T \bar{\bar{R}}_{\ell, lab}^{(m)} \mathbf{E}_L(z, \nu_L) \right|^2 dz, \quad (9.9)$$

which is more appropriate for thick or bulk samples where long-range Raman phase coherence is not physical. When *Incoherent intensity* is selected, PDielec also forces separate Raman-active layers to be combined as incoherent intensities, because there is then no well-defined layer amplitude to add coherently.

The mode intensity is then multiplied by the Stokes radiation and thermal factor,

$$(\nu_L - \nu_m)^4 \frac{n(\nu_m) + 1}{\nu_m} \quad (9.10)$$

where $n(\nu_m)$ is the Bose-Einstein occupation factor. The final spectrum is formed by applying Lorentzian broadening to each active mode,

$$I(\Delta\nu) = \sum_m I_m \frac{\sigma_m}{(\Delta\nu - \nu_m)^2 + \sigma_m^2} \quad (9.11)$$

where σ_m is the Lorentzian half-width at half maximum.

Crystal and powder calculations use the same scattered-frequency fourth-power weight. It is applied once per final phonon state, using its corrected frequency, including modal-pair, worker and contribution outputs. Raw optical amplitudes remain unweighted. This changes both the overall scale and relative line heights from the reduced crystal intensities produced before the September 2026 repair. Absolute experimental collection and cross-section constants are still omitted.

Only finite frequencies satisfying $0 < \nu_m < \nu_L$ and the acoustic cutoff contribute, independently of the scattered-field approximation. Negative frequencies remain available in phonon diagnostics as evidence of instability. Complex mechanical Raman tensors retain their phase through NAC mixing and EO addition; the EO susceptibility model remains the static electronic response.

9.6 Layer Phonon Frequencies

The optical field calculation and the active-layer vibrational calculation are separate parts of the model. The transfer or scattering matrix calculation determines the laser field and the reciprocal scattered field at optical frequencies. It does not, by itself, change the phonon frequencies. Any shift of an infrared-active Raman mode away from the bulk transverse-optic value must therefore be included in the phonon dynamical matrix used for the Raman-active layer.

For each Raman-active layer PDielec starts from the transverse-optic dynamical matrix, the mass-weighted Born charge tensor and the Raman tensor of the layer material. The phonon-induced polarisation is written as

$$\mathbf{P}_{ph} = \frac{1}{V} \bar{\bar{\mathbf{Z}}}^{mw} \mathbf{x} \quad (9.12)$$

where $\mathbf{x}$ is the mass-weighted normal-coordinate displacement vector and V is the unit-cell volume used for the Born charges. A macroscopic electrostatic field generated by this polarisation gives an additional restoring-force term,

$$\bar{\bar{\mathbf{D}}}^{eff} = \bar{\bar{\mathbf{D}}}^{TO} + \frac{1}{\epsilon_0 V} \left(\bar{\bar{\mathbf{Z}}}^{mw} \right)^T \bar{\bar{\mathbf{S}}}_{ph} \bar{\bar{\mathbf{Z}}}^{mw} \quad (9.13)$$

where $\bar{\bar{\mathbf{S}}}_{ph}$ is the phonon-field screening tensor. Different choices of $\bar{\bar{\mathbf{S}}}_{ph}$ define the phonon-frequency model for the active layer. After the correction has been applied, $\bar{\bar{\mathbf{D}}}^{eff}$ is diagonalised and the resulting frequencies and eigenvectors are used to form the Raman spectrum. The Raman tensors are transformed consistently with the corrected eigenvectors when mode mixing is introduced by the macroscopic field correction.

PDielec provides four levels of treatment for the active-layer phonon frequencies. These levels should be regarded as separate approximations for the phonon problem, not as alternatives for solving the optical propagation problem.

TO — bulk TO frequencies (GUI label: *TO*)

No macroscopic phonon-field correction is applied:

$$\bar{\bar{S}}_{ph} = \bar{0}, \quad \bar{\bar{D}}^{eff} = \bar{\bar{D}}^{TO} \quad (9.14)$$

This is appropriate for modes that are not infrared active, for calculations where the long-range electric field is deliberately neglected, or as a reference calculation. It is also the most direct comparison with a conventional Placzek Raman calculation using the bulk transverse-optic phonon frequencies.

Snell's law — travelling-wave NAC correction (GUI label: *Snell's law* or *Snell's law (EO)*)

The phonon is assigned the momentum carried by the Raman process. In the laboratory frame,

$$\mathbf{q}_{ph} = \mathbf{k}_L - \mathbf{k}_S \quad (9.15)$$

where $\mathbf{k}_L$ is the optical wave vector of the laser field in the active layer and $\mathbf{k}_S$ is the wave vector of the emitted Stokes photon in the same layer. The direction $\hat{\mathbf{q}}_{ph} = \mathbf{q}_{ph}/|\mathbf{q}_{ph}|$ is used in the usual non-analytic correction. After rotating the layer background permittivity into the laboratory frame, the screening tensor is

$$\bar{\bar{S}}_{ph}^{NAC} = \frac{\hat{\mathbf{q}}_{ph} \hat{\mathbf{q}}_{ph}^T}{\hat{\mathbf{q}}_{ph}^T \bar{\bar{\epsilon}}_{i,lab}^b \hat{\mathbf{q}}_{ph}} \quad (9.16)$$

This is the bulk non-analytic correction evaluated for the Raman scattering geometry. It is useful when the phonon is treated as a propagating long-wavelength bulk mode in the active material. In a multilayer calculation the optical fields may contain several forward and backward components. The implementation therefore uses the wave-vector branch associated with the selected incident and detected optical channels; if more than one branch is retained, the corresponding Raman contributions are evaluated and combined using the same coherent or incoherent choices as the optical Raman amplitudes.

Dominant mode — dominant Berreman eigenmode (GUI label: *Dominant mode* or *Dominant mode (EO)*)

Instead of the macroscopic Snell's-law estimate, the phonon wavevector direction is taken from the actual Berreman eigenvalue of the dominant forward-propagating optical mode in the active layer. The generalised transfer matrix (GTM) system is solved at the laser frequency ν_L to obtain the four Berreman k_z eigenvalues of the layer; the transmitted-p eigenvalue is used for p-polarised incidence and the transmitted-s eigenvalue for s-polarised incidence. A second GTM solve at the collection geometry gives the corresponding Stokes k_z . The phonon momentum transfer in the reduced wavevector $(\zeta, 0, q_z)$ space is then

$$\mathbf{q}_{ph} \propto \begin{pmatrix} \zeta_L - \zeta_S \\ 0 \\ q_{z,L} - q_{z,S} \end{pmatrix} \quad (\text{backscattering: } \zeta_L + \zeta_S, q_{z,L} + q_{z,S}) \quad (9.17)$$

where $\zeta = n \sin \theta$ is the in-plane reduced wavevector component. This accounts for optical anisotropy and birefringence that are neglected by the isotropic Snell's-law approximation of Level 2. The resulting unit vector $\hat{\mathbf{q}}_{ph}$ is used in the same NAC dynamical-matrix correction as above.

All modes — per Berreman q-channel-pair NAC (GUI label: *All modes* or *All modes (EO)*)

This level decomposes the incident and reciprocal scattered fields into Berreman propagation channels in each Raman-active layer. Numerically distinct Berreman eigenvectors that have the same propagation q_z within tolerance are first summed coherently into a single q-channel. This is important at normal incidence and in other degenerate or nearly-degenerate cases, where the individual Berreman eigenvectors are not unique but their summed field in a propagation subspace is unique.

The NAC correction is then evaluated separately for every combination of incident q-channel i_L and scattered q-channel j_S . For each pair a dedicated phonon wavevector $\hat{\mathbf{q}}_{ph}^{ij}$ is formed from the corresponding reduced in-plane wavevectors and q_z values,

$$\mathbf{q}_{ph}^{ij} \propto \begin{pmatrix} \zeta_L - \zeta_S \\ 0 \\ q_{z,L}^i - q_{z,S}^j \end{pmatrix}. \quad (9.18)$$

If $|\mathbf{q}_{ph}^{ij}|$ is negligible, that pair uses the uncorrected TO phonons. Otherwise the NAC dynamical matrix is solved for $\hat{\mathbf{q}}_{ph}^{ij}$, giving a q-resolved phonon frequency and Raman tensor for the pair. Since different q-channel pairs can give different corrected frequencies for the same original TO mode, one input mode can produce more than one spectral line.

The default modal-pair combination policy is *Group q channels*. Amplitudes whose phonon momentum and detected polarisation are the same are added coherently before squaring; different phonon-q final states are summed as intensities,

$$I_m \propto \sum_g \left| \sum_{(i_L, j_S) \in g} A_m^{ij} \right|^2, \quad (9.19)$$

where g labels a common phonon-q and detector-channel group. This is the physically recommended setting and is the one used for normal calculations. Two diagnostic alternatives are also available in the GUI: *Incoherent pairs*, which squares each q-channel-pair amplitude independently, and *Coherent all pairs*, which sums all modal-pair amplitudes before squaring and therefore mixes distinct phonon-momentum final states.

This is the most complete treatment available and reduces to the dominant-mode result when only one q-channel is significant in each geometry.

For incoherent depth integration, *Group q channels* has two explicit final-state models. *Bulk phase matched* retains each internal $\mathbf{q}_{ph}^{ij}$ and compares its direction with the externally selected momentum transfer $\mathbf{q}_{ext}$. The default 90 degree cutoff accepts the forward momentum hemisphere and rejects antiparallel pairs; accepted amplitudes share the final-state key defined by $\mathbf{q}_{ext}$. *Local incoherent* instead treats the internal Berreman products as components of one local optical field and assigns them all to the external final state before the local intensity is formed. The q-angle cutoff is not used in that model. Coherent-depth calculations always retain the pair-resolved q vectors and are unaffected by this choice.

The two momenta have distinct roles and must not be conflated: $\mathbf{q}_{ph}^{ij}$ describes the phase of an internal optical-pair contribution, while $\mathbf{q}_{ext}$ labels the externally selected scattering channel. A future finite-coherence implementation will replace the hard angular cutoff by a continuous phase-mismatch weight depending on $\mathbf{q}_{ph}^{ij} - \mathbf{q}_{ext}$ and the coherence length.

For an infrared-inactive mode $\bar{\bar{\mathbf{Z}}}^{mw}$ gives no macroscopic restoring-force correction, so all four levels reduce to the same transverse-optic frequency. For polar modes the corrected frequencies and Raman tensors depend on the surface orientation and the optical scattering geometry. The optical field enhancement and interference factors are still calculated separately at ν_L and ν_S ; they should not be interpreted as replacing the phonon-frequency correction described here.

9.7 Electro-Optic Correction to the Raman Tensor

For non-centrosymmetric polar crystals the Raman tensor also receives an electro-optic (EO) contribution that arises from the macroscopic electric field accompanying each infrared-active phonon. This field,

directed along $\hat{\mathbf{q}}$, modulates the optical susceptibility through the second-order non-linear susceptibility $\bar{\bar{\chi}}^{(2)}$. The result is a correction to the Raman tensor of each NAC-corrected mode that depends on the phonon wavevector direction.

The EO contribution to the Raman tensor of mode p is

$$\Delta \bar{\bar{R}}_{ij}^{(p)} = -\frac{8\pi f_{ij} s_p}{\hat{\mathbf{q}}^T \bar{\bar{\epsilon}}^b \hat{\mathbf{q}}} \quad (9.20)$$

where the electro-optic factor $\bar{f}$ is the contraction of $\bar{\bar{\chi}}^{(2)}$ with the phonon unit wavevector,

$$f_{ij} = \sum_l \chi_{ijl}^{(2)} \hat{q}_l \quad (9.21)$$

In these implementation equations $\chi^{(2)}$ denotes the internal cell-dependent quantity $\tilde{\chi}^{(2)}$. For a physical susceptibility expressed numerically in pm/V, the reader conversion is

$$\tilde{\chi}_{ijl}^{(2)} = \chi_{ijl}^{(2)} [\text{pm/V}] \frac{\sqrt{\text{amu}/m_e} e 10^8}{4\pi\epsilon_0 \sqrt{V_{\text{cell}} [\text{\AA}^3]}}.$$

The numerical 10^8 combines pm/V and $\text{\AA}$ displacement/volume units. Z^{mw} uses masses in electron-mass units. This conversion is not merely a change from picometres to $\text{\AA}$ or Bohr.

The explicit 4π in the correction is the Gaussian-unit electrostatic factor. PDielec's conversion of physical $\chi^{(2)}$ values from pm/V to the internal R_ϵ convention contains the reciprocal $1/(4\pi\epsilon_0)$ factor, so this is not an additional conversion of the reader Raman tensor. In particular, the separate 4π used by the Abinit reader converts its printed $d\chi/dQ$ transverse Raman tensor to $d\epsilon/dQ$ and is not applied again here.

and s_p is the projection of the Born-charge weighted polarisation onto the phonon eigenvector,

$$s_p = \left(\bar{\bar{Z}}^{mwT} \hat{\mathbf{q}} \right) \cdot \mathbf{x}_p \quad (9.22)$$

with $\mathbf{x}_p$ the mass-weighted eigenvector of mode p after NAC mixing, and $\hat{\mathbf{q}}^T \bar{\bar{\epsilon}}^b \hat{\mathbf{q}}$ the background permittivity along the phonon wavevector direction. The corrected Raman tensor for each mode is

$$\bar{\bar{R}}_{corrected}^{(p)} = \bar{\bar{R}}_{TO,mix}^{(p)} + \Delta \bar{\bar{R}}^{(p)} \quad (9.23)$$

where $\bar{\bar{R}}_{TO,mix}^{(p)}$ is the Raman tensor after mode mixing from the NAC dynamical-matrix correction.

The EO correction vanishes identically when:

- mode p is not infrared active (i.e. $\bar{\bar{Z}}^{mwT} \hat{\mathbf{q}}$ is orthogonal to $\mathbf{x}_p$),
- $\bar{\bar{\chi}}^{(2)} = 0$ (centrosymmetric crystals), or
- the TO level is selected (no NAC correction applied).

Availability of $\bar{\bar{\chi}}^{(2)}$ from DFT codes

PDielec reads $\bar{\bar{\chi}}^{(2)}$ automatically from the DFT output file when it is present. ABINIT, CASTEP and CRYSTAL report the d -tensor ($\bar{\mathbf{d}} = \frac{1}{2} \bar{\bar{\chi}}^{(2)}$); PDielec stores $\bar{\bar{\chi}}^{(2)} = 2\bar{\mathbf{d}}$ internally after converting the reported pm/V values to the same $\text{\AA}$ based convention used by the reader R_ϵ Raman tensors.

CASTEP writes a 3×6 Voigt-format block labelled `Nonlinear Optical Susceptibility (pm/V)` in the `.castep` output file. The Voigt column order is (11, 22, 33, 23, 13, 12), giving the full 3×3×3 tensor after symmetrisation of the last two indices.

Abinit writes a 27-row table labelled `Non-linear optical susceptibility tensor d` (pm/V) in the `.abo` output file, listing all index combinations (i_1, i_2, i_3) in Cartesian coordinates with 1-based integer indices.

CRYSTAL supplies $d(\text{MKS})$ in `CHI2.DAT` or its output table; the reader uses $\chi^{(2)} = 2d$ and expands the ten static components using Kleinman symmetry.

Quantum ESPRESSO supplies $\partial\epsilon/\partial E$ in Rydberg atomic units, from which the physical susceptibility in pm/V is $\chi^{(2)} = 0.5 \times 2.7502 \times \text{ELOP}$. XML values take precedence over the log. Each path then applies the common conversion above. The readers also average the tensor over the final unit-cell point group.

The required response is the electronic mixed optical/static susceptibility. Using a static or second-harmonic susceptibility assumes negligible relevant dispersion. A relaxed-ion Pockels tensor, the derivative of inverse permittivity, is a different quantity and must not be substituted directly. There is no universal requirement that the EO increment be small compared with the mechanical Raman tensor.

When the electro-optic option is enabled, $\bar{\chi}^{(2)}$ data are present, and any level other than *TO* is selected, PDielec applies the EO correction to every NAC-rotated Raman tensor before computing the Raman spectrum. The GUI labels for the three NAC levels then show an *(EO)* suffix (*Snell's law (EO)*, *Dominant mode (EO)*, *All modes (EO)*) to make clear that the electro-optic correction is active. A log message also confirms this at calculation time.

PHONON ANALYSIS

PDGui performs an analysis of a phonon mode in terms of the percentage of molecular centre-of-mass and rigid body rotational motion in each mode. Using the results of the analysis it is possible to differentiate between inter- and intra-molecular phonon modes (see for instance Jepsen et al [54]). In addition, for those systems with more than one formula unit in the unit-cell it is possible to break down the contribution of each molecule to a particular phonon mode.

10.1 Molecular Systems

It is common in molecular calculations of vibrational properties to construct a force constant matrix which enforces the requirements of zero energy change for centre-of-mass motion and rigid-body rotation. This can be achieved by using projection operators to transform the second derivative matrix to a set of coordinates that no longer include centre-of-mass motion or rigid-body rotation.

Defining a projection operator as;

$$\bar{\bar{P}} = \bar{\bar{1}} - \bar{V}\bar{V}^T \quad (10.1)$$

where $\bar{V}$ is an orthonormal column vector with length $3N$ (N is the number of atoms), the centre-of-mass motion can be described in Cartesian space as all atoms moving along the x, y or z axis with the same displacement. So considering the projection of the centre-of-mass motion along the x-axis, for each atom a we can write;

$$\bar{V}_a^x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad (10.2)$$

For the projection operator which will project out all components of translation along the x-axis for every atom in the molecule we have;

$$\bar{V}^x = \begin{pmatrix} \bar{V}_1^x \\ \bar{V}_2^x \\ \vdots \end{pmatrix} \quad (10.3)$$

There are two other projection operators describing translation along the y- and z-axis.

In a similar fashion, it is possible to describe an infinitesimal molecular rotation using a vector V and therefore constructing a projection operator to remove rigid-body rotation. If the coordinates of atom a relative to the centre-of-mass of the molecule are x, y and z , the component of the projection vector, $\bar{\bar{V}}$ representing rotation about the x-axis in the yz-plane is;

$$\bar{V}_a^{yz} = \begin{pmatrix} 0 \\ -z \\ +y \end{pmatrix} \quad (10.4)$$

In a similar fashion rotations about the y- and z-axis are respectively;

$$\bar{V}_a^{xz} = \begin{pmatrix} +z \\ 0 \\ -x \end{pmatrix} \quad (10.5)$$

and

$$\bar{V}_a^{xy} = \begin{pmatrix} -y \\ +x \\ 0 \end{pmatrix} \quad (10.6)$$

In a similar fashion to Equation (10.3) the complete projection operators for the 3 rotational modes can be assembled (using rotation about the x-axis as an example)

$$\bar{V}^{yz} = \begin{pmatrix} \bar{V}_1^{yz} \\ \bar{V}_2^{yz} \\ \vdots \end{pmatrix} \quad (10.7)$$

In practice, the projection operators will be defined using mass-weighted cartesian coordinates as this simplifies the expressions used later. As an example the transformation to mass-weighted coordinates for atom a translating along the x-axis is;

$$\sqrt{m_a} \bar{V}_a^x \quad (10.8)$$

10.2 Periodic Systems

The invariants of the energy in a periodic system are the three translational modes corresponding to motion of all the atoms in the same direction. The rotation of an infinite lattice is not invariant. However, it is proposed to construct projection operators for each molecule in the unit-cell, which will project out the motion of each molecular translational and rotational degree of freedom. Such projection operators will be used to separate the external from the internal modes and there will therefore be a projection operator for each external mode in the system.

Hug and Haesler have shown [55] that the vibrational kinetic energy can be decomposed into single centre atomic contributions by considering the kinetic energy, T_p , of a normal mode Q_p ;

$$T_p = \frac{1}{2} \dot{Q}_p^2 = \sum_a T_{a,p} = \frac{1}{2} \sum_a m_a \Delta_{\vec{x},p}^T \Delta_{\vec{x},p} = \frac{1}{2} \dot{Q}_p^2 \sum_a \bar{\mathbf{L}}_{a,p}^T \bar{\mathbf{L}}_{a,p} \quad (10.9)$$

Where the sum is over atoms, a , with mass m_a , $\Delta_{\vec{x},p}$ is the time derivative of the Cartesian displacement vector of atom a along the p^{th} normal mode and $\bar{\mathbf{L}}_p$ are the components of the orthogonal transformation relating the p^{th} normal mode and mass-weighted Cartesian displacements. Thus $\bar{\mathbf{L}}$ are the eigenvectors of the mass-weighted second derivative matrix. The atom subscript is used to indicate that only those atomic coordinates involving atom a are being considered. Derivation of this expression has made use of the following relationship between the mass-weighted normal mode;

$$\sqrt{m_a} \Delta_{\vec{x},p} = \bar{\mathbf{L}}_{a,p} Q_p \quad (10.10)$$

As proposed by Hug and Haesler [55], since the kinetic energy can be expressed as atomic contributions, consideration of the Virial theorem indicates that this is true also for the potential energy. Using mass-weighted Cartesian coordinates for convenience, an analysis of the phonon modes may therefore be

constructed in the following way. A given phonon mode, p , will have its total kinetic energy partitioned between the molecules in the unit-cell according to;

$$E_p^{total} = \sum_{mol} E_p^{mol} \quad (10.11)$$

$$E_p^{mol} = \dot{Q}_p^2 \sum_{a \in mol} \bar{\mathbf{L}}_{a,p}^T \bar{\mathbf{L}}_{a,p} \quad (10.12)$$

To calculate the contribution a particular mode has to the centre-of-mass kinetic energy we use the projection operators given in Equation (10.3) (now in mass-weighted Cartesian coordinates) where only the atoms in that molecule are used to construct the operator;

$$\bar{\bar{\mathbf{P}}}_{mol}^x = \bar{\mathbf{1}} - \bar{\mathbf{V}}_{mol}^x (\bar{\mathbf{V}}_{mol}^x)^T \quad (10.13)$$

This approach is used to define 6 projection operators for each molecule. The centre-of-mass energy contained in the p_{th} mode can be written as;

$$E_p^{cm} = \dot{Q}_p^2 \sum_{i=x,y,z} \sum_{mol} (\bar{\bar{\mathbf{P}}}_{mol}^i \bar{\mathbf{L}}_p)^T (\bar{\bar{\mathbf{P}}}_{mol}^i \bar{\mathbf{L}}_p) \quad (10.14)$$

and the rigid-body rotational energy associated with the mode is;

$$E_p^{rot} = \dot{Q}_p^2 \sum_{i=xy,yz,xz} \sum_{mol} (\bar{\bar{\mathbf{P}}}_{mol}^i \bar{\mathbf{L}}_p)^T (\bar{\bar{\mathbf{P}}}_{mol}^i \bar{\mathbf{L}}_p) \quad (10.15)$$

The molecular vibrational contribution, which can be used to classify the internal modes of the system, can be obtained by subtracting the centre-of-mass and rigid-body rotational; energies from the total;

$$E_p^{vib} = E_p^{total} - E_p^{cm} - E_p^{rot} \quad (10.16)$$

Since all of the energy terms depend in the same way on $\dot{Q}_p$, it is not required in the calculation of the relative contributions to the energy coming from the external (molecular centre-of-mass and rigid-body rotation) and the internal modes (vibrational contributions)

10.3 Implementation

In order to calculate the relative internal and external contributions to each phonon mode it is first necessary to identify the molecules in the crystal. In many cases, the unit-cell is packed with atoms in such a way that the cell is filled, rather than in a way reflecting the bondedness of the molecules. The program first replicates the atoms in all cells neighbouring the central unit-cell. Within this supercell the bonds are determined by calculating the distances between all atoms in the supercell. In practice, an order N method is used whereby instead of searching all the supercell only the space around each atom is searched for potentially bonded partners. The criterion of the presence of a bond between atoms i and j is given by the requirement that the distance between the atoms r_{ij} is less than the bonding requirement;

$$r_{ij} < scale(radius_i + radius_j) + toler \quad (10.17)$$

Here $radius_i$ is the covalent radius of atom i , $scale$ and $toler$ are factors that can be altered when running the program.

Once the bonding in the supercell has been determined the program starts with the first atom in the central cell and determines all atoms which are connected to the molecule that it is in. If there are any remaining atoms in the central cell which are not bonded yet then further molecules are added until all atoms in the central cell have been allocated to a molecule.

Finally a new cell with the same dimensions as the original is constructed. Where necessary each molecule is shifted into the cell so that the molecule's centre-of-mass lies within it.

The projection operators, Equation (10.13), can now be constructed and the relative energies calculated using Equations (10.14), (10.15) and (10.16).

10.3.1 Examples

10.4 Isoleucine

The example of a calculation on isoleucine using Castep is available in the Examples/Castep/Isoleucine directory of the distribution. The analysis was performed using PDGui. The program first finds the four molecules of isoleucine in the unit-cell of the crystal structure. The ordering of the atoms at this point is different to the ordering in the original cell and in the QM calculation. The visualiser shows the atoms in the new positions. These will not be the same as in the original calculation. The molecules and their centres of mass are summarised in [Table 10.1](#).

Table 10.1: Molecular constituents of the isoleucine unit-cell
(centre-of-mass is in fractional coordinates)

Molecule	Mass	CM _a	CM _b	CM _c
0	131.172	0.293600	0.051683	0.304906
1	131.172	0.706400	0.551683	0.695094
2	131.172	0.793231	0.588919	0.287125
3	131.172	0.206769	0.088919	0.712875

[Table 10.2](#) shows the results of the analysis of the phonon modes with frequencies below 100 cm⁻¹. The first 3 modes should have zero frequency as they are the translationally invariant modes of the lattice. The negative number shown here actually indicates that the mode has an imaginary frequency. As can be seen under the column %mol-cme they are almost completely associated with molecular centre-of-mass motion. This is true also of the lowest non-zero frequency at 32.81 cm⁻¹ and to some extent of the next mode at 39.73 cm⁻¹. However, for this latter mode there is a considerable contribution from rigid-body rotational motion (see the %mol-rot column). As the frequency of the phonon mode increases the contribution from rigid-body motion generally decreases and the contribution from vibrational modes (see the %vib column) increases. The total contribution from molecular motion is summarised in the last four columns.

Table 10.2: Isoleucine: percentage contributions of the centre-of-mass and the rotational molecular modes to each phonon mode

Freq(cm ⁻¹)	%cm	%rot	%vib	%mol-0	%mol-1	%mol-2	%mol-3
-0.07	99.5	0.5	0.0	26.9	26.9	23.1	23.1
-0.06	98.6	1.1	0.3	23.6	23.6	26.4	26.4
-0.04	99.7	0.2	0.1	23.8	23.8	26.2	26.2
32.81	92.0	5.9	2.1	26.8	26.8	23.2	23.2
39.73	77.9	19.6	2.5	22.1	22.1	27.9	27.9
49.42	10.4	81.9	7.7	19.0	19.0	31.0	31.0
52.12	80.7	17.3	2.0	17.6	17.6	32.4	32.4
54.15	73.4	19.0	7.5	21.2	21.2	28.8	28.8
59.59	37.7	56.6	5.7	23.9	23.9	26.1	26.1
62.05	77.3	17.2	5.5	23.4	23.4	26.6	26.6
69.91	9.6	69.4	21.0	17.6	17.6	32.4	32.4
73.19	7.3	61.1	31.6	6.6	6.6	43.4	43.4
76.00	13.6	61.7	24.7	31.6	31.6	18.4	18.4
76.25	3.2	48.1	48.7	10.6	10.6	39.4	39.4
80.49	31.5	43.8	24.7	19.3	19.3	30.7	30.7
87.54	37.9	35.4	26.7	26.8	26.8	23.2	23.2
89.31	12.2	58.1	29.6	27.5	27.5	22.5	22.5
90.49	16.9	53.9	29.2	22.4	22.4	27.6	27.6
94.51	49.1	36.9	14.0	39.4	39.4	10.6	10.6
94.79	47.0	16.9	36.1	23.0	23.0	27.0	27.0

10.5 BaTiO₃

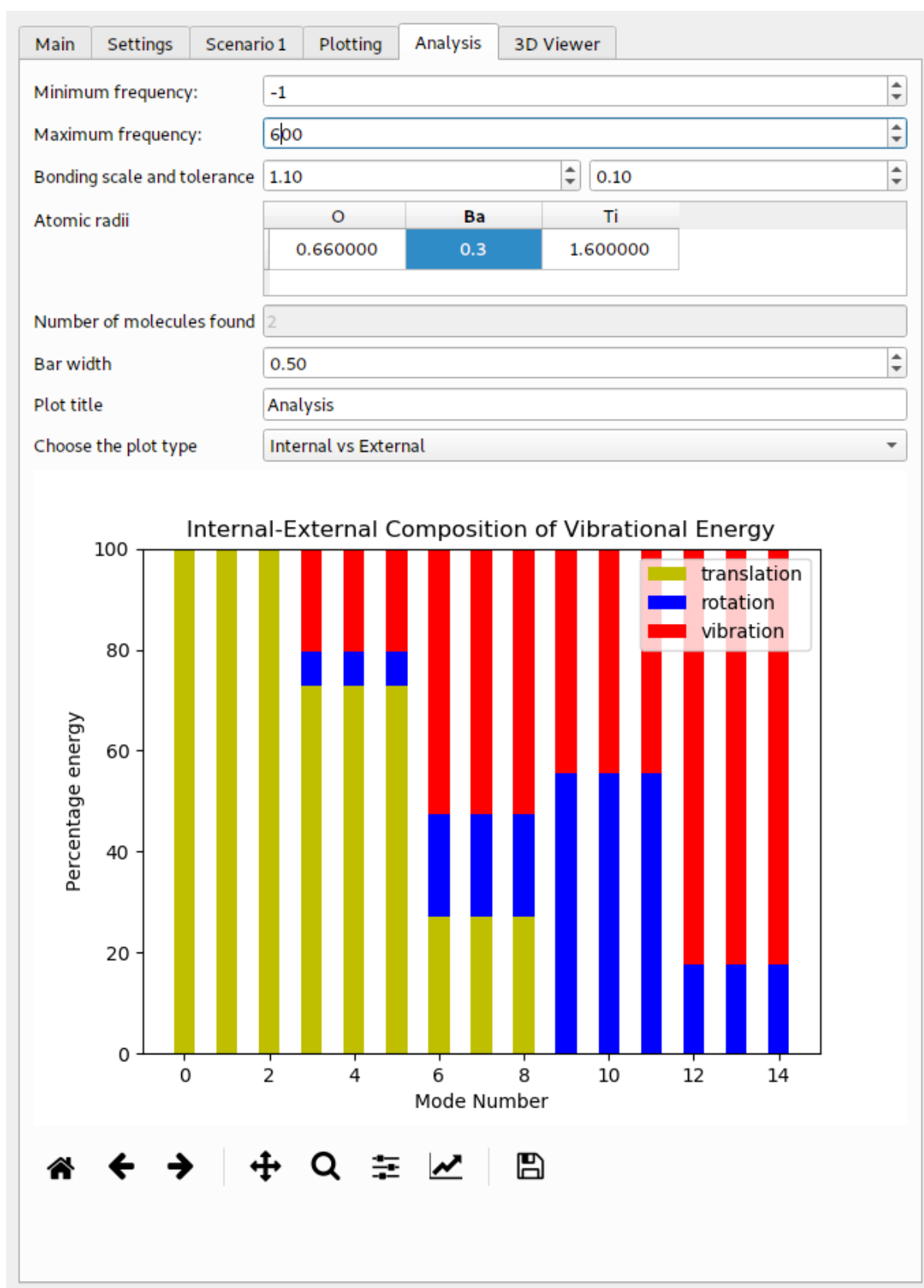
The example for BaTiO₃ can be found in Example/AbInit/BaTiO3 in the main directory of the PDielec distribution. The results of the Abinit calculation can be analysed using PDGui. For this example, the default covalent radius of the Barium atoms has been changed to 0.3 Å, so this atom will be treated as though it is not bonded to anything else in the cell. This results in 2 ‘molecules’ being found in the cell; a TiO₃ moiety and the Ba²⁺ ion.

The results of the analysis are shown in Table 10.3 below. In this example, the lowest 3 modes have been projected so there is no problem with translational invariance. They are purely centre-of-mass modes. The energy in these three modes seems to be mainly in molecule 1, which is the Ba²⁺ ion. The modes at 197.73 cm⁻¹ also have a large centre-of-mass component, this time mainly coming from molecule 0, which is the TiO₃ moiety. Above 200 cm⁻¹ there is a much smaller centre-of-mass contribution to the energy and all the modes are dominated by the TiO₃ group.

Table 10.3: BaTiO₃: percentage contributions of the centre-of-mass and the rotational molecular modes to each phonon mode

Freq(cm ⁻¹)	%cm	%rot	%vib	%mol-0	%mol-1
0.00	100.0	0.0	0.0	41.1	58.9
0.00	100.0	0.0	0.0	41.1	58.9
0.00	100.0	0.0	0.0	41.1	58.9
197.73	72.9	6.7	20.4	70.0	30.0
197.73	72.9	6.7	20.4	70.0	30.0
197.74	72.9	6.8	20.4	70.0	30.0
269.34	27.1	20.3	52.7	88.9	11.1
269.34	27.0	20.3	52.7	88.9	11.1
269.34	27.0	20.3	52.7	88.9	11.1
270.59	0.0	55.6	44.4	100.0	0.0
270.59	0.0	55.6	44.4	100.0	0.0
270.59	0.0	55.6	44.4	100.0	0.0
493.61	0.1	17.4	82.5	100.0	0.0
493.61	0.1	17.4	82.5	100.0	0.0
493.61	0.1	17.4	82.5	100.0	0.0

The visualisation of the internal and external contributions to the phonon modes is shown in the [Fig. 10.1](#)

Fig. 10.1: Internal and external contributions to the phonon modes of BaTiO₃

APPLICATIONS OF EFFECTIVE MEDIUM THEORY

Several examples are given to illustrate some applications of the package to the absorption of light in powdered dielectric materials. The calculations used to provide the data for the permittivities are sufficiently accurate to illustrate aspects of the theory. The examples are chosen to show the package being used with the QM packages CASTEP and VASP and with the MM package GULP. The data used to generate the figures shown below are available in the Examples/ directory. The application_note.py file is a script that will generate and perform the calculations used in each application note and the application_note.xlsx file is a spreadsheet containing the calculated data.

11.1 MgO using CASTEP

Magnesium oxide is an isotropic medium, the initial unit cell and the space group symmetry ($Fm\bar{3}m$) were taken from the Inorganic Crystal Structure Database (ICSD) [56] reference number ICSD-52026 [57]. The primitive cell was optimised using CASTEP. Norm-conserving pseudo-potentials were used to represent the core electrons of magnesium and oxygen. An energy cutoff of 1000 eV was used with the PBE [58] density functional and a k-point spacing for the Monkhorst-Pack grid of $0.04 \text{ \AA}^{-1}$. The primitive cell was optimised and a Density Functional Perturbation Theory (DFPT) calculation of the phonon spectrum at the gamma point was performed. The optimised lattice parameter was found to be $2.1234 \text{ \AA}$, compared with the experimental value of $2.107 \text{ \AA}$. Only 3 degenerate modes contribute to the permittivity. A summary of the results is presented in Table 11.1.

Table 11.1: Experimental and Calculated Properties of MgO

Property	Experimental	Calculated
Unit cell dimensions ($\text{\AA}$):	2.107	2.1234
Space group:	$Fm\bar{3}m$	
Optical permittivity:		3.1400
Static permittivity:		10.000
TO phonons	Frequency (cm^{-1})	Intensity ($\text{D}^2 \text{\AA}^{-2} \text{amu}^{-1}$)
T:	388.3	9.29
LO phonons		
T (001):	693.7	

Because MgO is isotropic with only a single frequency contributing to the permittivity, it makes a useful example application to illustrate several features of PDielec. The real and imaginary frequency-dependent permittivities of the perfect crystal are shown in Fig. 11.1, where a damping factor (σ) of 10 cm^{-1} has been used. The Figure shows one component of the diagonal, isotropic matrix. In the Figure the real permittivity at zero frequency corresponds to the static permittivity in Table 11.1, and at frequencies above the absorption at 388 cm^{-1} the permittivity tends to the optical permittivity as the frequency increases. The

real permittivity has zero values at 388.3 and 693.7 cm^{-1} which are the TO and LO frequencies respectively. The data for this calculation can be found by looking in the [Examples](#) section of the installation guide under the Castep/MgO example.

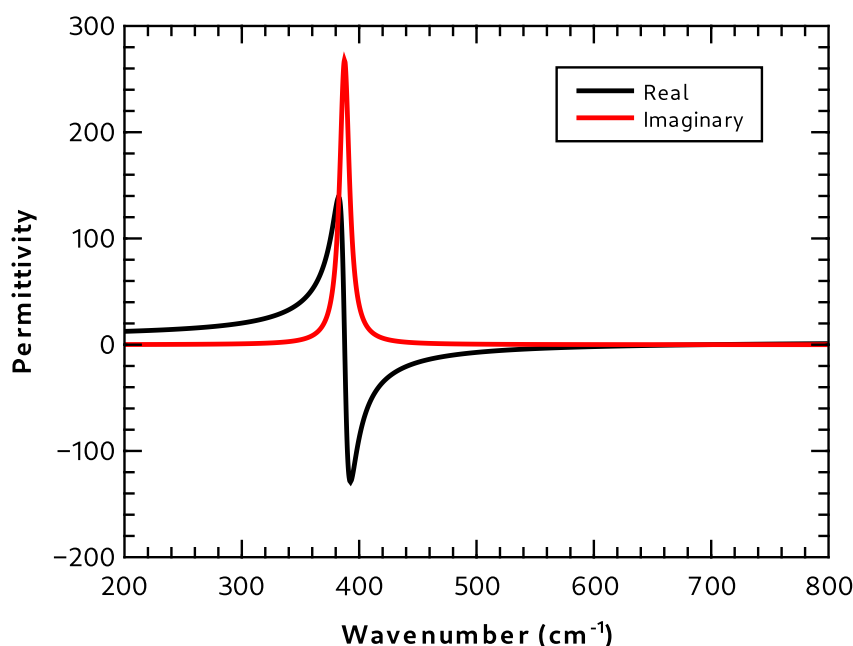


Fig. 11.1: Permittivity of MgO

Using the Maxwell-Garnett mixing rule, [Fig. 11.2](#) shows the calculated permittivities of a 1% volume fraction of MgO spheres in a supporting medium with a frequency-independent permittivity of 2.0, which would be typical of a material such as PTFE. Due to the dilution effect, the real component has shifted to a base-line value close to 2, and the absorption, as indicated by the maximum in the imaginary component has shifted by about 150 cm^{-1} to 550 cm^{-1} .

The effect of volume fraction on the predicted molar absorption coefficient, using the Maxwell-Garnett mixing rule, is shown in [Fig. 11.3](#). The lowest volume fraction of MgO gives the largest shift of the absorption peak to high frequency. As the volume fraction increases the mixing rule predicts a broadening of the absorption, whilst the peak in the molar absorption coefficient moves to lower frequency. At the highest loading ($f=0.9$) the maximum absorption occurs quite close to the TO frequency. The Maxwell-Garnett mixing rule is regarded as being appropriate for low-volume fractions and so should not be used for interpreting results in which higher-volume fractions of absorbing media have been used [33]. The data for this calculation can be found by looking in the [Examples](#) section of the installation guide under the Castep/MgO/application_note_mg example.

[Fig. 11.4](#) shows the same plot for the Bruggeman mixing rule. At low volume fractions the Bruggeman mixing rule predicts a similar absorption to the Maxwell-Garnett. Indeed as the volume fraction approaches zero the two rules predict the same absorption characteristics. However, even at the relatively low 1% loading, the Bruggeman mixing rule shows an additional broadening of the peak; the shape of the absorption peak has lost its Lorentzian characteristic shape as can be seen clearly in [Fig. 11.4](#). At 10% loading the Bruggeman predicted absorption is broad with the peak shifted to lower wavenumber. This broadening increases with increased loading until, at the higher loadings, the TO peak begins to dominate the absorption. The data for this calculation can be found by looking in the [Examples](#) section of the installation guide under the Castep/MgO/application_note_bruggeman example.

[Fig. 11.5](#) shows the effect of varying the permittivity of the supporting medium. The calculations were performed on spherical MgO particles with a 1% volume fraction. The lowest permittivity is that of a

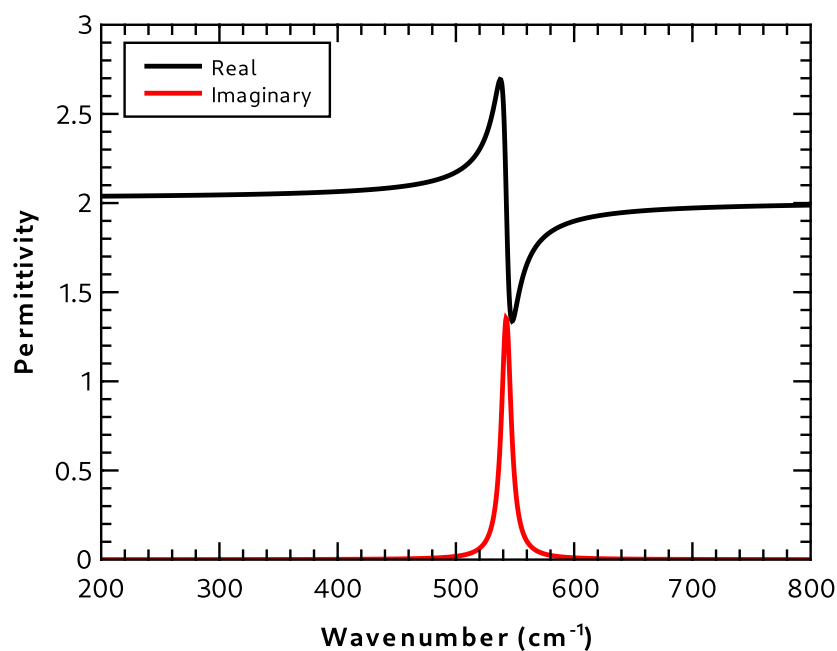


Fig. 11.2: Real and imaginary permittivities of a 1% volume fraction of MgO spheres in PTFE using the Maxwell-Garnett method

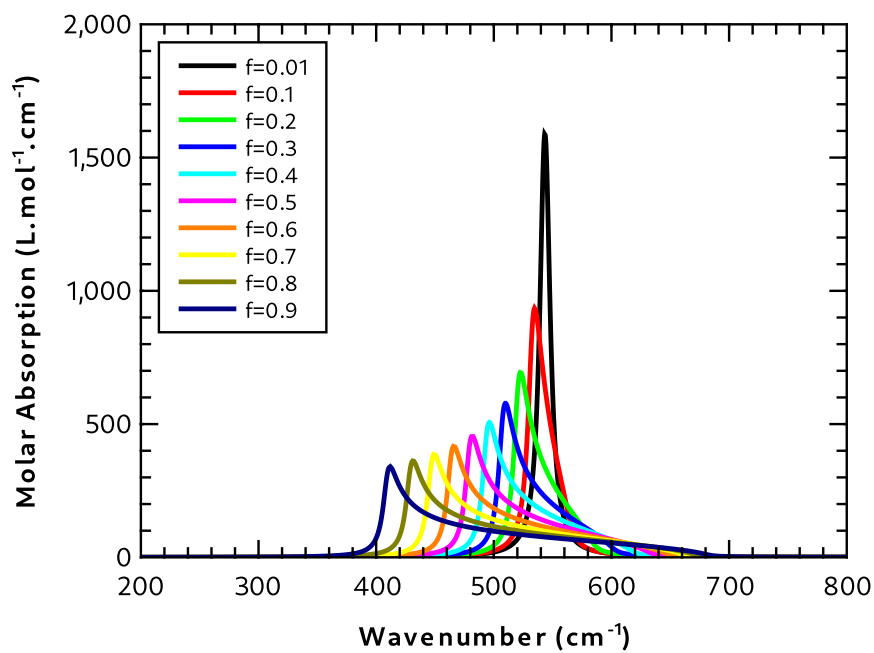


Fig. 11.3: Effect of volume fraction on the Maxwell-Garnett molar absorption coefficient of MgO spheres in PTFE

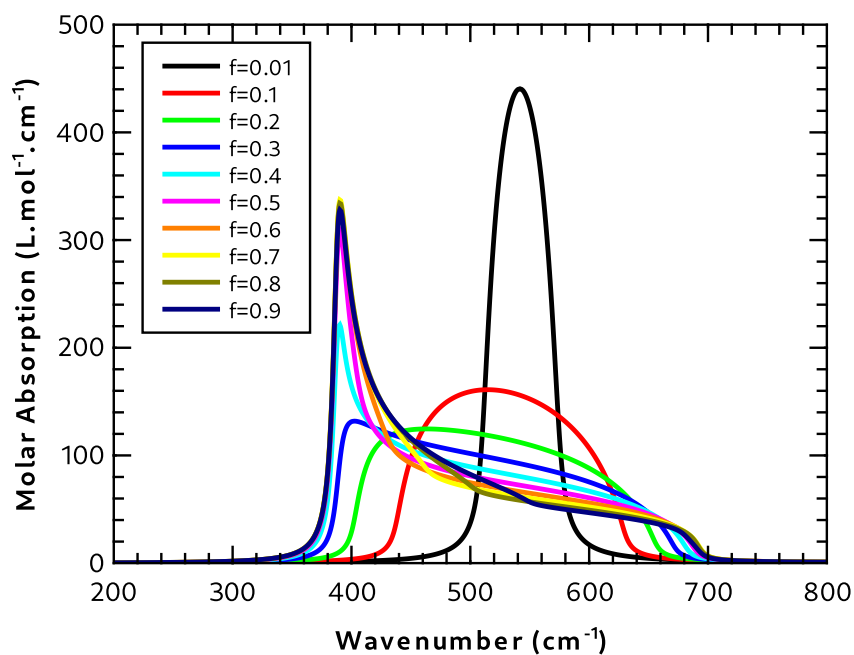


Fig. 11.4: Effect of volume fraction on the Bruggeman molar absorption coefficient of MgO spheres in PTFE

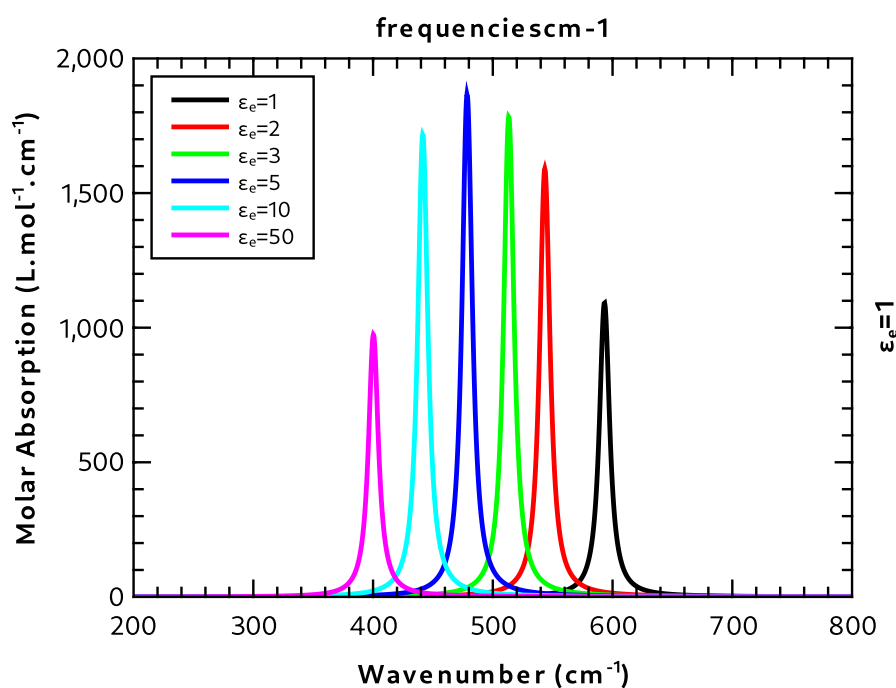


Fig. 11.5: The Maxwell-Garnett molar absorption of spherical MgO particles, 1% volume fraction embedded in a media of varying permittivity

vacuum (or air) and shows the highest shift of the absorption maximum to higher frequencies. Increasing the permittivity lowers the shift until it becomes quite small. A similar effect is seen in the Bruggeman mixing model. However, the absorption resulting from particles in a low dielectric medium is considerably broader than that seen in the Maxwell-Garnet case. This broadening reduces as the permittivity of the medium increases (see Fig. 11.6). The data for these calculations can be found by looking in the *Examples* section of the installation guide under the Castep/MgO/application_note_permittivity and application_note_br_permittivity example.

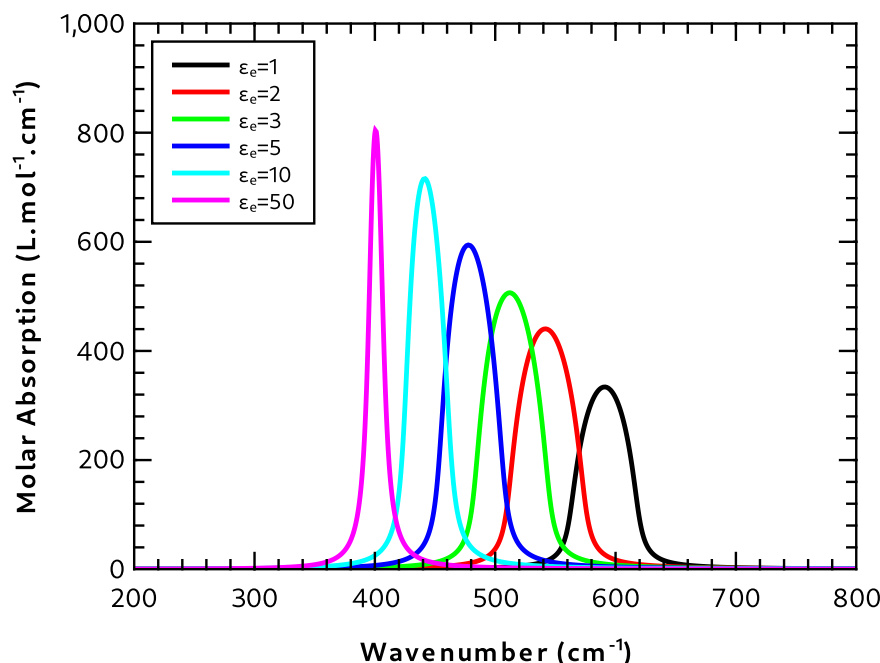


Fig. 11.6: The Bruggeman molar absorption of spherical MgO particles, 1% volume fraction, embedded in media of varying permittivities

11.2 ZnO using VASP

Zinc oxide crystallises in space group $P6_3mc$ (wurtzite). All calculations were performed by VASP [1] using projector augmented-wave PAW [58] pseudo-potentials, the PBE [59] density functional, an energy cutoff of 600 eV and a k-point resolution of approximately $0.1 \text{ \AA}^{-1}$. The initial unit cell was taken from the ICSD [56] with code ICSD-26170 [60]. The unit cell and atom positions were optimised using VASP and the permittivity was calculated using DFPT and the results are reported in Table 11.2. Only two of the bands showed any significant intensity, a doubly degenerate band (E) with a TO frequency of 372.1 cm^{-1} and a non-degenerate band (A) with a TO frequency of 350.0 cm^{-1} . The LO frequency of the non-degenerate band is shifted to 502.0 cm^{-1} for a wave-vector with direction (001), whilst the degenerate modes are unaffected. In the case of the (010) direction, the LO frequency of one of the E modes is shifted to 511.2 cm^{-1} . It is known that ZnO can crystallise with a plate morphology [61] with the (001) surface dominant. Calculations of the molar absorption were performed for a sphere, plate and needle-like shapes with the unique directions of the plate and the needle being normal to the (001) surface. A volume fraction of 1% was chosen for these calculations and the predicted molar absorption coefficients for the Maxwell-Garnett mixing rule are shown in Fig. 11.7.

Table 11.2: Experimental and Calculated Properties of ZnO

Property	Experimental	Calculated
Unit cell dimensions a,b (Å):	3.250	3.295
Unit cell dimensions a (Å):	5.207	5.285
Space group:	$P6_3mc$	
Optical permittivity xx,yy:		5.09
Optical permittivity zz:		6.00
Static permittivity xx, yy:		10.83
Static permittivity zz:		11.67
TO phonons	Frequency (lcm-1)	Intensity ($D^2\text{\AA}^{-2}\text{amu}^{-1}$)
A	350.0	17.1
E	372.1	16.4
LO phonons		
A (001):	502.0	
E (010):	511.2	

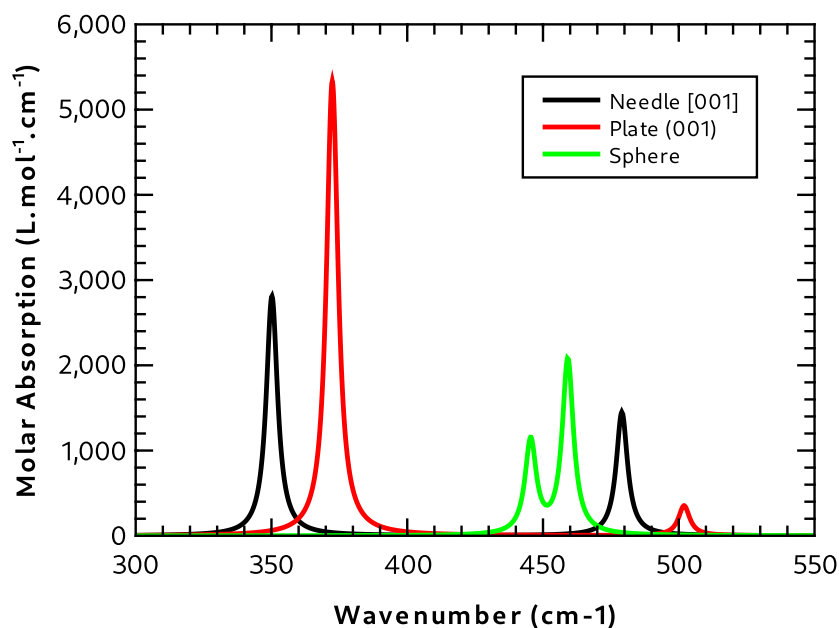


Fig. 11.7: ZnO: Effect of shape on absorption

For the Maxwell-Garnett mixing rule the sphere morphology results in the two absorption peaks shifting from their TO positions to a higher wavenumber by about 80 cm^{-1} . The plate morphology results in one of the peaks moving to higher wavenumber by about 130 cm^{-1} , whilst the other remains at the TO position. The Maxwell-Garnett results are in close accord with some experimental results by Yamamoto et al [61] who measured the infrared spectrum of ZnO smoke particles and observed peaks in the absorption at 380 , 530 and 550 cm^{-1} . Previous work [62, 63] has also used effective medium theory to explain the observed spectrum. The data for this calculation can be found by looking in the [Examples](#) section of the installation guide under the Vasp/ZnO/application_note example.

11.3 Calcite using GULP

Calcite is the most stable polymorph of calcium carbonate and the crystal structure belongs to the $R\bar{3}c$ space group. The force field and atomic structures used here are described in detail in the work by Fislir [64]. Briefly, the oxygen ions are described using a core-shell model [65]. The carbon - oxygen potential of the carbonate is taken to be a Morse potential and an additional 3 atom potential is used to maintain the O-C-O angle at 120° . The van der Waals interactions between non-bonded atoms are taken to be Buckingham potentials and the charges on the calcium, carbon and oxygen ions are +2, +1.3435 and -1.1145 respectively. The shell charge of the oxygen ion is -2.133 and the spring constant for the core-shell interaction is $52.74 \text{ eV/\AA}^2$. The unit cell was optimised using the primitive unit cell and the full space group symmetry. The calculation of the phonon spectrum was performed without symmetry but still using the primitive cell of the lattice. A summary of the calculated properties is given in Table 11.3. The data for this calculation can be found by looking in the [Examples](#) section of the installation guide under the Gulp/calcite example.

Table 11.3: Experimental and Calculated Properties of Calcite

Property	Experimental	Calculated
Unit cell dimensions a,b,c ($\text{\AA}$):	6.375	6.376
Unit cell angles:	46.1	46.1
Space group:	$R\bar{3}c$	
Optical permittivity xx,yy:		1.91
Optical permittivity zz:		2.00
Static permittivity xx, yy:		6.70
Static permittivity zz:		7.10
TO phonons	Frequency (cm^{-1})	Intensity ($\text{D}^2\text{\AA}^{-2}\text{amu}^{-1}$)
E:	114.8	2.39
A:	127.4	3.36
A:	249.3	1.23
E:	320.7	5.82
A:	338.1	4.14
E:	620.1	3.38
A:	732.0	26.89
E:	1463.6	16.97

Fig. 11.8 shows the resultant spectra after postprocessing with PDielec. The damping parameter used in the calculation was a value of 5 cm^{-1} . A 10% volume fraction was used with sphere and plate morphologies for the particles. The unique axis for the plate was taken to be the normal to the (211) surfaces in the primitive cell axes (or the (104) surfaces in the standard unit cell). Such surfaces define the rhombohedral faces commonly seen in calcite crystals [66]. Fig. 11.8 shows that the doubly degenerate TO absorption peak at 620 cm^{-1} is not significantly affected by spherical particles and there is a small shift to higher frequencies in the case of plate-like particles. The non-degenerate TO transition at 732 cm^{-1} , which corresponds to the motion of the carbon atom of the carbonate along the unique direction of the slab, shows a shift to 786 and 819 cm^{-1} for the sphere and plate respectively. The doubly degenerate peak at 1463 cm^{-1} is shifted to 1480 cm^{-1} by spherical particles and is split by plate-like particles with one component which shifts to 1491 cm^{-1} . The data for this calculation can be found by looking in the [Examples](#) section of the installation guide under the Gulp/calcite example.

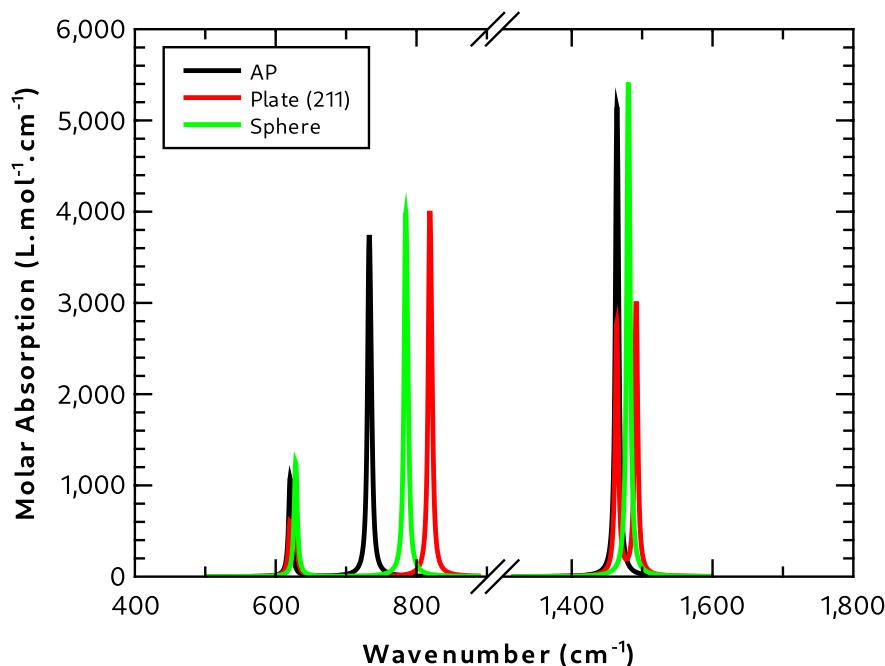


Fig. 11.8: Calculated Maxwell-Garnett absorption spectrum of 10% volume fraction of calcite in PTFE

11.4 Fluorapatite using VASP

The line shapes of the infrared absorption of apatite and fluorapatite were examined extensively by Balan *et al* [31]. Their calculations included the effect of crystallite habit on the spectrum and the results reported here are similar to their conclusions. The method used by Balan *et al.* is an infinitely dilute Maxwell-Garnett model, so the only difference between the methods used by them and those reported here using PDielec are the incorporation of the volume fraction into the theory and the use of an ellipsoidal shape for comparison with the other shapes.

All calculations were performed by VASP [1] using projector augmented-wave PAW [58] pseudo-potentials, the PBE [59] density functional, an energy cutoff of 600 eV and a k-point resolution of approximately $0.1 \text{ \AA}^{-1}$. Table 11.4 summarises the results of the calculations. Only the 3 highest frequency bands are reported and discussed. The TO intensity of the highest frequency band at 1038 cm^{-1} is low and will not be discussed further. The Bravais Friedel Donnay Harker (BFDH) [67] crystal habit of the optimised crystal is shown in Fig. 11.9. The habit was calculated using the Mercury software package. [68]. The BFDH crystal habit is often used to give an idea of the likely important faces of a crystal. It uses only the crystal lattice and space group to determine the crystal morphology. Fig. 11.9 shows that the {100} surfaces form a tube which are capped by the {011} surfaces. The effect of different particle shapes on the predicted spectrum is shown in Fig. 11.10. The calculations of the spectra were performed with a damping parameter (σ) of 2 cm^{-1} . The ellipsoid was chosen to have an aspect ratio, a/b , of 2 and a principle axis along [001], which was compatible with the morphology predicted by the BDFH method. The two TO absorption frequencies at 981 and 986 cm^{-1} have A and E symmetry respectively. Spherical crystallites result in three absorption peaks at around 1000 , 1010 and 1015 cm^{-1} . Needle shaped crystallites leave the A symmetry TO absorption peak at 981 cm^{-1} unaffected but shift and split the E symmetry TO peak to 1020 and 1046 cm^{-1} . A plate morphology with (100) surfaces results in the A and one component of the E TO absorption peak remaining at the TO frequencies, with the other component of the E shifting 85 cm^{-1} to 1075 cm^{-1} . The ellipsoidal morphology shows three shifted peaks at 1000 , 1018 and 1045 cm^{-1} . These results are consistent with those of Balan *et al.* [31], who gave detailed results for hydroxyapatite. The data for this calculation can be found by looking in the *Examples* section of the installation guide under the Castep/F-Apatite example.

Table 11.4: Experimental and Calculated Properties of Fluorapatite

Property	Experimental [69]	Calculated
Unit cell dimensions a,b (Å):	9.417	9.447
Unit cell dimensions c (Å):	6.875	6.926
Space group:	$P6_3m$	
Optical permittivity xx,yy:		2.891
Optical permittivity zz:		2.894
Static permittivity xx,yy:		12.081
Static permittivity zz:		8.841
TO phonons	Frequency (cm⁻¹)	Intensity (D²Å⁻²amu⁻¹)
A:	981.8	112.6
E:	986.3	101.0
E:	1038.1	7.92



Fig. 11.9: BFDH Morphology of fluorapatite

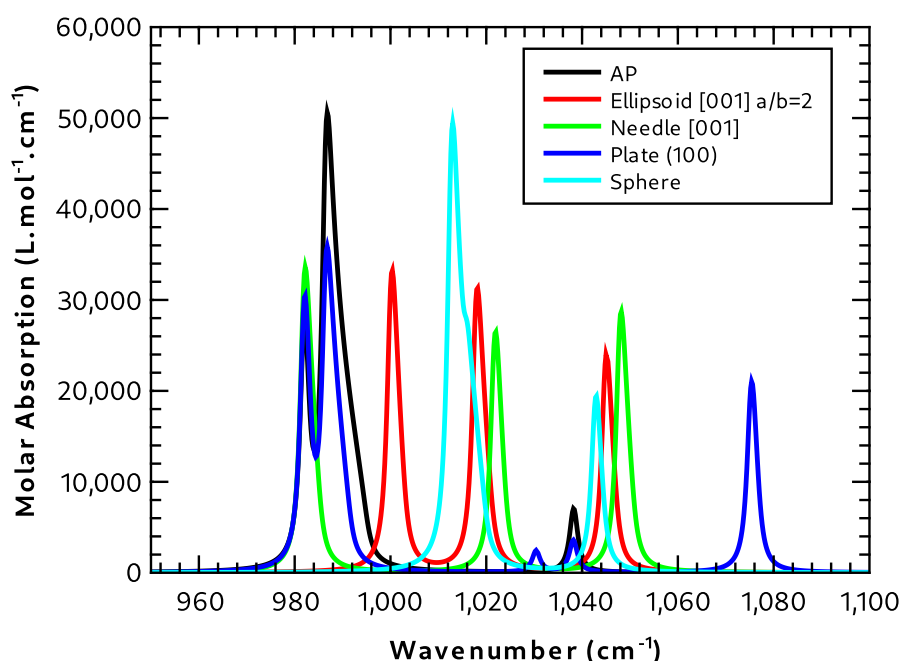


Fig. 11.10: Calculated Maxwell-Garnett absorption spectra of 10% fluorapatite in PTFE

11.5 L-aspartic Acid using CASTEP

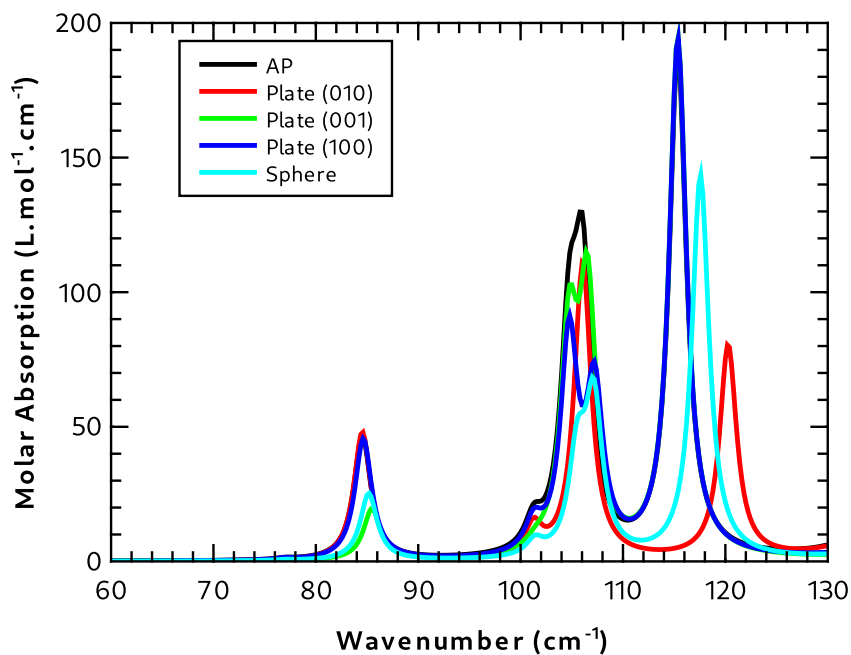
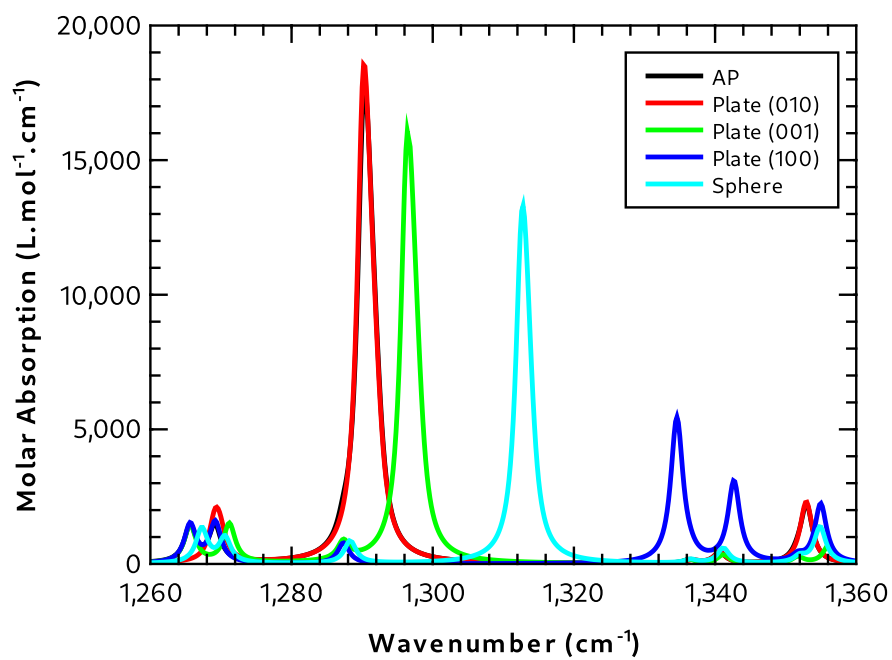
L-aspartic acid is a zwitterion in the solid state so the shape of the particles used in the measurement of IR and THz spectra may be important. The starting geometry for optimisation of the unit cell and molecular structure of L-aspartic acid was taken from Derissen et al [70]. The PBE [59] functional was used with a plane wave energy cutoff of 1000 eV and norm-conserving pseudo-potentials. A dispersion correction using the Tkatchenko-Scheffler scheme [71] available in CASTEP was applied for both the geometry

optimisation and the calculation of the phonon spectrum at the gamma point, with an S_6 scaling factor [72] of 1.0. A summary of the results of the calculations is shown in Table 11.5.

Table 11.5: Experimental and calculated properties of L-aspartic acid

Property	Experimental [70]	Calculated
Unit cell dimensions a (Å):	7.617	7.597
Unit cell dimensions b (Å):	6.982	7.028
Unit cell dimensions c (Å):	5.142	5.113
Unit cell angle β :	99.84	98.77
Space group:	$P2_1$	
Optical permittivity xx:		2.68
Optical permittivity yy:		2.20
Optical permittivity zz:		2.56
Static permittivity xx:		4.58
Static permittivity yy:		3.65
Static permittivity zz:		3.65
TO phonons	Frequency (cm⁻¹) Intensity (D²Å⁻²amu⁻¹)	
	84.5	0.120
	104.7	0.202
	106.0	0.243
	115.3	0.474
	137.3	0.617
	1290.0	55.0
	2945.9	102.8
	2947.3	48.2
	3053.7	44.1

The THz spectrum of L-aspartic acid has been reported by Juliano and Korter [72] in the frequency range 0-90 cm⁻¹. The infrared spectrum has been reported and assigned by Lopez *et al* [73]. Fig. 11.11, Fig. 11.12 and Fig. 11.13 show the calculated absorption spectra for L-aspartic acid for three frequency ranges. The calculation of the spectra used the Maxwell-Garnett mixing rule with a 10% volume fraction of L-aspartic acid in PTFE and for comparison the TO mixing rule. A damping factor of 2 cm⁻¹ was used. Spherical and a variety of plate-like inclusions were used to illustrate their effect on the absorption spectra. Fig. 11.11 shows the frequency range from 60-130 cm⁻¹ which is that covered by THz spectroscopy. The shifts observed for the different particle morphologies are not large, but the change in intensities is significant. The molecular motions associated with phonons at these frequencies tend to be whole molecule motion involving rotation. Fig. 11.12 shows the frequency range from 1260-1340 cm⁻¹. In this frequency range bending of the carboxylate anion contributes to the spectrum significantly. The three different plate morphologies show different and significant shifts in the TO absorption peak at 1290 cm⁻¹. The spherical morphology shows a shift of around 25 cm⁻¹ to higher wavenumber. Fig. 11.13 shows the spectrum in the frequency range 2900-3100 cm⁻¹, which corresponds to the motion of O-H (below 2980 cm⁻¹) and N-H (above 2980 cm⁻¹) stretching. The effect of the different possible crystal morphologies is large with shifts to higher frequency of up to 50 cm⁻¹. The spectrum below 3000 cm⁻¹ arises from two TO absorptions at 2946 and 2947 cm⁻¹. Because the motions associated with each mode interact differently with the internal field within each crystal they give rise to different shifts producing more complex spectra. The data for this calculation can be found by looking in the *Examples* section of the installation guide under the Castep/AsparticAcid example.

Fig. 11.11: Aspartic Acid: Frequency range 60-130 cm^{-1} Fig. 11.12: Aspartic Acid: Frequency range 1260-1340 cm^{-1}

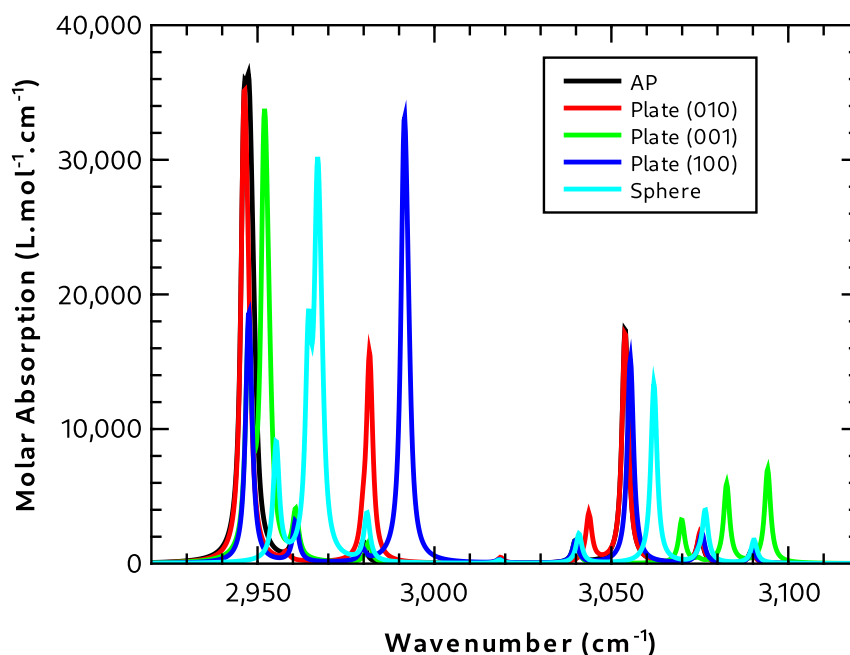


Fig. 11.13: Aspartic Acid: Frequency range 2900-3100 cm^{-1}

11.6 MgO Example using Mie Scattering

Fig. 11.14 compares a Mie scattering calculation with the results from Maxwell-Garnett effective medium theory. The same data set was used for the CASTEP, MgO example. A volume fraction of 1% was used with a small sphere radius ($0.1 \mu\text{m}$) and a broadening of 5 cm^{-1} embedded in a matrix of PTFE. A power expansion in the size parameter of the Mie expressions indicates that for small sizes of particles, the Mie and the Maxwell-Garnett methods should be the same. This is verified in Figure. The data for this calculation can be found by looking in the *Examples* section of the installation guide under the Mie/MgO example.

To better understand what makes particles large or small Table 11.6 shows the dimensionless size parameter, x , as a function of wavenumber and of sphere radius. It has been assumed that the supporting medium is PTFE. Since the power expansion of the size parameters leads to terms that are quadratic in x , it should be expected that when x is less than about $0.1 \mu\text{m}$, the particles can be considered small. It can be seen that particles less than $0.01 \mu\text{m}$ are small over the range of frequencies considered. But while $0.1 \mu\text{m}$ particles are small in the THz regime, and low-frequency infrared, they should not be considered small over the more extended infrared frequencies. $1 \mu\text{m}$ particles should be considered large for both the THz and the extended infrared.

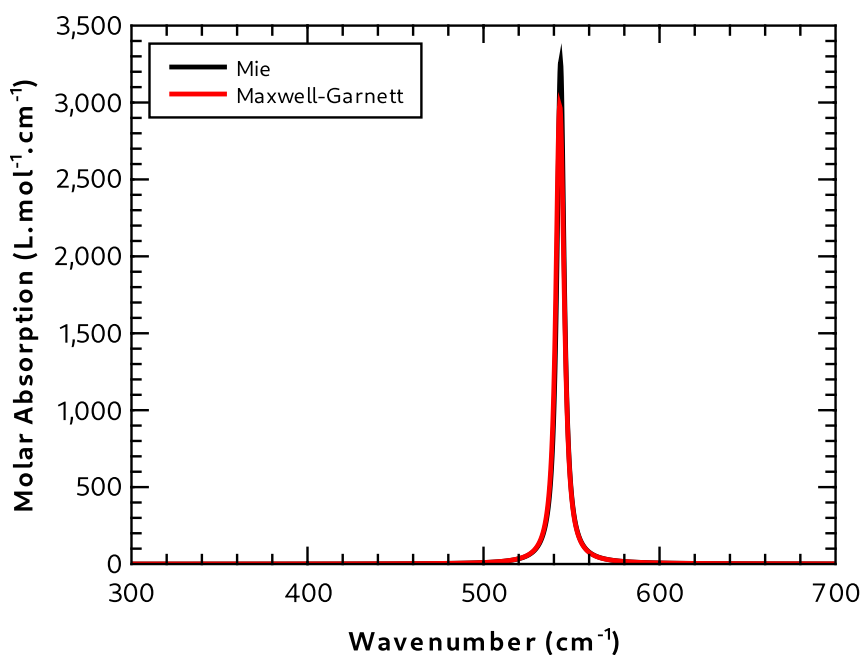


Fig. 11.14: Comparison of Mie and Maxwell methods. 1% volume fraction of MgO in PTFE, sphere radius of $0.1 \mu\text{m}$ and a broadening of 5 cm^{-1}

Table 11.6: Variation of size parameter with wavenumber and radius of sphere

Wavenumber (cm-1)	Radius of sphere (μm)				
	0.001	0.01	0.1	1	10
100	0.0009	0.0089	0.0888	0.8884	8.8841
200	0.0018	0.0178	0.1777	1.7768	17.7682
300	0.0027	0.0267	0.2665	2.6652	26.6523
400	0.0036	0.0355	0.3554	3.5536	35.5364
500	0.0044	0.0444	0.4442	4.4420	44.4204
600	0.0053	0.0533	0.5330	5.3305	53.3045
700	0.0062	0.0622	0.6219	6.2189	62.1886
800	0.0071	0.0711	0.7107	7.1073	71.0727
900	0.0080	0.0800	0.7996	7.9957	79.9568
1000	0.0089	0.0888	0.8884	8.8841	88.8409

Fig. 11.15 shows how the Mie predictions change as the particle radius changes from 0.2 to $1.6 \mu\text{m}$. As the particle size increases the peak above 500 cm^{-1} splits into two. One broader peak moves to a lower frequency as the particle size increases and the peak at about 550 cm^{-1} loses intensity as the particle size increases. There is also the onset of absorption at 388 cm^{-1} which corresponds to the bulk TO modes.

Fig. 11.16 shows the effect of increasing the particle size further. More structure appears in the absorption, with increasing absorption around the bulk TO frequency. Above $4.0 \mu\text{m}$ there is more low frequency structure appearing, below 300 cm^{-1} . The data for this calculation can be found by looking in the [Examples](#) section of the installation guide under the Mie/MgO example.

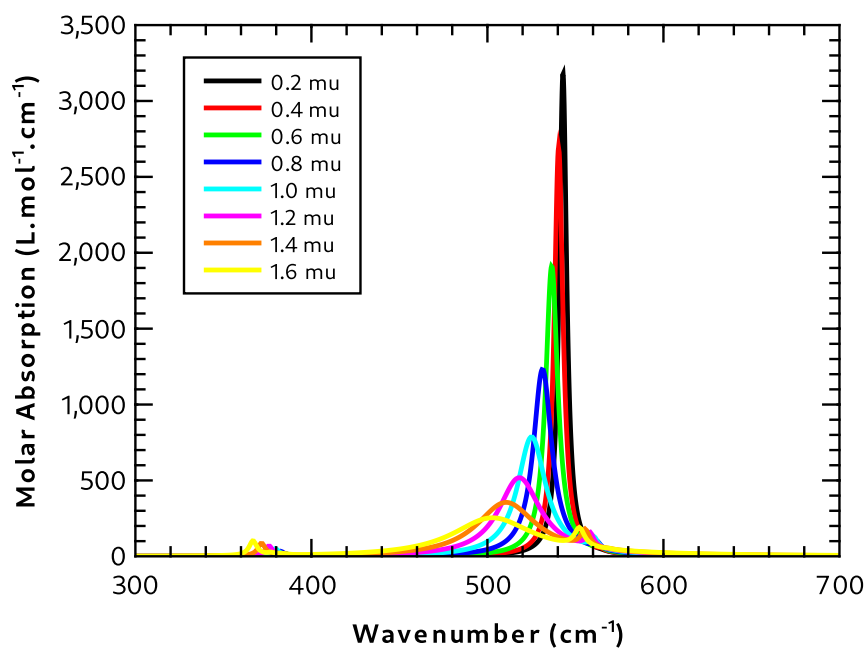


Fig. 11.15: Variation in absorption calculated by the Mie method for different radii of spheres. 1% volume fraction of MgO in PTFE and a broadening of 5 cm^{-1}

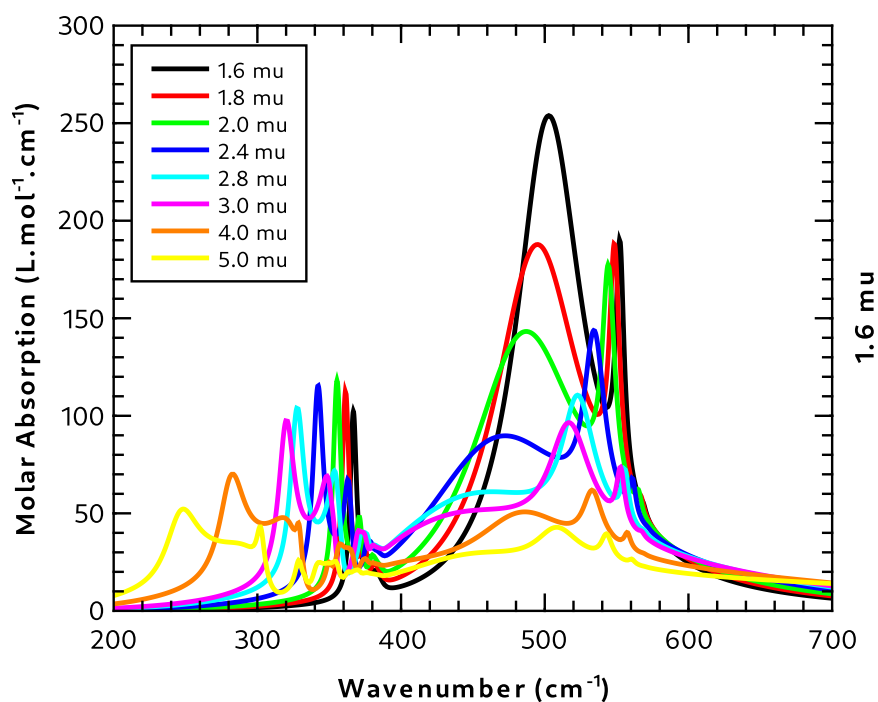


Fig. 11.16: Variation in absorption calculated by the Mie method for larger sphere radii. 1% volume fraction of MgO in PTFE and a broadening of 5 cm^{-1}

11.7 ZnO Example using Mie Scattering

ZnO is an anisotropic material, so the treatment described here using Mie scattering is an approximation. However, the permittivity constant tensor is diagonal due to the space group symmetry of the crystal.

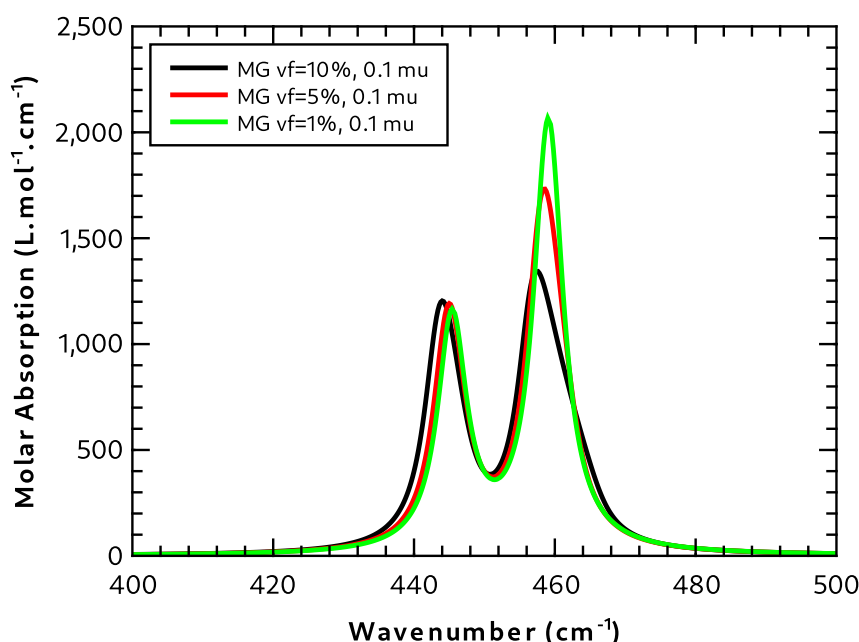


Fig. 11.17: ZnO spheres in PTFE using Maxwell Garnett

Figures Fig. 11.17 and Fig. 11.18 compare the capabilities of the Maxwell-Garnett and Mie methods for describing volume fraction effects. Fig. 11.17 shows that Maxwell-Garnett predicts a lowering of intensity and frequency of the high-frequency peak as the volume fraction is increased. Fig. 11.18 illustrates that Mie theory shows no effect of the change in volume fraction. This is to be expected as the theory assumes that each sphere is isolated and does not affect the other spheres around it. It should be pointed out that the figures are plotting molar absorption coefficients. The actual absorption would increase with volume fraction of ZnO.

Fig. 11.19 shows that the variation of the Mie scattering with sphere radius follows a similar pattern to that observed in MgO, though slightly more complex. The initial peaks at about 440 and 460 cm^{-1} broaden and shift to lower frequencies as the particle size increases. Bulk bands around 350 and 372 cm^{-1} can be seen which grow in intensity and shift to lower frequencies as the particle size increases. The data for this calculation can be found by looking in the [Examples](#) section of the installation guide under the Vasp/ZnO/application_note_mie example.

11.8 Fitting the ATR Spectrum of $\text{Na}_2(\text{SO}_4)_2$

To calculate the expected Attenuated Total Reflectance spectrum of $\text{Na}_2(\text{SO}_4)_2$ a VASP calculation was performed of the optimised structure followed by the dynamical matrix and the phonon spectrum at the Γ point. After reading in the OUTCAR with PDGui, the scenario tab is modified to reflect the materials used in an ATR experiment. The support matrix is set to air and the volume fraction of the dielectric is set 80%. This value needs to be chosen to reflect the particle shape and size distribution. When looking at the plot with the plotting tab, ensure that the 'Powder ATR' plot type is used. This ensures that the fitter tab uses that plot for the fitting. For the case of non-polarised radiation the choice of 0.5 for the 'ATR S polarisation fraction' is appropriate. The settings tab will look like the tab shown below.

The initial fitter tab will look something like what is shown below. A file has been specified holding the

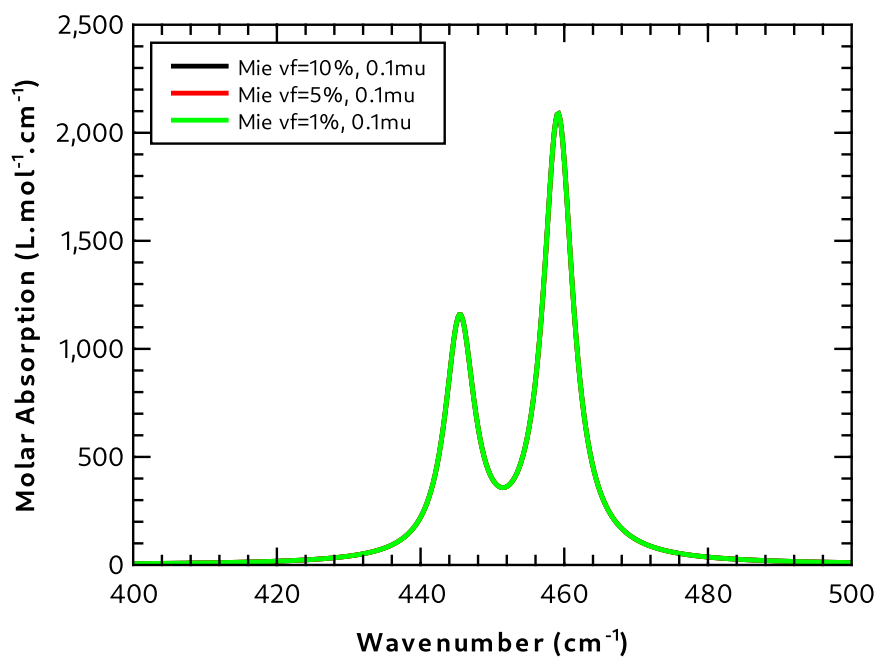


Fig. 11.18: ZnO spheres using Mie

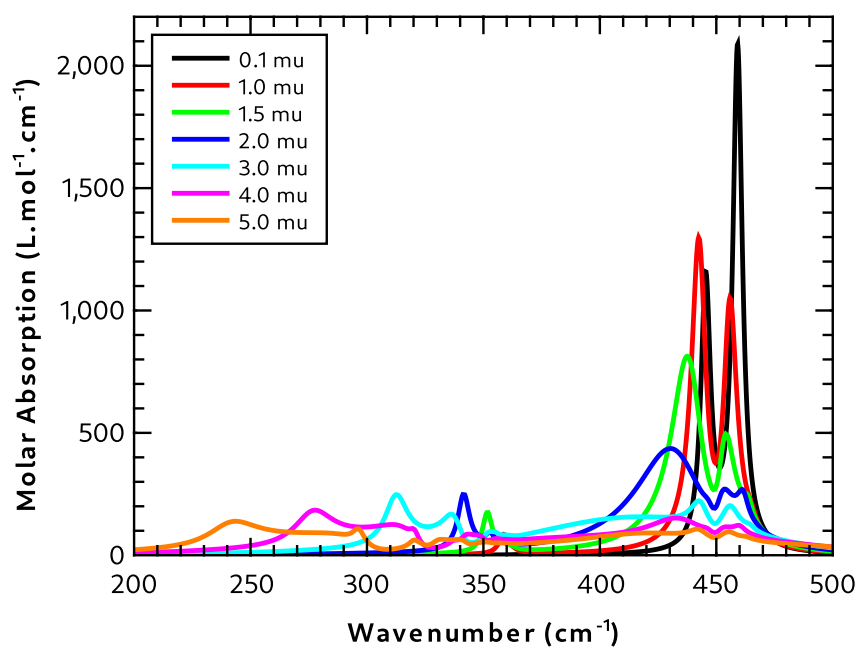


Fig. 11.19: ZnO spheres using Mie Mie scattering of 10% volume fraction ZnO spheres in PTFE using a line broadening factor of 5 cm^{-1} .

PDGui - Using PDIELEC library 7.0.1

Main Settings Scenario 1 Plotting Analysis 3D Viewer Fitter

Support matrix air

Support density 0.001

Support permittivity 1.000

% Air void volume fraction 0.0

Air void radius (μm) 30.000

% Mass fraction of dielectric 99.988237

% Volume fraction of dielectric 80.000000

Method Maxwell-Garnett

Particle radius (μm) 0.000100

Particle sigma (μm) 0.000000

Particle shape Sphere

Unique direction [abc] 0 0 1

Ellipsoid a/b eccentricity 1.000000

ATR material refractive index 4.000

ATR incident angle 45.0

ATR S polarisation fraction 0.500

Powder scenario legend Calculated ATR

Add another scenario Delete this scenario Switch to crystal scenario

Fig. 11.20: $\text{Na}_2(\text{SO}_4)_2$ Scenario Tab for an ATR scenario

experimental ATR spectrum. There is a list of Lorentzian widths for each phonon transition. The *Replot* button re-plots the graph after any changes to the sigma widths of the transition. Changes to the phonon transition widths made in the *Fitter Tab* are reflected in the *Settings tab*.

If changes are made to the widths a re-plot will cause all the information needed (crystal permittivity and powder permittivity) to be recalculated. The progress of the calculation is shown below the three buttons. After a recalculation four items of data are show;

- **X-correlation**

This is the maximum in the cross-correlation function calculated between the experimental and simulated spectrum. A value of 1.0 means that there is perfect agreement between the shapes of the calculated and experimental curves, a value of 0.0 indicates that they are not correlated at all.

- **frequency scale**

This shows the frequency scaling factor being used to scale the calculated frequency. This can be set using the Settings at the top of the *Fitter tab*. It is usually 1.0, but where a DFT method is known to show systematic errors in its calculated frequencies, it is possible to compensate for these using this value.

- **shift**

This is the frequency at which the cross correlation function is a maximum, it is sometimes called the ‘lag’. By default the plot is shown without the frequency shift, but this can be changed by using the *Replot with frequency shift* button.

- **rmse**

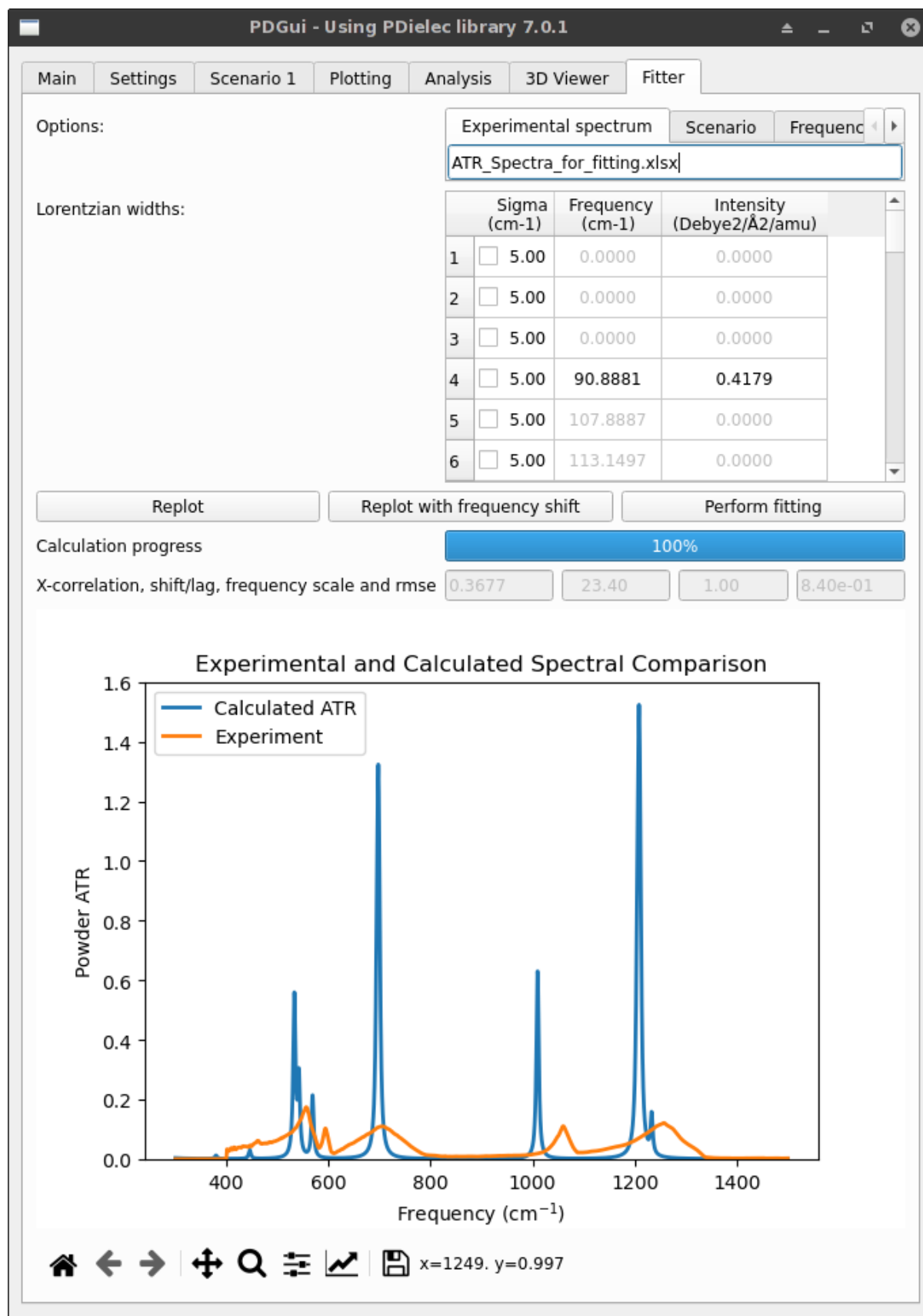
This is the root mean squared difference between the experimental and calculated spectra. If the units of the two spectra are different then this is not a very useful measure of their similarity and any fitting should be done using the maximum in the cross-correlation function.

In the situation above the calculated and experimental spectra have been plotted on the same scale, but very often the units used for absorption are confusing and if necessary a separate scale for the experimental and calculated spectra can be used by toggling ‘Independent y-axes’ on the Fitter settings tab.

Visual inspection of the spectra above indicates that the transition widths are too narrow and need broadening. This is simply achieved by choosing a frequency range and editing the sigma values. Around 1200 cm^{-1} there are two transitions with similar intensities. The values of both sigmas should be modified until the calculated peak height is similar to the experimental one. Where possible the same values of sigma should be used for several transitions and this process repeated for the important frequency ranges. The table below shows the changes that can be made to improve the fit.

Table 11.7: Modified sigma parameters for $\text{Na}_2(\text{SO}_4)_2$ after fitting

Mode	Frequency (cm-1)	Starting sigma	Final sigma
21	445.2	5	10
22	528.1	5	200
24	532.4	5	10
28	663.9	5	100
31	1000.9	5	35
33	1189.0	5	130
35	120.6	5	100

Fig. 11.21: Na₂(SO₄)₂ Fitter Tab for fitting ATR spectra

The cross-correlation of this fit is 0.7369 at a frequency shift of 21.20 cm^{-1} . The plot below shows the comparison of the calculated and experimental curves and the plot was made incorporating the frequency shift. It is possible to further improve the quality of the agreement of the fit by selecting some of the sigmas for optimisation using the tick box next to each frequency and clicking on the ‘Perform fitting’ button. If the ‘Optimise scaling’ option is selected in the Settings tab, then very often the value of the shift parameter is reduced considerably. The problem with doing the fitting automatically is that maximising the correlation function may not lead to comparisons that ‘look’ as good as simply making sure the peak heights of the experimental and calculated spectra are similar. The data for this calculation can be found by looking in the [Examples](#) section of the installation guide under the ATR/Na₂SO₄2_fit example.

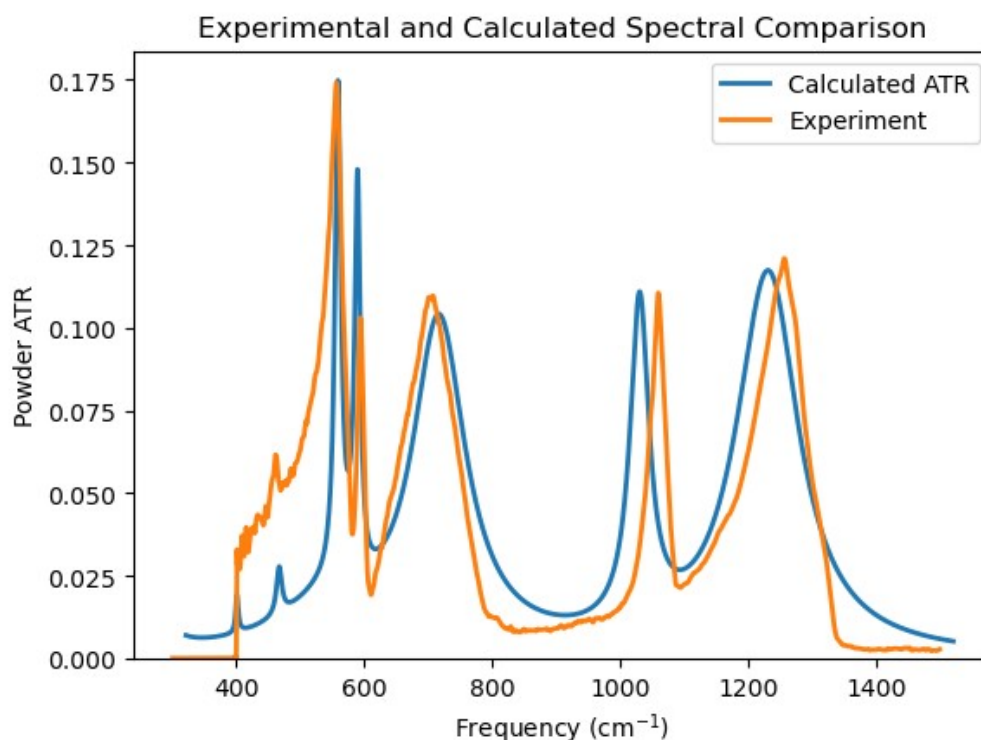


Fig. 11.22: Na₂(SO₄)₂ Fitted Spectra

SINGLE CRYSTAL APPLICATIONS

The transmission, reflection and absorption of infrared light through single crystals can be simulated using PDielec and PDGui. Incoherence of all dielectric layers is modelled by considering the transmission matrix for the intensity of light rather than the amplitude or by averaging the propagation phase of the internally reflected light.

12.1 Single Crystal Study of Thickness Effects in MgO

This example looks at an isotropic system. The results of a CASTEP calculation of MgO have provided the frequency-dependent permittivity and this is used to look at the effect of crystal thickness on the reflectance, transmittance and absorbance. The examples shown here were calculated using the scripts `mgo_coherent_size_effects.py` in the `Examples/SingleCrystal/MgO` directory of the GitHub distribution. A 45° incident beam on the (001) surface of MgO is used to calculate the transmittance, reflectance and absorbance of thin films with thickness of 0.1, 1, 10 and 100 microns. The results of the coherent calculations are shown in Fig. 12.1

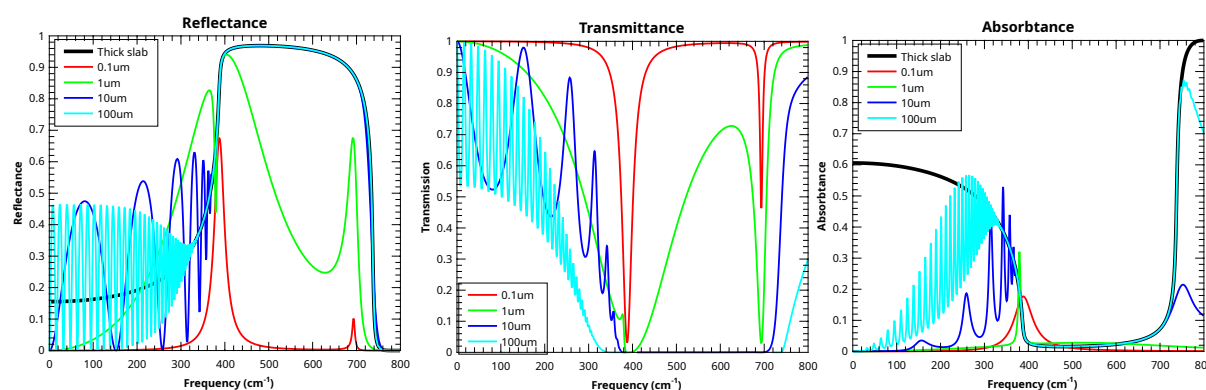


Fig. 12.1: Effects of Film Thickness of MgO in Coherent Light, a 45° incident beam with p- polarisation

The results for the 'thick slab' case are shown only for reflectance and absorbance. It can be seen that the reflection in the restrahlen region between the TO and LO frequencies (388 and 693 cm^{-1}) is almost total with also very little absorbance. The 'thick slab' designation refers to a situation where the dielectric material is used as the substrate. As such there is no internal reflection and no transmittance. The $0.1\mu\text{m}$ thickness shows peaks in reflectance and corresponding drops in transmittance at both the TO and LO frequencies, whilst showing only absorbance at the TO frequency. The $1\mu\text{m}$ thickness shows considerable broadening of these signals, with the TO reflectance splitting into a doublet. At $10\mu\text{m}$, below the TO frequency there are oscillations in the reflectance, transmittance and absorbance which are a result of interference between the internally reflected waves. They occur because the light is coherent. The oscillations continue until they meet the 'thick slab' limit and they then follow this limit until frequencies above the LO frequency. At $100\mu\text{m}$, the oscillations below the TO frequency are very

fast and they reach the ‘thick slab’ limiting case at lower frequencies than for the $10\mu\text{m}$ case. Similarly at higher frequencies they follow the ‘thick slab’ limiting case.

The effect of thickness on the “Incoherent (intensity)” method is shown in Fig. 12.2. This method uses intensities rather than amplitudes for the transfer matrix and as a result, all the phase information is lost, resulting in no interferences after internal reflections. The calculations shown are for p- polarised incident radiation.

The incoherent method deviates from the ‘thick slab’ results at very low frequencies, but as the film thickness increases the deviations become smaller. Similarly at frequencies above the LO frequency, the deviations from the ‘thick slab’ results are less for the thicker films. The user needs to be aware that for large thicknesses the calculation of the propagation matrix can lead to numerical overflows. These manifest themselves as sudden changes in reflectance or transmittance. In such cases it is recommended to use the ‘Incoherent (non-reflective)’ method or use the scattering matrix method with Incoherent (phase averaging).

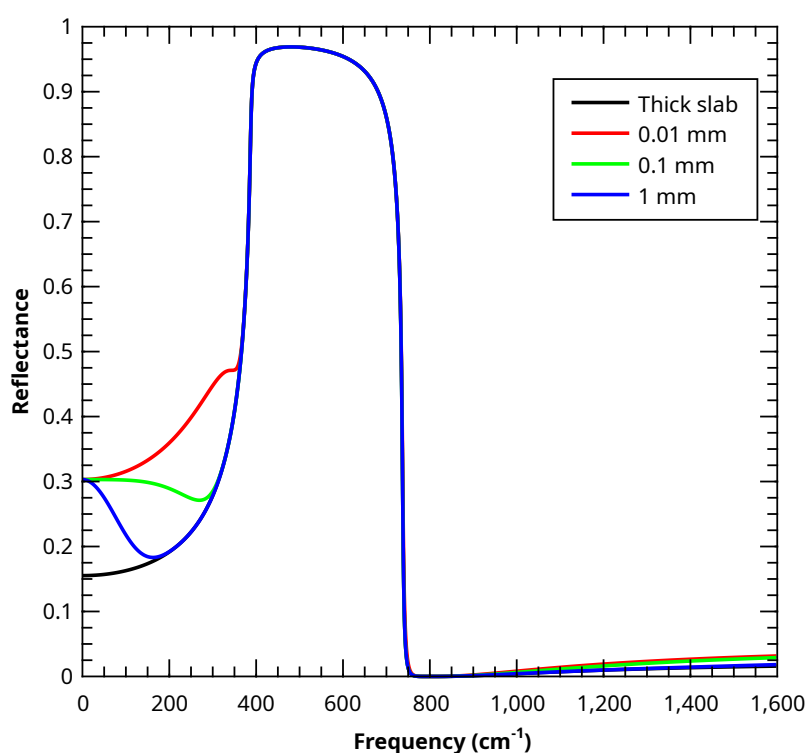


Fig. 12.2: Incoherent (intensity) mode - a 45° incident beam with p- polarisation

Komandin et al [74] have reported terahertz spectra of $0.5\mu\text{m}$ thick MgO crystals which are dominated by interference at low frequencies. In Fig. 12.3 a digitised spectrum obtained from this publication is compared with calculations. Considering only calculated information (refractive index, LO and TO frequencies, etc.) is being used, it can be seen that, despite the rather poor digitisation, the agreement is very good. Agreement with the number of interference bands and the amplitude of the bands is also good.

Finally, we compare a digitised spectrum of the reflectance measured over the range $0 - 800\text{ cm}^{-1}$ with the spectrum calculated using PDGui and Castep. The ‘thick slab’ approximation has been used as no oscillations are detected in the low-frequency experimental results. The calculated and experimental results are in excellent agreement. The shoulder on the experimental result is due to two phonon processes that are not considered in the present theoretical work.

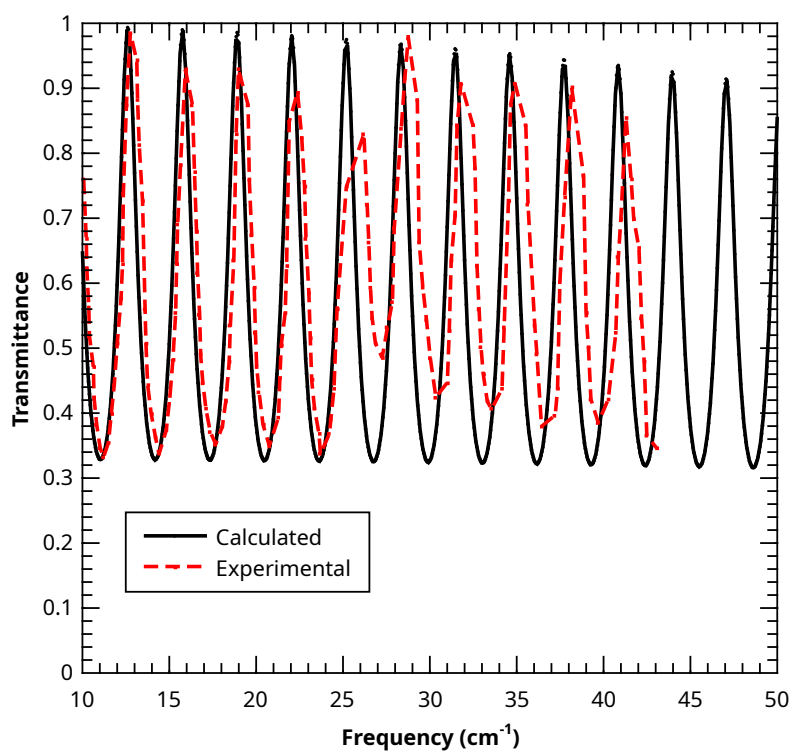


Fig. 12.3: Comparison of calculated and experimental transmittance terahertz spectra

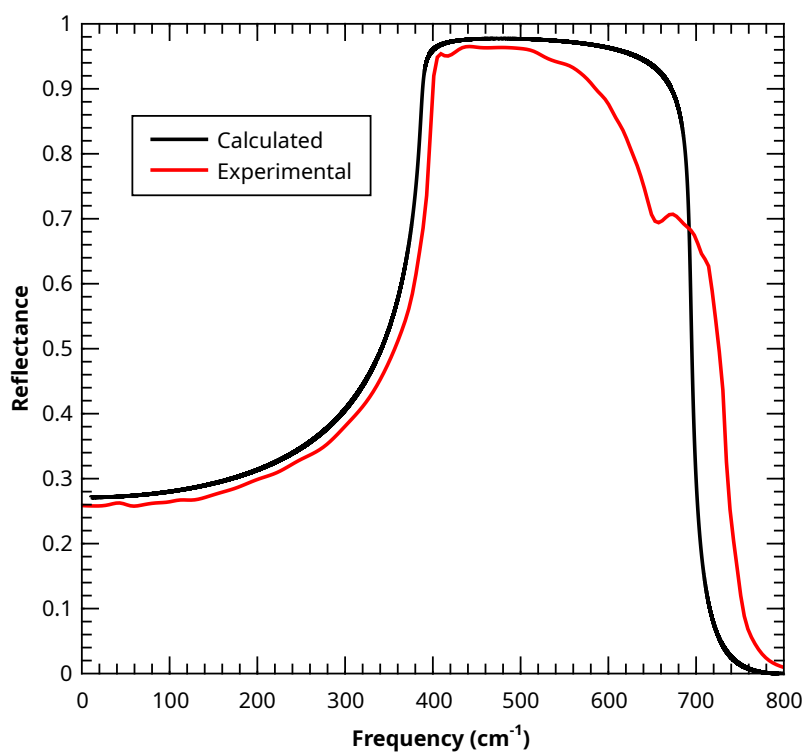


Fig. 12.4: Comparison of calculated and experimental reflectance infrared spectra

12.2 Single Crystal Study of Forsterite

In a publication on the interpretation of the experimental spectrum of forsterite [75] a Four Parameters Semi-Quantum (FPSQ) model for the single crystal infrared spectrum of this material was presented. In this application note that we use the experimental file format of PDielec to define the FPSQ model parameters from the paper and use it to calculate the reflectance spectrum for comparison with that published in the paper. The FPSQ model is defined by the following equation.

$$\epsilon(\omega) = \epsilon_{\infty} \prod_j \frac{\Omega_{LOj}^2 - \omega^2 - i\gamma_{LOj}\omega}{\Omega_{TOj}^2 - \omega^2 - i\gamma_{TOj}\omega} \quad (12.1)$$

Forsterite is an orthosilicate belonging to an orthorhombic space group. The permittivity matrix is diagonal when the x, y and z axes of the crystal are aligned with a, b and c axes of the unit cell. The parameters for the FPSQ model are summarised in the tables below.

Table 12.1: Optical Permittivity

	X	Y	Z
X	2.83	0.0	0.0
Y	0.0	2.69	0.0
Z	0.0	0.0	2.76

Table 12.2: FPSQ Model Parameters for Forsterite

EPS(xx)				EPS(yy)				EPS(zz)			
TO	TO	LO	LO	TO	TO	LO	LO	TO	TO	LO	LO
Freq	sigma	Freq	sigma	Freq	sigma	Freq	sigma	Freq	sigma	Freq	sigma
202.6	0.18	202.9	0.28	144.9	0.09	145.5	0.21	278.3	0.50	278.9	0.54
276.6	0.59	277.1	0.84	278.6	0.68	278.9	0.79	292.7	0.90	305.6	0.36
296.1	0.79	300.6	1.18	290.8	1.28	311.3	0.68	306.3	0.39	316.7	0.53
321.1	0.60	322.5	0.88	351.7	2.77	376.7	1.02	411.6	2.03	414.6	1.97
383.2	1.52	390.5	1.52	397.3	2.15	412.4	2.30	418.5	3.19	456.4	1.88
405.4	1.58	468.9	2.90	419.8	2.17	444.1	2.75	425.0	4.22	424.5	3.83
476.6	4.46	478.0	3.69	458.1	4.39	489.2	2.59	478.9	3.22	487.2	3.99
504.7	3.81	532.6	5.65	507.8	2.77	513.7	4.16	506.2	2.71	580.8	8.79
532.3	5.67	552.3	9.41	529.5	4.62	575.0	8.33	874.4	4.17	996.8	4.32
606.6	4.77	651.6	12.10	838.6	6.56	844.5	7.63				
840.8	6.75	841.5	7.03	872.9	5.02	965.2	3.33				
959.9	2.41	965.5	3.30	986.3	4.80	994.8	2.96				
978.5	3.83	1081.1	5.76								

The publication presents experimental polarised reflectance infrared spectra of single crystals of forsterite. The incident beam angle to the sample was 10°. The s-polarised geometry was used and the measurements were made with the electric field parallel to the a-axis, b-axis and c-axis.

In PDGui the laboratory frame is defined by XYZ, with the incident and reflected light in the surface in the XZ plane, with the surface-normal aligned with the Z-axis. The surface is therefore in the XY plane. The azimuthal angle can be used to rotate the crystal about the normal until the crystal axis of interest is parallel to the Y axis of the laboratory frame. With this geometry the s-polarised wave field will be oscillating parallel to the chosen axis. The arrangement described is shown in the Figure below where E_s and E_p are the field directions of the incident light.

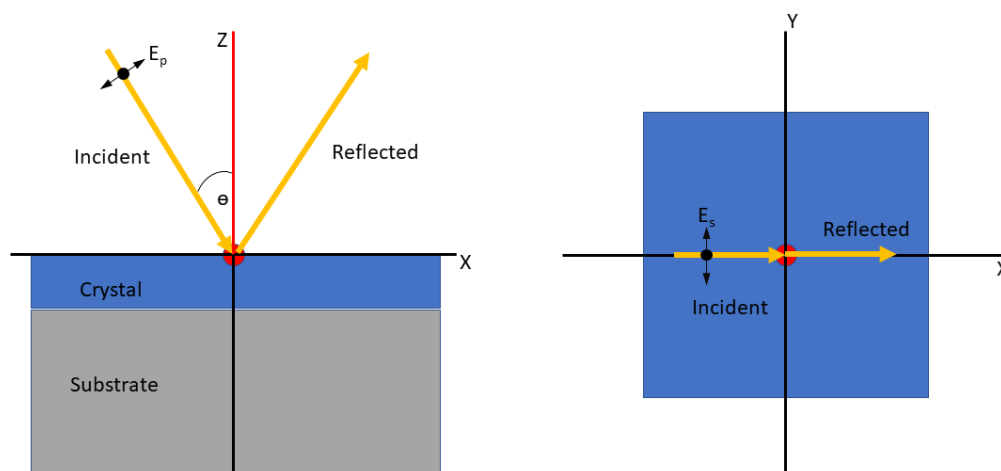


Fig. 12.5: Definition of the single crystal laboratory coordinates in PDGui

As can be seen E_s is parallel to the laboratory Y-axis, so it is necessary to line up the crystal unit cell so that the cell direction being investigated is parallel to the Y-axis. In the single crystal scenario tab PDGui shows the crystal axes in terms of the laboratory coordinates so it is relatively straightforward to make sure that the cell direction of interest is aligned along the Y-axis. It is possible to set each axis in one of two ways, depending on the surface being used. For example to make sure the a-axis is aligned with the laboratory Y-axis, we can use either the [001] or the [010] surfaces and simply choose the appropriate azimuthal angle.

The following geometries can be used by PDGui to ensure the relevant axis is aligned with the laboratory Y-axis;

- **For the a-axis;**
 - the [001] surface with azimuthal angle = 90
 - the [010] with azimuthal angle = 90
- **For the b-axis**
 - the [001] surface and azimuthal angle = 0
 - the [100] surface and azimuthal = 90
- **for the c-axis**
 - the [100] surface azimuthal angle = 0
 - the [010] surface azimuthal angle = 0

The two figures below show the reflectance of s-polarised light along the a-axis and these figures show very good agreement with Figure 1 in the published experimental work [75]. The data for this calculation can be found by looking in the [Examples](#) section of the installation guide under the Experimental/forsterite example.

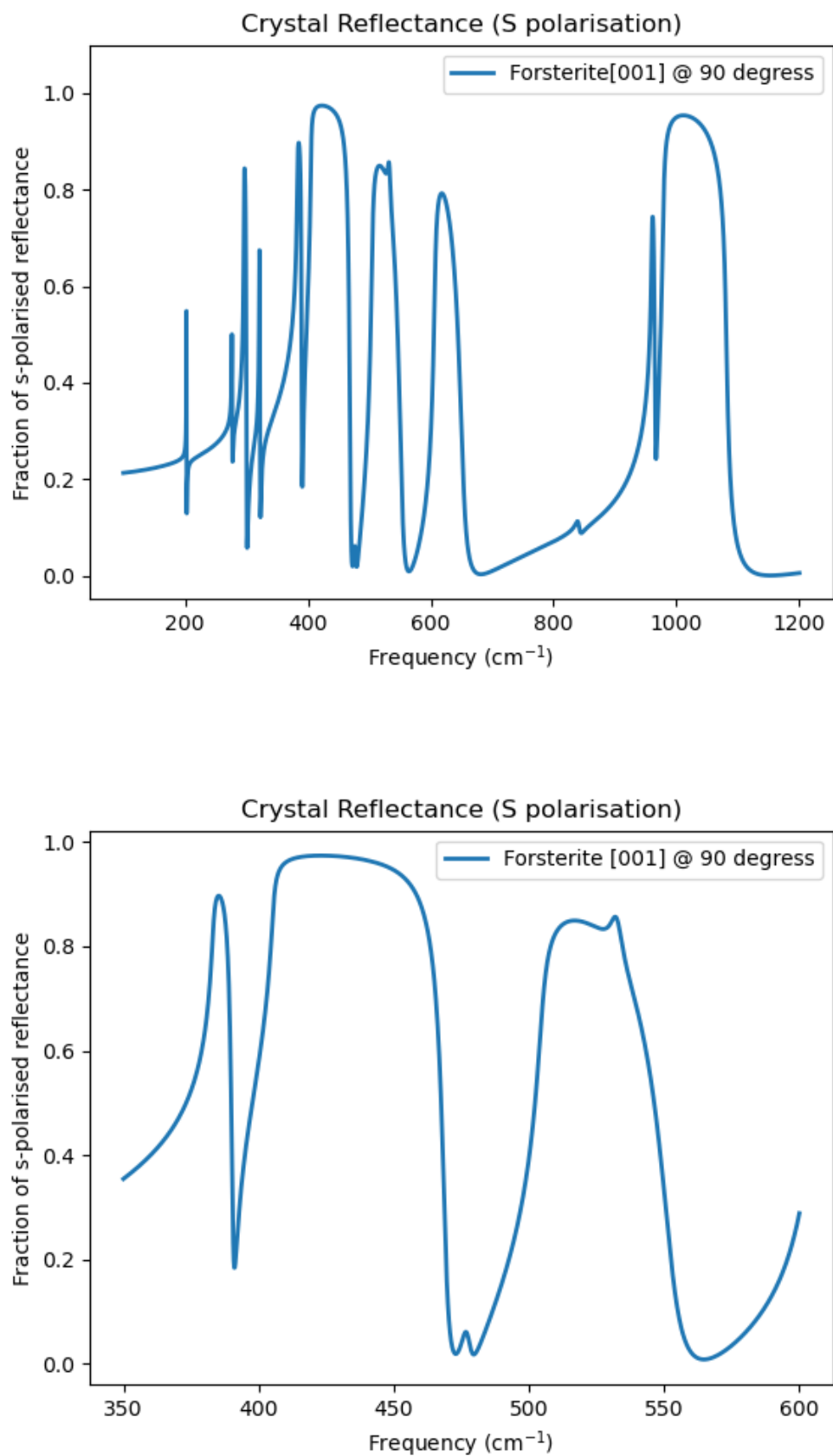


Fig. 12.6: Forsterite a-axis reflectance (upper figure 100 - 1200 cm⁻¹, lower figure 350 - 600 cm⁻¹)

12.3 Single Crystal Study of L-Alanine

In an experimental and computational study of the vibrational modes of single crystals of L-alanine, the experimental transmittance was reported for polarized radiation along each of the principal axes, a-, b- and c-. [76]

The directory Examples/SingleCrystal/L-alanine contains files for the calculation of the transmittance and reflectance of single crystals of L-alanine based on similar calculations to those reported in the paper, using Crystal17 to perform the DFT calculations, and using PDGui to calculate the transmittance and reflectance properties of the crystal. Details of the DFT calculation using the B97-3c functional can be found in phonon.dl2. The output can be found in phonon.out. Other files produced by the Crystal17 calculation are BORN.DAT and HESSFREQ.DAT. These are used by PDGui to calculate the theoretical spectrum.

The experimental results for the measurement of terahertz transmittance were kindly provided by the authors of the experimental paper: J. L. Allen, T. J. Sanders, J. Horvat, R. A. Lewis and K. C. Rule [76]. We would like to thank the authors of the paper for providing the experimental data and for helpful discussions. Their measurements can be found in the spreadsheets a-axis-experimental.xlsx, b-axis-experimental.xlsx and c-axis-experimental.xlsx. Each of these spreadsheets has two columns, the first is a column of frequencies in cm^{-1} and the second is the transmittance.

Scripts used to generate the crystal transmittance for the electric field along the respective crystal axes from the calculated phonon modes can be found in a-axis.py, b-axis.py and c-axis.py. These scripts use the *Fitter Tab* to compare the calculated spectrum with the experimental one. Changes have been made to the film thickness and to the widths of the peaks to improve the agreement between the two.

To run the scripts use a command such as,:

```
pdgui -script a-axis.py
```

In the following, the steps used to get to the scripts are explained. We will concentrate on the a-axis spectrum but the approach is similar for each axis considered.

L-alanine crystallises in an orthorhombic form, with 3 non-equivalent axes. To initialise PDGui with the results of the DFT calculation and starting the GUI interface with the *Scenario tab* in its single crystal mode use the command below,:

```
pdgui phonon.out -scenario crystal
```

The *Main Tab* shows the calculated unit cell and the frequencies as calculated by Crystal17.

The *Settings Tab* (see below) shows those settings which affect the calculation of the permittivity. By default, the frequencies are modified by projecting out any translational degrees of freedom. This is shown by the three zero frequencies in the table of frequencies and is a result of selecting *Apply Eckart conditions?* The atomic masses are selected by PDGui and maybe different to those used by the DFT calculation. A Lorentzian line width is applied to all phonon vibrations by default. In this case, we have chosen a large value by default and applied smaller and more realistic values for those phonon modes important to transmission for the a-polarised electric field.

The contribution to the permittivity arising from electronic polarisation is shown in this Tab. In the case of Crystal17 and the B9703c functional the CPHF calculations is not possible because the exchange/correlation functional is not supported and the ϵ_∞ calculation is not performed and a zero matrix will be presented here. In this example, we have typed in the contribution from a Crystal17 calculation using PBE-D3. The resulting spectrum is not very sensitive to these settings.

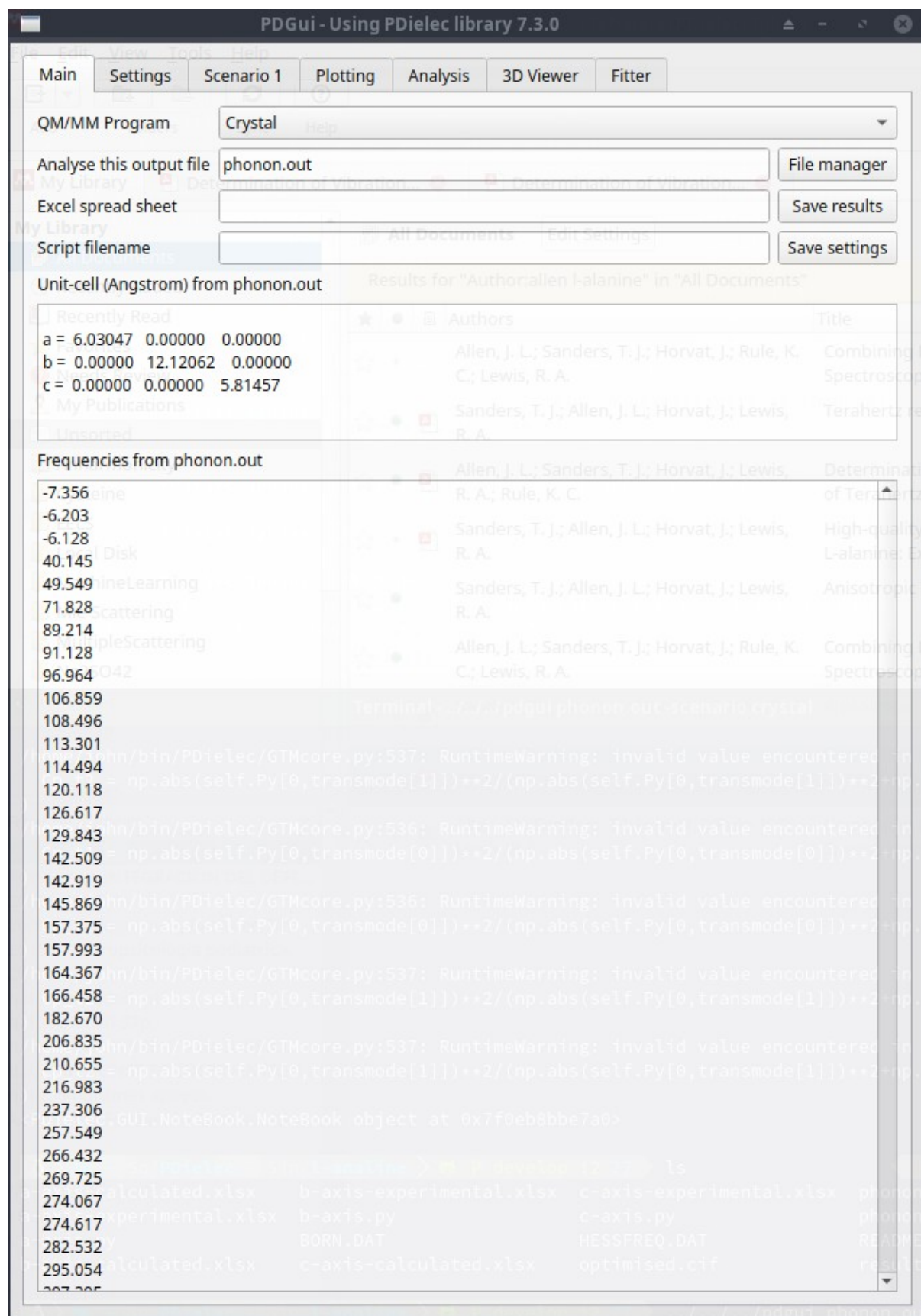


Fig. 12.7: L-alanine Main Tab

PDGui - Using PDIElec library 7.3.0

Main Settings Scenario 1 Plotting Analysis 3D Viewer Fitter

Apply Eckart conditions? ☒

Apply Born charge neutrality? ☐

Atomic mass definition: Average natural abundance

Atomic masses

	O	N	C	H
	15.999400	14.006700	12.010700	1.007940

Lorentzian width (σ): 30.0

Optical permittivity:

	x	y	z
x	2.4420	0.0000	0.0000
y	0.0000	2.3052	0.0000
z	0.0000	0.0000	2.5068

	Sigma (cm-1)	Frequency (cm-1)	Intensity (Debye ² /Å ² /amu)	Integrated Molar Absorption (L/mole/cm ²)	Absorption maximum (L/mole/cm)
1	☐ 30.0	-0.0000	0.0000	0.00	0.00
2	☐ 30.0	-0.0000	0.0000	0.08	0.00
3	☐ 30.0	0.0000	0.0000	0.17	0.00
4	☐ 30.0	40.1348	0.0000	0.00	0.00
5	☐ 30.0	49.5362	0.0000	0.00	0.00
6	☒ 30.0	71.8099	0.0270	114.19	2.42
7	☒ 30.0	89.1234	0.0656	277.18	5.88
8	☒ 2.0	91.1028	0.0054	22.73	7.23
9	☒ 30.0	96.9378	0.0042	17.85	0.38
10	☒ 2.0	106.7650	0.0183	77.50	24.67
11	☐ 30.0	108.4740	0.0000	0.00	0.00
12	☒ 2.0	113.2633	0.0190	80.41	25.60
13	☒ 30.0	114.4571	0.1728	730.07	15.49
14	☒ 30.0	120.0925	0.0022	9.11	0.19
15	☐ 30.0	126.5981	0.0000	0.00	0.00
16	☒ 30.0	129.8078	0.0374	158.17	3.36
17	☒ 2.0	142.4744	0.0709	299.50	95.33
18	☒ 2.0	142.8883	0.0993	419.79	133.62

Fig. 12.8: L-alanine Settings Tab

The next tab is the *Scenario tab* which is used to reflect the experimental parameters that are possible. In this case, we are looking at the single crystal scenario. The mode of operation in the example shown below is the *Coherent* mode. In this case, the surface that the THz beam is incident on is the (001) surface, so the c-axis will be pointing along the Z-axis of the laboratory coordinate frame. The permittivity of the superstrate and substrate is set to 1.0, consistent with the experimental conditions. The geometry of the incident beam relative to the crystal is controlled by the azimuthal angle. A zero value for the incident angle means the beam is at 90° to the surface. A non-zero angle defines the laboratory coordinates as the beam lies in the XZ plane. For this example, the incident beam angle will be set to 0° . Once this is done information about the position of the crystal axes relative to the crystal frame can be seen in the text box of the *Settings tab*. Changes to the azimuthal angle will result in alterations to the definitions of the crystal axes in terms of the laboratory coordinates. We are interested in radiation polarised with the electric field active along the a-axis of the crystal, which, according to the *Lab frame information* lies along the Y-axis of the laboratory coordinate. This corresponds to s-polarised light, so we will be looking at the s-component of the transmittance.

An important parameter to consider is the thickness of the film and the method by which we will calculate the transmission and reflectivity. To start with we will use the *Coherent* mode and a thickness of 0.2mm. Experimentally [76], the crystal thickness was measured to be 0.5 ± 0.1 mm, but with this thickness the calculation showed little transmission in the chosen frequency range.

The resulting spectrum is shown below. It can be seen that there are regular oscillations in the transmittance, especially at low frequencies. These are due to interference of the directly transmitted light and light that has undergone internal reflection and generated a series of decaying etalons.

To see if we can remove this we will change the option for this layer in the *Scenario tab* to *Incoherent (intensity)*. Re-plotting the spectrum and looking at the results in the *Fitter tab* is shown in the plot below. The calculated spectrum is compared directly with the experimental one. Both the calculated and experimental spectra show little transmittance above about 140cm^{-1} . The strong, very sharp peaks at 225cm^{-1} and above are probably numerical problems that occur with thicker films when transmission tends to zero. These numerical problems are exacerbated by the *Incoherent (intensity)* method as it uses intensities in the calculation of the transfer matrix, which requires the square of the electric field and the propagation matrices. The fact that these are numerical problems can be verified by looking at the *Coherent* method and seeing if they are present there. Even if the experimental measurements are designed to minimise etalons, it is generally sensible to compare both incoherent and coherent calculations before comparing with experiment.

A major advantage of using the *Fitter tab* to understand the calculated spectrum is that the user can modify the Lorentzian widths and see the result immediately, while also comparing them with the experimental spectrum. This also shows the cross-correlation between experiment and calculation, for this example the value is 0.9379 (where 1 is a perfect match).

Plots for the b-axis and c-axis electric fields are shown below. The b-axis example is given by using the same (001) surface but using an azimuthal angle of 0° . The thickness used for the b-axis example is 0.1mm, smaller than the 0.2mm used for the a-axis example. The c-axis field transmittance example uses the (010) surface with an azimuthal angle of 0° and a thickness of 0.1mm.

12.4 AlN on Silicon and Silicon Carbide

This is an example of a multilayer system where two layers are anisotropic. In an experimental and theoretical study of the infrared reflectance of aluminium nitride on silicon and silicon carbide, MacMillan, Devaty and Choyke [77] used a four parameter semi-quantum (FPSQ) model to describe their experimental results on aluminium nitride on various substrates. In the Examples/Experiment/AlN directory the file AlN.exp provides an experimental file that describes their FPSQ model in a format the PDGui can use.

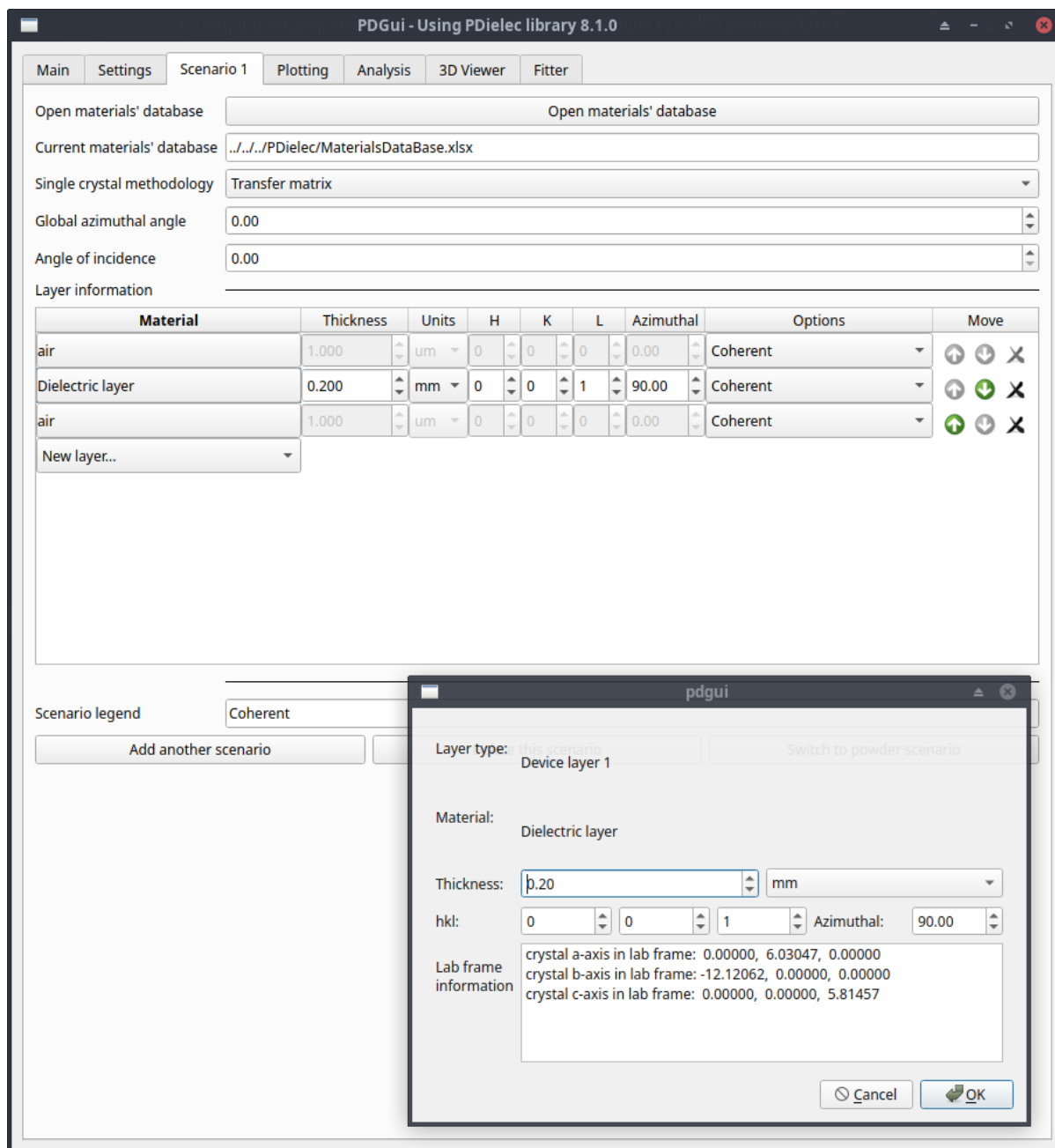


Fig. 12.9: L-alanine Scenario Tab

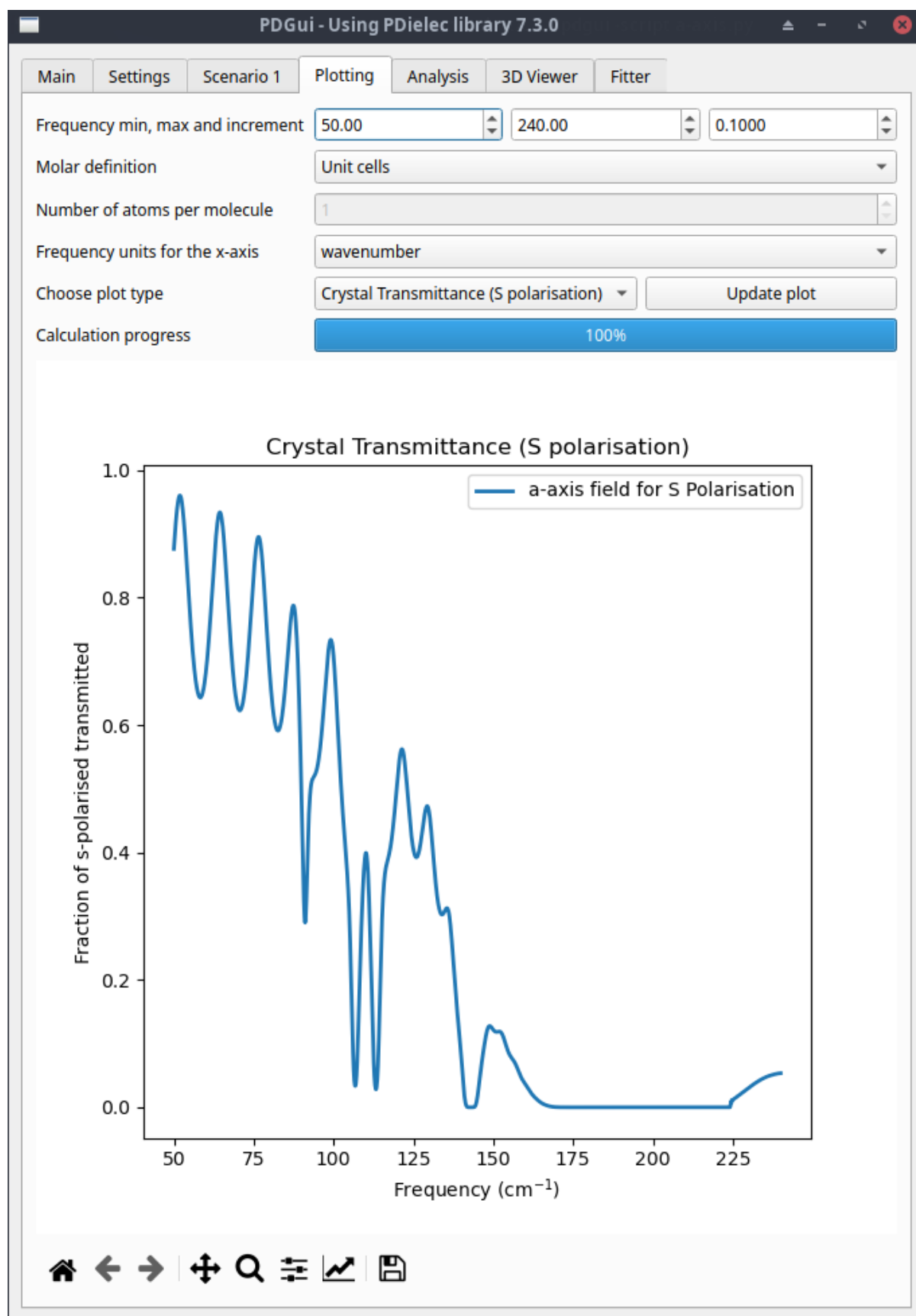


Fig. 12.10: L-alanine Plotter Tab showing coherent light transmittance

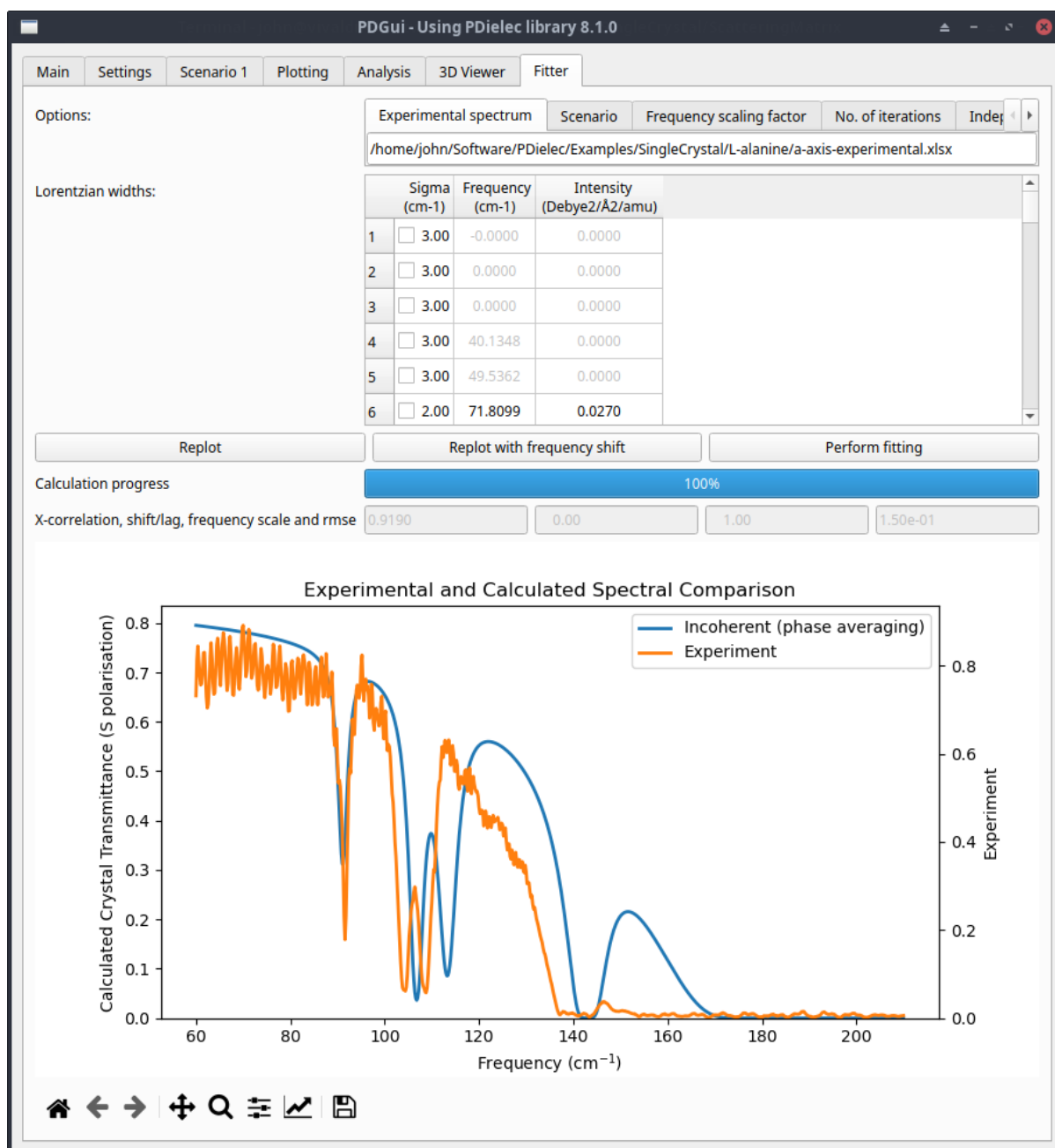


Fig. 12.11: L-alanine Plotter Tab showing incoherent light transmittance for a-axis polarised light

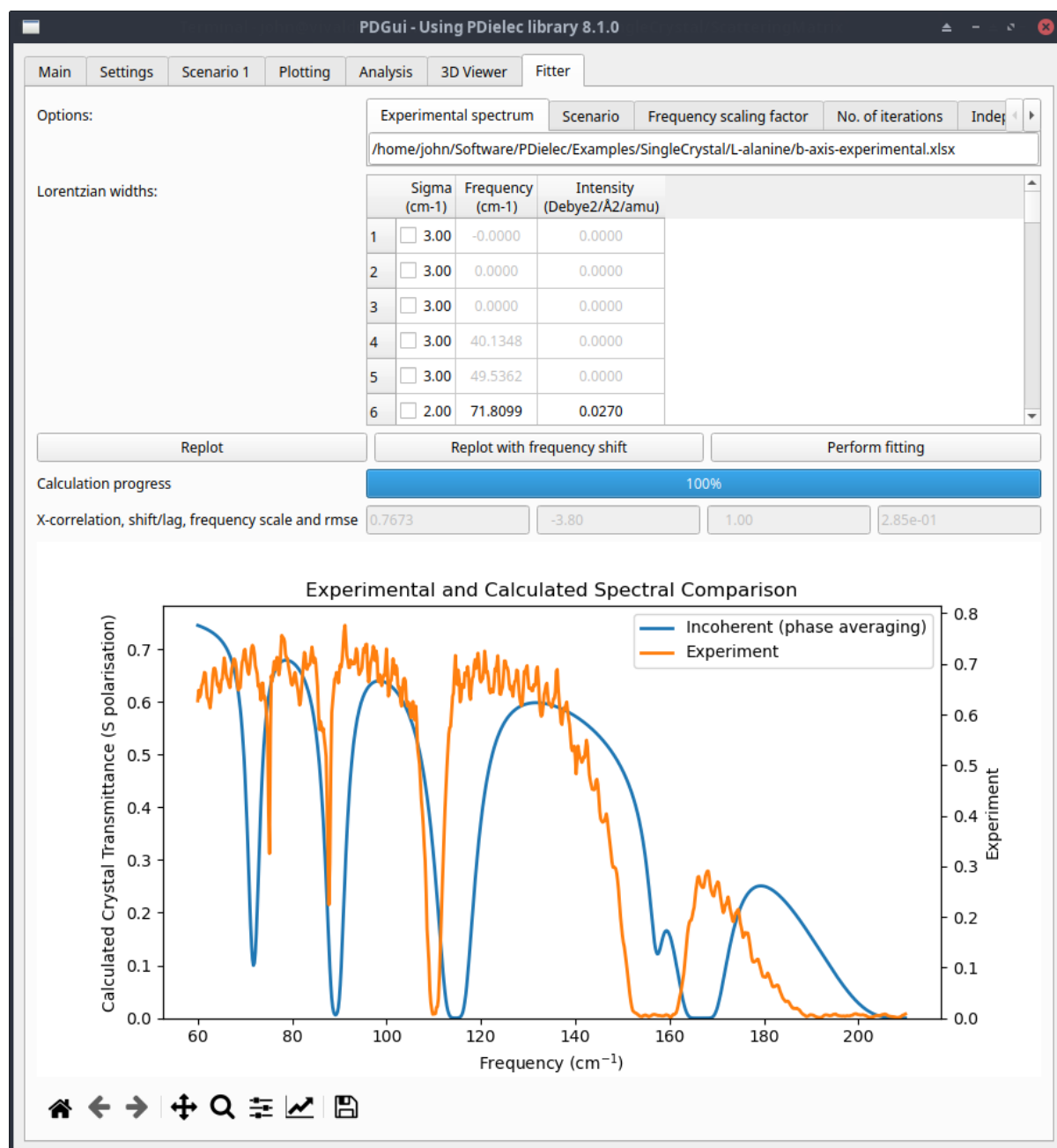


Fig. 12.12: L-alanine Plotter Tab showing incoherent light transmittance for b-axis polarised light

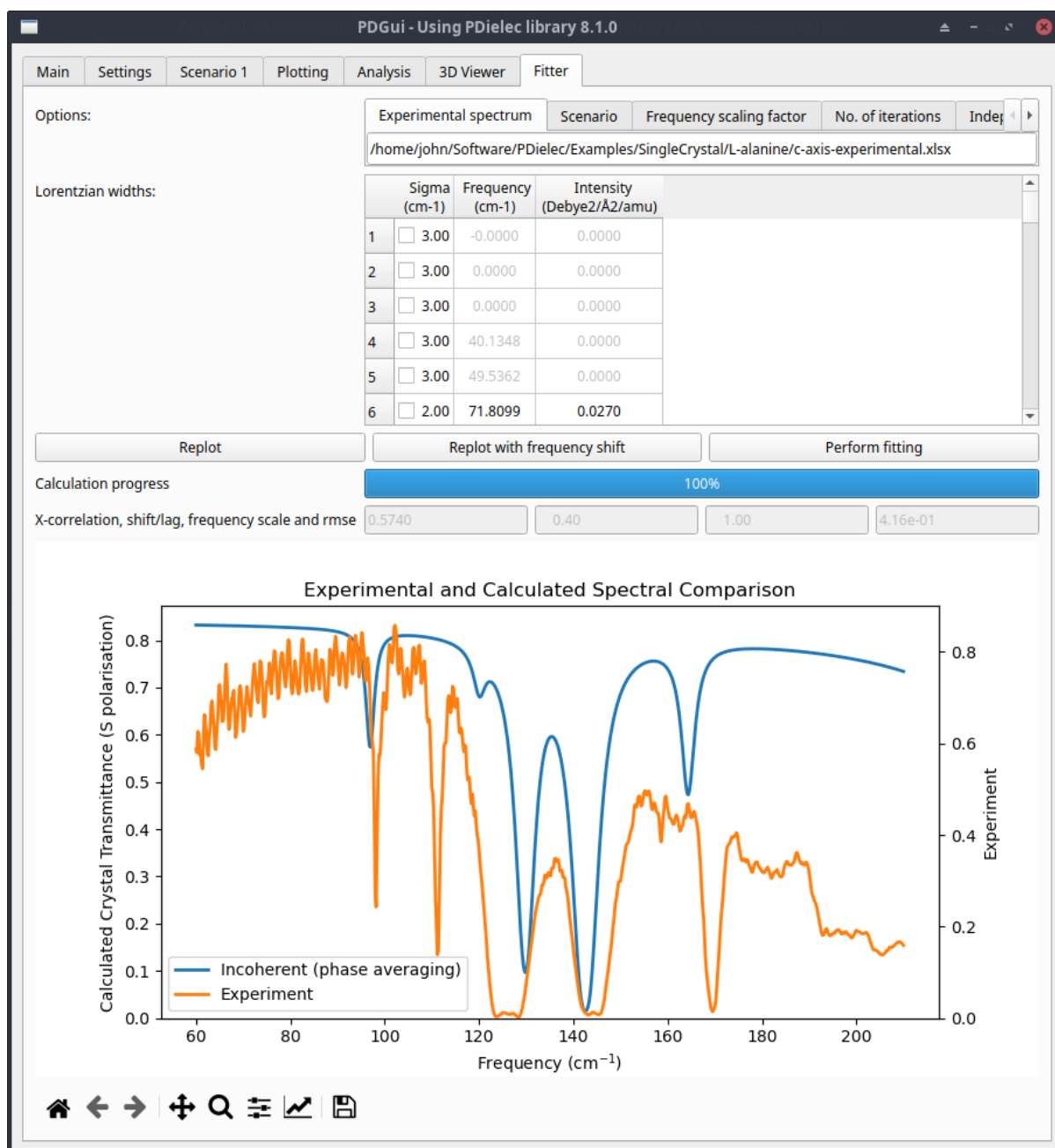


Fig. 12.13: L-alanine Plotter Tab showing incoherent light transmittance for c-axis polarised light

In addition, parameters for the permittivities of silicon and 6H-SiC are given in the TestDataBase.xlsx file in the same directory. The 6H-SiC permittivities are provided by a Drude-Lorentz model and the silicon permittivities are calculated from an experimental refractive index. References for the origins of the models and experimental data are given in the spreadsheet.

The first experimental/calculated system considered [77] was the reflectance of a 0.92μ AlN on a Si substrate. This was modelled using PDGui after reading in the experimental file, AlN.exp, the *Single Crystal Scenario Tab* was used to define the system as described in the paper. The superstrate material was chosen as air, an AlN dielectric film of 0.92μ was specified on top of a 1.0μ m film of silicon, and an angle of incidence of 7.2° was used. The (001) surface of AlN was defined, so the perpendicular to the surface aligns with the laboratory Z-axis. The bottom layer is treated as a semi-infinite layer, so the size of the silicon layer specified in the GUI is irrelevant. The scenario is shown below:

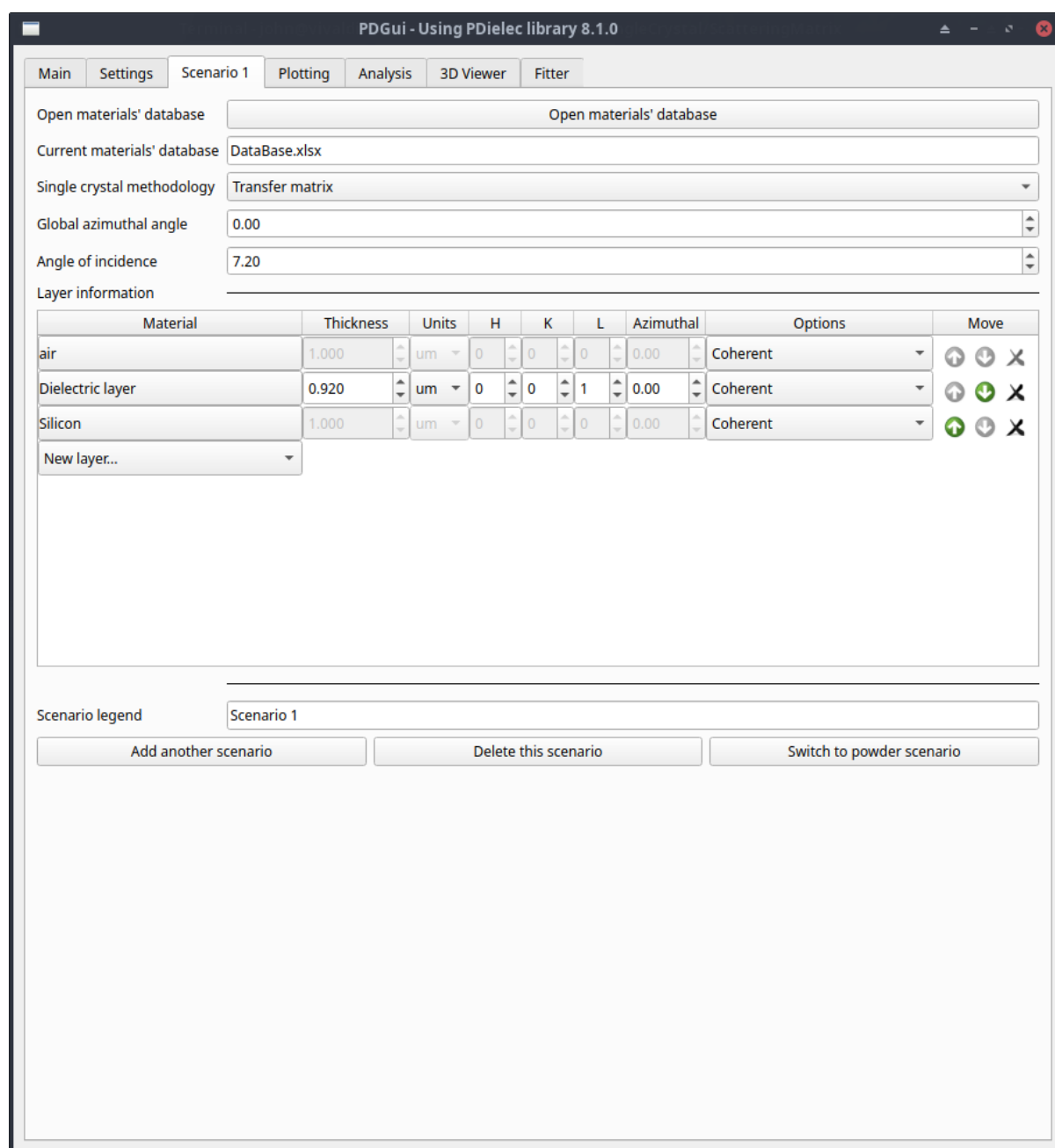


Fig. 12.14: AlN on Silicon Scenario Tab

Two comparisons are made with the calculated results of the published paper after digitising Figures 1 and 2 of the paper. In the first, the range of the plot extends for 0 to 6000 cm^{-1} and clearly shows the etalons associated with the interference. In the second the plot examines the region between 400 and 1200 cm^{-1} . The agreement is excellent.

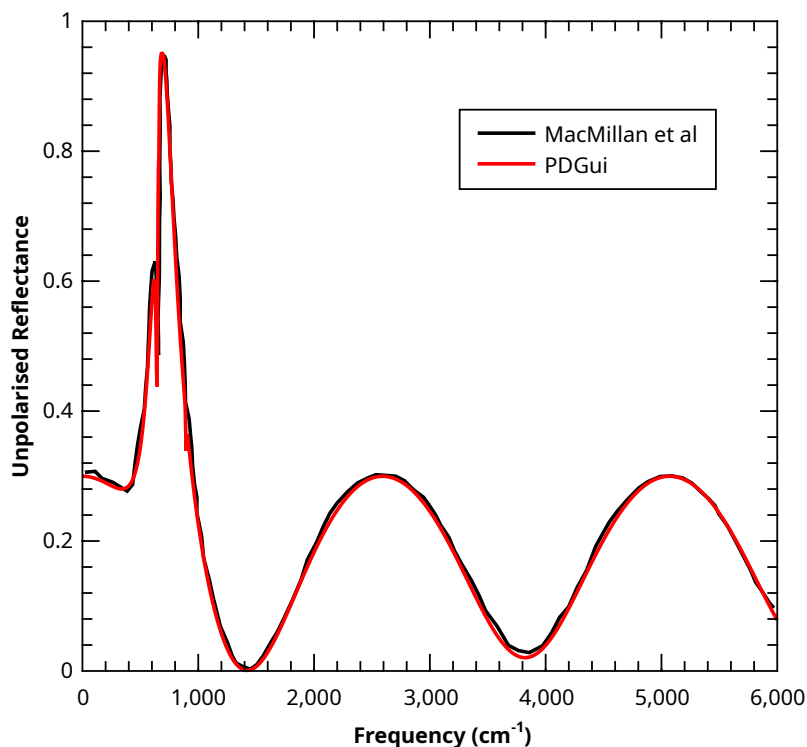


Fig. 12.15: AlN on Silicon reflectance comparison with experiment

Figure 3 of the paper shows the reflectance of a $0.98\mu\text{m}$ film of AlN on the (0001) surface of 6H-SiC. To model this with PDGui two layers have been created, one for the dielectric layer, which has been created from an experimental file, and one for SiC, which has been taken from MaterialsDatabase.xlsx. The layer editor has been used to show that both layers are (001) surfaces with the c-axis of the crystals pointing along the Z-laboratory axis. The top layer (the superstrate) is air below which is the *Dielectric layer* (AlN in this case).

A comparison is given below between the calculated spectrum reported in the paper and that produced by PDGui.

A final comparison is made between Figure 4 of the paper and the results of PDGui. In this case, a $0.24\mu\text{m}$ film of SiC is supported on a $0.56\mu\text{m}$ film of AlN, which is deposited on a substrate of Silicon. For this example, the *Coherent* mode is used. The ScenarioTab for this system is shown below.

A comparison is given below between the calculated spectrum reported in the paper and that produced by PDGui.

12.5 Sapphire Orientation Dependence of Polarized Infrared Light

Sapphire and ruby are $\alpha\text{Al}_2\text{O}_3$, which belongs to an hexagonal crystal system. Due to the complexity of vibrations in the infrared red region and to the anisotropy of the permittivity, the polarized IR reflectance is sensitive to the crystal plane and to its orientation. Lee et. al. [78] measured the infrared reflectance response of sapphire and fitted an FPSQ model to the experimental observations. As confirmation of the correct behaviour of PDGui for the generation of crystal surfaces and for the calculation of reflectance,

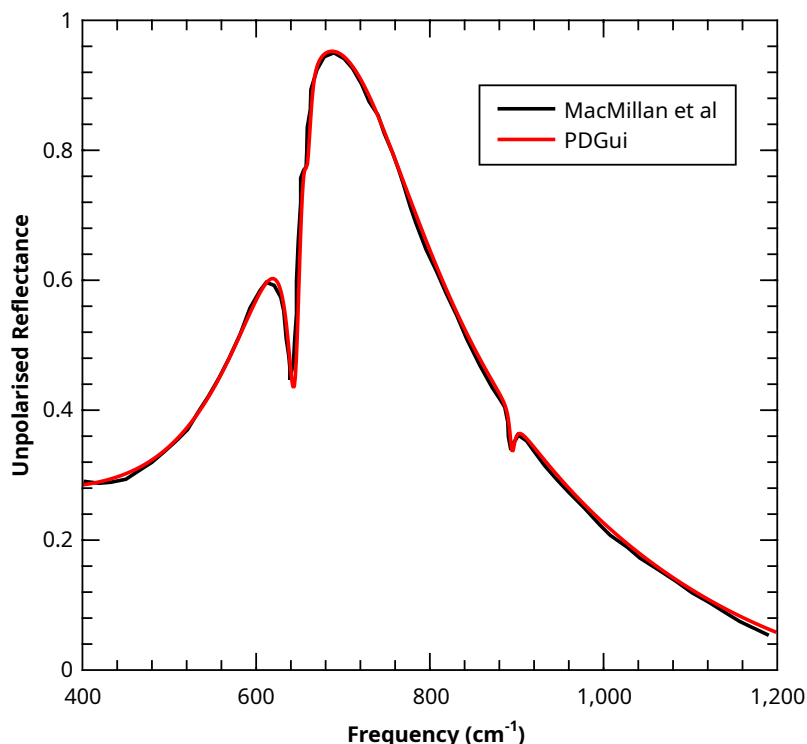


Fig. 12.16: AlN on Silicon reflectance low frequency comparison with experiment

the FPSQ model proposed by Lee et. al. has been incorporated into the materials' database of PDGui and the calculations reported in their paper have been reproduced. To achieve this, a simple experiment file for air has been used as the dielectric medium. Files with the information necessary to reproduce the results presented here are available on the PDielec GitHub in the `Examples/Experiment/Sapphire` directory.

The paper of Lee et. al. [78] gives very clear descriptions of the c-plane, a-planes and r-planes of sapphire. This note will describe in detail how the planes have been established in PDGui. The c-plane is easy to set up. For hexagonal crystals the Miller-Bravais index is (hkil) which represents a plane that intersects the 4 axes of hexagonal crystal coordinate system. By definition the third integer, i, is equal to $-(h+k)$ and is therefore redundant but useful in elucidating the families of faces which are equivalent. For the c-plane the only other related surface to (0001) would be (000-1) and the Miller indices for these surfaces would be (001) and (00-1).

The figure shows the settings in the *Single Crystal Scenario Tab*. The *DataBase.xlsx* file is being used as the materials database and the last layer in the list of layers (Silicon) will be treated as a semi-infinite material. The *Angle of incidence* is set to 7.2° as specified in the paper. The 'dielectric' film in this case is air, which has a constant unit permittivity, its thickness is arbitrary. The second layer is a $1.0\mu\text{m}$ film of sapphire taken from the database with (hkl) set to (001). The layer was added using the *Layer Editor*.

The a-plane family of surfaces are described by (11-20) and its permutations: (1-210), (-2110) (-1-120), (-12-10) and (2-1-10). This is equivalent to the Miller notation (110), (1-20), (-210), (-1-10), (-120) and (2-10). The paper used the (11-20) plane but in this work, the equivalent (-2110) is used with an azimuthal angle set for the layer of 90° . The *Layer Editor* for this is shown below, where it can be seen that the c-axis of the crystal lies along the X-laboratory axis and the a-axis of the crystal lies parallel with the Z-laboratory axis, as described in Figure 1 of the paper. Once the correct orientation has been chosen the azimuthal angle of the complete system can be changed by altering the *Global azimuthal angle*.

The r-plane family of surfaces are described by (1-102), (10-12), (01-12), (-110-2), (-101-2) and (0-11-

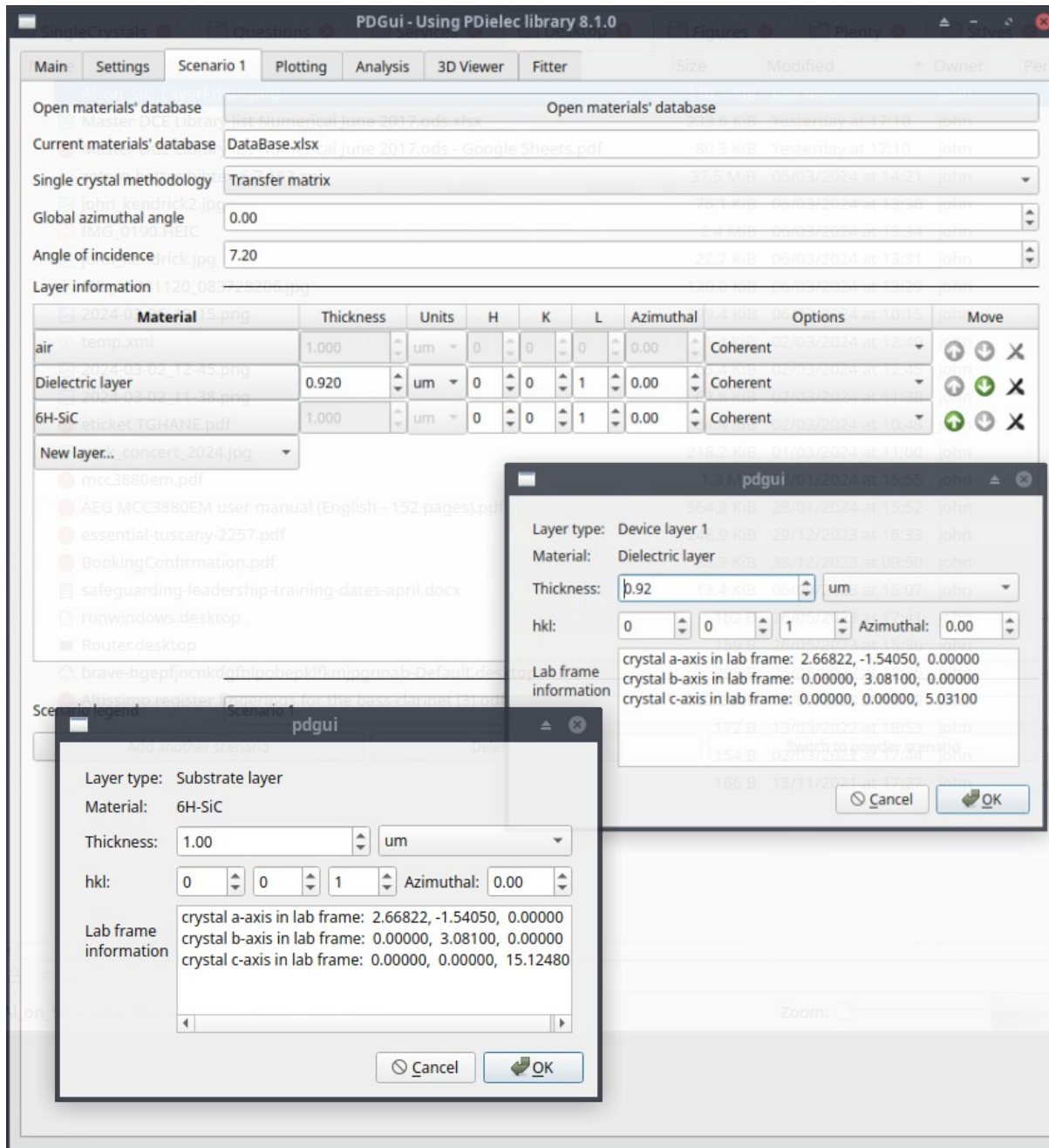


Fig. 12.17: AlN on SiC Scenario Tab

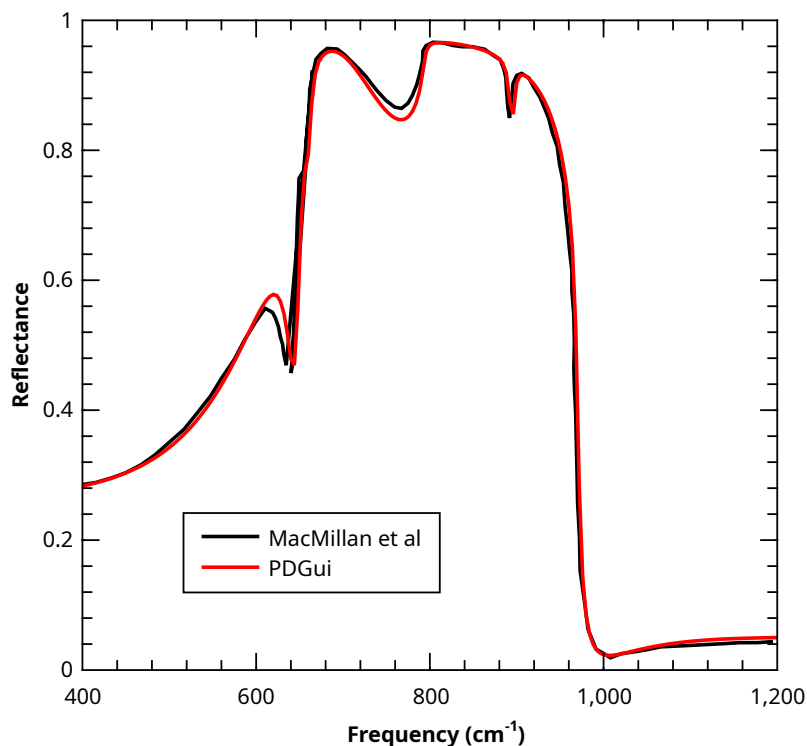


Fig. 12.18: AlN on SiC reflectance compared with experiment

2). The equivalent Miller index descriptions are (1-12), (102), (012), (-11-2), (-10-2) and (0-1-2). In this case, the (1-12) surface was used with a layer azimuthal angle of 90° . The *Layer editor* for this surface is shown below which shows the c-axis of the crystal lying in the XZ plane of the laboratory coordinate system as described in the paper. Again the *Global azimuthal angle* can be used to rotate the whole system relative to the laboratory Z-axis.

The results are shown below and can be compared with Figures, 4, 5 and 6 of the paper.

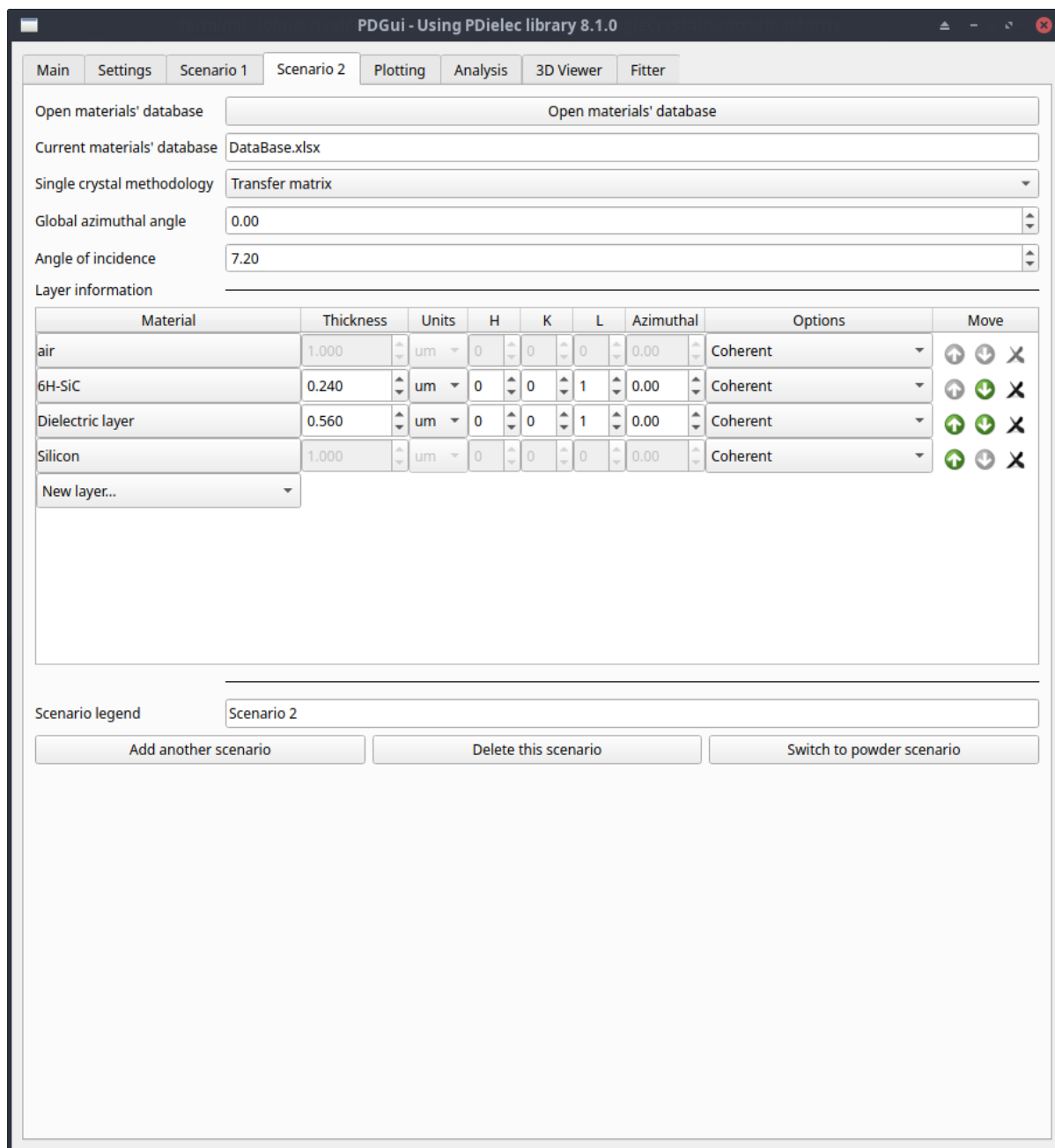


Fig. 12.19: SiC on AlN on Silicon Scenario Tab

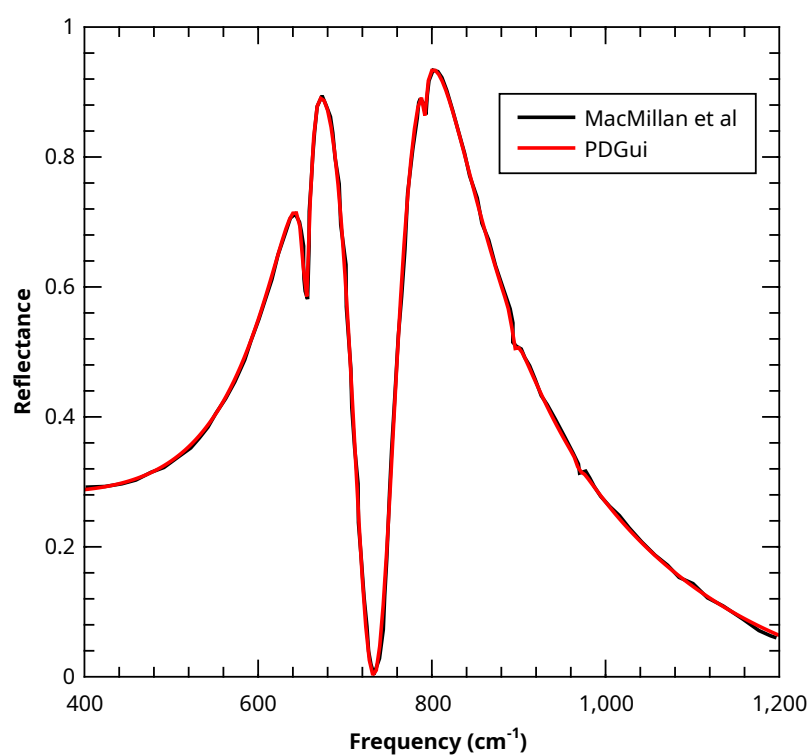


Fig. 12.20: SiC on AlN on Silicon reflectance comparison with experiment

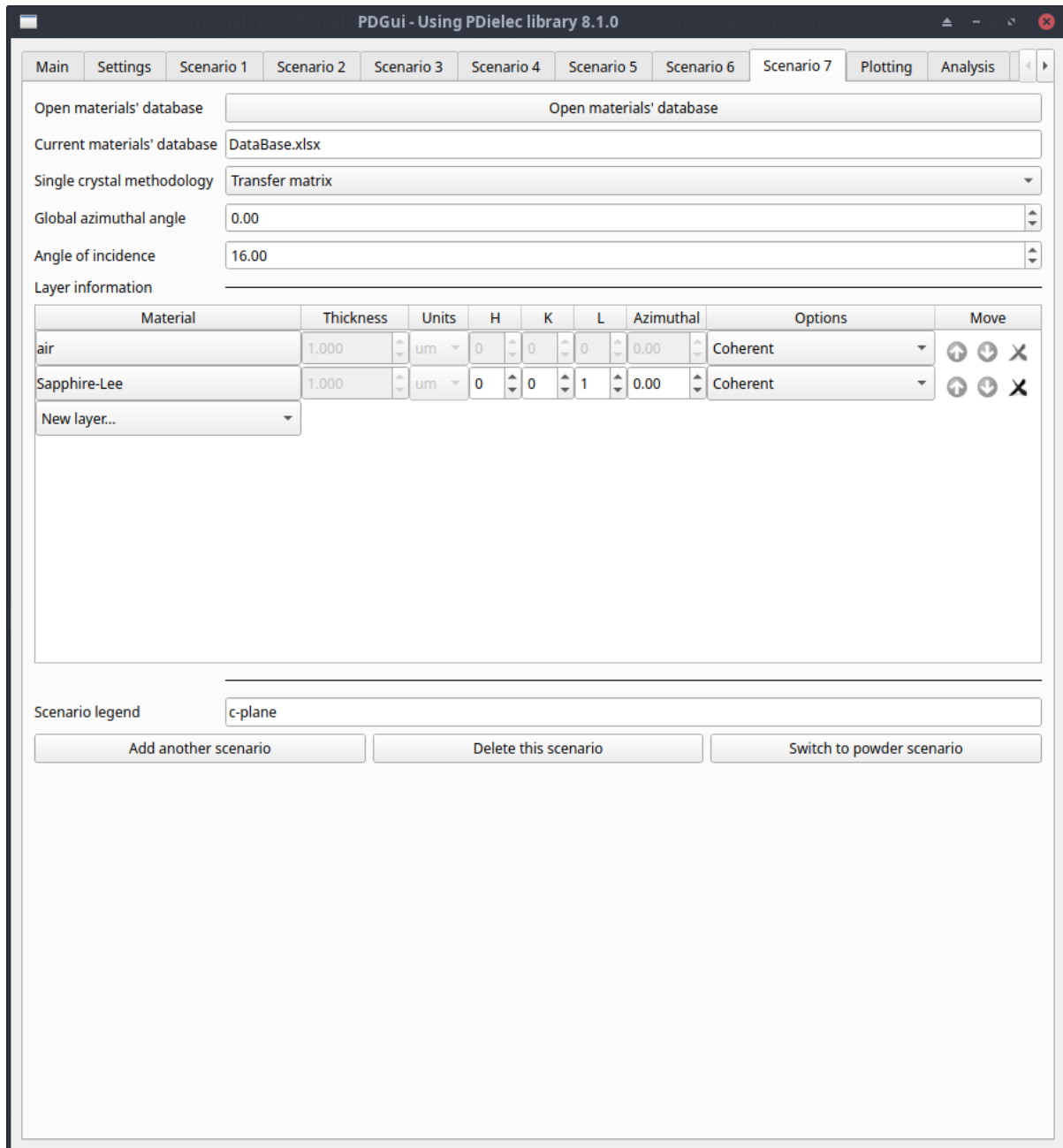


Fig. 12.21: Sapphire scenario tab

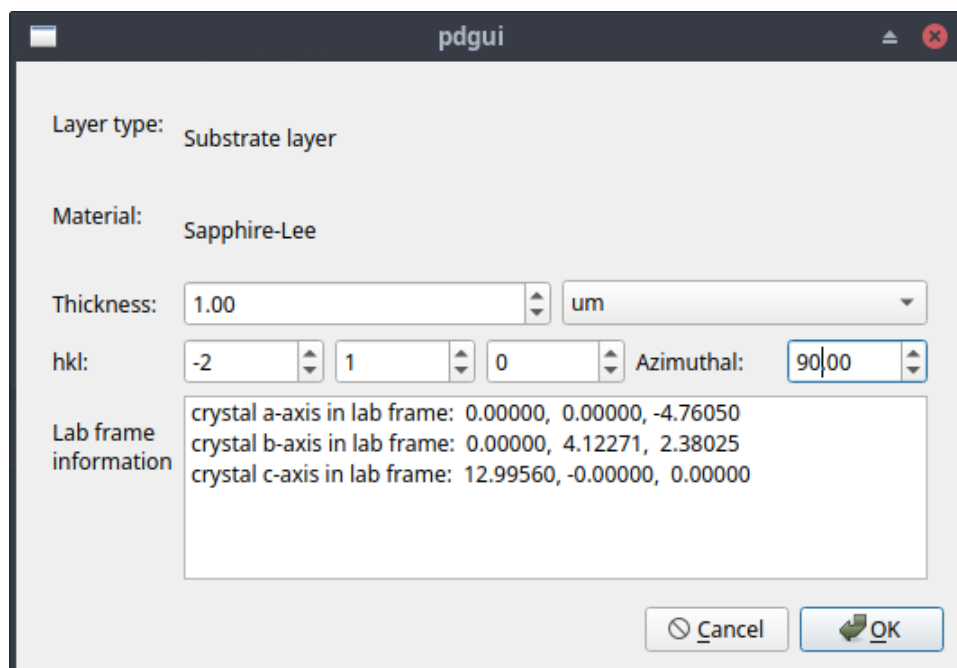
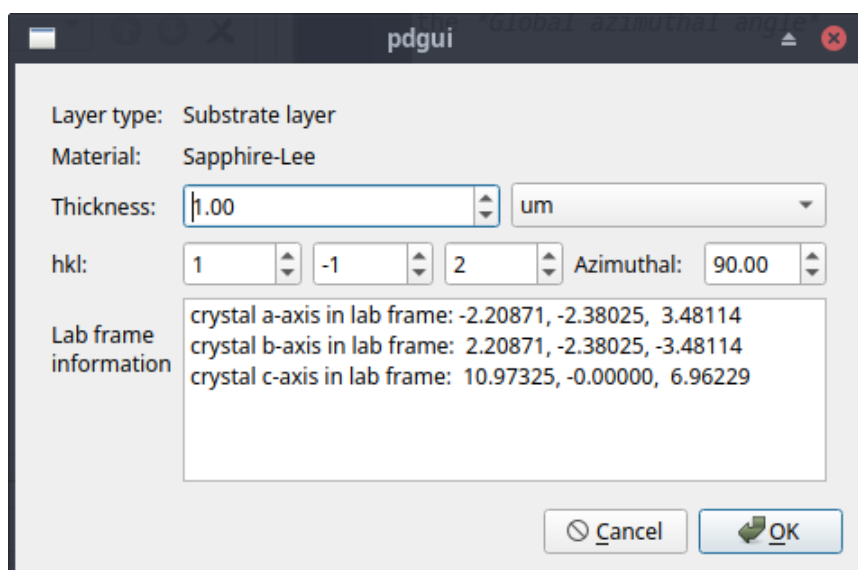


Fig. 12.22: Sapphire layer editor



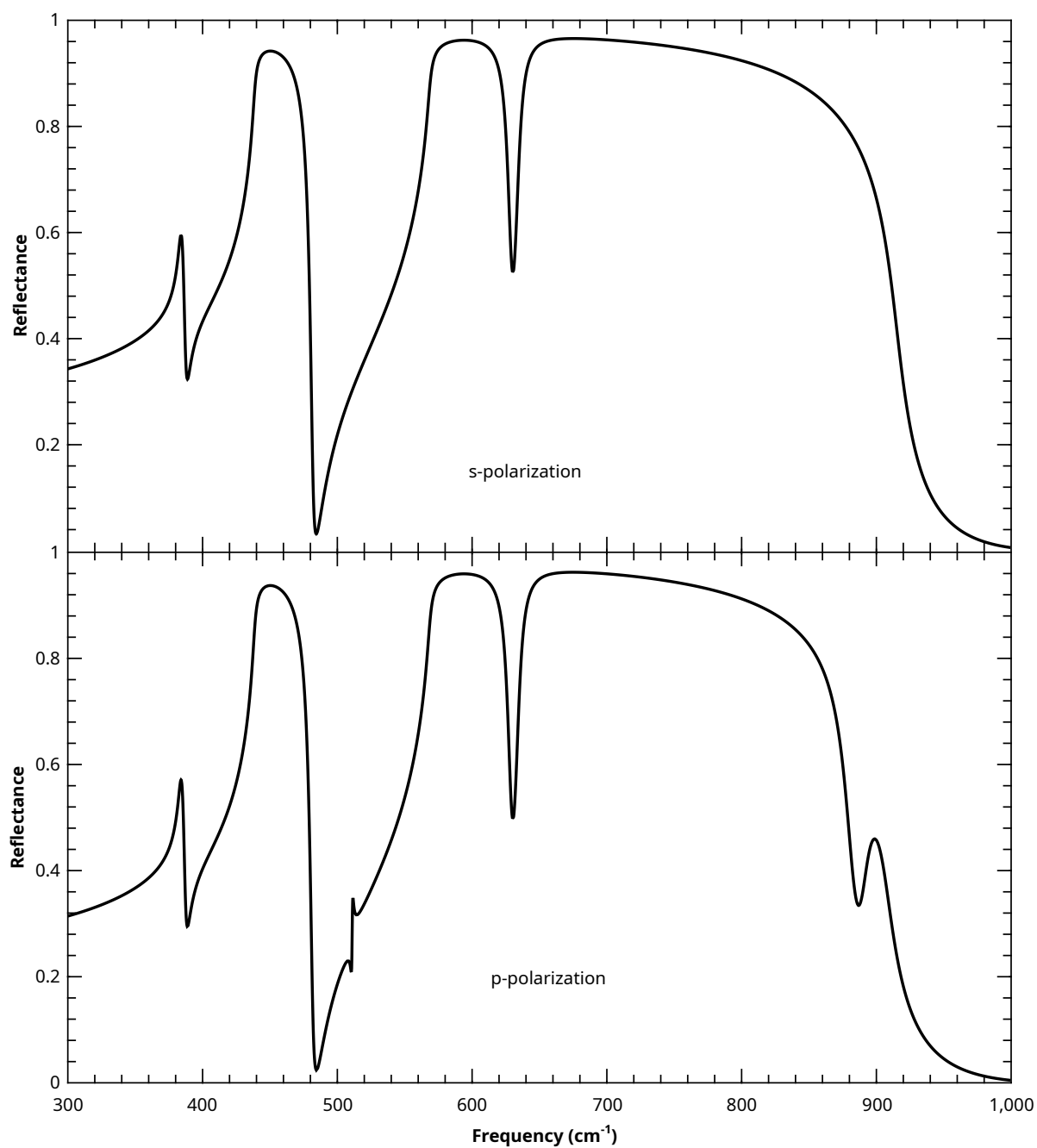


Fig. 12.23: c-plane reflectance

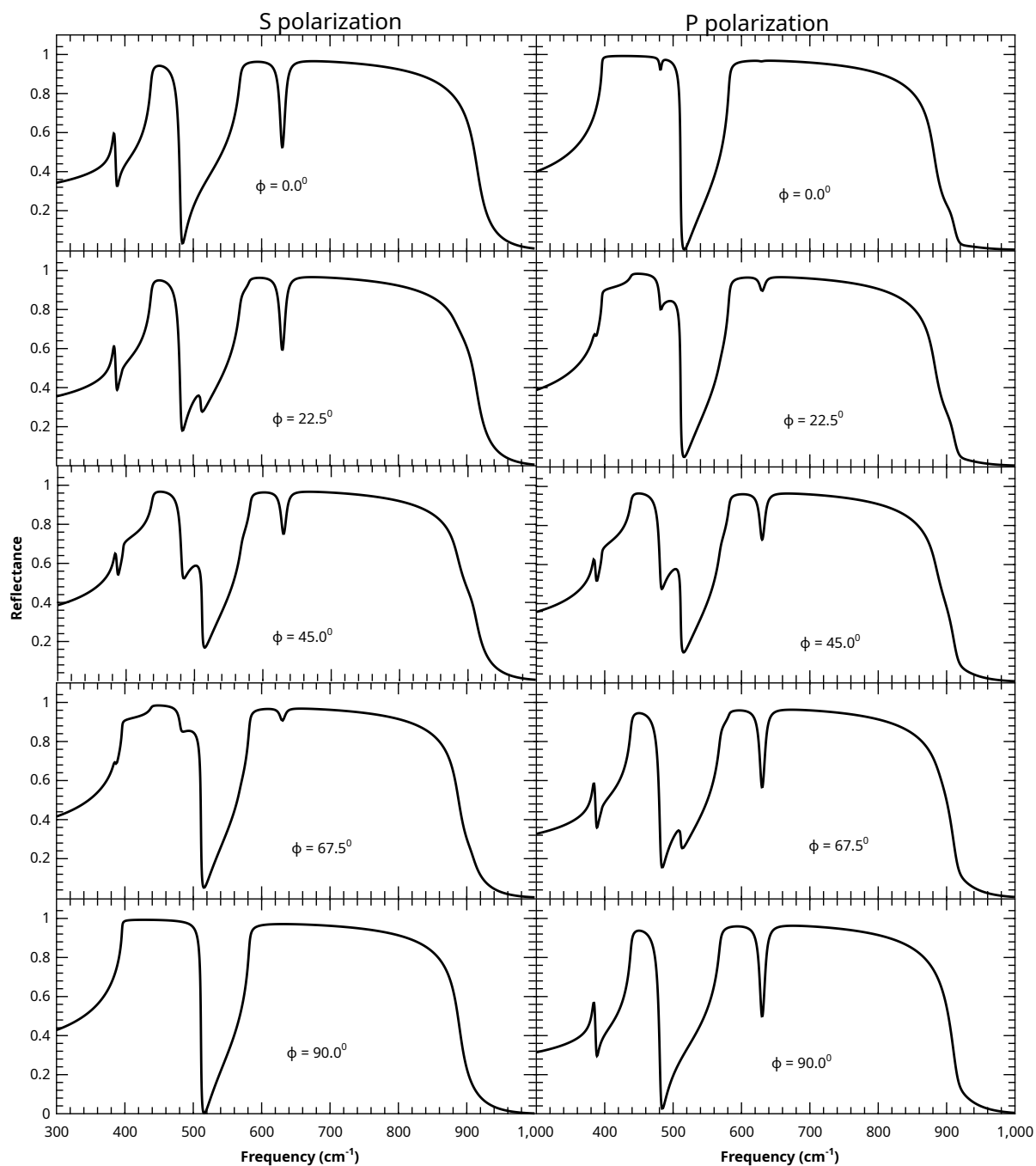


Fig. 12.24: a-plane reflectance

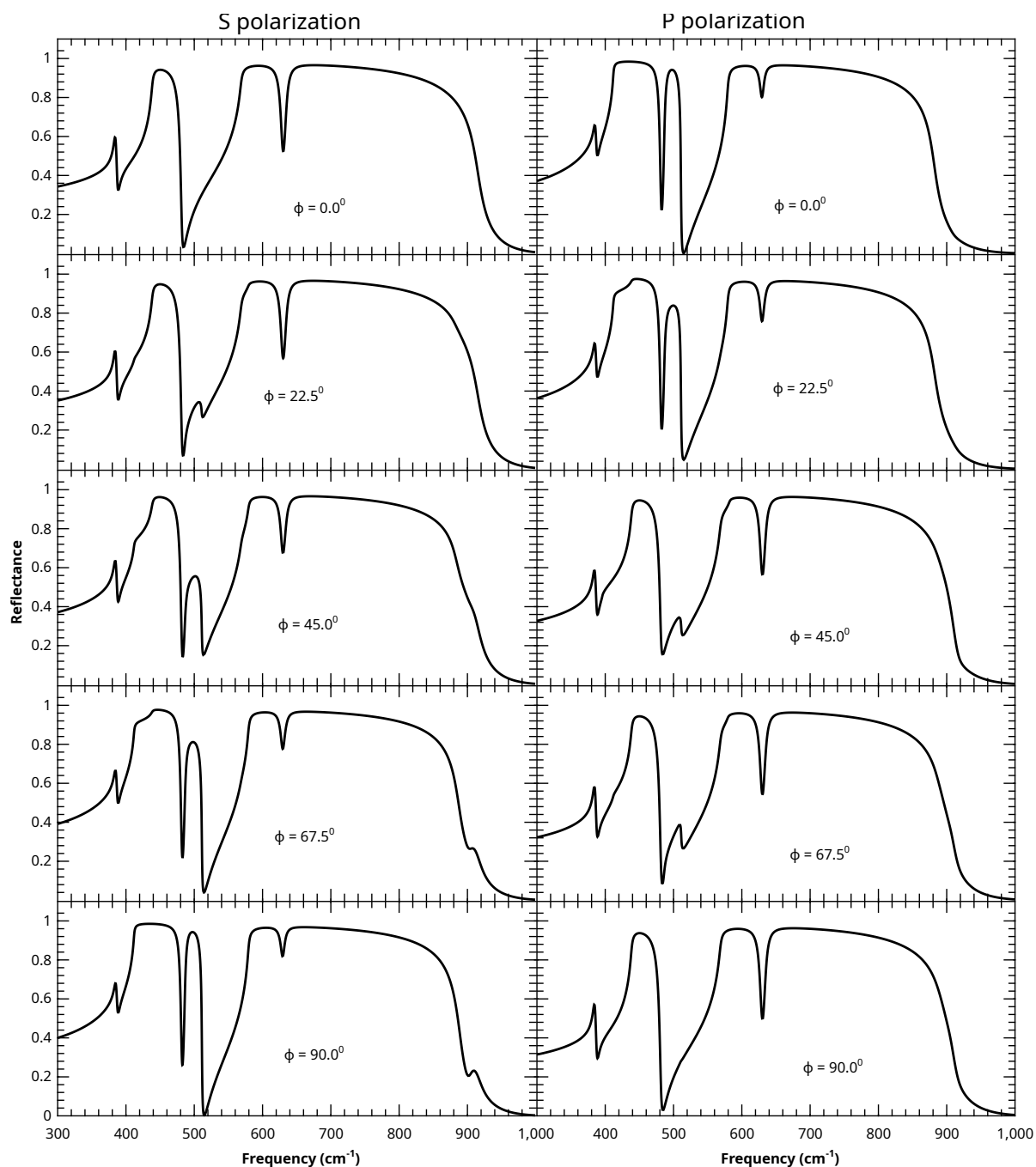


Fig. 12.25: r-plane reflectance

PHONOPY EXAMPLES

Phonopy [7] is an open source package for the calculation of phonon's using the harmonic and quasi-harmonic approximations. Calculations of the phonon spectrum at the Γ point in the Brillouin-zone using Phonopy can be used by PDielec to calculate the absorption spectrum of powdered and crystalline materials. The only additional requirement for performing calculations using PDielec with Phonopy is the provision of ϵ_∞ and the Born charges. These are also required by Phonopy if the non-analytical correction is necessary. In this case, Phonopy provides a script for several DFT packages which converts the DFT calculation of Born charges to a file BORN, which contains the ϵ_∞ and the Born charges. For VASP, the script is called phonopy-vasp-born. The BORN file stores only the symmetry unique charges and PDielec provides a command, phonopy-pdielec-born, which reads the BORN file and phonopy.yaml and generates all of the Born charges. The command only uses the Phonopy API, so it is necessary to use it in an environment that has Phonopy installed.

13.1 α -Al₂O₃

This example is based on the example/Al2O3 given in the Phonopy distribution. There may be some differences between the present calculations and that in Phonopy's distribution as the INCAR, KPOINTS and POTCAR files are not present in the distribution.

α -Al₂O₃ or corundum belongs to the $R\bar{3}c$ space group. The standard cell has 6 formula units (30 atoms) in the cell, and the primitive cell has 2 formula units (10 atoms).

In the PDielec's Examples/Phonopy/Al2O3 directory there are two directories, Standard_cell/ and Primitive_cell, containing results of Phonopy calculations for the standard and primitive cells respectively.

It is assumed that the optimisation of the structure has already been done as indicated below

13.1.1 Geometry Optimisation

The INCAR file for used for the optimisation was;

```
PREC = accurate
ALGO = Normal
ADDGRID = T
NELMIN = 5
IBRION = 2
POTIM = 2
NSW = 50
EDIFF = 1E-8
EDIFFG = -1E-3
ENCUT = 500.000000000
```

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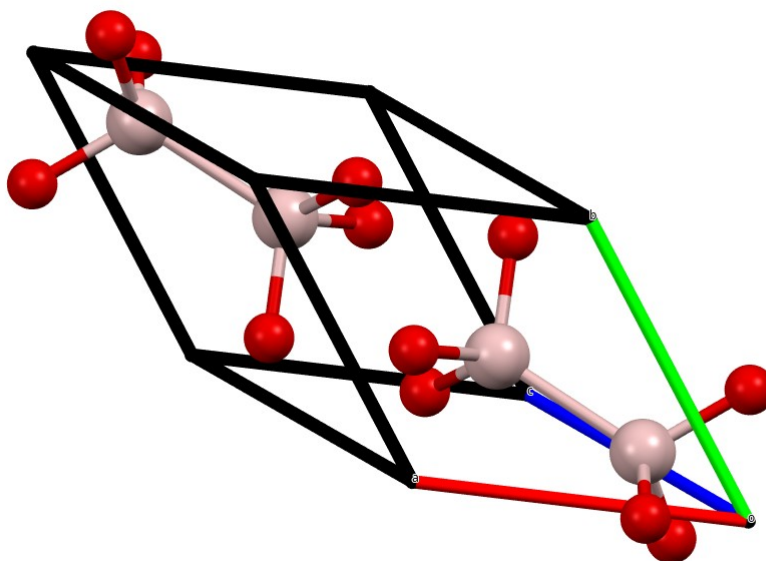


Fig. 13.1: The primitive cell of Corundum

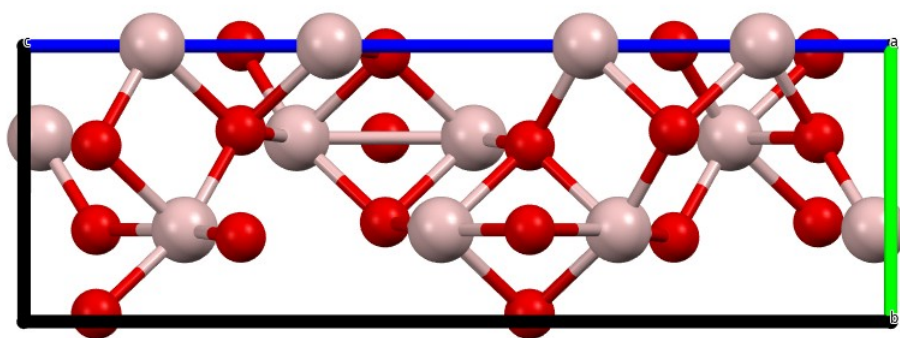


Fig. 13.2: The standard cell of Corundum

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```
LREAL = F
ISMEAR = 0
SIGMA = 0.010000000
GGA = PS
LMAXMIX = 4
```

The KPOINTS file is;

```
Regular k-point mesh
0
Monkhurst-Pack
3 3 1
0 0 0.5
```

The optimised POSCAR is;

```
generated by phonopy
1.0000000000000000
 4.7742263452989944  0.0000000000000000  -0.0000000000000000
-2.3871131726494972  4.1346012984458653  0.0000000000000000
-0.0000000000000000  0.0000000000000000  13.0113593913272219
Al    O
 12    18
Direct
0.3333333333333357  0.6666666666666643  0.0187603745624614
0.3333333333333357  0.6666666666666643  0.3145729587708743
0.0000000000000000  0.0000000000000000  0.1479062921042100
0.6666666666666643  0.3333333333333357  0.1854270412291257
0.0000000000000000  0.0000000000000000  0.3520937078957900
0.0000000000000000  0.0000000000000000  0.6479062921042100
0.6666666666666643  0.3333333333333357  0.4812396254375386
0.3333333333333357  0.6666666666666643  0.5187603745624614
0.6666666666666643  0.3333333333333357  0.6854270412291257
0.6666666666666643  0.3333333333333357  0.9812396254375386
0.3333333333333357  0.6666666666666643  0.8145729587708743
0.0000000000000000  0.0000000000000000  0.8520937078957900
0.3058086537880129  0.0000000000000000  0.2500000000000000
0.3608580128786514  0.3333333333333357  0.0833333333333357
0.0000000000000000  0.3058086537880129  0.2500000000000000
0.6666666666666643  0.0275246795453228  0.0833333333333357
0.6941913462119871  0.6941913462119871  0.2500000000000000
0.9724753204546772  0.6391419871213486  0.0833333333333357
0.9724753204546772  0.3333333333333357  0.5833333333333357
0.0275246795453228  0.6666666666666643  0.4166666666666643
0.6666666666666643  0.6391419871213486  0.5833333333333357
0.3333333333333357  0.3608580128786514  0.4166666666666643
0.3608580128786514  0.0275246795453228  0.5833333333333357
0.6391419871213486  0.9724753204546772  0.4166666666666643
0.6391419871213486  0.6666666666666643  0.9166666666666643
0.6941913462119871  0.0000000000000000  0.7500000000000000
```

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0.33333333333333357	0.9724753204546772	0.91666666666666643
0.00000000000000000	0.6941913462119871	0.75000000000000000
0.0275246795453228	0.3608580128786514	0.91666666666666643
0.3058086537880129	0.3058086537880129	0.75000000000000000

13.1.2 Calculation of the Born charges and ϵ_{∞}

Phonopy allows for the calculation of the phonons in either the primitive or standard basis. Starting the calculation in the standard cell basis will allow later calculations in the primitive cell basis.

In the Standard_cell/Born directory a calculation of the Born charges and ϵ_{∞} was performed. The inputs and outputs can be found in the distribution.

The optimised results (KPOINTS, INCAR, POSCAR, CONTCAR, POTCAR) have been copied to the Standard_cell/Born/ directory making sure that CONTCAR has been copied to POSCAR. The KPOINTS file has been edited to increase the number of k-points to allow for the calculation of converged Born Charges.

The INCAR for a single point, with the calculation of epsilon infinity and Born charges(LEPSILON) is as follows:

```
PREC = accurate
ALGO = Normal
ADDGRID = T
NELMIN = 5
  PREC = Accurate
  IBRION = -1
  NELMIN = 5
  ENCUT = 500
  EDIFF = 1.0000000e-08
  ISMEAR = 0
  SIGMA = 1.0000000e-02
  IALGO = 38
  LREAL = .FALSE.
  LWAVE = .FALSE.
  LCHARG = .FALSE.
LEPSILON = .TRUE.
LMAXMIX = 4
GGA = PS
```

The KPOINTS file is:

```
Regular k-point mesh
0
Gamma
6 6 2
0 0 0.5
```

Once the calculation is completed, the symmetry unique charges are calculated using the Phonopy script “phonopy-vasp-born”. Later on these will be converted to PDIElec format, which has all of the atoms in the file, not just the symmetry unique ones.

```
phonopy-vasp-born > BORN
```

13.1.3 Standard Cell Phonopy calculation

From the Standard_cell/Born directory copy the BORN, POSCAR, POTCAR and KPOINTS files to Standard_cell/. The higher density of k-points is no longer needed, indeed the density may be reduced if a supercell is used.

Modify the INCAR file to do single point calculations with no storage of the charge density or the wavefunction:

```
PREC = Accurate
ALGO = Normal
IBRION = -1
NELMIN = 5
ENCUT = 500
EDIFF = 1.0000000e-08
ISMear = 0
SIGMA = 1.0000000e-02
LREAL = .FALSE.
LWAVE = .FALSE.
LCHARG = .FALSE.
LMAXMIX = 4
GGA = PS
ADDGRID = T
```

Since we will change the size of the cell, we can modify the KPOINTS file:

```
Regular k-point mesh
0
Monkhurst-Pack
2 2 1
0 0 0.5
```

Using Phonopy to create the perturbed and unperturbed structures.

```
phonopy -d --dim 2 2 1
mkdir SuperCell
cp SPOSCAR SuperCell/POSCAR
(cd SuperCell; cp ../INCAR .; ln -s ../POTCAR; ln -s ../KPOINTS)
```

An unperturbed supercell calculation will be performed to calculate the wavefunction and charge density. The perturbed geometries are stored in POSCAR-??? files and the supercell in SPOSCAR. The size of the supercell will be 2,2,1.

Once the supercell POSCAR file has been created, links are made to POTCAR and KPOINTS in the SuperCell/ directory. The INCAR file is copied as it will be edited to save the wavefunction and charge density files.

Edit the SuperCell/INCAR file so that the wavefunction and charge are saved. The WAVECAR and CHGCAR files will restart the perturbed calculations. After running the VASP job in SuperCell/. Create links for the WAVECAR and CHGCAR files

```
ln -s SuperCell/WAVECAR; ln -s SuperCell/CHGCAR
```

Move the POSCAR-??? files for each displacement to a directory (POSCARS) where the calculations will be performed. POTCAR, KPOINTS and INCAR files are the same for every calculation, so link to the original SuperCell files. An advantage of using links for these files is that changes to the INCAR or KPOINTS files apply to all calculations. WAVECAR and CHGCAR links need to be made in each of the perturbed geometry directories.

```
mkdir POSCARS
for f in POSCAR-*
do
    mkdir POSCARS/$f
    cp $f POSCARS/$f/POSCAR
    ( cd POSCARS/$f; ln -s ../../INCAR; ln -s ../../POTCAR; ln -s ../../
    ↪KPOINTS; ln -s ../../WAVECAR; ln -s ../../CHGCAR )
done
```

Submit all the jobs to a batch queue for processing.

```
cd POSCARS
for d in *
do
    (cd $d; runvasp $d 8 8 )
done
```

Where runvasp is a script used to create and run a job script to the batch queue

Calculate the FORCES_SETS

```
phonopy -f POSCARS/POSCAR-{001..005}/vasprun.xml
```

Calculate the Dynamical Matrix

The dynamical matrix is calculated in the standard basis, not the primitive.

```
phonopy --dim="2 2 1" --qpoints="0 0 0" --writedm
```

Calculate the full set of Born charges

PDielec needs the Born charges for every atom, not just the symmetry unique ones. The full set of charges is generated in file "BORN_PDIELEC" by the phonopy-pdielec-born command provided by PDielec. Note that this command only uses the Phonopy API, no feature of the PDielec API is used.

```
phonopy-pdielec-born > BORN_PDIELEC
```

For comparison the BORN file for provided in the distribution of Phonopy is;

```
# epsilon and Z* of atoms 1 13
 3.27649624 -0.00000000 0.00000000 -0.00000000 3.27649624 0.
↪00000000 0.00000000 0.00000000 3.21878866
 2.96008813 -0.02948662 -0.00000000 0.02948662 2.96008813 -0.
↪00000000 -0.00000000 0.00000000 2.92614183
 -2.07328119 0.00000000 0.00000000 -0.00000000 -1.87350298 0.
↪25811770 -0.00000000 0.35087130 -1.95076122
```

Whilst the BORN file calculated by the present calculation is;

```
# epsilon and Z* of atoms 1 13
 3.27607426 -0.00000000 0.00000000 0.00000000 3.27607426 0.
↪00000000 0.00000000 0.00000000 3.23683513
 2.95996414 -0.02948839 -0.00000000 0.02948839 2.95996414 0.
↪00000000 0.00000000 -0.00000000 2.92951772
 -2.07319123 0.00000000 0.00000000 0.00000000 -1.87342763 0.
↪25818161 0.00000000 0.35167360 -1.95301181
```

The agreement is accurate to 3 decimal places.

Standard Cell - PDGui

A pdgui calculation of the infrared intensities and the phonon frequencies is now possible.

```
pdgui phonopy.yaml
```

PDGui can be given any yaml file on the command line, the actual files read are always phonopy.yaml, qpoints.yaml and BORN_PDIELEC.

13.2 Primitive Cell Phonopy

A primitive cell calculation of the dynamical matrix will be performed in the Primitive_cell/ directory. Copy over files from the following files from the Standard_cell/ directory.

```
cd Standard_cell
cp BORN FORCE_SETS phonopy_disp.yaml phonopy.yaml ../Primitive_cell
```

The primitive cell has fewer atoms (10) in it than the standard cell (30), so the BORN_PDIELEC file needs to be calculated as it stores the Born charges for all atoms. Before doing this the phonopy.yaml has to be updated with the primitive cell.

This is carried out by during a calculation of the dynamical matrix in the primitive basis. After this the phonopy.yaml file should have the primitive cell and the BORN_PDIELEC file can be calculated. It is important that the BORN_PDIELEC file is always updated after a change to phonopy.yaml.

```
phonopy --dim="2 2 1" --qpoints="0 0 0" --writedm --pa auto
phonopy-pdielec-born > BORN_PDIELEC
```

At this point it possible to calculate the absorption using pdgui, in the same way as was done previously, but now the primitive cell basis is used.

13.3 Comparison of results

In the following figures the results of the two different Phonopy calculations are compared. In addition results are shown for DFPT calculations on the same systems using VASP. The first figure shows a comparison over the full frequency range. The second figure compares the results over a limited frequency range to highlight the small differences between the calculations.

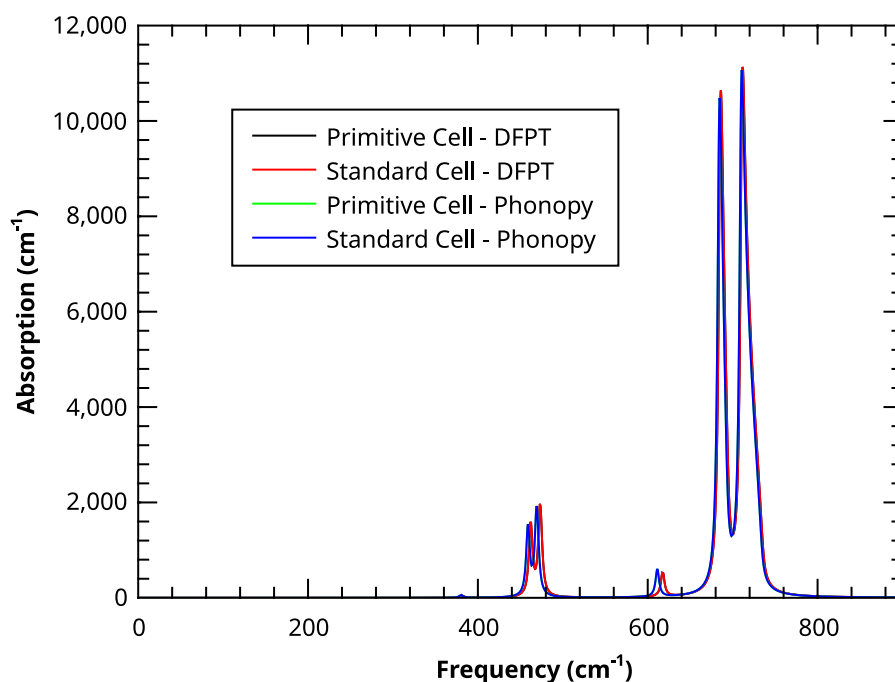


Fig. 13.3: Comparison of absorption predicted by Phonopy and DFPT (full frequencies)

The results of the two DFPT calculations are indistinguishable from one another, as are those of the two Phonopy calculations. There is a small shift in the frequency peaks predicted by DFPT and Phonopy but this is to be expected.

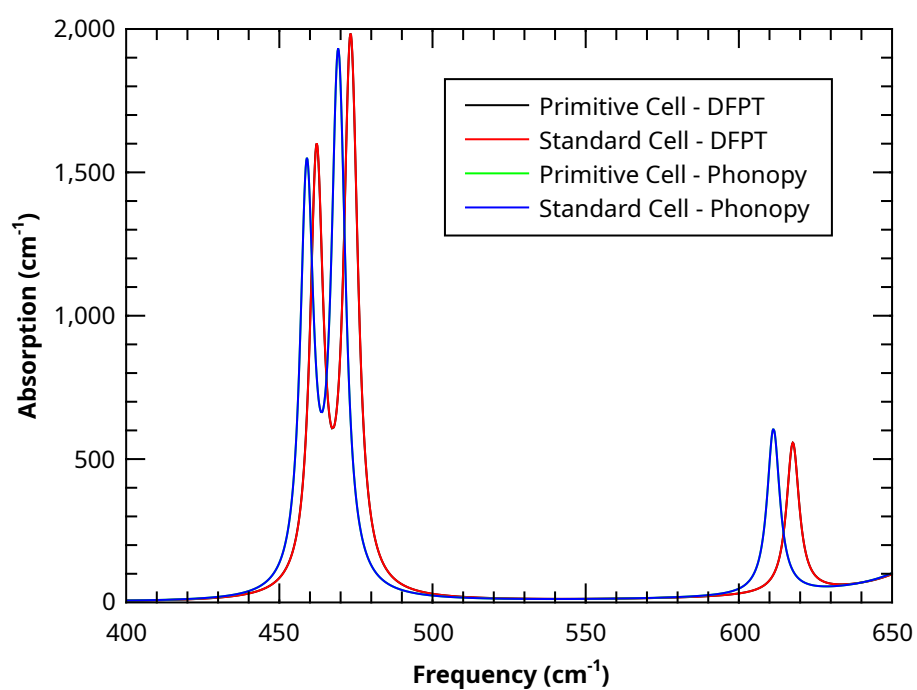


Fig. 13.4: Comparison of absorption predicted by Phonopy and DFPT (restricted frequencies)

GAN: FROM POWDER RAMAN SPECTRA TO CRYSTAL MODAL PAIRS

This application note uses one CRYSTAL23 calculation of wurtzite GaN to explore three questions: how particle boundary conditions change a powder Raman spectrum, why the internal optical modes matter in a birefringent crystal, and what is assumed when Raman emission is incoherent through the sample depth. The working files are in `Examples/ApplicationNotes/GaN`.

Two single-crystal experiments motivate the crystal calculations. Irmer et al. [GaN-Irmer2013] measured phonon-polariton dispersion using displaced collection apertures. Mina, Alhaddad and Pagès [GaN-Mina2026] used near-forward scattering and interchanged the incident and detected polarisations. These experiments make a demanding illustration of optical momentum selection. The powder calculation is an additional teaching example, not a powder measurement reported in either paper.

Important

PDGui's *All modes* treatment resolves internal optical-mode pairs for the directional, zone-centre non-analytical correction (NAC). It does not calculate finite-wavevector phonon-polariton dispersion. The separate `polariton_response.py` script illustrates that additional physics. Agreement between PDGui options is a consistency check; it is not by itself validation against the measured polariton spectrum.

14.1 Preparing the calculation

Use a PDIELEC environment with the current Crystal Raman and Powder Raman capabilities. Run commands below from the indicated directory. The PDGui session files are loaded with `-script`; they are not ordinary standalone Python programs.

The starting structure is the Schulz–Thiemann refinement [GaN-Schulz1977], transcribed in `GaN_Schulz_Thiemann_1977.cif`: space group $P6_3mc$, $a = 3.190 \text{ \AA}$, $c = 5.189 \text{ \AA}$ and nitrogen internal coordinate $u = 0.377$. The supplied `Optimisation/optimise.d12` uses PBE-D3 and POB-TZVP-REV2, with full cell and coordinate optimisation. `Raman/raman.d12` requests frequencies, infrared response and Raman response at that optimised geometry. The dense SHRINK 16 32 mesh, TOLDEE 10, tight integral thresholds and XXL integration grid are choices made for this example; convergence tests have not been performed, and they are not a substitute for convergence tests when preparing a different material.

Keep the contents of `Raman/` together. In particular, `TENS_RAMAN.DAT` supplies the Raman derivatives, `HESSFREQ.DAT` the force-constant information and `BORN.DAT` the Born charges used alongside `raman.log`. Opening only a copied log can lose information needed for the Raman tensors and polar-mode corrections. Reproducing the PDGui examples does not require rerunning CRYSTAL23.

From Examples/ApplicationNotes/GaN, start with:

pdgui crystal Raman/raman.log

On the **Main Tab**, check that the program is *Crystal*. On the **Settings Tab**, select *Powder Raman* or *Crystal Raman* as required below. Use average atomic masses, enable the Eckart projection, leave *Neutral Born charges* off, and retain the optical permittivity read from the calculation. Set the Lorentzian half-width σ to 5 cm^{-1} for every mode, corresponding to a 10 cm^{-1} full width. This common width makes the comparisons easy to read; it is not a fit to measured lifetimes.

PDGui automatically selects the active modes for the spectroscopy type. Modes 4, 5, 7, 8, 9, 10 and 11 should be selected. Always keep both members of a degenerate E pair. The following assignments come from the raw CRYSTAL output; the average-mass recalculation in PDGui changes them slightly, so these are not exact plotted peak positions.

Table 14.1: Raman-active modes in the supplied calculation

Mode numbers	Assignment	Raw frequency cm^{-1}	Role in the example
4–5	E_2 (low)	137.05	Non-polar powder reference
7	A_1 (TO)	529.30	Polar, displacement along c
8–9	E_1 (TO)	554.95	Polar, basal-plane displacement
10–11	E_2 (high)	562.67	Non-polar frequency reference

The corresponding reference TO frequencies used by Irmer et al. are 531.8 and 558.8 cm^{-1} for A_1 and E_1 ; their E_2 values are 144 and 567.6 cm^{-1} [GaN-Irmer2013]. The few-wavenumber differences are material-model differences. Do not shift each calculated line independently to obtain apparent agreement.

If select **Apply NAC** you can explore the effect of the NAC correction to the LO frequency. The $\bar{q}$ direction is supplied by the GUI. Generally speaking, there is little change in activity unless the EO correction is included. This correction does not change the LO mode frequency but it can affect its activity.

14.1.1 Intensities and the temperature factor

Set the scenario temperature to 298 K throughout. Both powder and crystal spectra now weight each final phonon state by

$$W_m(T) = (\tilde{\nu}_L - \tilde{\nu}_m)^4 \frac{n_m(T) + 1}{\tilde{\nu}_m}, \quad n_m(T) = \left[\exp\left(\frac{hc\tilde{\nu}_m}{k_B T}\right) - 1 \right]^{-1}.$$

Here all wavenumbers are in consistent units, and $\tilde{\nu}_L = 10^7/\lambda_L$ in cm^{-1} when λ_L is in nm. The implementation uses the corrected phonon frequency and applies this weight once before Lorentzian broadening. Temperature changes the harmonic Stokes population; it does not introduce thermal expansion, anharmonic frequency shifts or temperature-dependent linewidths.

Choose *Spectrum renormalisation = none* on the **Plotting Tab**. The *Raman activity units* selector on the Settings Tab controls the displayed activity convention, not the normalisation used by the scenario calculation. Activities remain in arbitrary units: omitted experimental collection factors and the different sample models preclude comparing the absolute powder and crystal ordinates as calibrated cross sections.

14.2 1. Powder Raman: separate orientation averaging from particle effects

Start a fresh session from GaN/Powder using:

```
pdgui -script gan_powder.py
```

Alternatively, select *Powder Raman* on the Settings Tab and configure four scenarios. Set the laser wavelength to 488 nm, temperature to 298 K, volume fraction to 0.1 and particle shape to *Sphere*. Disable *Include electro-optic term*. Use the following matrix and polarisation combinations:

Table 14.2: Powder scenario matrix

Legend	Matrix	Raman laser polarisation	Purpose
none VV	none	VV	Parallel, uncorrected TO tensor average
none VH	none	VH	Crossed, uncorrected TO tensor average
air VV	air	VV	Parallel, particle response in air
air VH	air	VH	Crossed, particle response in air

The *none* matrix is a deliberate diagnostic: it suppresses both the particle frequency correction and optical local-field correction. It is not equivalent to *air*, whose permittivity is approximately one but which still imposes a dielectric boundary around GaN. Keeping the same volume fraction in all four scenarios isolates these corrections. The particle model is the small-particle, quasistatic limit; the stored particle-size setting is not a measured GaN size or a finite-size Raman resonance.

For spheres, the orientational average uses analytic tensor invariants. There is no orientation-sampling convergence parameter to tune for this case. For a symmetric Raman tensor, VV contains $45\alpha^2 + 4\gamma^2$ and VH contains $3\gamma^2$, where α is the isotropic part and γ measures anisotropy. Thus a traceless isolated mode has $VH/VV = 3/4$. This is useful for the E_2 modes, but overlapping lines and Lorentzian tails can change the ratio measured at a point on a broadened spectrum.

For particles in air, two optical local-field factors act on each Raman tensor. The phonon depolarisation field also changes the restoring force of polar modes. Consequently, particle shape and host can change frequencies as well as intensities. This calculation uses the electronic permittivity for the optical fields and does not model visible optical dispersion in detail. See *Theory for Powder Raman* for the equations.

On the Plotting Tab choose *Powder Raman*, 100–800 cm^{-1} , with an increment of 0.5 cm^{-1} . On the Main Tab set the Excel destination to `Data/gan_powder.xlsx` and press *Save results*. Create `Data` first if it is absent. *Save settings* writes a reusable PDGui session; use a new filename when experimenting so the supplied reference script is preserved.

The regenerated TO-reference spectrum has its strongest high-frequency feature at 562.0 cm^{-1} and the low E_2 feature at 136.5 cm^{-1} . In air these non-polar features remain, while a polar feature appears near 677.0 cm^{-1} . The VV peak near 562 falls from about 4.07×10^{14} to 9.92×10^{12} in the common arbitrary units. These changes demonstrate boundary and local-field effects, not finite-q polariton dispersion.

14.3 2. Crystal Raman: near-forward scattering and modal pairs

14.3.1 The geometry and the first scenario

Start from GaN/ModalPairs with:

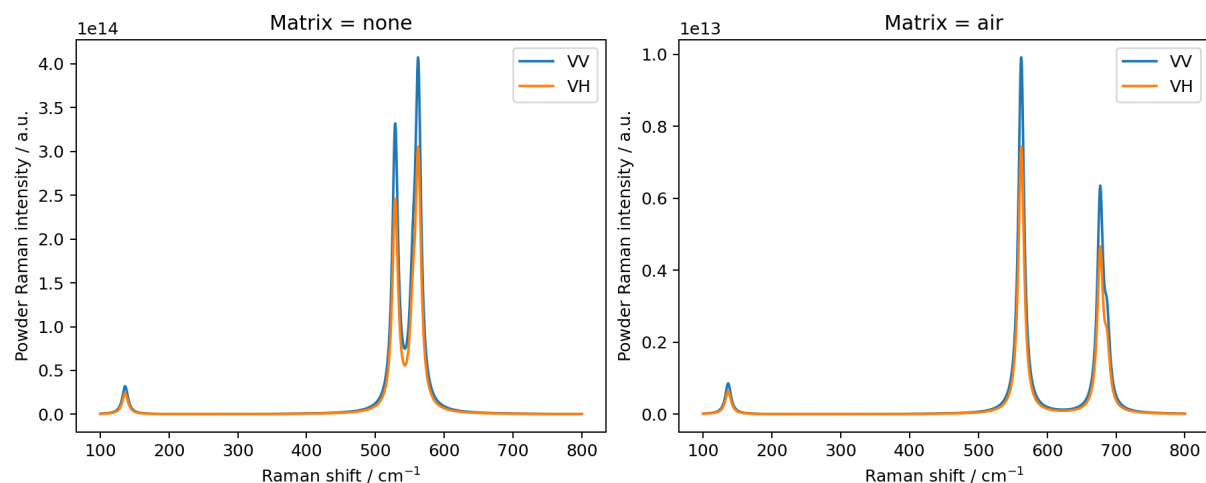


Fig. 14.1: GaN powder comparison. The two panels have different ordinate scales; individual spectra have not been normalised. Generate with `python regenerate.py powder` from the GaN directory.

```
pdgui -script gan_modal_pairs.py
```

This loads 24 scenarios. To construct the first one manually, select *Crystal Raman* and configure the following settings on the Scenario Tab. The dielectric layer is the material read from the CRYSTAL calculation.

Table 14.3: Common settings for the Mina-inspired comparison

Control	Value	Reason
Optical method	Scattering matrix	Stable layered optical-field calculation
Layer stack	air / Dielectric layer / air	Transmission through a GaN edge
GaN surface normal and thickness	(100), 10 micrometres	m-plane orientation and a common sampling depth
Layer and global azimuths	0 degrees	c perpendicular to the incidence plane
Optical layer incoherence	Coherent for each layer	Retain optical interference in the local fields
Laser wavelength; temperature	488 nm; 298 K	Excitation used for the principal comparison
Angle of incidence	0 or 2 degrees	External-angle sensitivity test
Collection side; angle	substrate; 0 degrees	Fixed normal forward collection
Integration density	10 points per micrometre	Reference depth quadrature
Depth coherence	Incoherent intensity	Integrate local Raman intensities through the depth
Layer combination	Incoherent intensities	Required by the selected depth treatment
Include electro-optic term	Off	Isolate the lattice Raman and NAC comparison

In this orientation, incident/detected s corresponds to extraordinary light with electric field along c , and p to ordinary light. Build the four channels $e \rightarrow e$, $o \rightarrow o$, $e \rightarrow o$, $o \rightarrow e$ as s/s , p/p , s/p and p/s respectively. Repeat these for each incidence angle and each NAC choice below.

The scripts retain the default `Approximate ES = False`, so the reciprocal field is calculated at the Stokes frequency. This is a saved calculation setting rather than an additional visible control to find.

PDGui angles are external angles. The internal angles in the papers include refraction, so entering 2 degrees here does not reproduce an experimental internal 2-degree trace. The 10-micrometre layer is a controlled sampling model, not the experimental slab thickness or a model of the full multiple-reflection enhancement. Incoherent Raman depth integration also does not remove interference from the optical fields. For quantitative work, vary integration density, sampling depth and optical parameters separately; no numerical convergence claim is made for every geometry by choosing the reference density above.

14.3.2 Choosing the NAC treatment

Table 14.4: GUI names and saved-script values

Layer PDGui	NAC mode	in Script value	Momentum construction
Snell's law		geometry	One direction from the macroscopic optical geometry
Dominant mode		dominant_mode	One direction from the dominant internal optical mode
All modes		modal_pairs	Resolve incident and scattered internal mode pairs

For *All modes*, use *Modal pair combination = Group q channels*, *Modal-pair final-state model = Bulk phase matched* and *Modal pair q-angle tolerance = 90 degrees*. Keep the EO option off; otherwise the GUI may append (EO) to the NAC label. The Settings Tab NAC diagnostic does not replace the Scenario Tab's *Layer NAC mode*.

The optical-pair momentum is

$$\mathbf{q}_{ij} = \mathbf{k}_{L,i}(\omega_L) - \mathbf{k}_{S,j}(\omega_L - \Omega_m).$$

In a birefringent material, interchanging ordinary and extraordinary polarisations changes these internal wavevectors even when the external rays are unchanged. *All modes* retains this information. *Group q channels* adds amplitudes leading to the same phonon final state before squaring, and adds intensities for distinct final states. It does not simply add the intensities of every optical pair independently.

Use a plot range of 400–800 cm^{-1} with a 0.5 cm^{-1} increment and save to `Data/gan_modal_pairs.xlsx`. The `Crystal Raman` worksheet contains the broadened spectra with scenario legends. The supplied analysis integrates 450–780 cm^{-1} . Although called the polar-band interval in the scripts, this window also includes E_2 (high); it is not a projection onto polar modes alone.

14.3.3 Reading the results against Mina et al.

Table 14.5: Regenerated integrated intensity, All modes / Snell's law

External incidence	$e \rightarrow e$	$o \rightarrow o$	$e \rightarrow o$	$o \rightarrow e$
0 degrees	0.717	0.873	5.31×10^{-12}	6.20×10^{-12}
2 degrees	0.842	0.942	0.809	0.870

For this homogeneous geometry, *Dominant mode* and *Snell's law* agree numerically. This does not make them interchangeable in a general stack. The grouped modal-pair crossed signals are nearly extinguished

at exact forward scattering. At 2 degrees they recover, with integrated $I_{e \rightarrow o}/I_{o \rightarrow e} \simeq 2.75$ and a strong feature near 730 cm^{-1} . Tiny residual signals in the zero-angle crossed channels should not be assigned as observable Raman peaks.

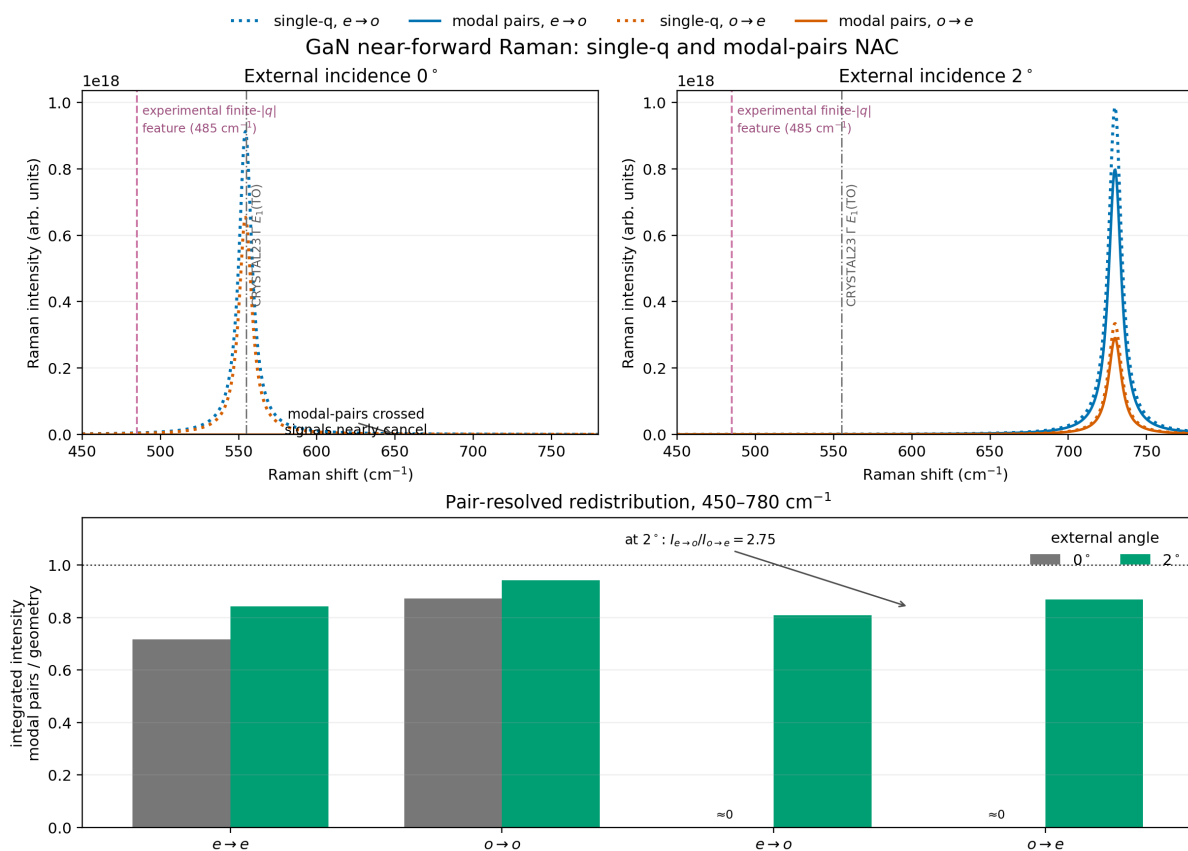


Fig. 14.2: Near-forward optical-pair comparison. The experimental 485 cm^{-1} marker identifies physics outside the PDGui directional-NAC calculation.

Mina et al. report a strong, sharp E_1 -derived phonon-polariton near 485 cm^{-1} under near-forward 488-nm excitation, with pronounced sensitivity to polarisation interchange and incidence angle [GaN-Mina2026]. The PDGui calculation demonstrates why pair identity matters, but its zero-angle suppression is not a reproduction of that enhanced experimental signal. Its approximately 730 cm^{-1} feature is a directional LO-like phonon result.

The NAC dynamical-matrix contribution contains

$$D^{\text{NAC}} \propto \frac{(\mathbf{q} \cdot \mathbf{Z}^*)(\mathbf{q} \cdot \mathbf{Z}^*)^T}{\mathbf{q} \cdot \boldsymbol{\varepsilon}_\infty \cdot \mathbf{q}}.$$

The magnitude of $\mathbf{q}$ cancels. It gives a directional $q \rightarrow 0$ limit, whereas a phonon-polariton frequency depends on the finite magnitude as well. Changing a linewidth or selecting a different final-state option cannot supply that missing dispersion.

14.4 3. Irmer et al.: an aperture scan and a finite-q comparison

From GaN/ModalPairs/Irmer2013, load:

```
pdgui -script irmer_pdgui.py
```

The 26 scenarios represent the central rays of the aperture scans in Figs. 9–11 of Irmer et al. [GaN-Irmer2013]. Keep the same layer, width, temperature, depth and modal-pair settings as above, but use a 514.5-nm laser and normal incidence. Set the collection side to *substrate*. For aperture displacement d at screen distance $L = 80$ mm, enter the external collection angle

$$\alpha_{\text{ext}} = \tan^{-1}(d/L).$$

For example, $d = 2$ mm gives 1.4321 degrees externally, and about 0.63 degrees internally for the paper’s ordinary index. PDGui performs the optical refraction; do not enter the refracted angle in its external angle control.

Table 14.6: Mapping the paper’s coordinate convention to PDGui

Paper figure and configuration	Displacements / mm	Global azimuth	az- Incident / detected
Fig. 9, $x(zy)x$	Y = 2, 3.5, 4.5, 6, 8, 10, 12.5, 15	0 degrees	s / p
Fig. 10, $x(yz)x$	Same Y values	0 degrees	p / s
Fig. 11a, $x(zz)x$	Z = 3, 4, 5, 7, 10	90 degrees	p / p
Fig. 11b, $x(yy)x$	Same Z values	90 degrees	s / s

The coordinate symbols in this table follow the publication: x is the surface normal and z the crystal c axis. Use the explicit side, angle and polarisation controls for these oblique collection scans. In particular, do not paste the paper’s $x(zy)x$ label into PDGui’s Porto field: PDGui interprets the repeated-axis shorthand as backscattering. Its explicit forward notation uses $+x$; the supplied scripts use the angle controls directly.

Plot 150–780 cm^{-1} at a 0.5 cm^{-1} increment and save `Data/irmer_pdgui.xlsx`. Then, in the same directory, run:

```
python polariton_response.py --model both
python analyse_verification.py
python plot_verification.py
```

The complementary script solves a projected, damped Maxwell response along a frequency-dependent Raman momentum. With $\mathbf{Q} = \mathbf{q}/(2\pi)$ in cycles/cm and Raman shift $\tilde{\nu}$ in cm^{-1} , it evaluates

$$M = \frac{Q^2 \mathbf{1} - \mathbf{Q}\mathbf{Q}^T}{\tilde{\nu}^2} - \epsilon(\tilde{\nu}), \quad R = \left| \text{Im} \left[\mathbf{e}^\dagger M^{-1} \mathbf{e} \right] \right|.$$

It compares a one-oscillator dielectric model using the paper’s parameters with PDielec’s dielectric response from the local CRYSTAL data. The paper-model electronic permittivities are 5.20 perpendicular and 5.31 parallel to c ; its E1 TO/LO frequencies are 558.8/741.0 and A1 TO/LO frequencies 531.8/734.0 cm^{-1} . The script uses a controlled damping parameter of 5 cm^{-1} [GaN-Irmer2013]. This damping enters the oscillator denominator; it should not be identified automatically with the HWHM used to plot the PDGui lines.

 Table 14.7: Regenerated central-ray peak ranges, in cm^{-1}

Scan	Finite-q response using DFT permittivity	Strongest PDGui NAC feature
Fig. 9	446.5–533.0	730.0
Fig. 10	179.8–526.2	730.0
Fig. 11a	293.0–497.0	722.5
Fig. 11b	290.5–497.0	562.0

The finite- q peaks move with aperture displacement, while the strongest PDGui features in these scans remain near parent-phonon frequencies. The latter column reports a spectrum maximum, not a branch assignment; the approximately 562 cm^{-1} feature can be dominated by non-polar E_2 scattering. At $Y = 6\text{ mm}$ in the Fig. 9 geometry, the DFT Maxwell response has a peak at about 488.8 cm^{-1} . Its proximity to 485 illustrates the frequency scale accessible to a polariton; it is not a reproduction of the different-wavelength Mina experiment.

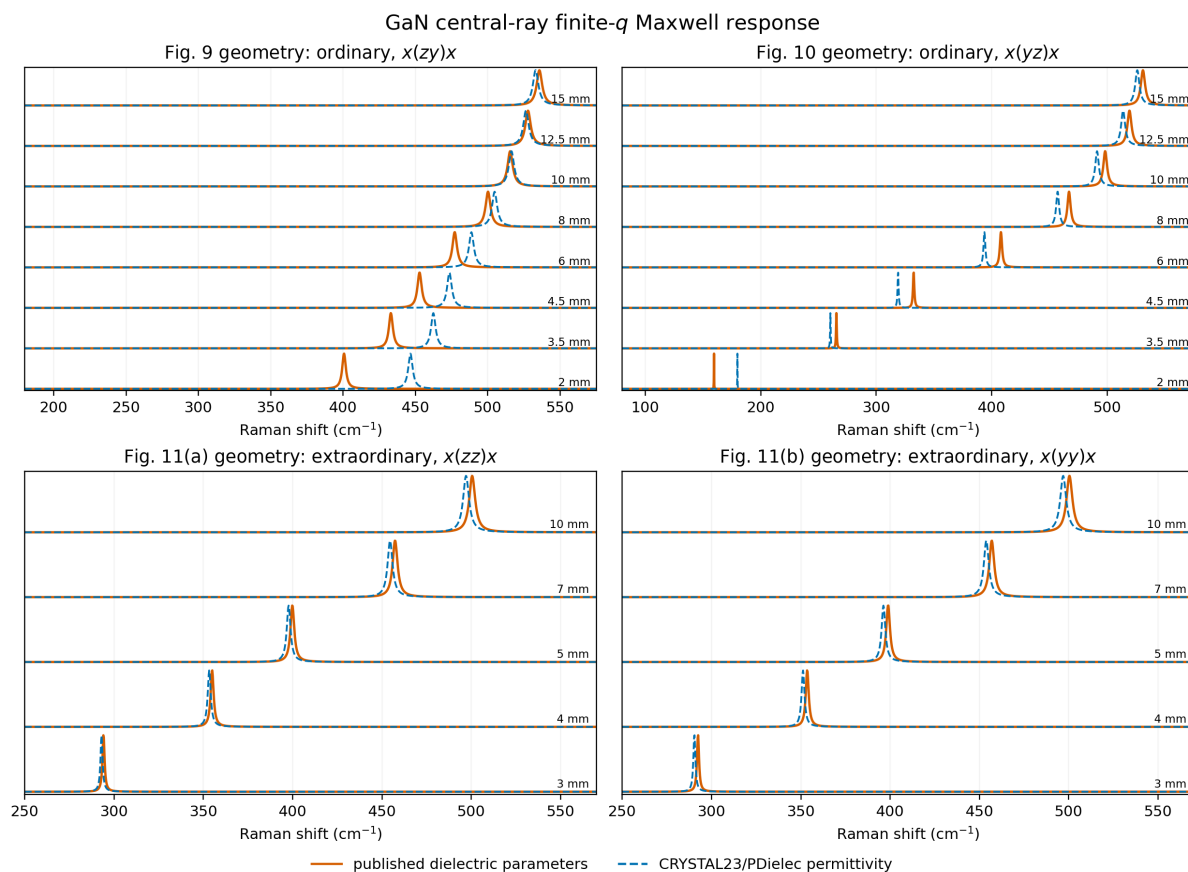


Fig. 14.3: Finite- q response curves are normalised separately and offset for readability. Compare peak positions and trends, not their relative heights as experimental Raman efficiencies.

Irmer et al. compare dispersion and Raman efficiency using a fuller scattering treatment and finite apertures. The supplied Maxwell script uses a generic transverse source, approximate optical indices and central rays. It omits the microscopic Raman source, Faust–Henry interference, aperture integration and experimental collection response. Its pole trends explain the distinction from directional NAC; they do not establish quantitative agreement with the measured band shapes or channel intensities.

14.5 4. What does an incoherent final state mean?

Load `gan_incoherent_models.py` from `GaN/ModalPairs/IncoherentModels`. The 32 scenarios keep *All modes*, *Group q channels* and *Depth coherence = Incoherent intensity*. They repeat the four polarisations at 0 and 2 degrees for two final-state models and two q -angle tolerances, 0 and 90 degrees.

Bulk phase matched retains the internal pair momentum and accepts its direction only within the selected tolerance relative to the external momentum transfer. **Local incoherent** assigns the local optical components to the externally selected final state before forming the local intensity; its q -angle tolerance is inactive. These controls specify how optical contributions are assigned to phonon final states. They do not turn the phonon dispersion into a finite- q polariton model.

Three different coherence choices must be kept separate. Optical layer coherence controls the fields in the stack. Raman depth coherence controls whether sources at different depths add as amplitudes or intensities. The modal-pair final-state model controls which internal optical contributions belong together locally. In particular, *Local incoherent* is not *Incoherent pairs*: the latter squares individual optical-pair contributions as a diagnostic alternative to *Group q channels*.

Table 14.8: Integrated final-state sensitivity at 2 degrees external incidence

Channel	Bulk (90 degrees) / local	Bulk (0 degrees) / bulk (90 degrees)
$e \rightarrow e$	0.842	2.54×10^{-11}
$o \rightarrow o$	0.942	0.546
$e \rightarrow o$	0.809	6.04×10^{-12}
$o \rightarrow e$	0.869	1.44×10^{-11}

The local spectra are identical for the two tolerances in all eight angle/channel combinations. In the bulk model the strict 0-degree tolerance nearly removes three channels at 2-degree incidence. This is a sensitivity test of a hard directional cutoff, not a fit parameter that should be selected to recover an experimental intensity.

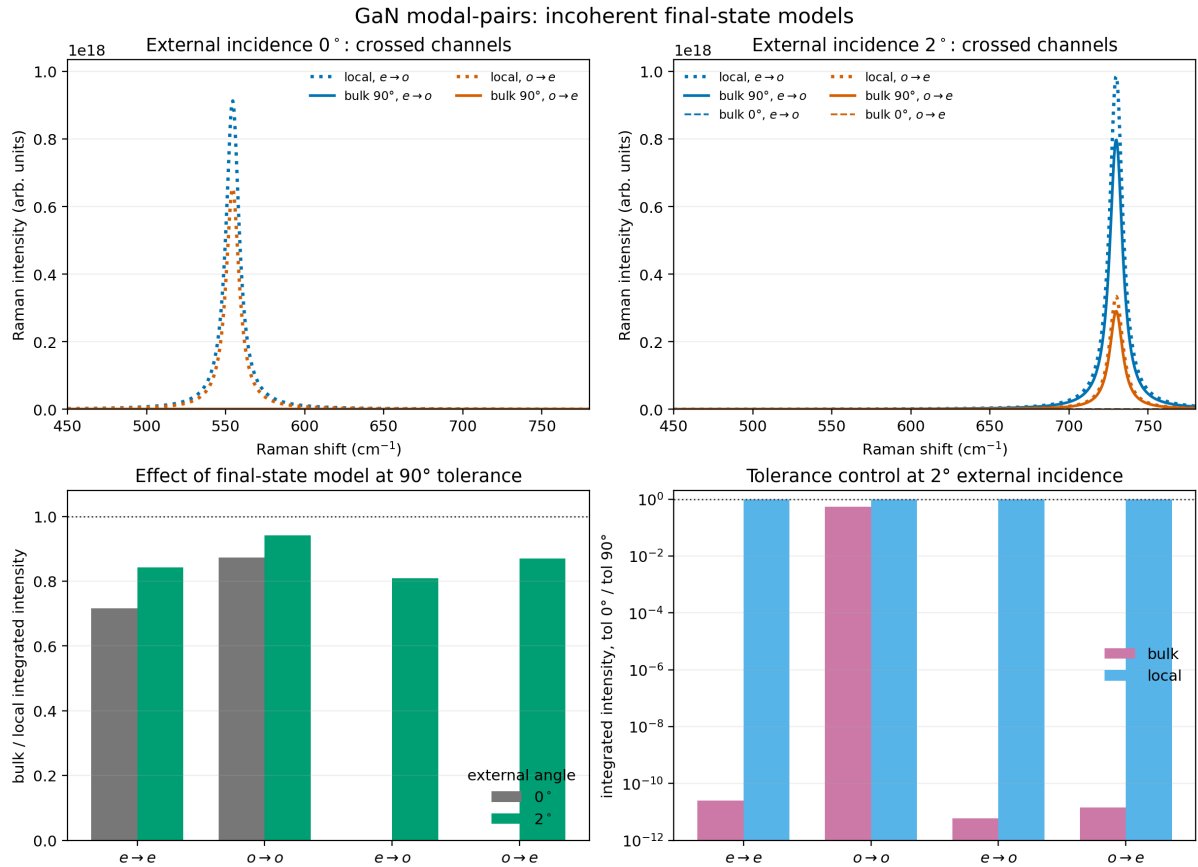


Fig. 14.4: Changing the final-state acceptance changes intensities substantially, even with the same material input and Raman depth integration.

For the ordered crystals and angle-selective experiments considered here, bulk momentum selection is a reasonable starting assumption. Use the local treatment as a contrasting, momentum-relaxed limit unless there is evidence for strong localisation or dephasing. Neither option is a complete finite-coherence the-

ory, nor are their intensities proven upper and lower bounds. A physical crossover depends on a quantity such as $|\Delta\mathbf{q}|L_{\text{coh}}$. The present angular cutoff does not use a measured phonon coherence length or the full magnitude of the momentum mismatch. Strong sensitivity to these options limits the quantitative conclusion that can be drawn from this example.

14.6 Reproducing the note and preparing documentation

The documentation includes copies of the four figures in `Sphinx/_static/figures/gan`, so building the manual does not require running the example calculations. After regenerating the plots, copy the four PNG files used above from the examples' `Figures` directories into that directory to update the documentation.

From a source checkout, activate the PDIELEC Python environment and run:

```
python Examples/ApplicationNotes/GaN/regenerate.py
```

This driver creates the output directories, runs all four PDGui sessions, and regenerates the summaries and PNG/PDF plots. It works from any directory when given the appropriate path to the driver. Existing generated products are overwritten; the CRYSTAL calculation is not rerun. Select a subset with, for example, `regenerate.py powder modal`. Use `--cpus 2` to set the PDGui worker count explicitly.

For a single batch calculation from `GaN/ModalPairs`, the equivalent commands are:

```
mkdir -p Data
pdgui -script gan_modal_pairs.py -spreadsheet "$PWD/Data/gan_modal_pairs.xlsx"
↪ -nosplash -exit
python analyse_results.py
python plot_results.py
```

On a machine without a display, set `QT_QPA_PLATFORM=offscreen`; the driver does this automatically. An interactive session omits `-nosplash -exit` and the spreadsheet argument, then uses the Main Tab's *Save results* button. If Python cannot import PDIELEC, install the checkout into the active environment or put the checkout on `PYTHONPATH`. The driver handles the source-checkout path itself.

After regeneration, run the supplied checks from the repository root:

```
python -m pytest -q Examples/ApplicationNotes/GaN/ModalPairs
```

These check scenario completeness, channel interchange, local-model tolerance invariance and finite-q response trends. Passing them tests the example's internal consistency, not the missing experimental scattering physics.

The Git inputs are the CRYSTAL source data, PDGui sessions, analysis and plotting scripts, checks, and this reStructuredText note. The local `GaN/.gitignore` excludes workbooks (including backups), generated `Data/` and `Figures/` directories, generated `RESULTS.md` files and caches. No experimental figures or digitised spectra are bundled. The illustrations above are original plots produced by these scripts.

Regenerate the figures before building this documentation from a clean checkout. To preview without publishing generated HTML into `docs/`, run from the repository root:

```
sphinx-build -b html Sphinx /tmp/pdielec-gan-docs
```

The usual `make html` in Sphinx also publishes the built HTML to `docs/`. For further controls and theory, see *Crystal Raman Scenarios*, *Theory for Crystal Layer Raman* and *Theory for Powder Raman*.

14.7 References

CHAPTER
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